Projekat prediktovanja dijabetesa

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##Instalacija paketa

```
#install.packages("caret", dependencies = TRUE)
#install.packages("paletteer")
#install.packages("rpart.plot")
```

##Potrebne biblioteke

```
library(ggplot2)
library(paletteer)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(caret)
```

```
## Loading required package: lattice
```

```
library(rpart)
library(randomForest)
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
library(boot)
```

```
##
## Attaching package: 'boot'
```

```
## The following object is masked from 'package:lattice':
##
##     melanoma
```

```
library(rpart.plot)
colorS = c("#88A0DCFF", "#381A61FF", "#7C4B73FF", "#ED968CFF", "#AB3329FF", "#E78429F
F", "#F9D14AFF","#73652DFF")
```

##Funkcije

```
cramer_v <- function(var1, var2) {
  tabela <- table(var1, var2)
  chi2 <- chisq.test(tabela)$statistic
  n <- sum(tabela)
  phi2 <- chi2 / n
  r <- nrow(tabela)
  k <- ncol(tabela)
  v <- sqrt(phi2 / min(r - 1, k - 1))
  return(as.numeric(v))
}
tukey_fun <- function(numericka_var, grupna_var) {
  model <- aov(numericka_var ~ as.factor(grupna_var))
  rezultat <- TukeyHSD(model)
  return(rezultat)
}
chi_sq_test <- function(var1, var2) {
  tabela <- table(var1, var2)
  rezultat <- chisq.test(tabela)
  return(rezultat)
}
anova_test <- function(numericka_var, grupna_var) {
  model <- aov(numericka_var ~ as.factor(grupna_var))
  return(summary(model))
}
pearson_funkcija <- function(x, y) {
  if(length(x) != length(y)) stop("Vektori moraju biti iste dužine")
  mean_x <- mean(x)
  mean_y <- mean(y)
  brojilac <- sum((x - mean_x) * (y - mean_y))
  imenilac <- sqrt(sum((x - mean_x)^2) * sum((y - mean_y)^2))
  return(brojilac / imenilac)
}
izvuci_sve_metrike <- function(pred, stvarni, ime_modela) {

  cm <- confusionMatrix(pred, stvarni, mode = "everything")


  izvestaj <- as.data.frame(cm$byClass[, c("Precision", "Recall", "F1")])
  izvestaj$Model <- ime_modela
  izvestaj$Klasa <- nivoi

  return(izvestaj)
}
```

##Učitavanje podataka

```
radni_direktorijum <- dirname(rstudioapi::getActiveDocumentContext()$path)
setwd(radni_direktorijum)
cat("Radni direktorijum je postavljen na:", getwd(), "\n")
```

```
## Radni direktorijum je postavljen na: C:/Users/BARON/Desktop/unp/пројекат/Seminarski
-rad-Diabetes012
```

```
data <- read.csv("diabetes_dataset.csv")
cat("Skup podataka mozete pogledati u promenljivoj data koja se automatski otvorila u
radnom okruzenju", View(data))
```

```
## Skup podataka mozete pogledati u promenljivoj data koja se automatski otvorila u ra
dnom okruzenju
```

##Osnovne informacije o skupu

```
dimenzije_skupa = dim(data)
cat("Broj featura skupa: ", dimenzije_skupa[2], "\n", "Broj opservacija: ", dimenzije_
skupa[1])
```

```
## Broj featura skupa:  22
##  Broj opservacija:  253680
```

```
str(data)
```

```
## 'data.frame':    253680 obs. of  22 variables:
##  $ Diabetes_012        : num  0 0 0 0 0 0 0 0 2 0 ...
##  $ HighBP              : num  1 0 1 1 1 1 1 1 1 0 ...
##  $ HighChol            : num  1 0 1 0 1 1 0 1 1 0 ...
##  $ CholCheck           : num  1 0 1 1 1 1 1 1 1 1 ...
##  $ BMI                 : num  40 25 28 27 24 25 30 25 30 24 ...
##  $ Smoker              : num  1 1 0 0 0 1 1 1 1 0 ...
##  $ Stroke              : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ HeartDiseaseorAttack: num  0 0 0 0 0 0 0 0 1 0 ...
##  $ PhysActivity        : num  0 1 0 1 1 1 0 1 0 0 ...
##  $ Fruits              : num  0 0 1 1 1 1 0 0 1 0 ...
##  $ Veggies             : num  1 0 0 1 1 1 0 1 1 1 ...
##  $ HvyAlcoholConsump   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ AnyHealthcare       : num  1 0 1 1 1 1 1 1 1 1 ...
##  $ NoDocbcCost         : num  0 1 1 0 0 0 0 0 0 0 ...
##  $ GenHlth             : num  5 3 5 2 2 2 3 3 5 2 ...
##  $ MentHlth            : num  18 0 30 0 3 0 0 0 30 0 ...
##  $ PhysHlth            : num  15 0 30 0 0 2 14 0 30 0 ...
##  $ DiffWalk            : num  1 0 1 0 0 0 0 1 1 0 ...
##  $ Sex                 : num  0 0 0 0 0 1 0 0 0 1 ...
##  $ Age                 : num  9 7 9 11 11 10 9 11 9 8 ...
##  $ Education           : num  4 6 4 3 5 6 6 4 5 4 ...
##  $ Income              : num  3 1 8 6 4 8 7 4 1 3 ...
```

```
summary(data)
```

```
##   Diabetes_012        HighBP         HighChol        CholCheck    
##  Min.   :0.0000   Min.   :0.000   Min.   :0.0000   Min.   :0.0000  
##  1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:1.0000  
##  Median :0.0000   Median :0.000   Median :0.0000   Median :1.0000  
##  Mean   :0.2969   Mean   :0.429   Mean   :0.4241   Mean   :0.9627  
##  3rd Qu.:0.0000   3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:1.0000  
##  Max.   :2.0000   Max.   :1.000   Max.   :1.0000   Max.   :1.0000  
##       BMI            Smoker           Stroke        HeartDiseaseorAttack
##  Min.   :12.00   Min.   :0.0000   Min.   :0.00000   Min.   :0.00000     
##  1st Qu.:24.00   1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:0.00000     
##  Median :27.00   Median :0.0000   Median :0.00000   Median :0.00000     
##  Mean   :28.38   Mean   :0.4432   Mean   :0.04057   Mean   :0.09419     
##  3rd Qu.:31.00   3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:0.00000     
##  Max.   :98.00   Max.   :1.0000   Max.   :1.00000   Max.   :1.00000     
##   PhysActivity        Fruits          Veggies       HvyAlcoholConsump
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   
##  1st Qu.:1.0000   1st Qu.:0.0000   1st Qu.:1.0000   1st Qu.:0.0000   
##  Median :1.0000   Median :1.0000   Median :1.0000   Median :0.0000   
##  Mean   :0.7565   Mean   :0.6343   Mean   :0.8114   Mean   :0.0562   
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.0000   
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   
##  AnyHealthcare     NoDocbcCost         GenHlth         MentHlth     
##  Min.   :0.0000   Min.   :0.00000   Min.   :1.000   Min.   : 0.000  
##  1st Qu.:1.0000   1st Qu.:0.00000   1st Qu.:2.000   1st Qu.: 0.000  
##  Median :1.0000   Median :0.00000   Median :2.000   Median : 0.000  
##  Mean   :0.9511   Mean   :0.08418   Mean   :2.511   Mean   : 3.185  
##  3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:3.000   3rd Qu.: 2.000  
##  Max.   :1.0000   Max.   :1.00000   Max.   :5.000   Max.   :30.000  
##     PhysHlth         DiffWalk           Sex              Age        
##  Min.   : 0.000   Min.   :0.0000   Min.   :0.0000   Min.   : 1.000  
##  1st Qu.: 0.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 6.000  
##  Median : 0.000   Median :0.0000   Median :0.0000   Median : 8.000  
##  Mean   : 4.242   Mean   :0.1682   Mean   :0.4403   Mean   : 8.032  
##  3rd Qu.: 3.000   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:10.000  
##  Max.   :30.000   Max.   :1.0000   Max.   :1.0000   Max.   :13.000  
##    Education        Income     
##  Min.   :1.00   Min.   :1.000  
##  1st Qu.:4.00   1st Qu.:5.000  
##  Median :5.00   Median :7.000  
##  Mean   :5.05   Mean   :6.054  
##  3rd Qu.:6.00   3rd Qu.:8.000  
##  Max.   :6.00   Max.   :8.000  
```

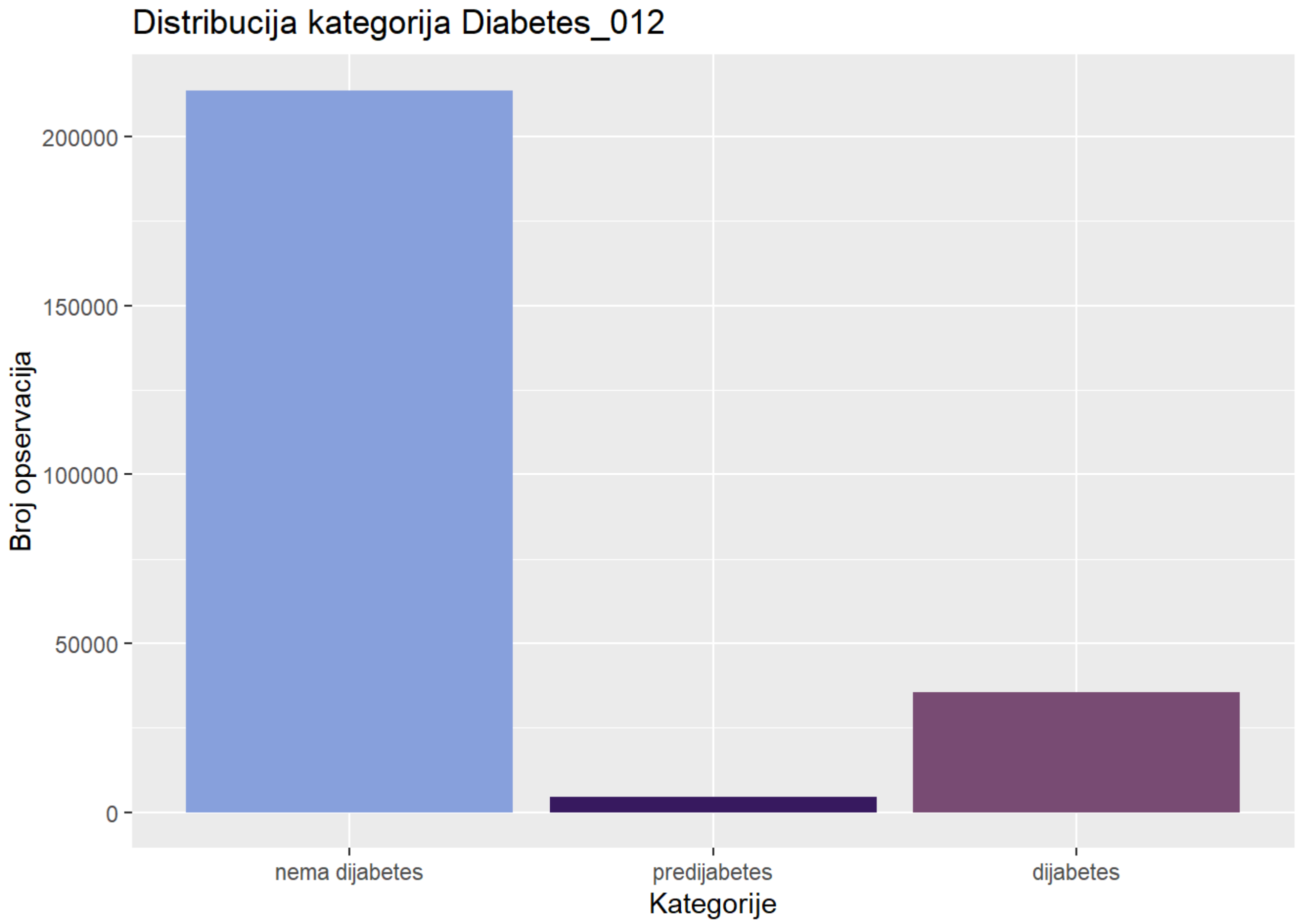
##Pripreme za EDA

```
daNe_kategorije = c("HighBP", "HighChol", "CholCheck", "Smoker", "Stroke",
                 "HeartDiseaseorAttack", "PhysActivity", "Fruits", "Veggies",
                 "HvyAlcoholConsump", "AnyHealthcare", "NoDocbcCost", "DiffWalk")
ordinalne_kategorije = c("GenHlth", "Education", "Income")
data[daNe_kategorije] = lapply(data[daNe_kategorije], factor, levels = c(0,1), labels
= c("Ne","Da"))
data$Sex = factor(data$Sex, levels = c(0,1), labels = c("zensko","musko"))

nivoi_Diabetes_012 = sort(unique(data$Diabetes_012))
data$Diabetes_012 = factor(data$Diabetes_012,
                            levels = nivoi_Diabetes_012,
                            labels = c("nema dijabetes", "predijabetes", "dijabetes"))

nivoi_GenHlth = sort(unique(data$GenHlth), decreasing = TRUE)
data$GenHlth = factor(
  data$GenHlth,
  levels = nivoi_GenHlth,
  labels = c("Odlično", "Vrlo dobro", "Dobro", "Zadovoljavajuće", "Loše"),
  ordered = TRUE
)

nivoi_Income = sort(unique(data$Income))
data$Income = factor(data$Income, levels = nivoi_Income,
                            ordered = TRUE,
                            labels = c("<10.000$",
                                       "10.000-14.999$",
                                       "15.000-19.999$",
                                       "20.000-24.999$",
                                       "25.000-34.999$",
                                       "35.000-49.999$",
                                       "50.000-74.999$",
                                       ">=75.000$"))

nivoi_Education = sort(unique(data$Education))
data$Education<- factor(data$Education, levels = nivoi_Education,
                     ordered = TRUE,
                     labels = c("bez skole/isao u vrtic",
                                "osnovna skola",
                                "3-god srednja",
                                "4-god srednja",
                                "3-god fakultet",
                                "4-god fakultet"))
str(data)
```

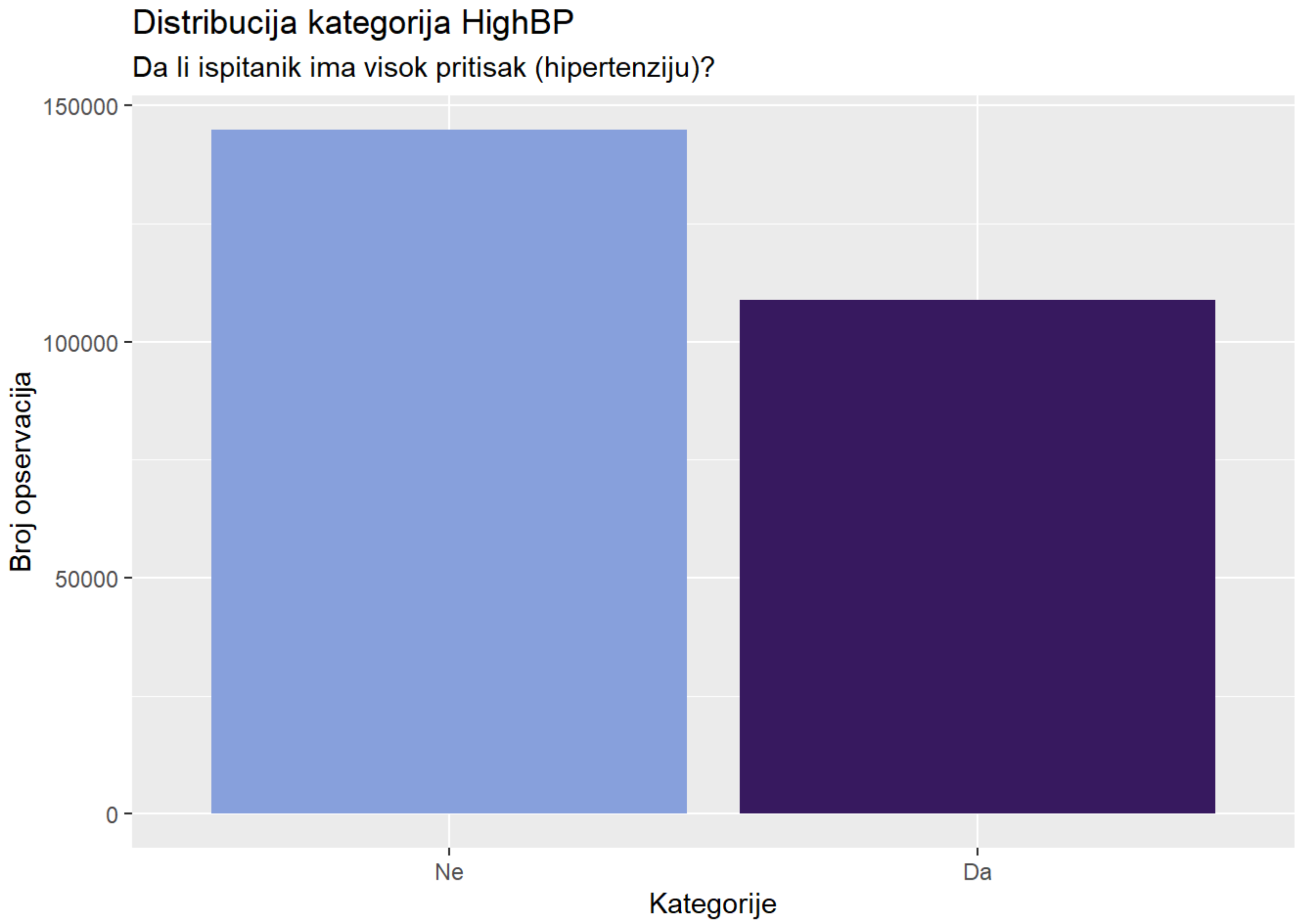
```
## 'data.frame':    253680 obs. of  22 variables:
##  $ Diabetes_012        : Factor w/ 3 levels "nema dijabetes",..: 1 1 1 1 1 1 1 1 3
1 ...
##  $ HighBP              : Factor w/ 2 levels "Ne","Da": 2 1 2 2 2 2 2 2 2 1 ...
##  $ HighChol            : Factor w/ 2 levels "Ne","Da": 2 1 2 1 2 2 1 2 2 1 ...
##  $ CholCheck           : Factor w/ 2 levels "Ne","Da": 2 1 2 2 2 2 2 2 2 2 ...
##  $ BMI                 : num  40 25 28 27 24 25 30 25 30 24 ...
##  $ Smoker              : Factor w/ 2 levels "Ne","Da": 2 2 1 1 1 2 2 2 2 1 ...
##  $ Stroke              : Factor w/ 2 levels "Ne","Da": 1 1 1 1 1 1 1 1 1 1 ...
##  $ HeartDiseaseorAttack: Factor w/ 2 levels "Ne","Da": 1 1 1 1 1 1 1 1 2 1 ...
##  $ PhysActivity        : Factor w/ 2 levels "Ne","Da": 1 2 1 2 2 2 1 2 1 1 ...
##  $ Fruits              : Factor w/ 2 levels "Ne","Da": 1 1 2 2 2 2 1 1 2 1 ...
##  $ Veggies             : Factor w/ 2 levels "Ne","Da": 2 1 1 2 2 2 1 2 2 2 ...
##  $ HvyAlcoholConsump   : Factor w/ 2 levels "Ne","Da": 1 1 1 1 1 1 1 1 1 1 ...
##  $ AnyHealthcare       : Factor w/ 2 levels "Ne","Da": 2 1 2 2 2 2 2 2 2 2 ...
##  $ NoDocbcCost         : Factor w/ 2 levels "Ne","Da": 1 2 2 1 1 1 1 1 1 1 ...
##  $ GenHlth             : Ord.factor w/ 5 levels "Odlično"<"Vrlo dobro"<..: 1 3 1 4
4 4 3 3 1 4 ...
##  $ MentHlth            : num  18 0 30 0 3 0 0 0 30 0 ...
##  $ PhysHlth            : num  15 0 30 0 0 2 14 0 30 0 ...
##  $ DiffWalk            : Factor w/ 2 levels "Ne","Da": 2 1 2 1 1 1 1 2 2 1 ...
##  $ Sex                 : Factor w/ 2 levels "zensko","musko": 1 1 1 1 1 2 1 1 1 2
...
##  $ Age                 : num  9 7 9 11 11 10 9 11 9 8 ...
##  $ Education           : Ord.factor w/ 6 levels "bez skole/isao u vrtic"<..: 4 6 4
3 5 6 6 4 5 4 ...
##  $ Income              : Ord.factor w/ 8 levels "<10.000$"<"10.000-14.999$"<..: 3 1
8 6 4 8 7 4 1 3 ...
```

##Univarijantna

```
ggplot(data, aes(x = Diabetes_012, fill = Diabetes_012)) +
      geom_bar() +
      labs(title = "Distribucija kategorija Diabetes_012", x = "Kategorije", y = "Broj
opservacija") +
      scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



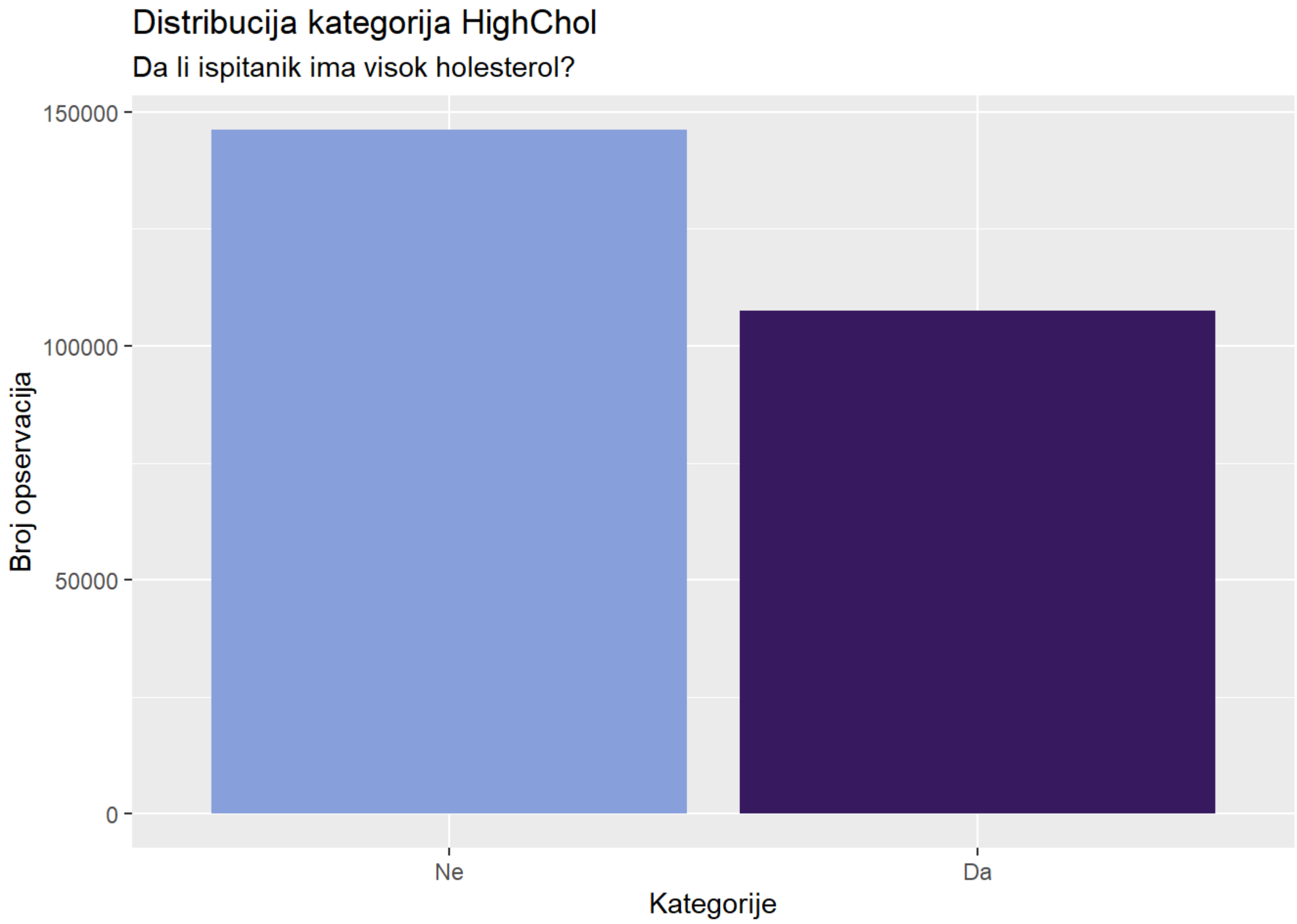

```
ggplot(data, aes(x = HighBP , fill = HighBP )) +
      geom_bar() +
      labs(title = "Distribucija kategorija HighBP", subtitle = "Da li ispitanik ima v
isok pritisak (hipertenziju)?", x = "Kategorije", y = "Broj opservacija") +
      scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_highBP = var(as.numeric(data$HighBP))
cat("Varijansa HighBP: ", varijansa_highBP, "od maksimmalne moguce 0.25\n")
```

```
## Varijansa HighBP:  0.2449601 od maksimmalne moguce 0.25
```

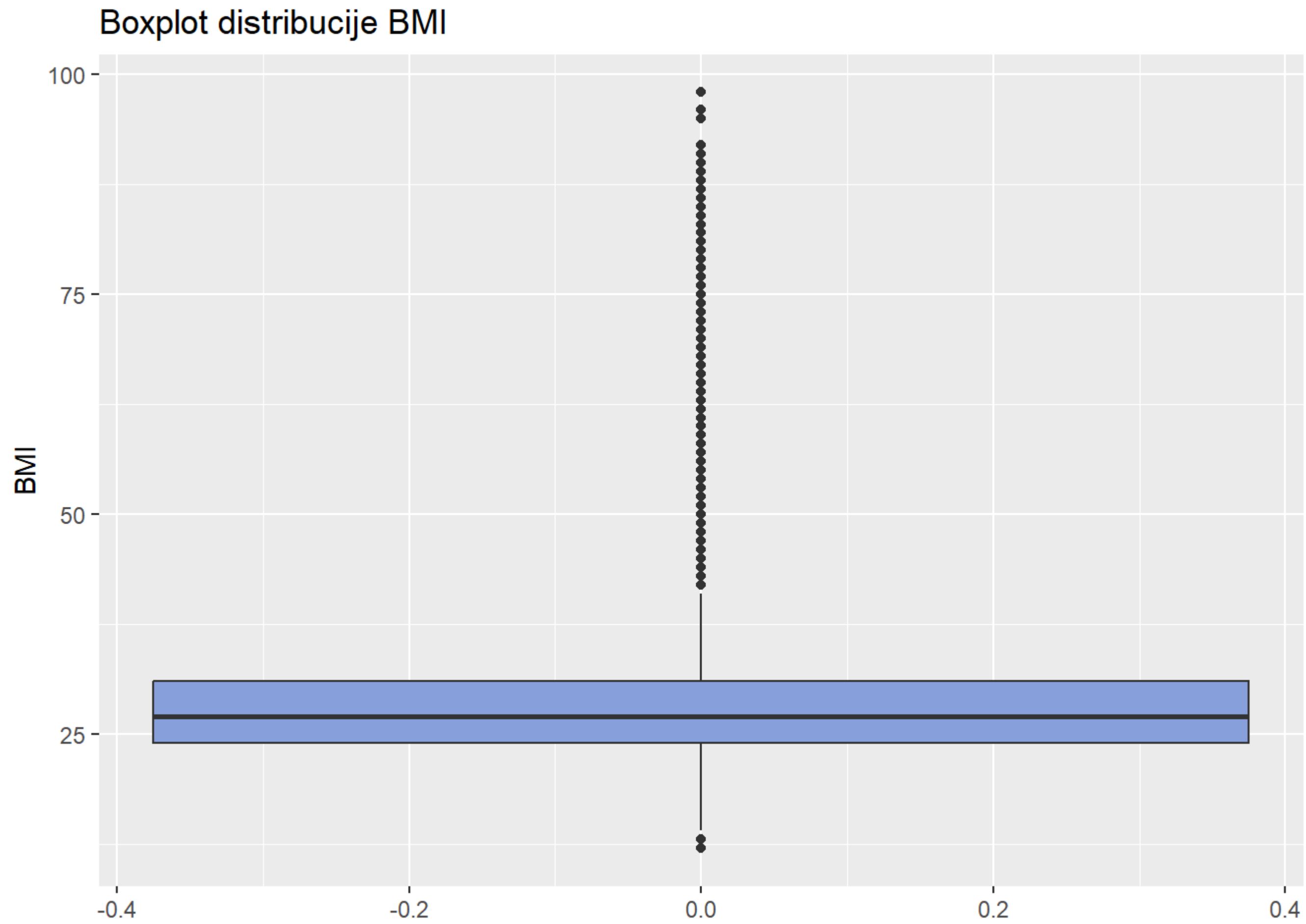
```
ggplot(data, aes(x = HighChol , fill = HighChol )) +
      geom_bar() +
      labs(title = "Distribucija kategorija HighChol", subtitle = "Da li ispitanik ima
visok holesterol?", x = "Kategorije", y = "Broj opservacija") +
      scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



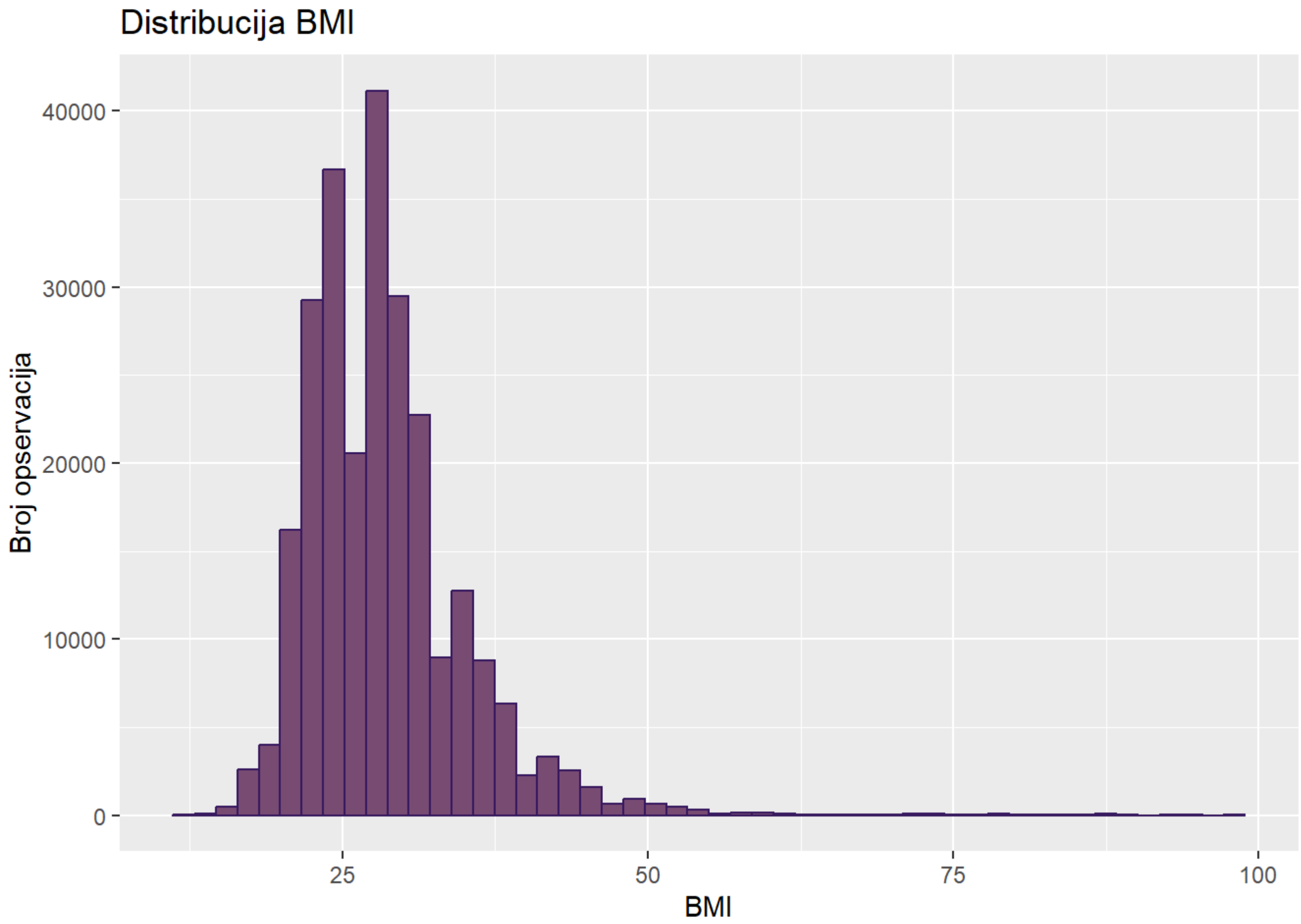

```
varijansa_highChol = var(as.numeric(data$HighChol))
cat("Varijansa HighChol: ", varijansa_highChol, "od maksimmalne moguce 0.25\n")
```

```
## Varijansa HighChol:  0.2442433 od maksimmalne moguce 0.25
```

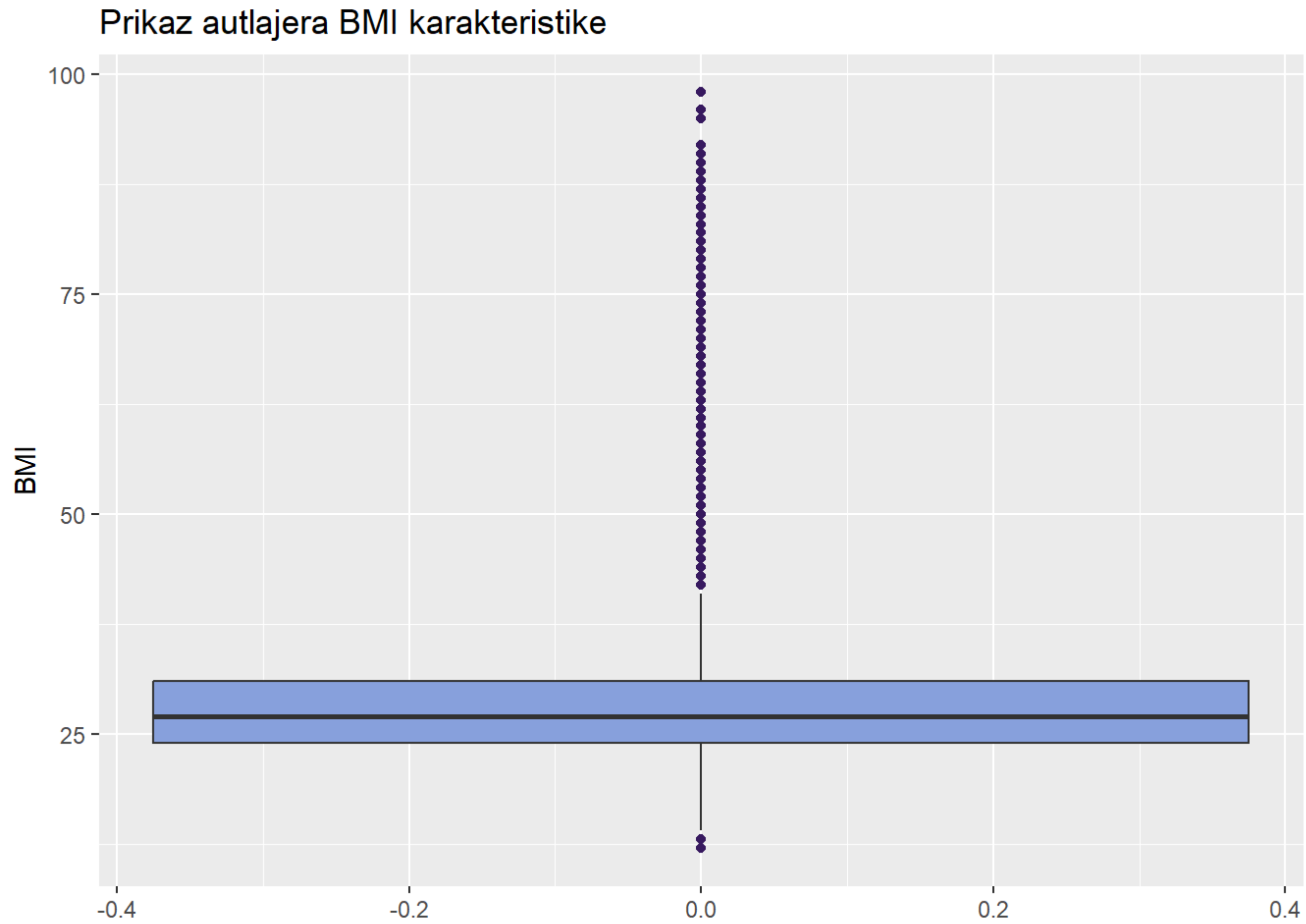
```
ggplot(data, aes(y = BMI)) +
  geom_boxplot(fill= "#88A0DCFF") +
  labs(title = "Boxplot distribucije BMI", y = "BMI") +
  scale_fill_paletteer_d("MetBrewer::Archambault")
```




```
ggplot(data, aes(x = BMI)) +
      geom_histogram(bins = 50, fill = "#7C4B73FF", color="#381A61FF") +
      labs(title = "Distribucija BMI", x = "BMI", y = "Broj opservacija")
```




```
ggplot(data, aes(y = BMI)) +
  geom_boxplot(
    fill = "#88A0DCFF",
    outlier.color = "#381A61FF") +
  labs(
    title = "Prikaz autlajera BMI karakteristike",
    y = "BMI"
  )
```



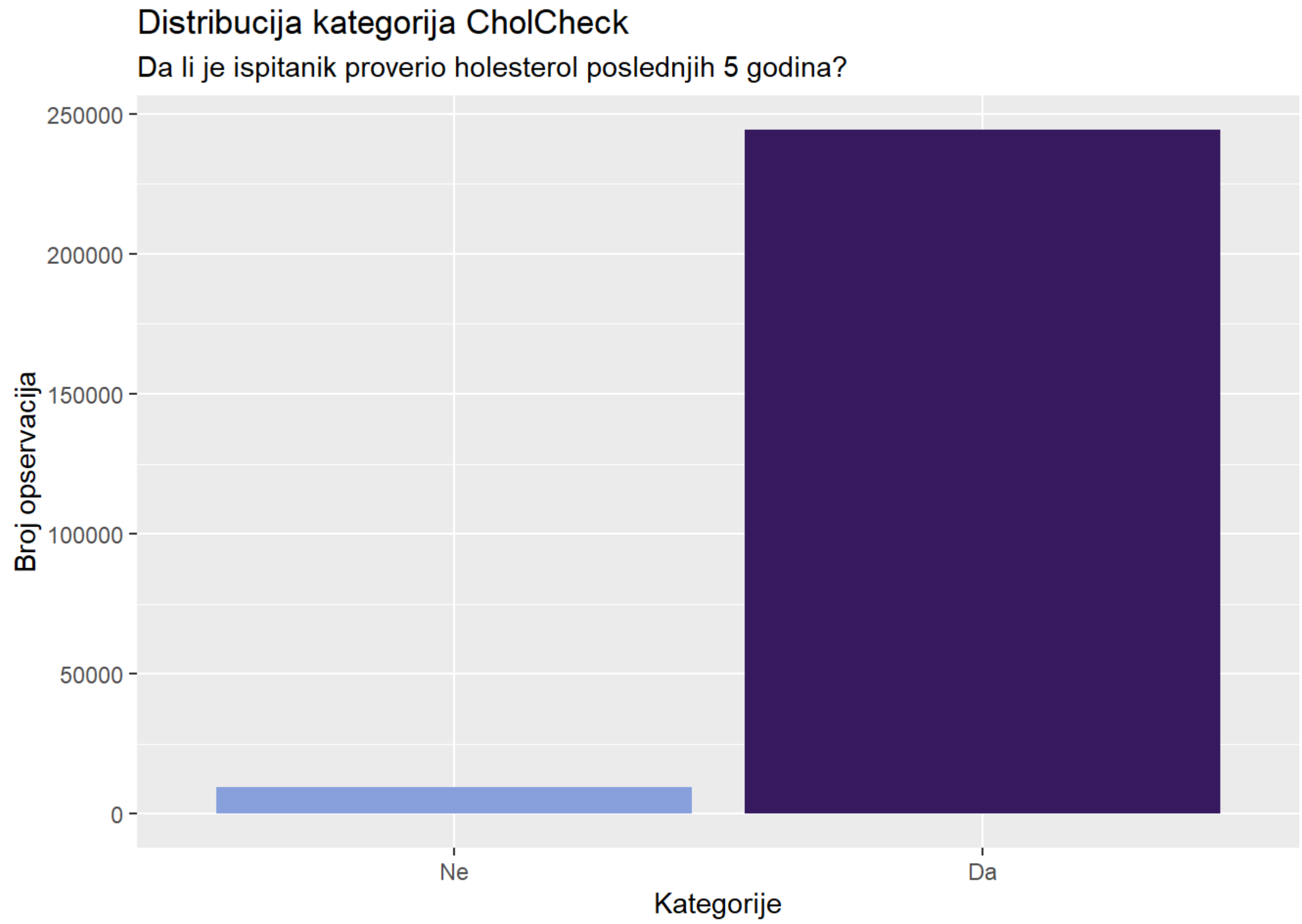

```
donja_granica = quantile(data$BMI, 0.25) - 1.5 * IQR(data$BMI)
gornja_granica = quantile(data$BMI, 0.75) + 1.5 * IQR(data$BMI)

autlajeri_statisticki = data %>% filter(BMI < donja_granica | BMI > gornja_granica)
cat("Po IQR metodu statističkih autlajera ima " , nrow(autlajeri_statisticki), ".Što j
e", nrow(autlajeri_statisticki)/nrow(data)*100,"% ukupnog broja opservacija\n")
```

```
## Po IQR metodu statističkih autlajera ima  9847 .Što je 3.881662 % ukupnog broja ops
ervacija
```

```
autlajeri_domenski = data %>% filter(BMI < 12 | BMI > 70)
cat("Po domenskom znanju autlajera ima " , nrow(autlajeri_domenski), ".Što je", nrow(a
utlajeri_domenski)/nrow(data)*100,"% ukupnog broja opservacija\n")
```

```
## Po domenskom znanju autlajera ima  584 .Što je 0.2302113 % ukupnog broja opservacij
a
```

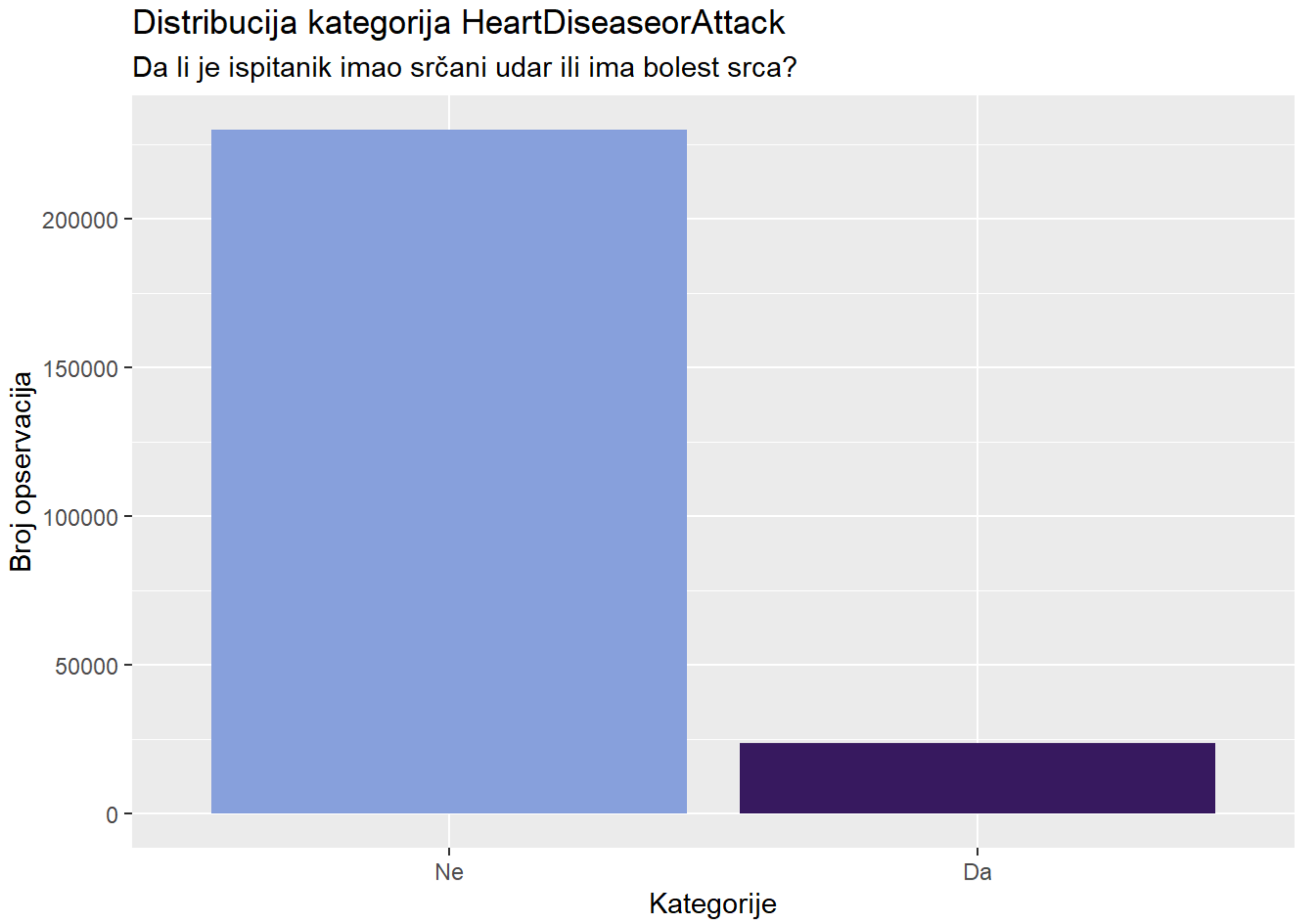
```
ggplot(data, aes(x = CholCheck , fill = CholCheck )) +
  geom_bar() +
  labs(title = "Distribucija kategorija CholCheck", subtitle = "Da li je ispitanik pro
verio holesterol poslednjih 5 godina?", x = "Kategorije", y = "Broj opservacija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_CholCheck = var(as.numeric(data$CholCheck))
cat("Varijansa CholCheck: ", varijansa_CholCheck, "od maksimmalne moguce 0.25\n")
```

```
## Varijansa CholCheck:  0.03593707 od maksimmalne moguce 0.25
```

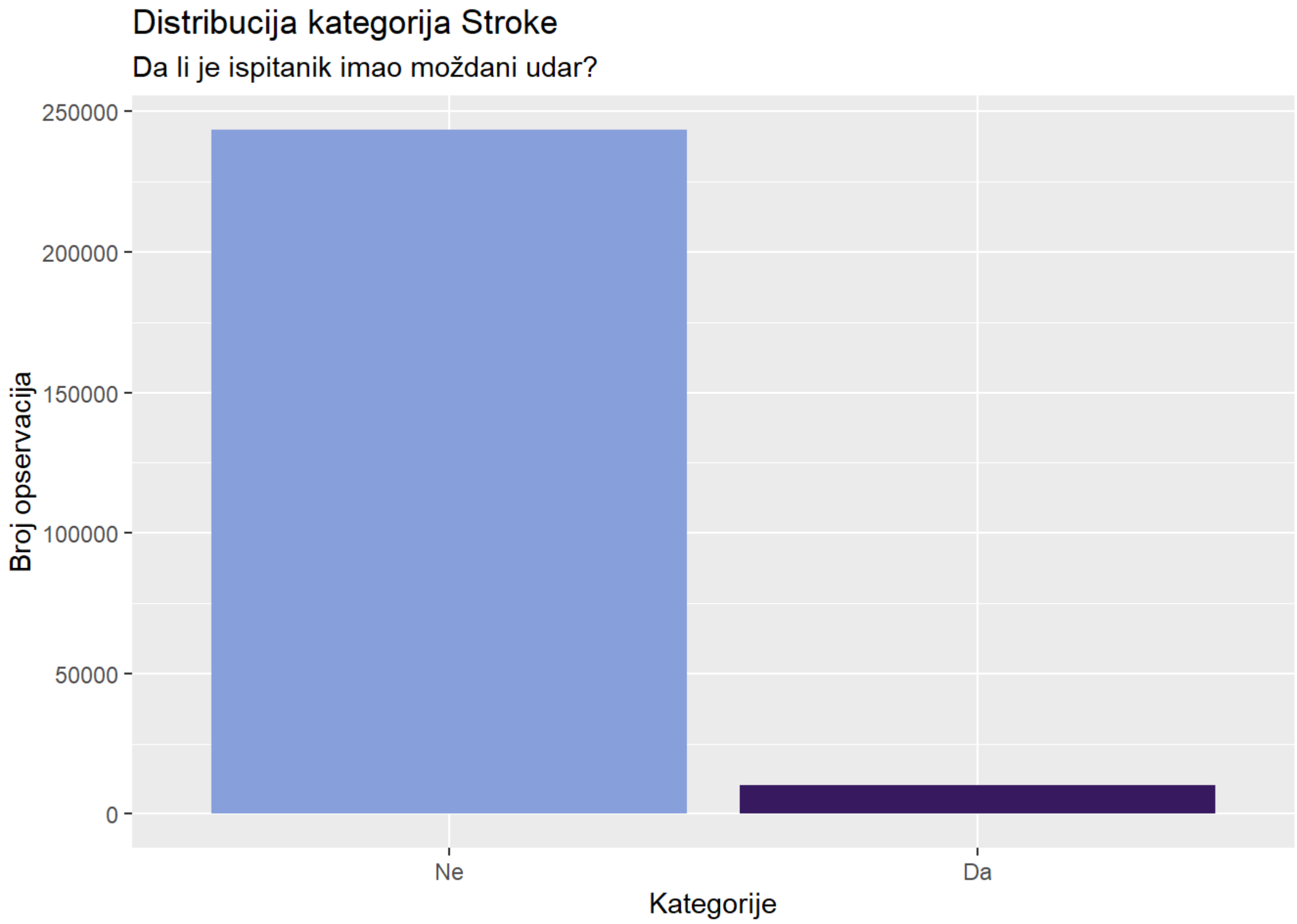
```
ggplot(data, aes(x = HeartDiseaseorAttack  , fill = HeartDiseaseorAttack  )) +
  geom_bar() +
  labs(title = "Distribucija kategorija HeartDiseaseorAttack ", subtitle = "Da li je i
spitanik imao srčani udar ili ima bolest srca?", x = "Kategorije", y = "Broj opservaci
ja") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_HeartDiseaseorAttack  = var(as.numeric(data$HeartDiseaseorAttack ))
cat("Varijansa HeartDiseaseorAttack : ", varijansa_HeartDiseaseorAttack , "od maksimma
lne moguce 0.25\n")
```

```
## Varijansa HeartDiseaseorAttack :  0.085315 od maksimmalne moguce 0.25
```

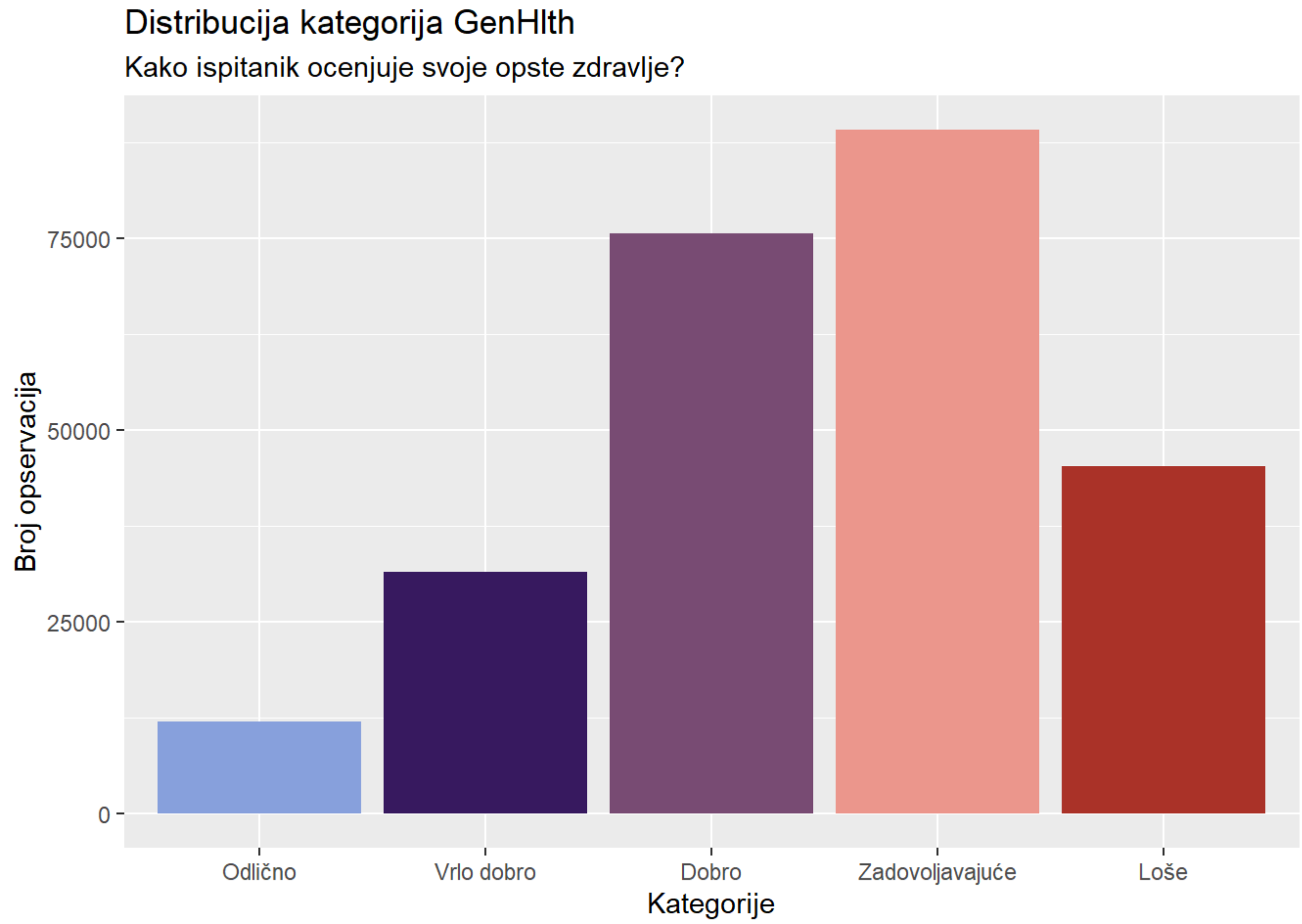
```
ggplot(data, aes(x = Stroke  , fill = Stroke  )) +
  geom_bar() +
  labs(title = "Distribucija kategorija Stroke ", subtitle = "Da li je ispitanik imao
moždani udar?", x = "Kategorije", y = "Broj opservacija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_Stroke  = var(as.numeric(data$Stroke ))
cat("Varijansa Stroke : ", varijansa_Stroke , "od maksimmalne moguce 0.25\n")
```

```
## Varijansa Stroke :  0.03892496 od maksimmalne moguce 0.25
```

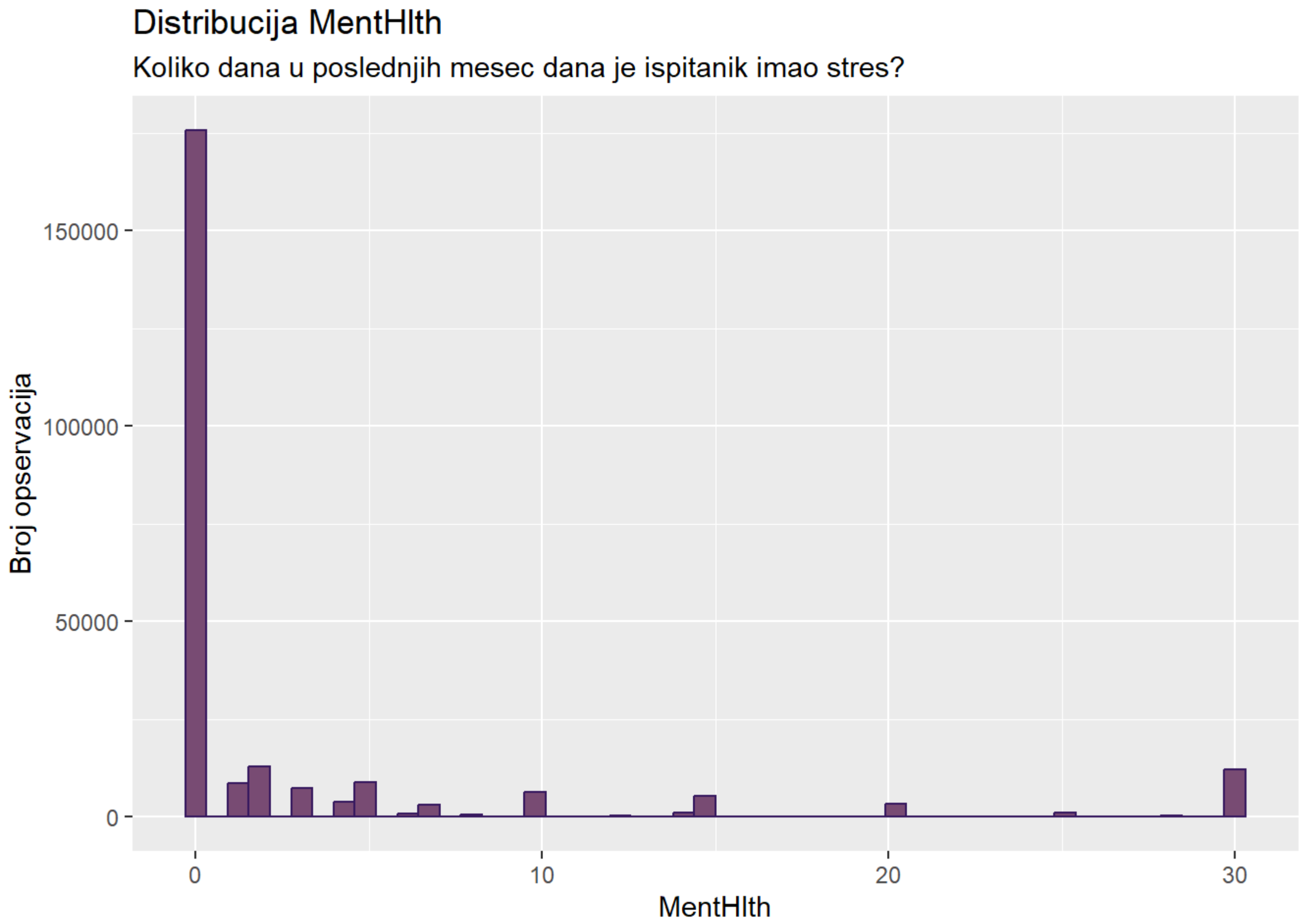
```
ggplot(data, aes(x = GenHlth , fill = GenHlth )) +
  geom_bar() +
  labs(title = "Distribucija kategorija GenHlth", subtitle = "Kako ispitanik ocenjuje
svoje opste zdravlje?", x = "Kategorije", y = "Broj opservacija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



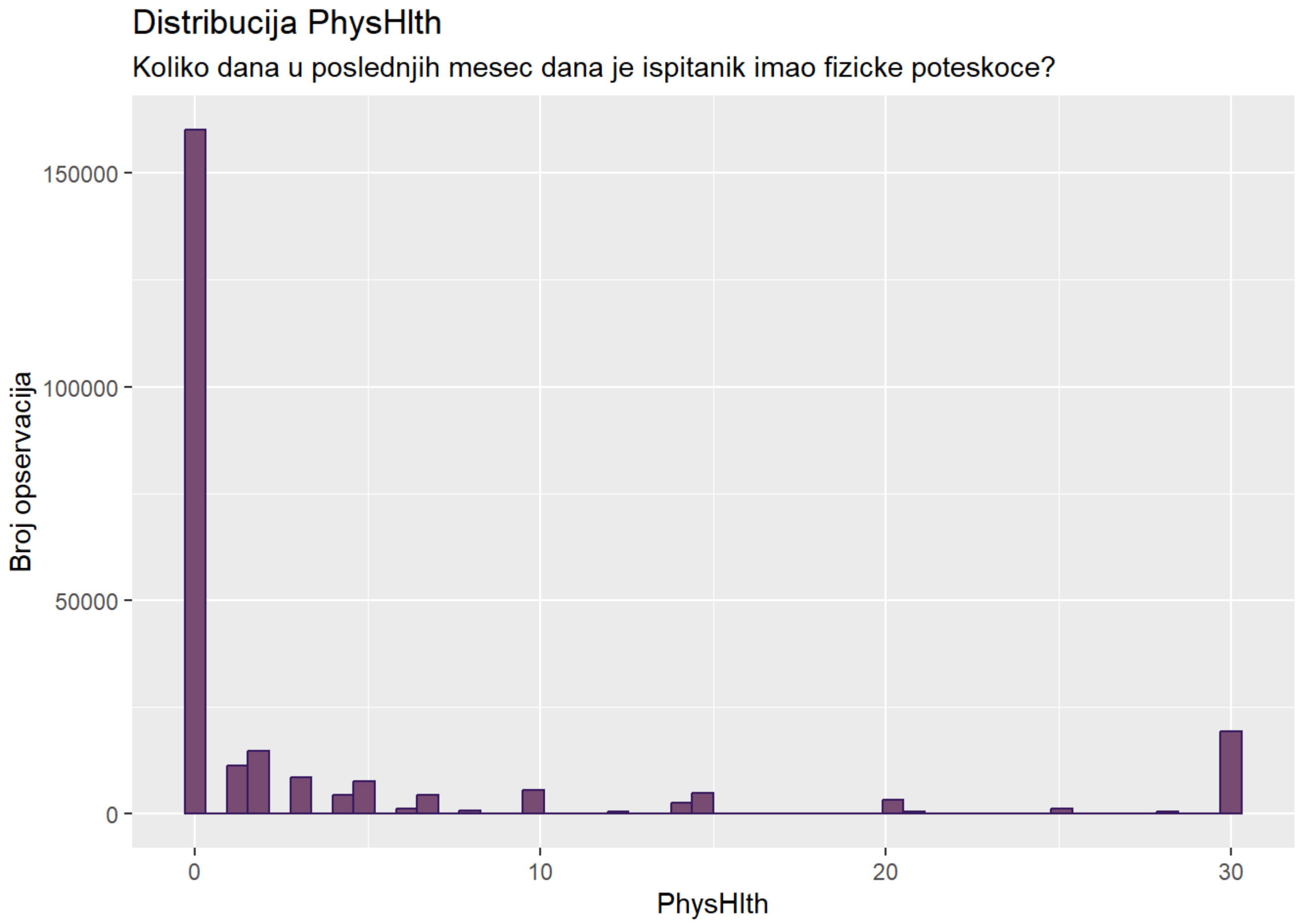

```
data %>% group_by(GenHlth) %>%
summarise(
          Broj_opservacija = n(),
          Procenat_opservacija = round(n() / nrow(data) * 100, 2)
)
```

```
## # A tibble: 5 × 3
##   GenHlth         Broj_opservacija Procenat_opservacija
##   <ord>                      <int>                <dbl>
## 1 Odlično                    12081                 4.76
## 2 Vrlo dobro                 31570                12.4
## 3 Dobro                      75646                29.8
## 4 Zadovoljavajuće            89084                35.1
## 5 Loše                       45299                17.9
```

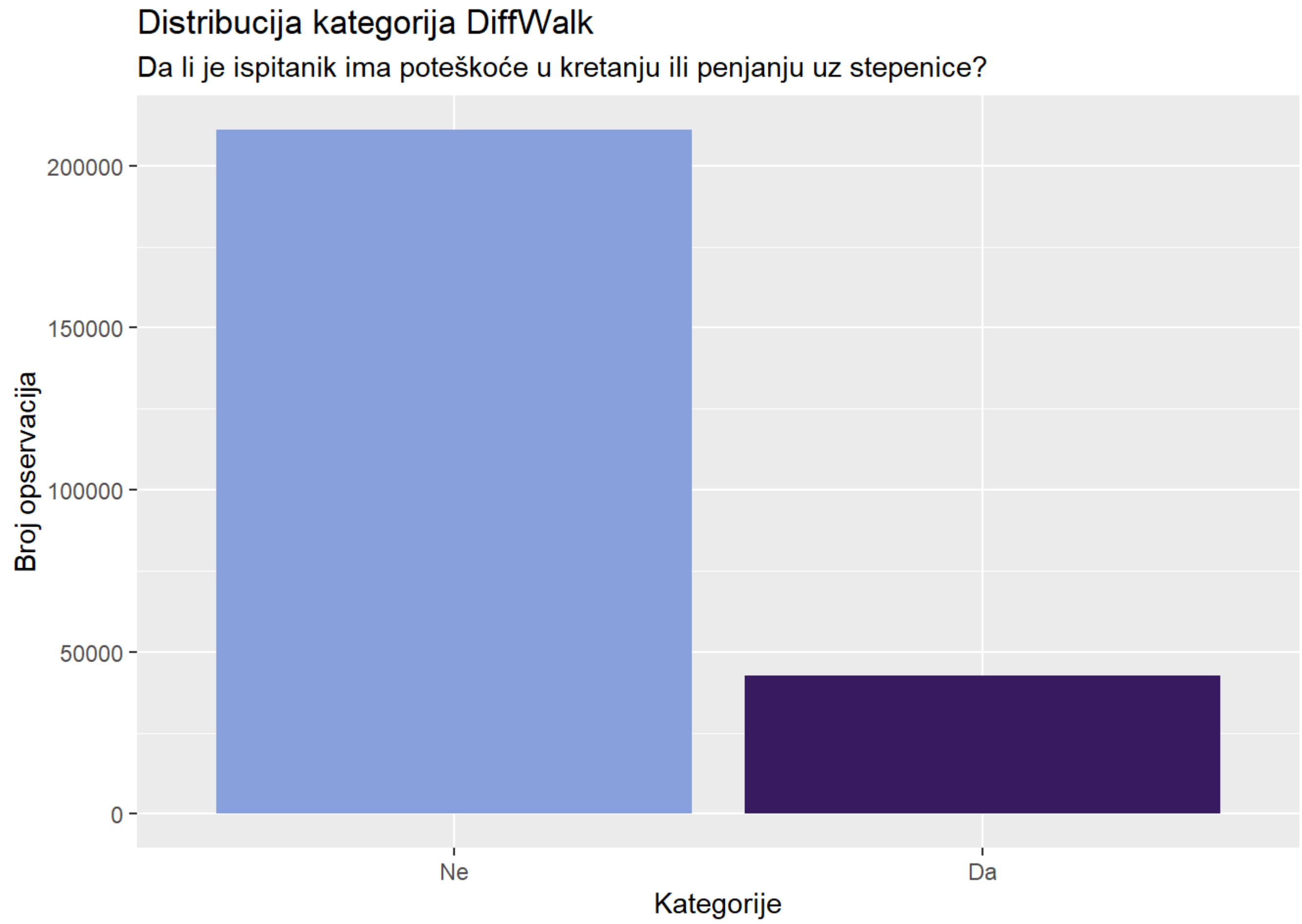
```
ggplot(data, aes(x = MentHlth)) +
      geom_histogram(bins = 50, fill = "#7C4B73FF", color="#381A61FF") +
      labs(title = "Distribucija MentHlth", x = "MentHlth", y = "Broj opservacija",
      subtitle = "Koliko dana u poslednjih mesec dana je ispitanik imao stres?")
```




```
ggplot(data, aes(x = PhysHlth)) +
      geom_histogram(bins = 50, fill = "#7C4B73FF", color="#381A61FF") +
      labs(title = "Distribucija PhysHlth", x = "PhysHlth", y = "Broj opservacija",
      subtitle = "Koliko dana u poslednjih mesec dana je ispitanik imao fizicke potesk
oce?")
```



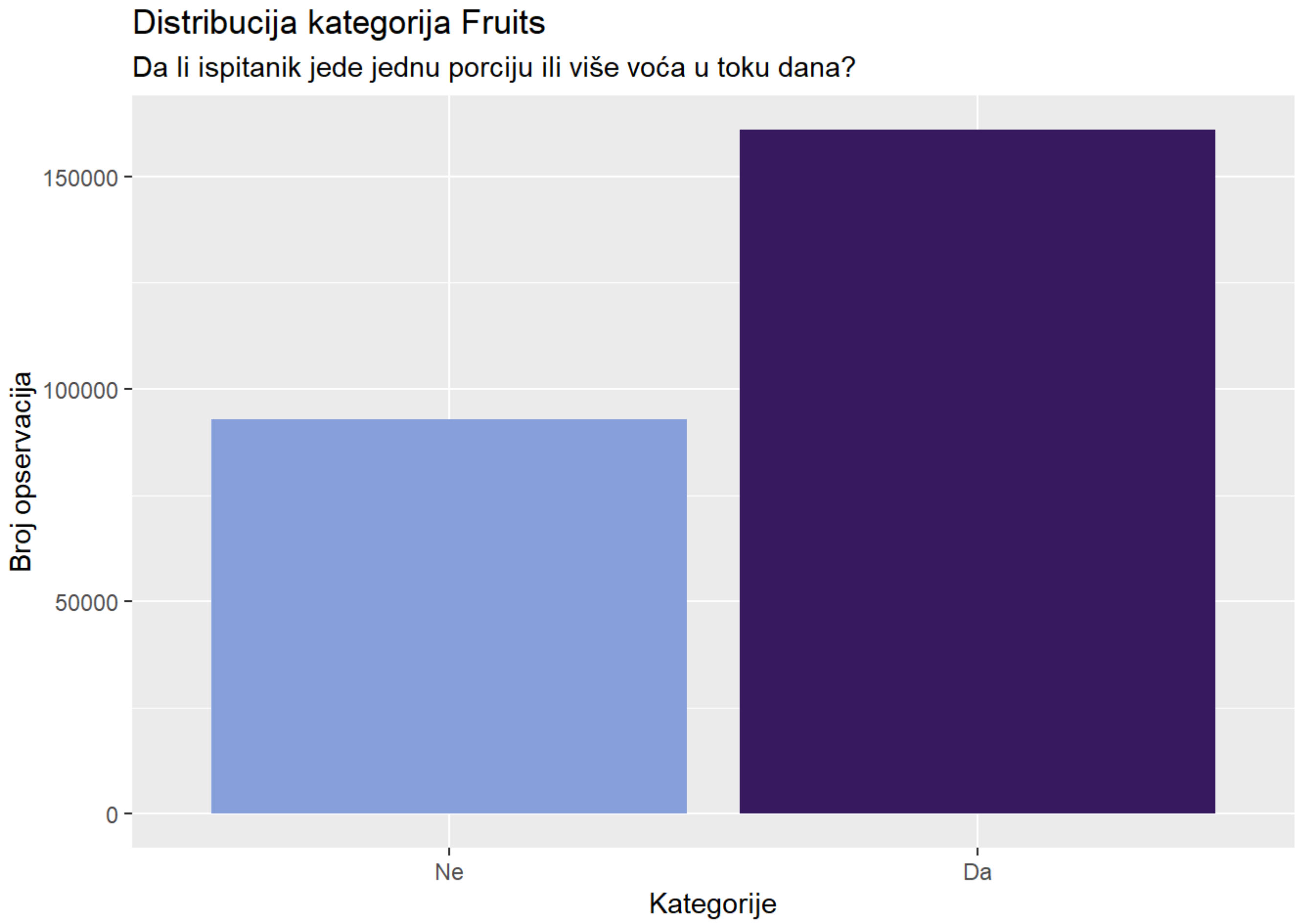

```
ggplot(data, aes(x = DiffWalk  , fill = DiffWalk  )) +
  geom_bar() +
  labs(title = "Distribucija kategorija DiffWalk ", subtitle = "Da li je ispitanik ima
poteškoće u kretanju ili penjanju uz stepenice?", x = "Kategorije", y = "Broj opservac
ija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_DiffWalk  = var(as.numeric(data$DiffWalk ))
cat("Varijansa DiffWalk : ", varijansa_DiffWalk , "od maksimmalne moguce 0.25\n")
```

```
## Varijansa DiffWalk :  0.1399251 od maksimmalne moguce 0.25
```

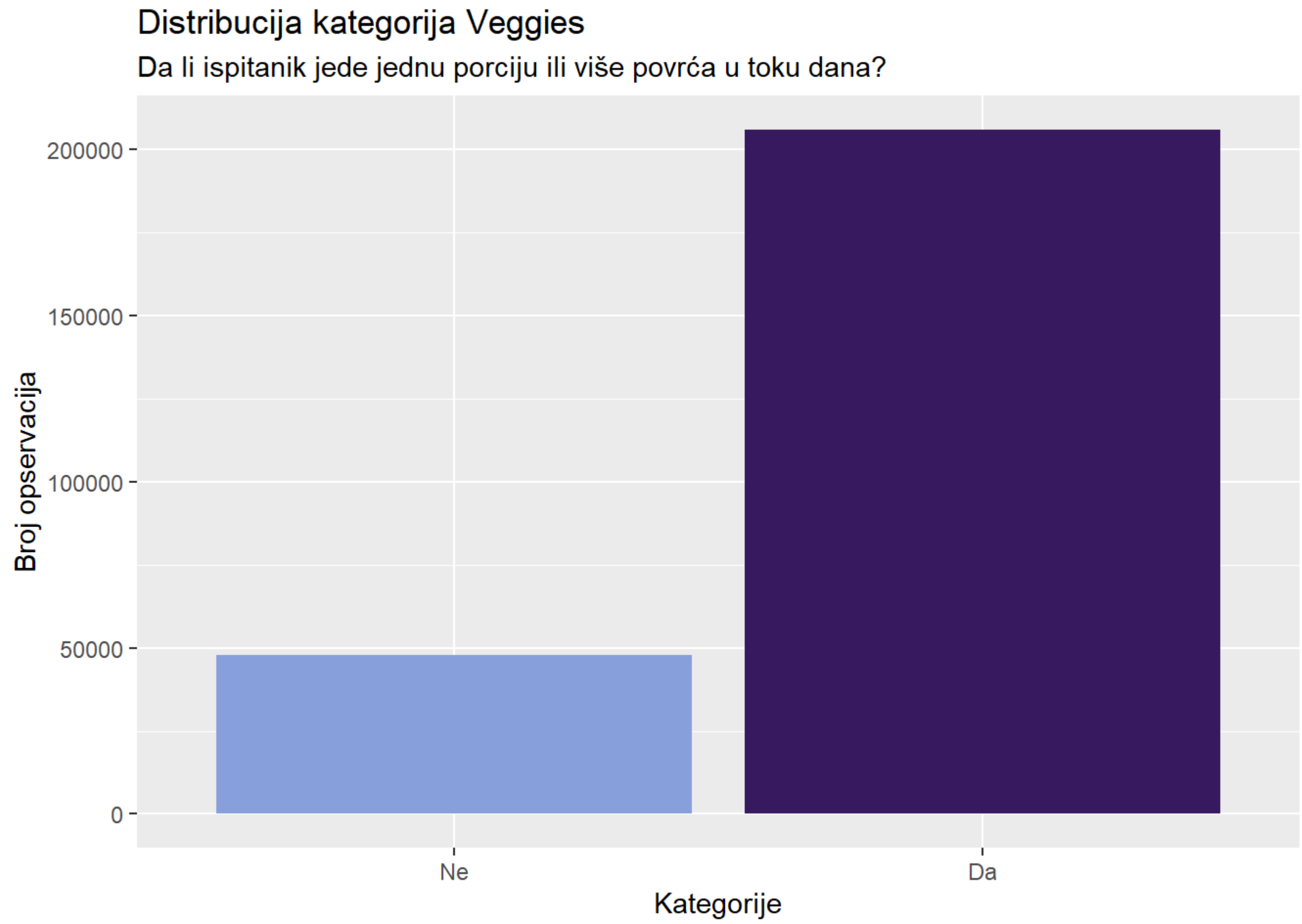
```
ggplot(data, aes(x = Fruits , fill = Fruits )) +
      geom_bar() +
      labs(title = "Distribucija kategorija Fruits", subtitle = "Da li ispitanik jede
jednu porciju ili više voća u toku dana?", x = "Kategorije", y = "Broj opservacija") +
      scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_fruits = var(as.numeric(data$Fruits))
cat("Varijansa Fruits: ", varijansa_fruits, "od maksimmalne moguce 0.25\n")
```

```
## Varijansa Fruits:  0.2319763 od maksimmalne moguce 0.25
```

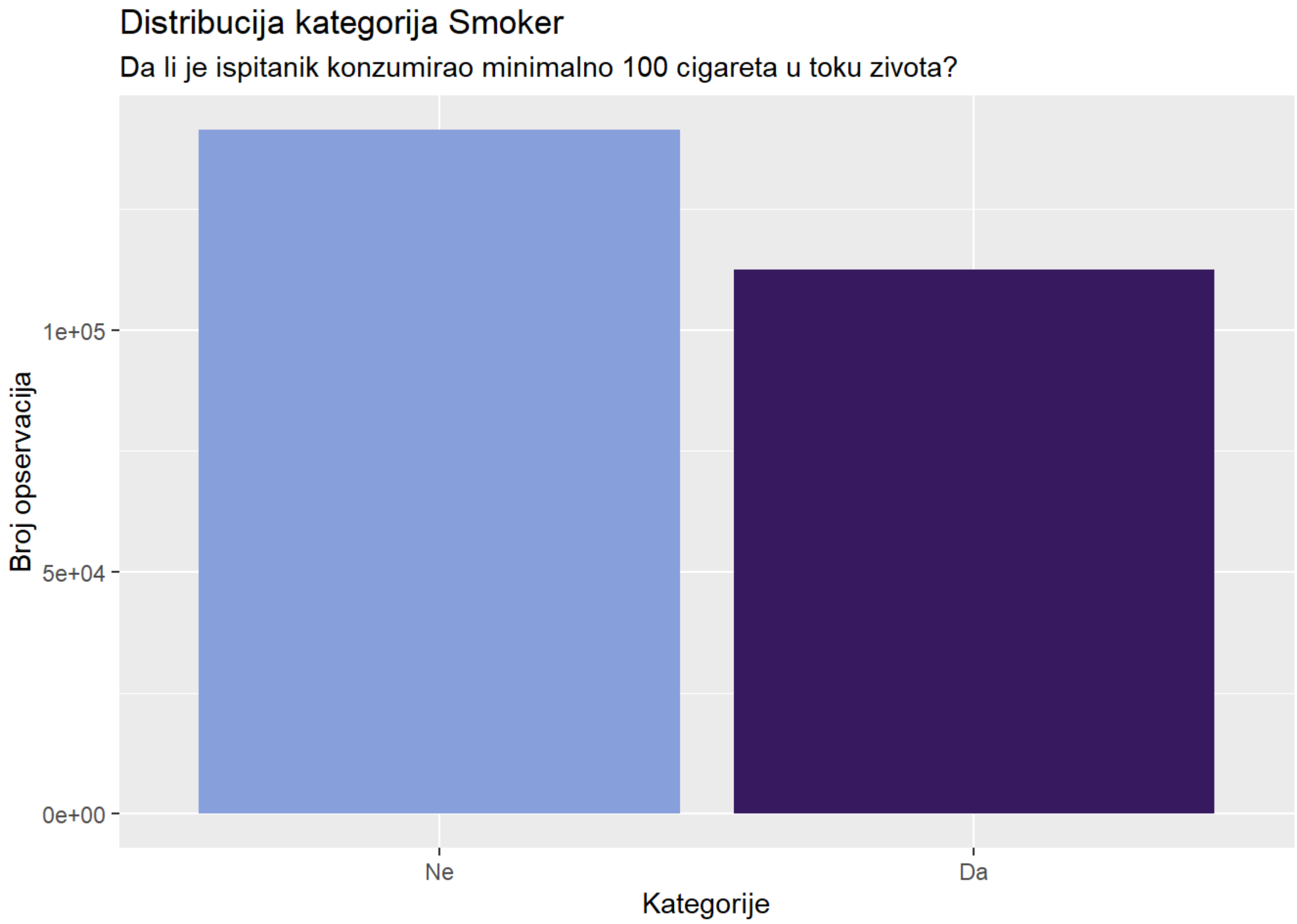
```
ggplot(data, aes(x = Veggies , fill = Veggies )) +
      geom_bar() +
      labs(title = "Distribucija kategorija Veggies", subtitle = "Da li ispitanik jede
jednu porciju ili više povrća u toku dana?", x = "Kategorije", y = "Broj opservacija")
+
      scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_veggies = var(as.numeric(data$Veggies))
cat("Varijansa Veggies: ", varijansa_veggies, "od maksimalne moguce 0.25\n")
```

```
## Varijansa Veggies:  0.1530182 od maksimalne moguce 0.25
```

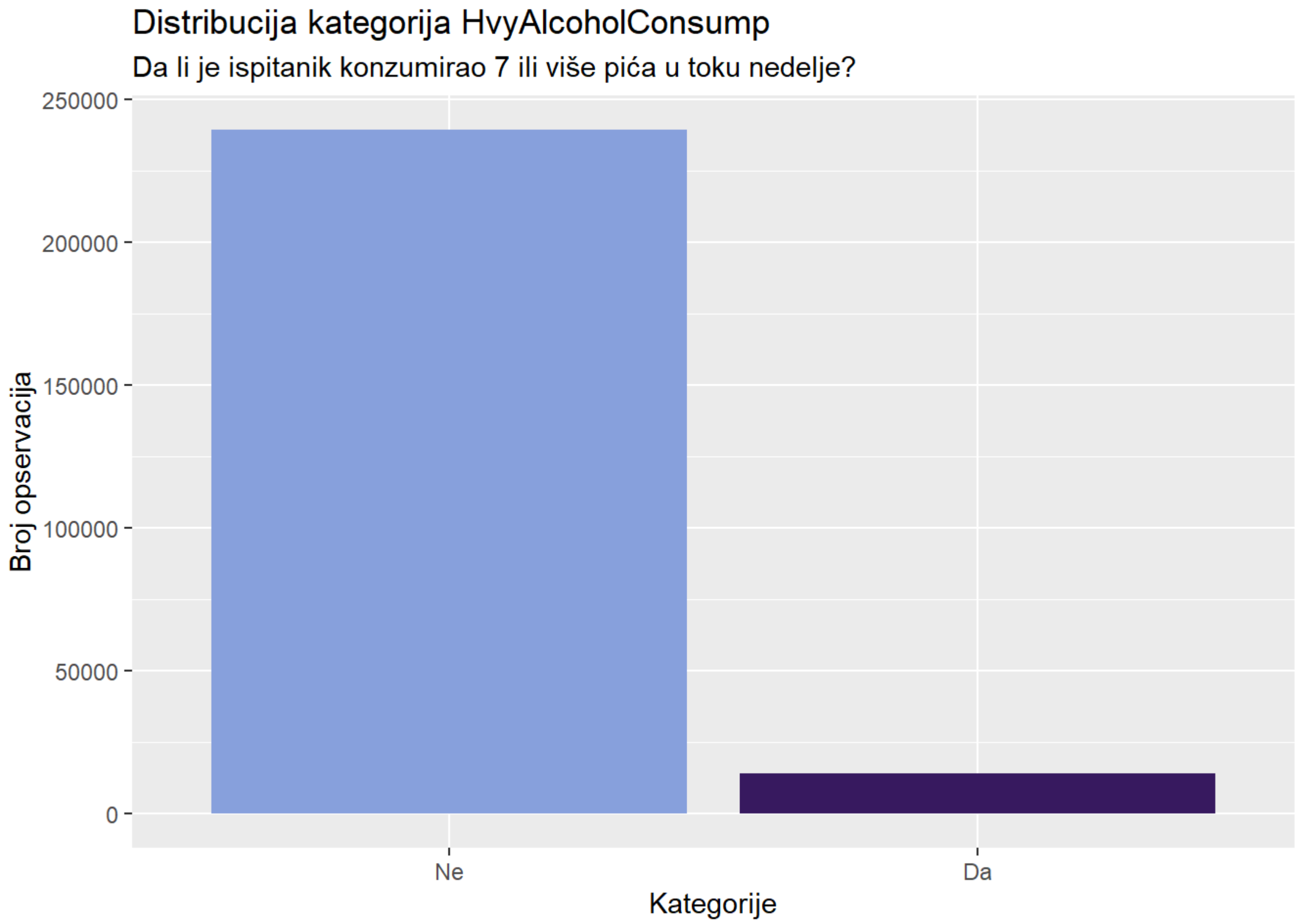
```
ggplot(data, aes(x = Smoker , fill = Smoker )) +
      geom_bar() +
      labs(title = "Distribucija kategorija Smoker", subtitle = "Da li je ispitanik ko
nzumirao minimalno 100 cigareta u toku zivota?", x = "Kategorije", y = "Broj opservaci
ja") +
      scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_smoker = var(as.numeric(data$Smoker))
cat("Varijansa Smoker: ", varijansa_smoker, "od maksimmalne moguce 0.25\n")
```

```
## Varijansa Smoker:  0.2467712 od maksimmalne moguce 0.25
```

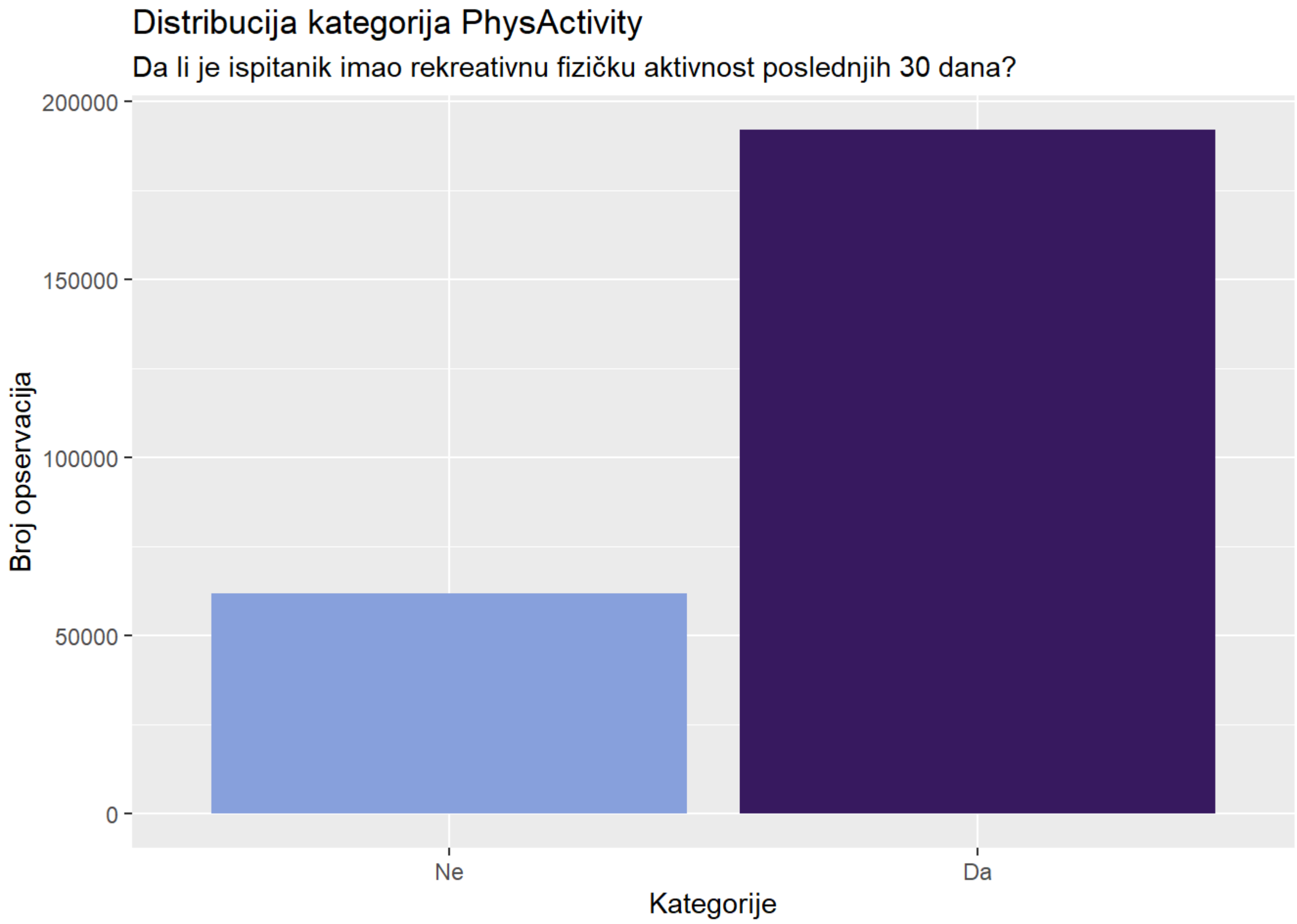
```
ggplot(data, aes(x = HvyAlcoholConsump , fill = HvyAlcoholConsump )) +
      geom_bar() +
      labs(title = "Distribucija kategorija HvyAlcoholConsump", subtitle = "Da li je i
spitanik konzumirao 7 ili više pića u toku nedelje?", x = "Kategorije", y = "Broj opse
rvacija") +
      scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_hvyAlcoholConsump = var(as.numeric(data$HvyAlcoholConsump))
cat("Varijansa HvyAlcoholConsump: ", varijansa_hvyAlcoholConsump, "od maksimmalne mogu
ce 0.25\n")
```

```
## Varijansa HvyAlcoholConsump:  0.05303891 od maksimmalne moguce 0.25
```

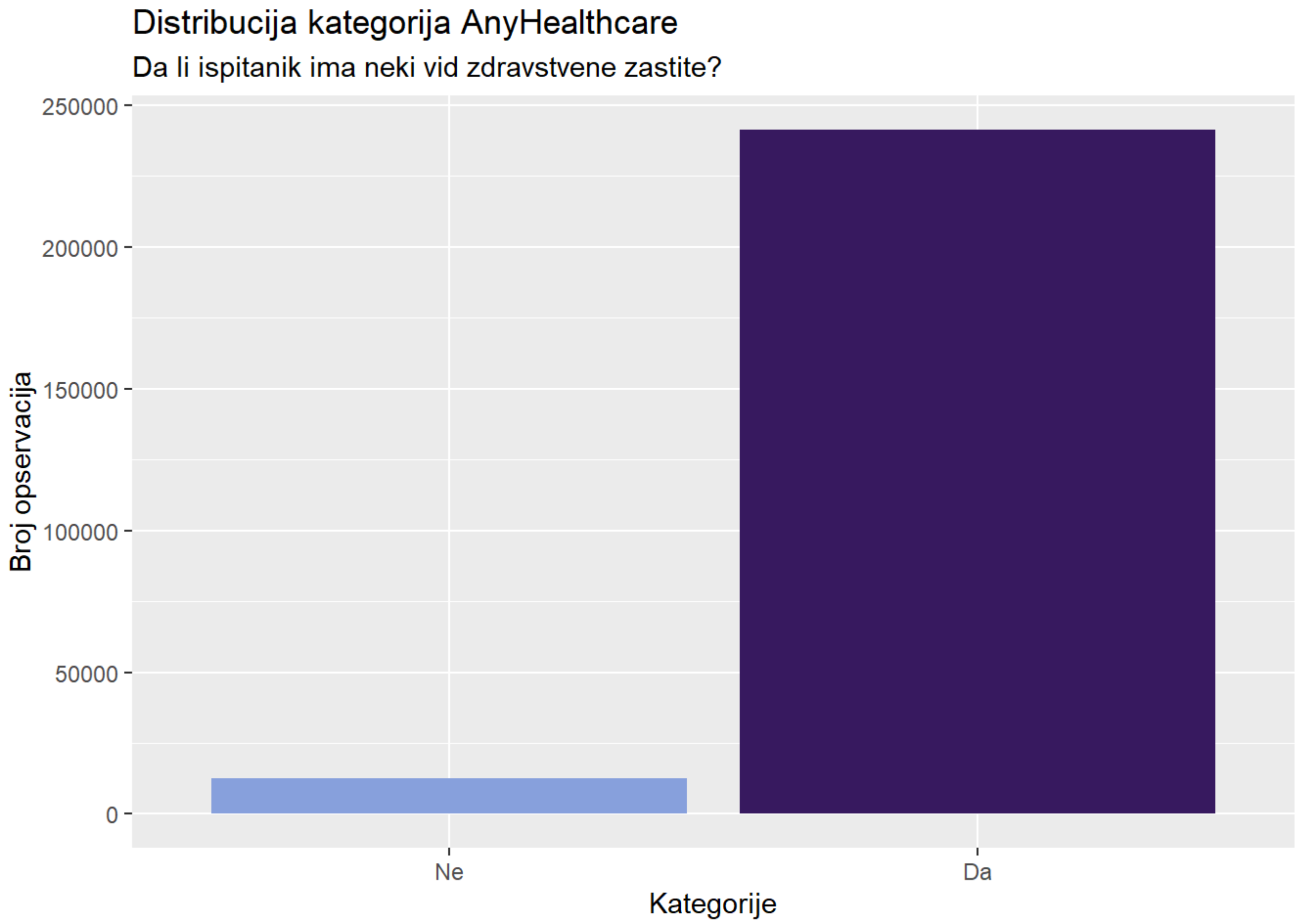
```
ggplot(data, aes(x = PhysActivity  , fill = PhysActivity  )) +
  geom_bar() +
  labs(title = "Distribucija kategorija PhysActivity ", subtitle = "Da li je ispitanik
imao rekreativnu fizičku aktivnost poslednjih 30 dana?", x = "Kategorije", y = "Broj o
pservacija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_PhysActivity  = var(as.numeric(data$PhysActivity ))
cat("Varijansa PhysActivity : ", varijansa_PhysActivity , "od maksimmalne moguce 0.25
\n")
```

```
## Varijansa PhysActivity :  0.1841861 od maksimmalne moguce 0.25
```

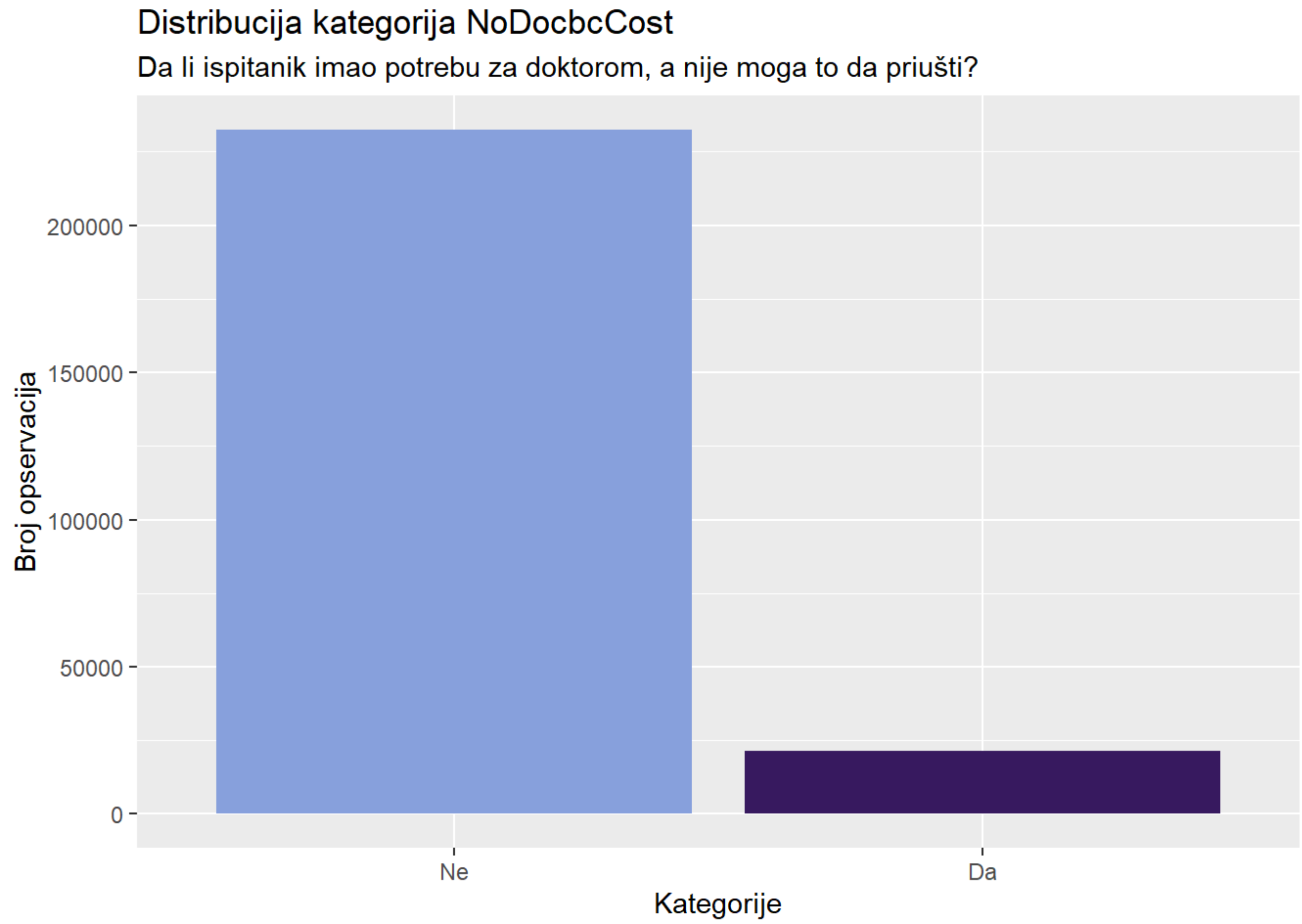
```
ggplot(data, aes(x = AnyHealthcare , fill = AnyHealthcare )) +
  geom_bar() +
  labs(title = "Distribucija kategorija AnyHealthcare", subtitle = "Da li ispitanik im
a neki vid zdravstvene zastite?", x = "Kategorije", y = "Broj opservacija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
varijansa_anyHealthcare = var(as.numeric(data$AnyHealthcare))
cat("Varijansa AnyHealthcare: ", varijansa_anyHealthcare, "od maksimmalne moguce 0.25
\n")
```

```
## Varijansa AnyHealthcare:  0.04655182 od maksimmalne moguce 0.25
```

```
ggplot(data, aes(x = NoDocbcCost , fill = NoDocbcCost )) +
  geom_bar() +
  labs(title = "Distribucija kategorija NoDocbcCost", subtitle = "Da li ispitanik imao
potrebu za doktorom, a nije moga to da priušti?", x = "Kategorije", y = "Broj opservac
ija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```

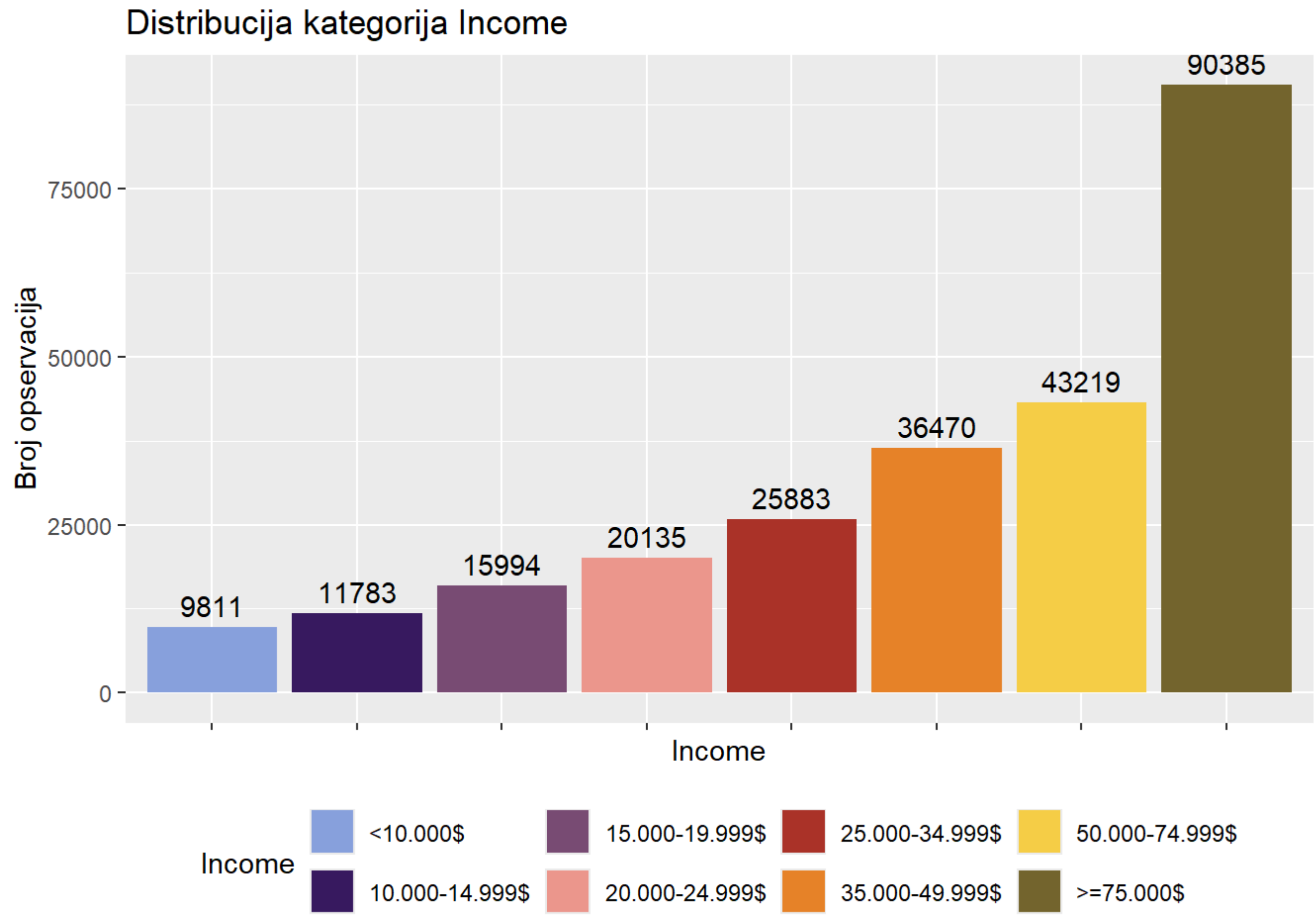



```
varijansa_noDocbcCost = var(as.numeric(data$NoDocbcCost))
cat("Varijansa NoDocbcCost: ", varijansa_noDocbcCost, "od maksimalne moguce 0.25\n")
```

```
## Varijansa NoDocbcCost:  0.07709147 od maksimalne moguce 0.25
```

```
ggplot(data, aes(x = Income, fill = Income)) +
  geom_bar() +
  labs(
    title = "Distribucija kategorija Income",
    y = "Broj opservacija",
  ) +
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5) +
  scale_fill_manual(values = colorS)+
  theme(legend.position = "bottom",
        axis.text.x  = element_blank())
```

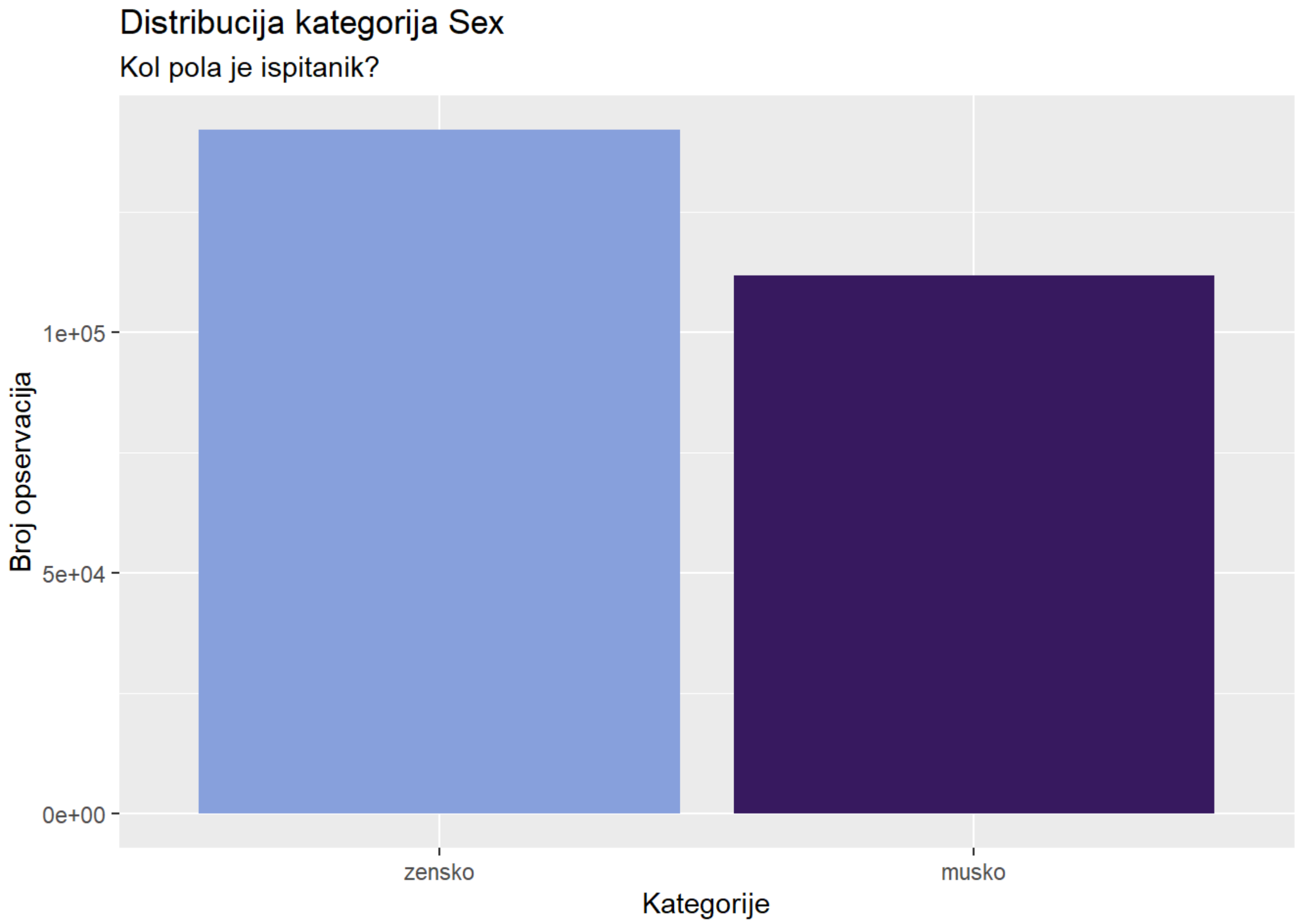
```
## Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
## ℹ Please use `after_stat(count)` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
prop.table(table(data$Income))*100
```

```
##
##        <10.000$ 10.000-14.999$ 15.000-19.999$ 20.000-24.999$ 25.000-34.999$
##        3.867471       4.644828       6.304793       7.937165      10.203012
## 35.000-49.999$ 50.000-74.999$       >=75.000$
##      14.376380      17.036818      35.629533
```

```
ggplot(data, aes(x = Sex , fill = Sex )) +
  geom_bar() +
  labs(title = "Distribucija kategorija Sex", subtitle = "Kol pola je ispitanik?", x =
"Kategorije", y = "Broj opservacija") +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



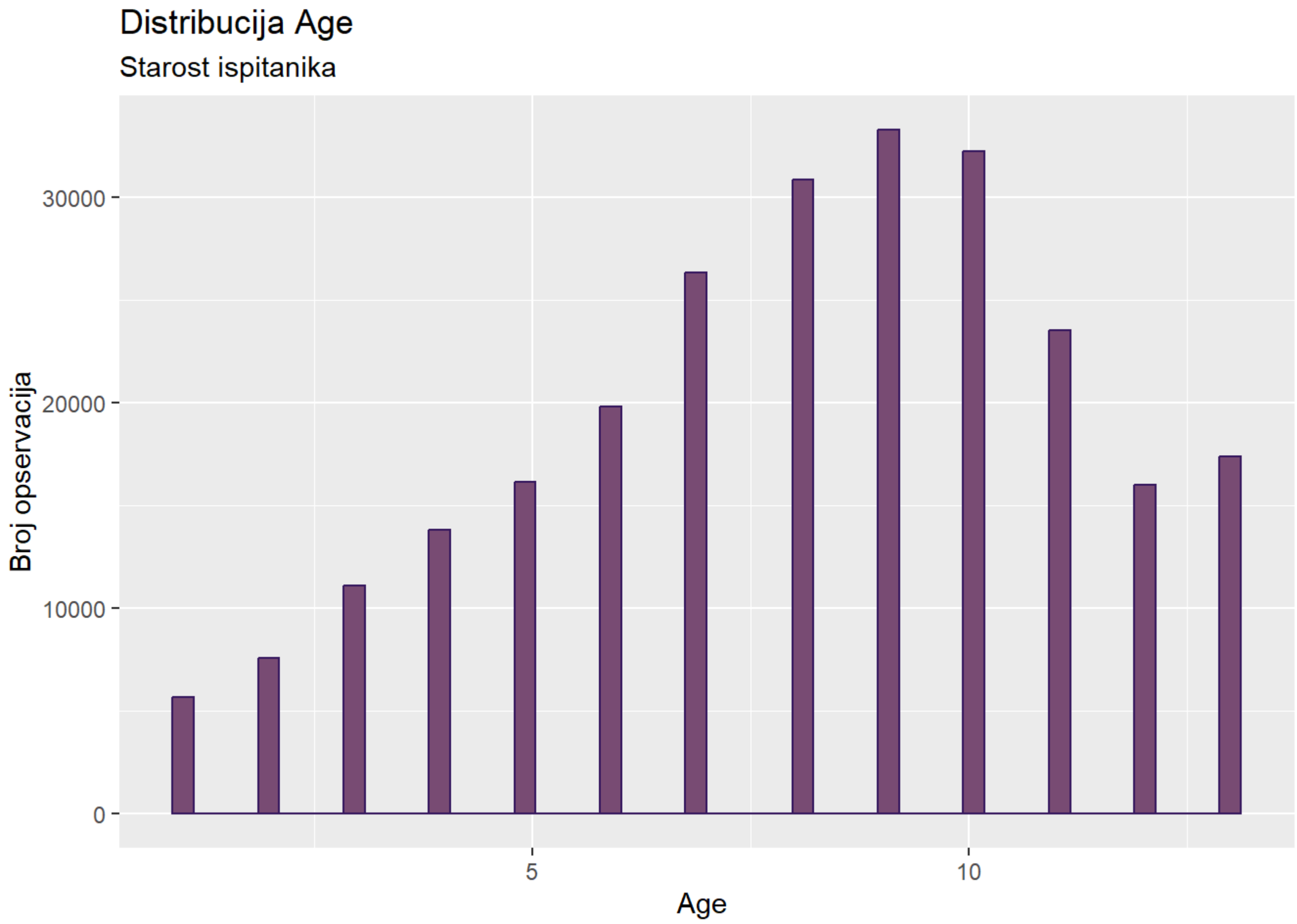

```
varijansa_sex = var(as.numeric(data$Sex))
cat("Varijansa Sex: ", varijansa_sex, "od maksimmalne moguce 0.25\n")
```

```
## Varijansa Sex:  0.2464419 od maksimmalne moguce 0.25
```

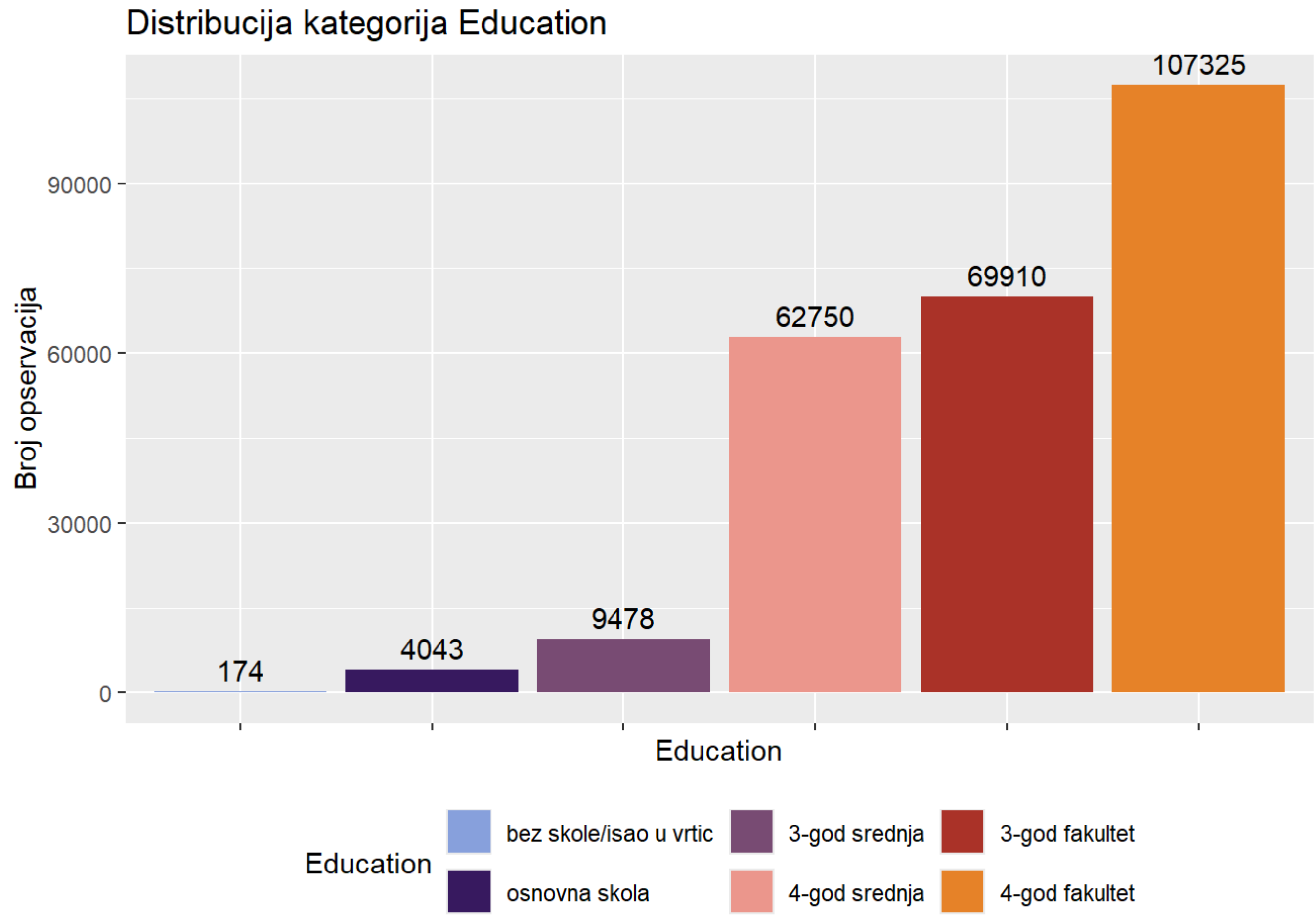
```
prop.table(table(data$Age))*100
```

```
##
##         1         2         3         4         5         6         7         8
##  2.246925  2.995112  4.384658  5.448991  6.369048  7.812599 10.372911 12.153895
##         9        10        11        12        13
## 13.104699 12.690792  9.276648  6.299275  6.844450
```

```
ggplot(data, aes(x = Age )) +
  geom_histogram(bins = 50, fill = "#7C4B73FF", color="#381A61FF") +
  labs(title = "Distribucija Age ", x = "Age ", y = "Broj opservacija",
       subtitle = "Starost ispitanika")
```

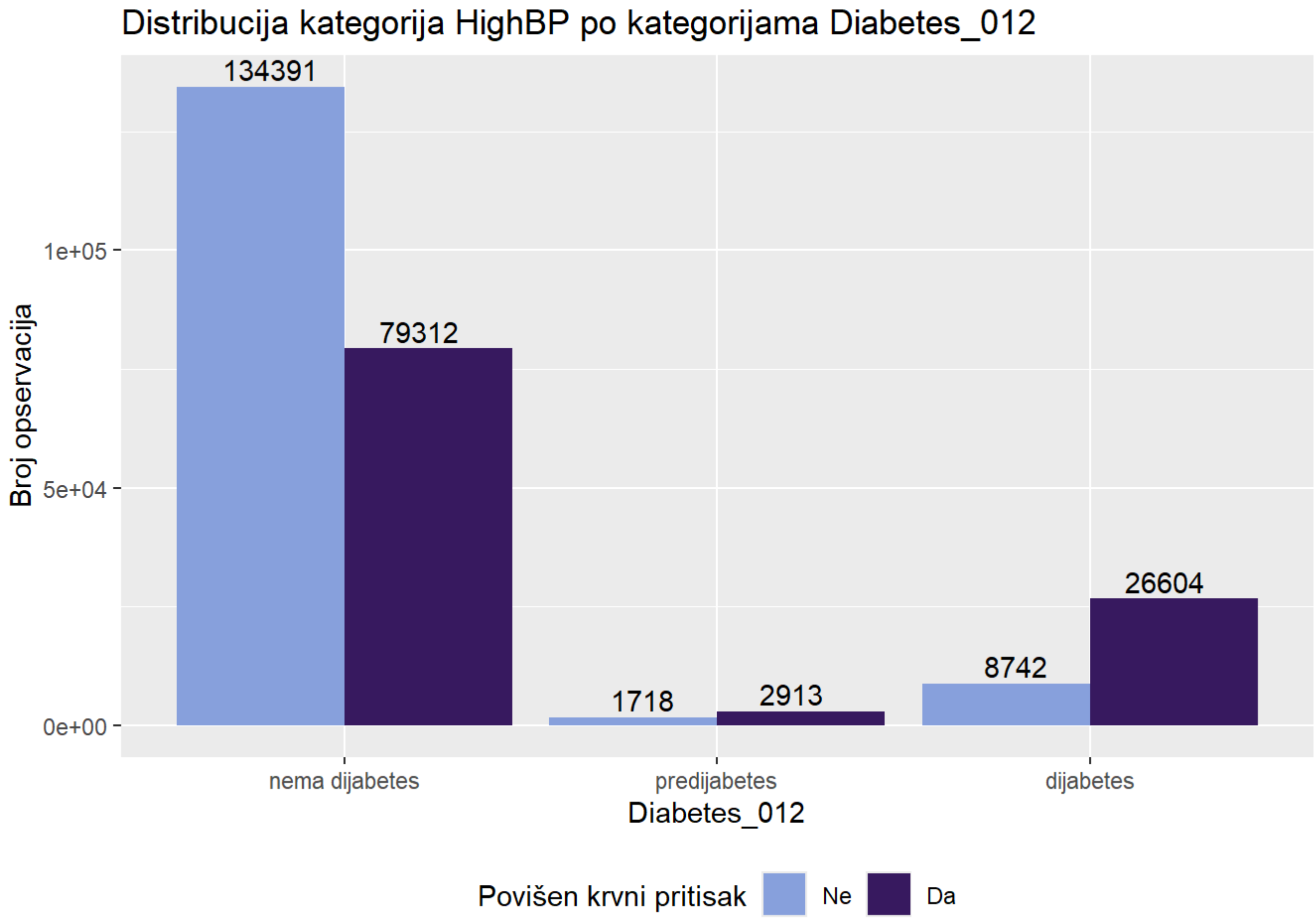



```
ggplot(data, aes(x = Education, fill = Education)) +
  geom_bar() +
  labs(
    title = "Distribucija kategorija Education",
    y = "Broj opservacija",
  ) +
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5) +
  scale_fill_manual(values = colorS)+
  theme(legend.position = "bottom",
        axis.text.x  = element_blank())
```

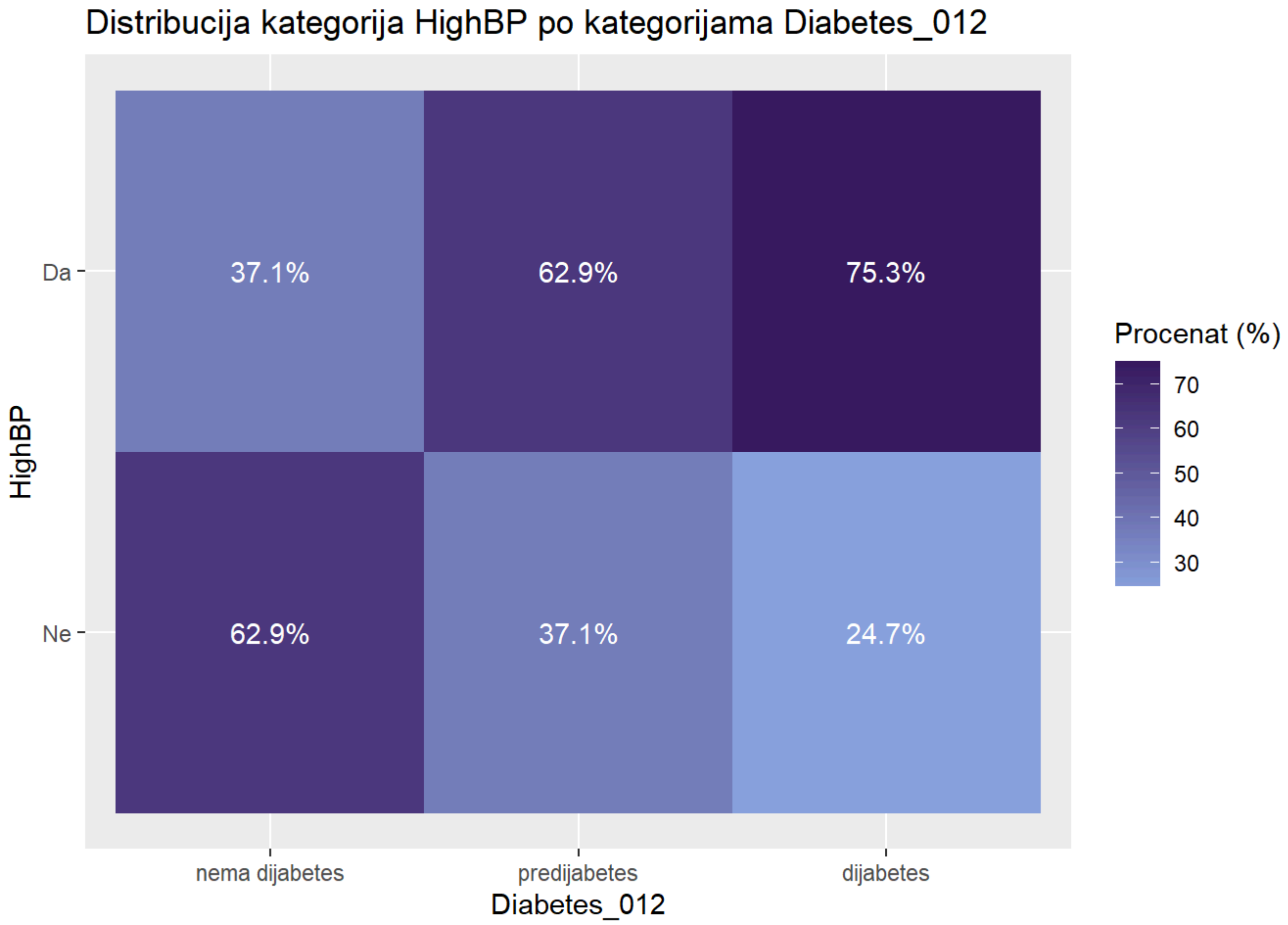


##Bivarijanta

```
ggplot(data, aes(x = Diabetes_012, fill = HighBP))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija HighBP po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Povišen krvni pritisak"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```




```
data %>%
  count(Diabetes_012, HighBP) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = HighBP, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija HighBP po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "HighBP",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



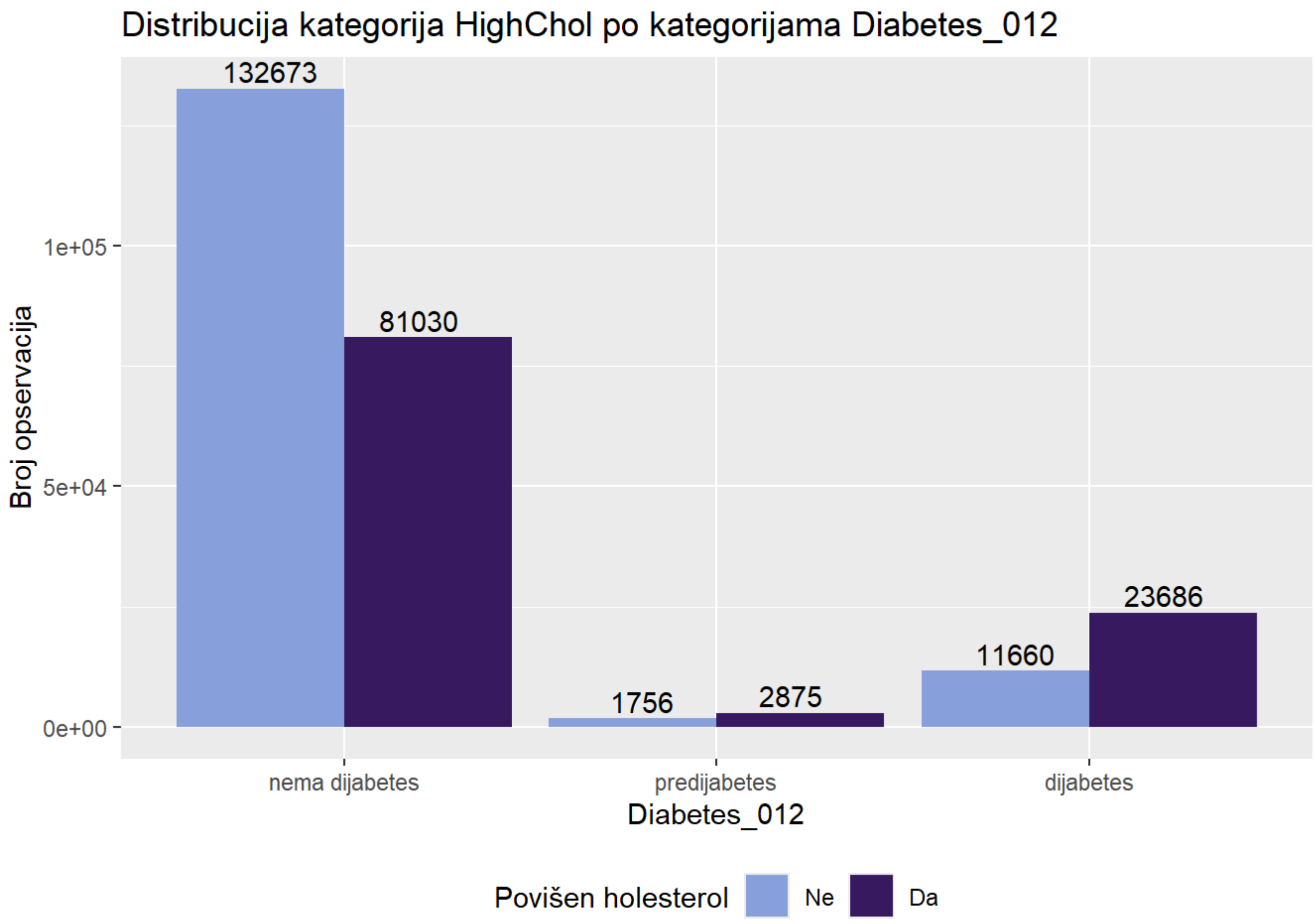

```
chi_sq_test(data$HighBP, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 18795, df = 2, p-value < 2.2e-16
```

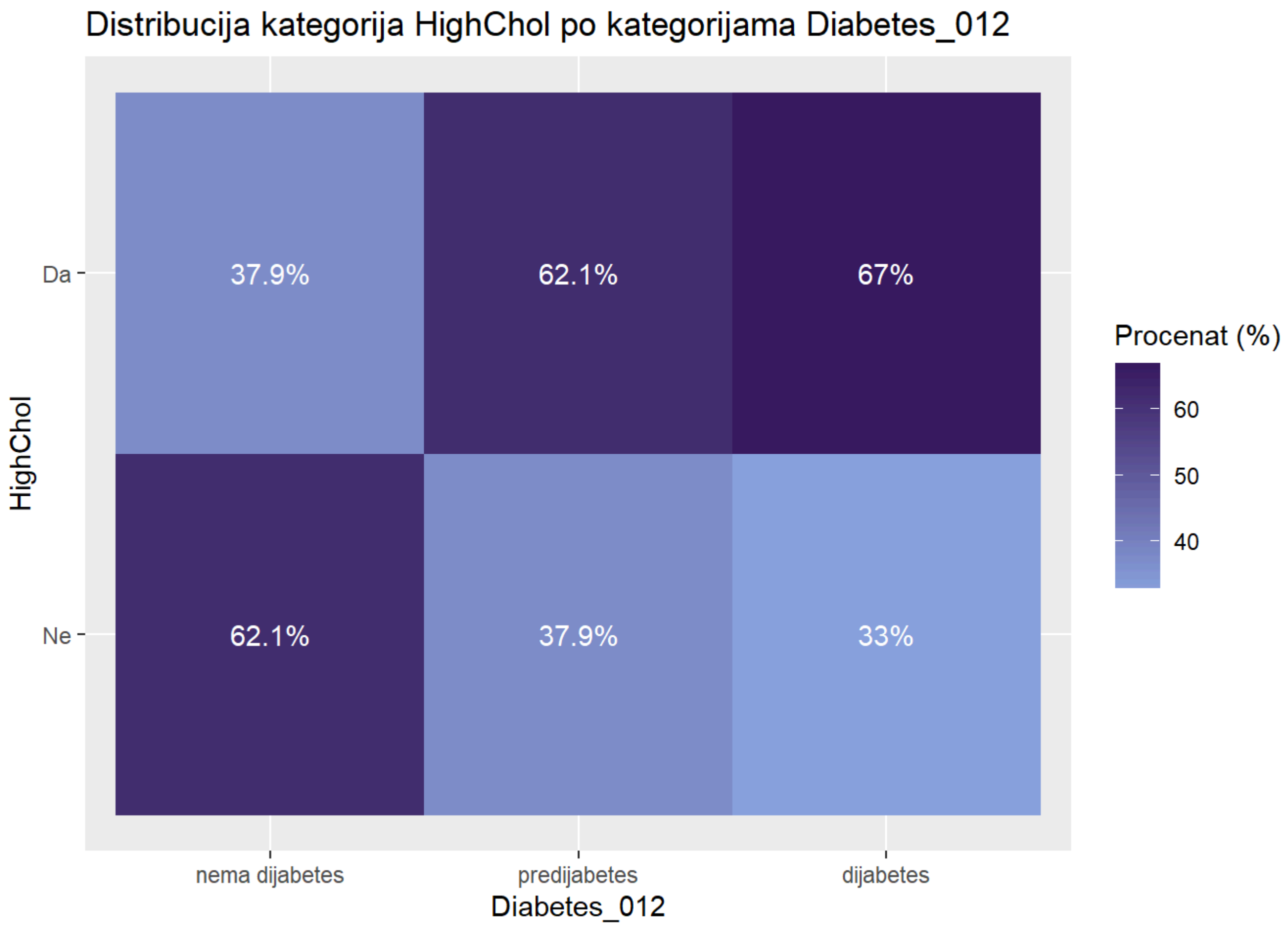
```
cramer_v(data$HighBP, data$Diabetes_012)
```

```
## [1] 0.2721911
```

```
ggplot(data, aes(x = Diabetes_012, fill = HighChol))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija HighChol po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Povišen holesterol"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  )+
  theme(legend.position="bottom")
```



```
data %>%
  count(Diabetes_012, HighChol) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = HighChol, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija HighChol po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "HighChol",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



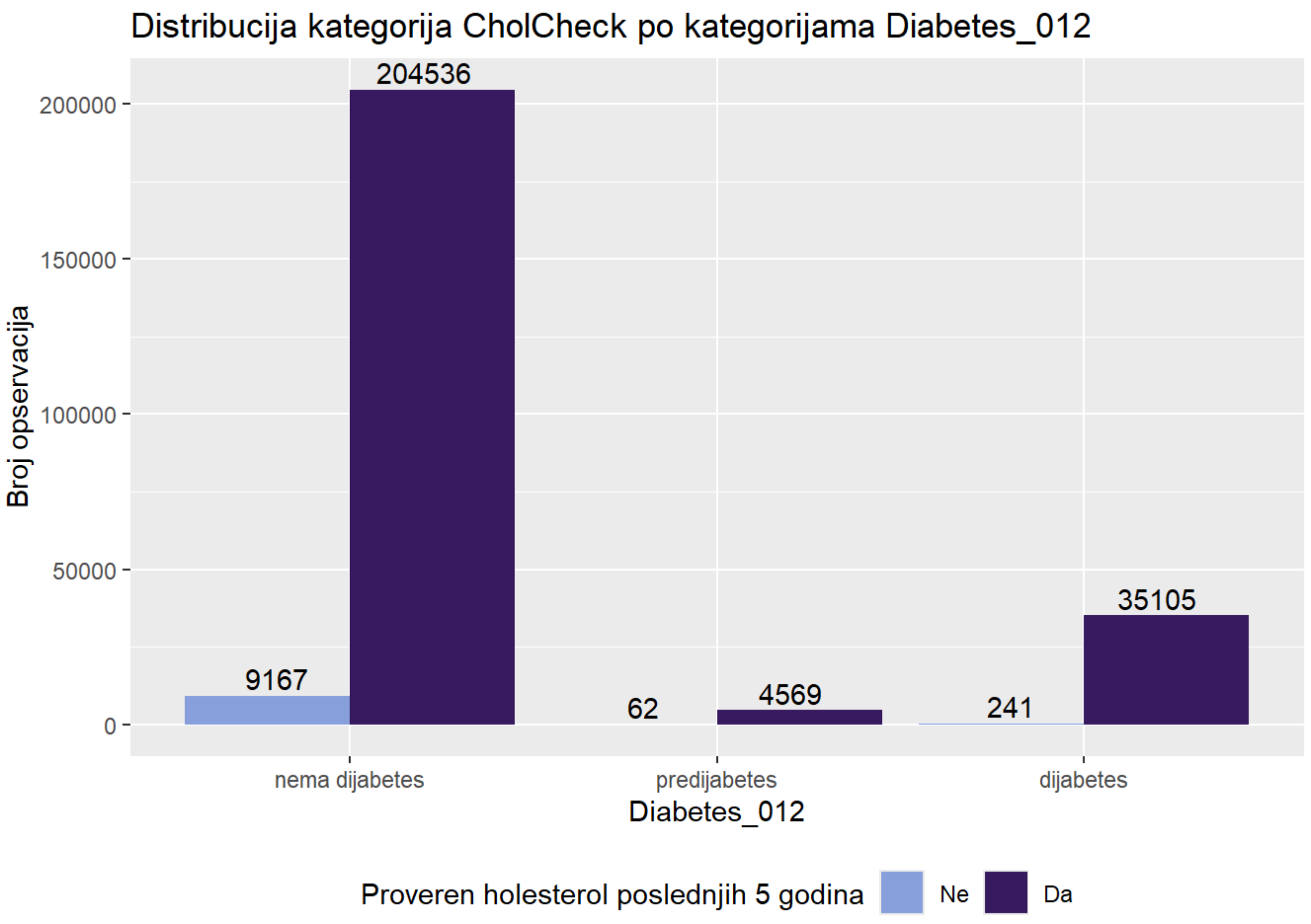

```
chi_sq_test(data$HighChol, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 11259, df = 2, p-value < 2.2e-16
```

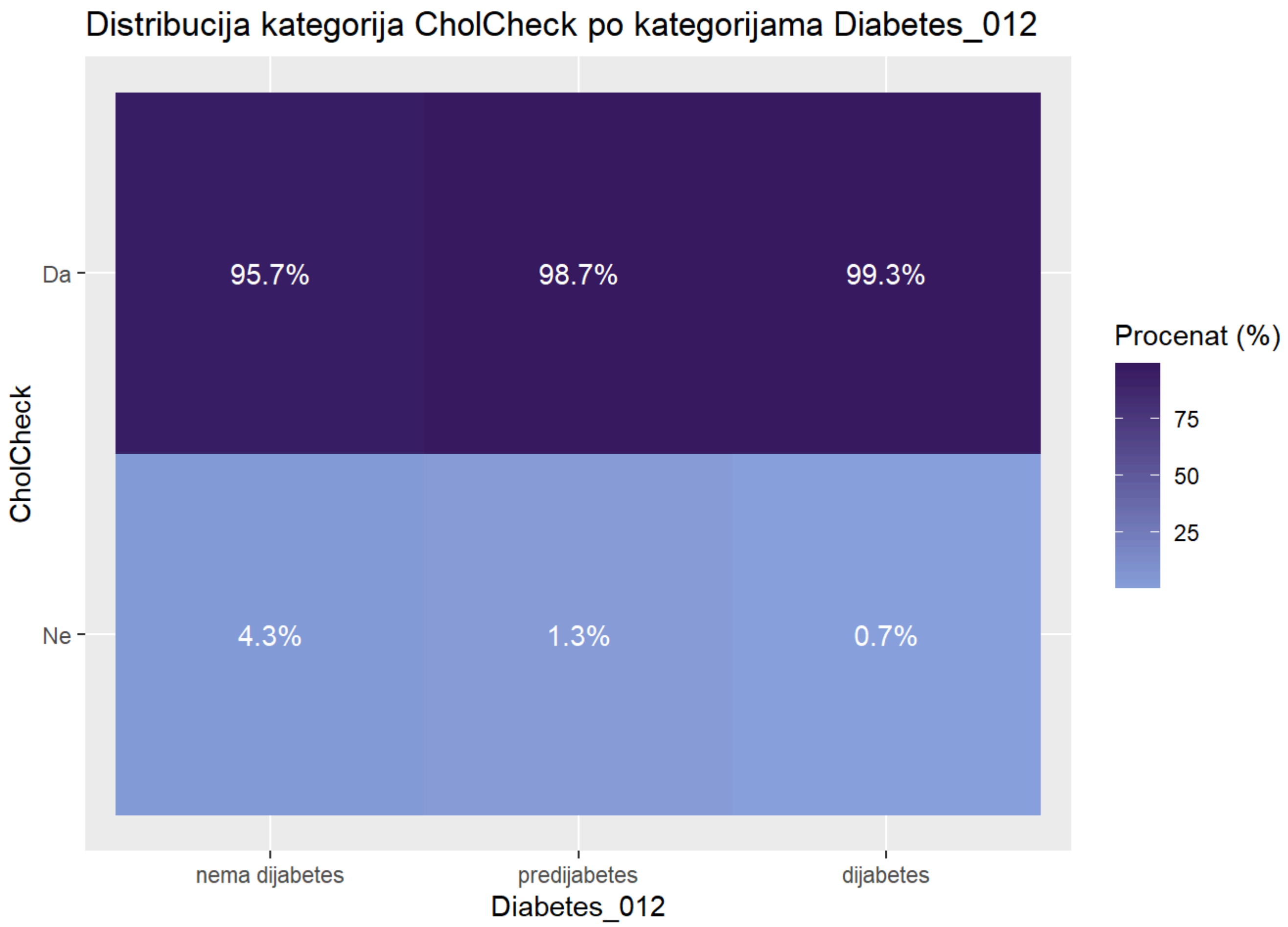
```
cramer_v(data$HighChol, data$Diabetes_012)
```

```
## [1] 0.2106712
```

```
ggplot(data, aes(x = Diabetes_012, fill = CholCheck))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija CholCheck po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Proveren holesterol poslednjih 5 godina"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```



```
data %>%
  count(Diabetes_012, CholCheck) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = CholCheck, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija CholCheck po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "CholCheck",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



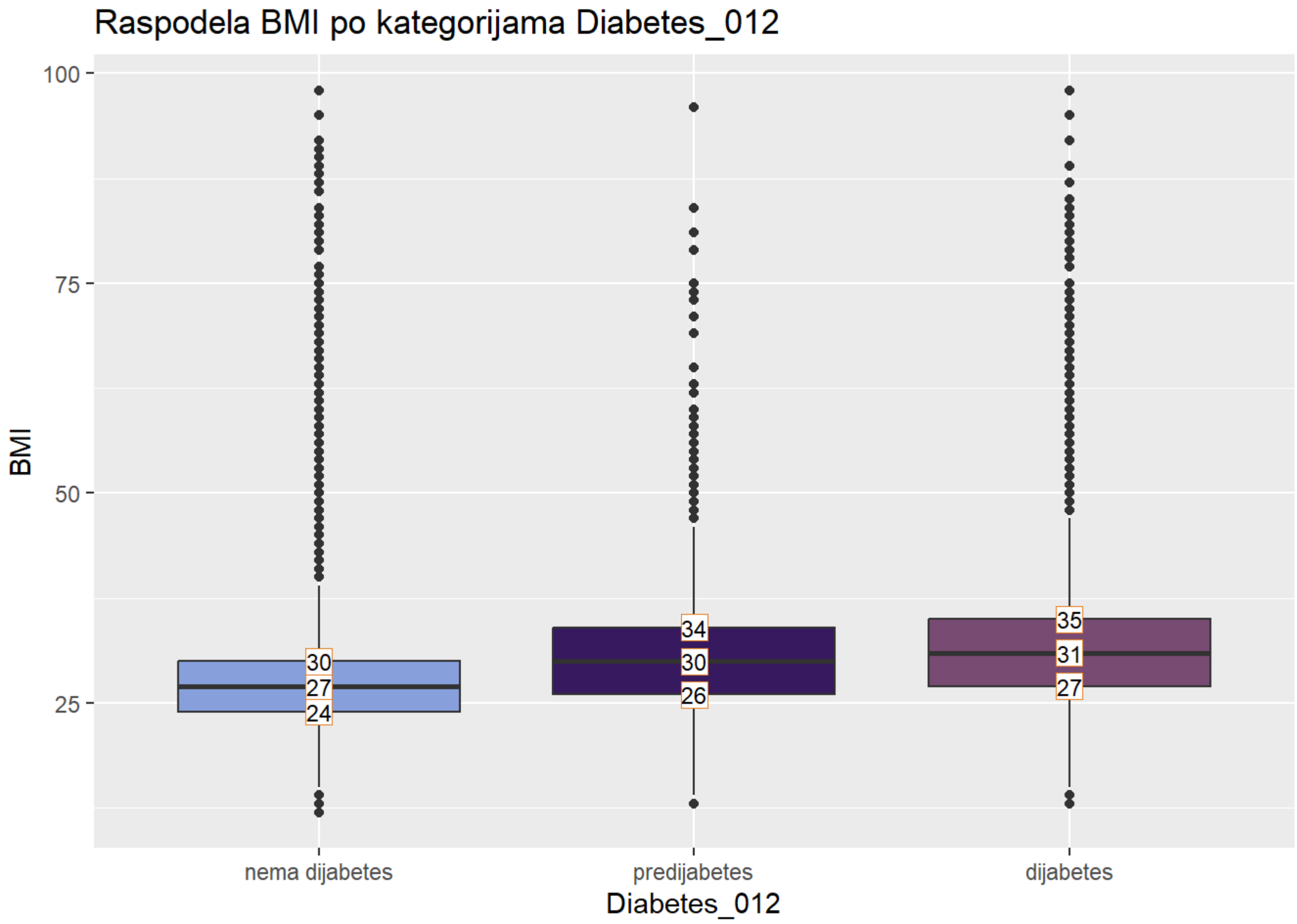

```
chi_sq_test(data$CholCheck, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 1173.7, df = 2, p-value < 2.2e-16
```

```
cramer_v(data$CholCheck, data$Diabetes_012)
```

```
## [1] 0.06802124
```

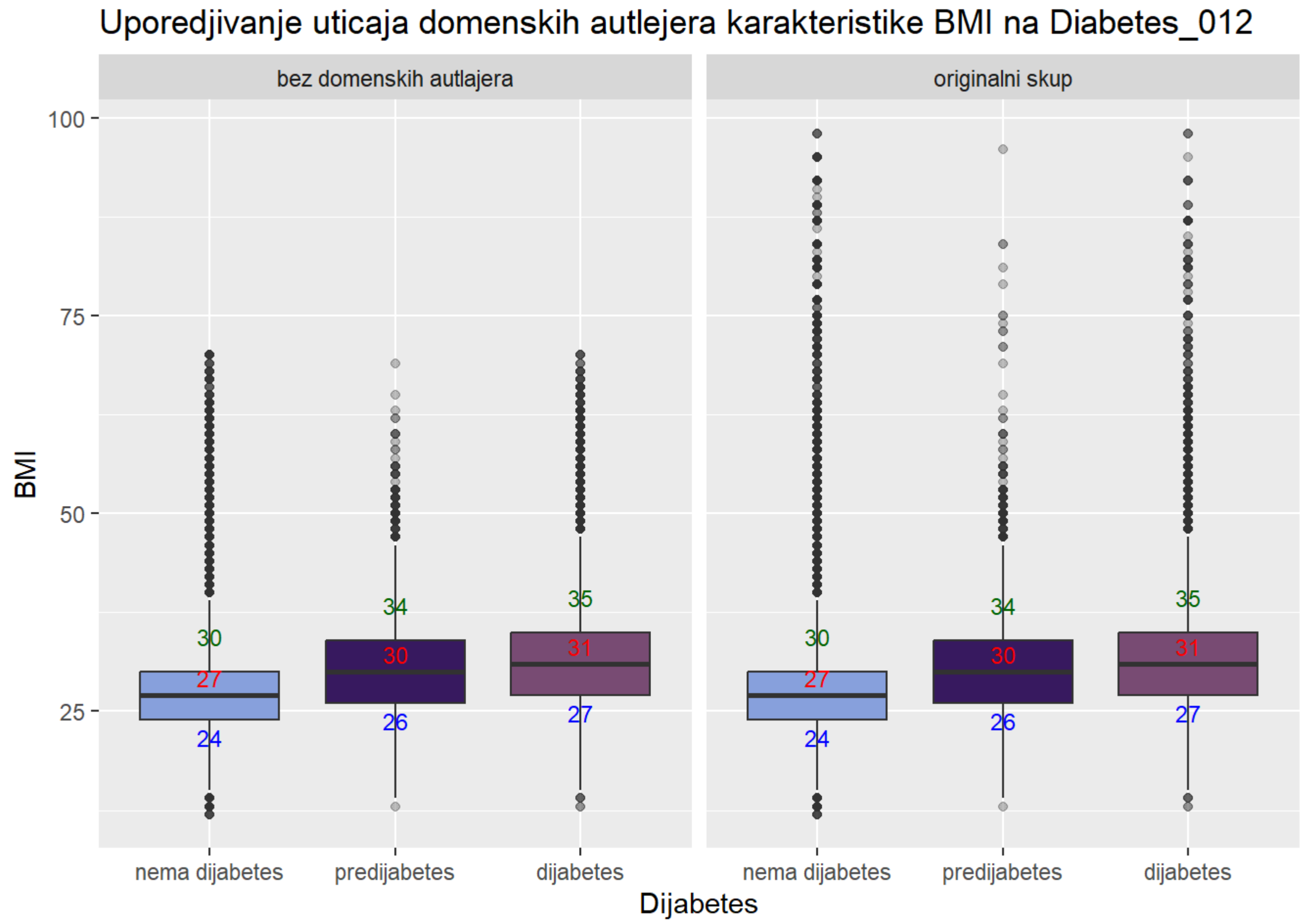
```
vrenostiBoxPlot <- data%>%
  group_by(Diabetes_012)%>%
  summarise(
    Q1 = quantile(BMI, 0.25),
    median = median(BMI),
    Q3 = quantile(BMI, 0.75),
  )%>%
  pivot_longer(
    cols = c(Q1, median, Q3)
  )

ggplot(data, aes(x = Diabetes_012, y = BMI, fill = Diabetes_012))+
  geom_boxplot()+
  geom_point(
    data = vrenostiBoxPlot,
    aes(x = Diabetes_012, y = value),
    shape = 22,
    size = 5,
    fill = "white",
    color = "#E78429FF"
  )+
  geom_text(
    data = vrenostiBoxPlot,
    aes(x = Diabetes_012, y = value, label = round(value, 1)),
    size = 3
  )+
  labs(
    title = "Raspodela BMI po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "BMI"
  )+
  scale_fill_paletteer_d("MetBrewer::Archambault")+
  theme(legend.position = "none")
```




```
BMI_bezdomeskihAutlajera = data %>%
  filter(BMI >= 12 & BMI <= 70)

BMI_bezStatistickihAutlajera = data %>%
  filter(BMI >= donja_granica & BMI <= gornja_granica)

BMI_bezdomenskih_sabrani = bind_rows(
  data %>%
    select(BMI, Diabetes_012) %>%
    mutate(Skup = "originalni skup"),

  BMI_bezdomeskihAutlajera %>%
    select(BMI, Diabetes_012) %>%
    mutate(Skup = "bez domenskih autlajera"),
)
statistickeVrednostiBezDomenskih = BMI_bezdomenskih_sabrani %>%
  group_by(Skup, Diabetes_012) %>%
  summarise(
    Q1 = quantile(BMI, 0.25),
    Median = quantile(BMI, 0.5),
    Q3 = quantile(BMI, 0.75)
  )
```

```
## `summarise()` has grouped output by 'Skup'. You can override using the
## `.groups` argument.
```

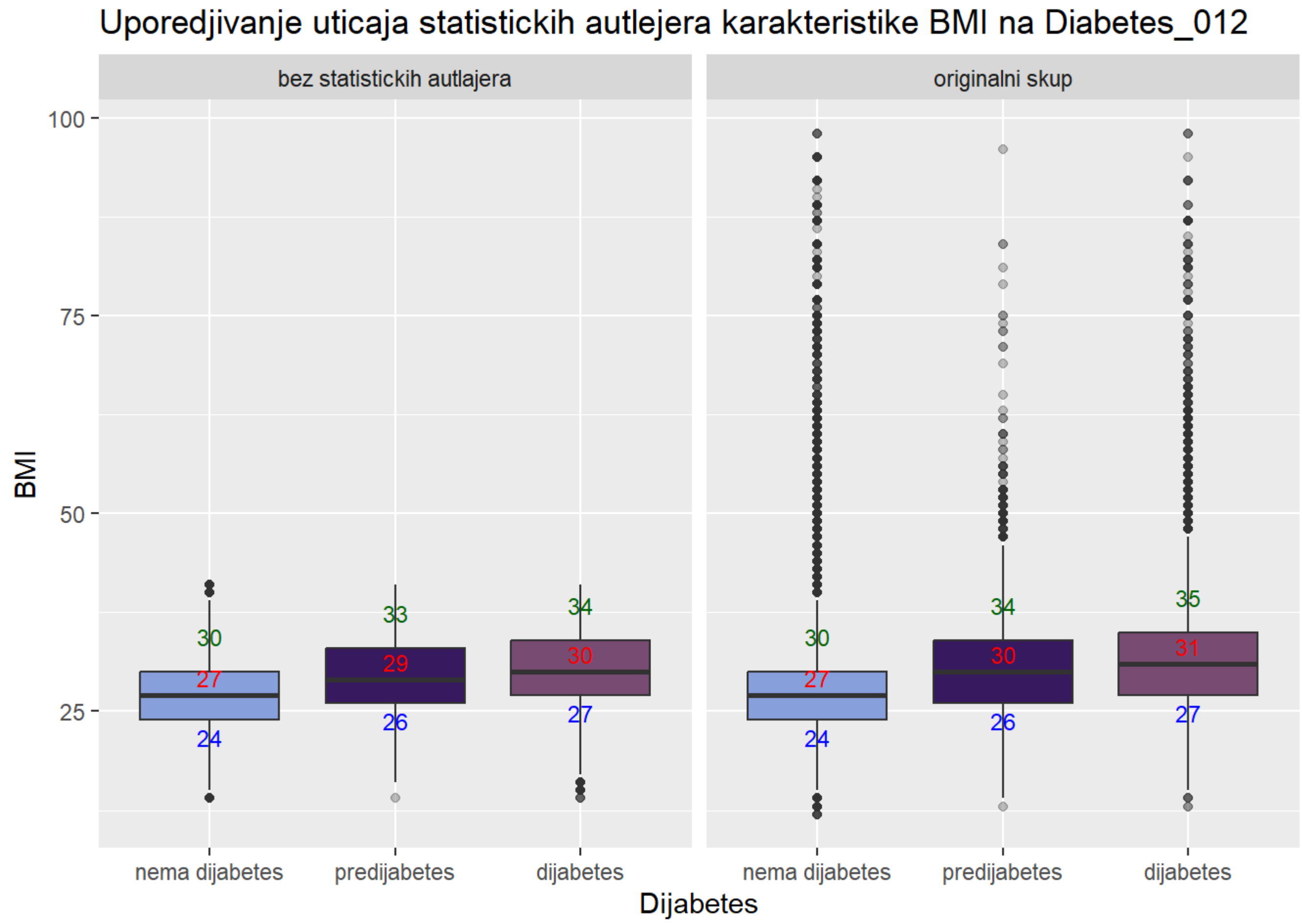
```
ggplot(BMI_bezdomenskih_sabrani, aes(x = Diabetes_012, y = BMI, fill = Diabetes_012))
+
  geom_boxplot(outlier.alpha = 0.3) +
  facet_wrap(~ Skup)+
  geom_text(
    data = statistickeVrednostiBezDomenskih,
    aes(x = Diabetes_012, y = Q1, label = round(Q1, 1)),
    vjust = 1.5, color = "blue", size = 3,
  ) +
  geom_text(
    data = statistickeVrednostiBezDomenskih,
    aes(x = Diabetes_012, y = Median, label = round(Median, 1)),
    vjust = -0.5, color = "red", size = 3
  ) +
  geom_text(
    data = statistickeVrednostiBezDomenskih,
    aes(x = Diabetes_012, y = Q3, label = round(Q3, 1)),
    vjust = -1.5, color = "darkgreen", size = 3
  ) +
  labs(
    title = "Uporedjivanje uticaja domenskih autlejera karakteristike BMI na Diabetes_
012",
    y = "BMI",
    x = "Dijabetes"
  )+
  theme(legend.position = "none")+
  scale_fill_paletteer_d("MetBrewer::Archambault")
```




```
BMI_bezstatistickih_sabrani <- bind_rows(
  data %>%
    select(BMI, Diabetes_012) %>%
    mutate(Skup = "originalni skup"),

  BMI_bezStatistickihAutlajera %>%
    select(BMI, Diabetes_012) %>%
    mutate(Skup = "bez statistickih autlajera"),
)
statistickeVrednostiBezStatistickih <- BMI_bezstatistickih_sabrani %>%
  group_by(Skup, Diabetes_012) %>%
  summarise(
    Q1 = quantile(BMI, 0.25),
    Median = quantile(BMI, 0.5),
    Q3 = quantile(BMI, 0.75)
  )
```

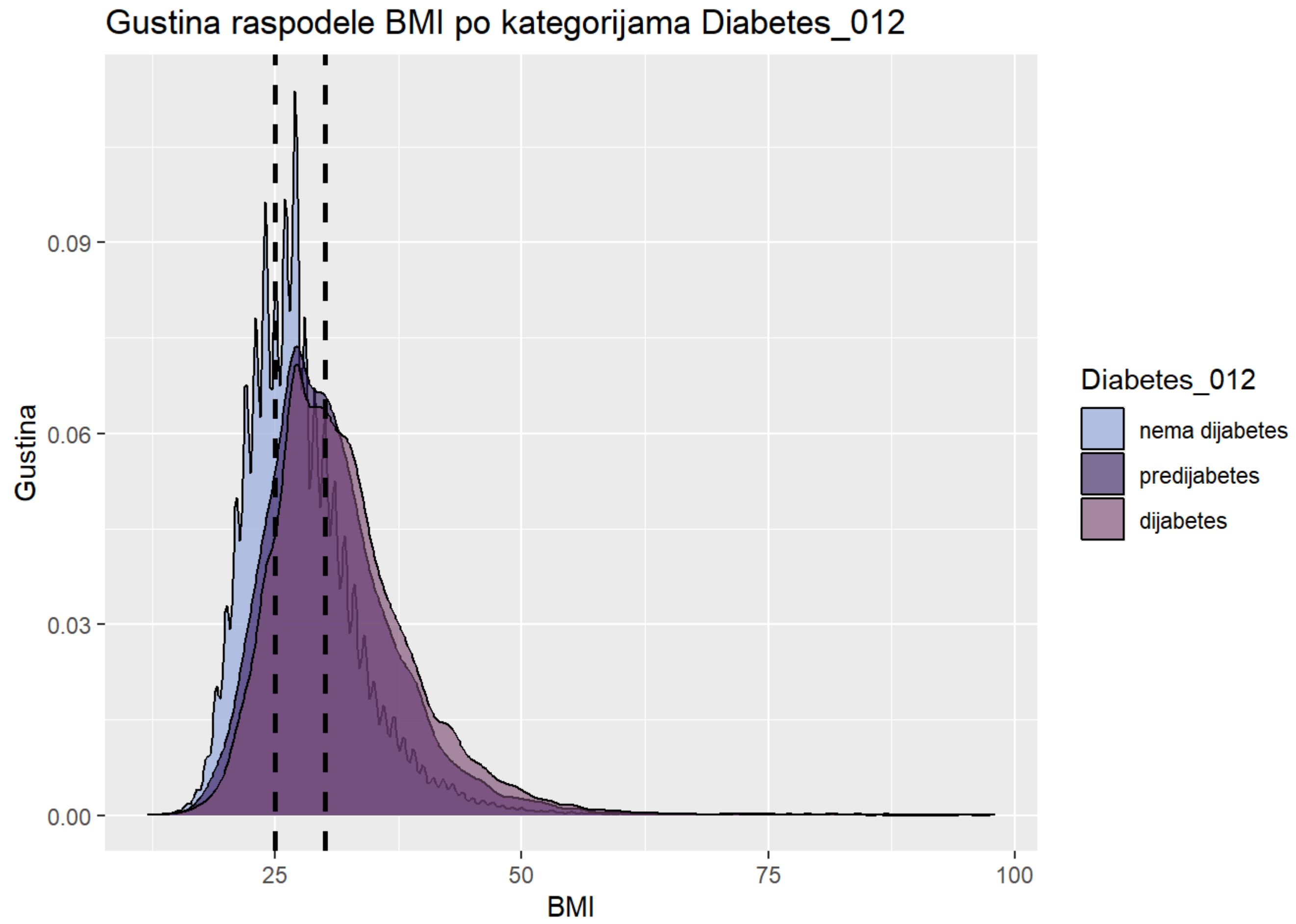
```
## `summarise()` has grouped output by 'Skup'. You can override using the
## `.groups` argument.
```

```
ggplot(BMI_bezstatistickih_sabrani, aes(x = Diabetes_012, y = BMI, fill = Diabetes_01
2)) +
  geom_boxplot(outlier.alpha = 0.3) +
  facet_wrap(~ Skup)+
  geom_text(
    data = statistickeVrednostiBezStatistickih,
    aes(x = Diabetes_012, y = Q1, label = round(Q1, 1)),
    vjust = 1.5, color = "blue", size = 3,
  ) +
  geom_text(
    data = statistickeVrednostiBezStatistickih,
    aes(x = Diabetes_012, y = Median, label = round(Median, 1)),
    vjust = -0.5, color = "red", size = 3
  ) +
  geom_text(
    data = statistickeVrednostiBezStatistickih,
    aes(x = Diabetes_012, y = Q3, label = round(Q3, 1)),
    vjust = -1.5, color = "darkgreen", size = 3
  ) +
  labs(
    title = "Uporedjivanje uticaja statistickih autlejera karakteristike BMI na Diabet
es_012",
    y = "BMI",
    x = "Dijabetes"
  )+
  theme(legend.position = "none")+
  scale_fill_paletteer_d("MetBrewer::Archambault")
```




```
ggplot(data, aes(x = BMI, fill = Diabetes_012)) +
  geom_density(alpha = 0.6) +
  labs(
    title = "Gustina raspodele BMI po kategorijama Diabetes_012",
    x = "BMI",
    y = "Gustina",
    fill = "Diabetes_012"
  ) +
  scale_fill_paletteer_d("MetBrewer::Archambault")+
  geom_vline(xintercept = c(25, 30), linetype = "dashed", color = "black", size = 1)
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



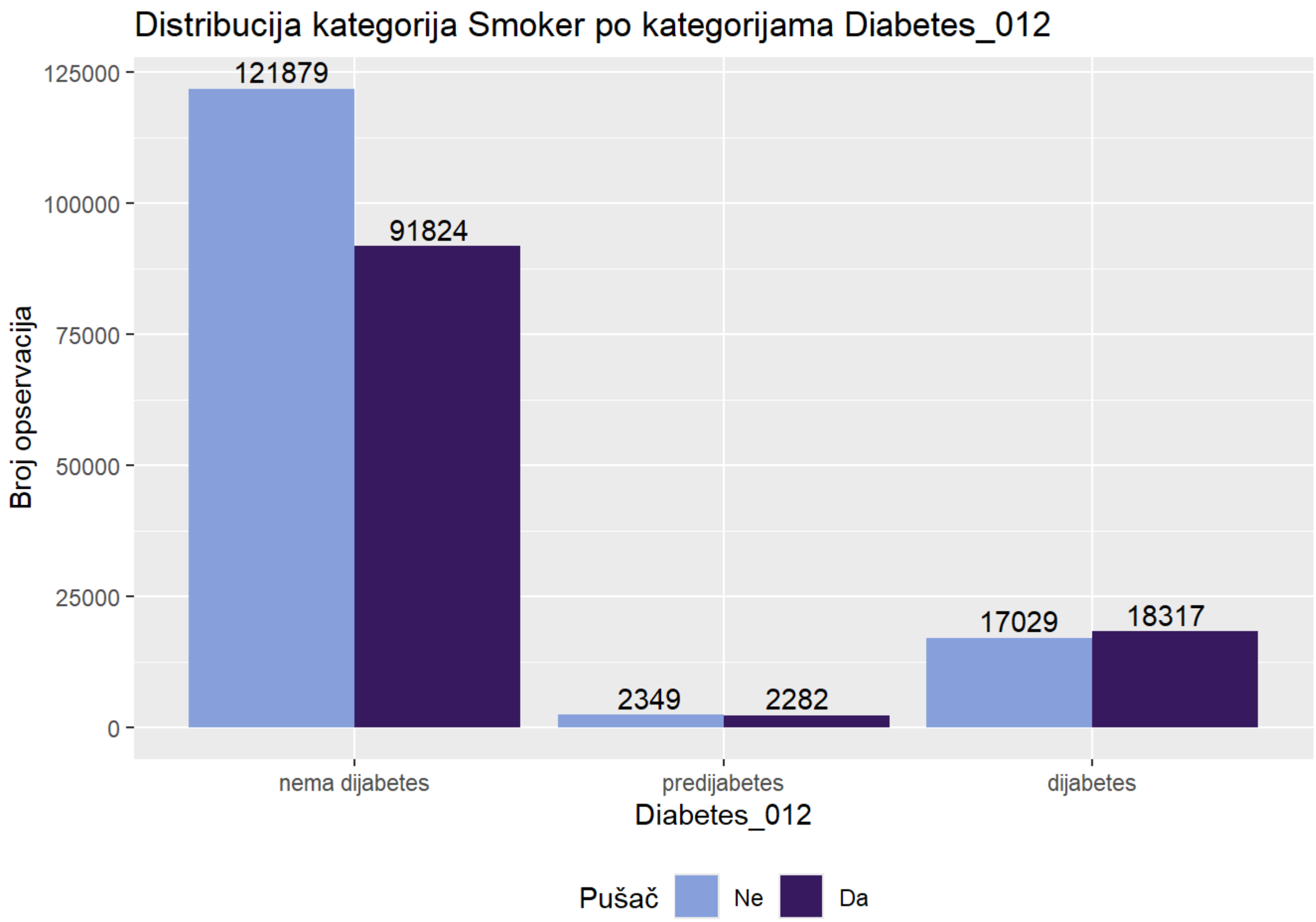

```
anova_test(data$BMI, data$Diabetes_012)
```

```
##                            Df   Sum Sq Mean Sq F value Pr(>F)    
## as.factor(grupna_var)       2   561268  280634    6768 <2e-16 ***
## Residuals              253677 10518121      41                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

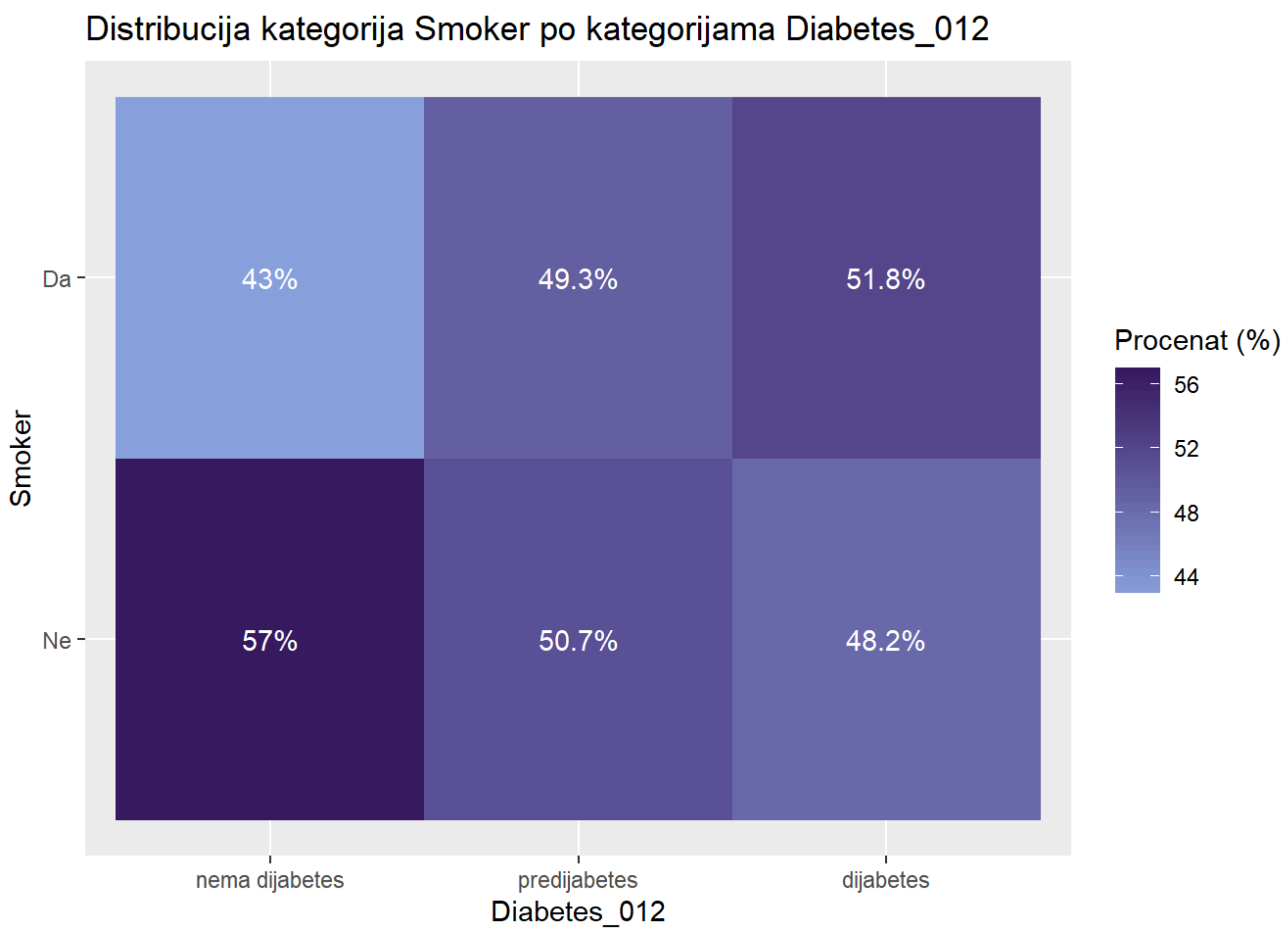
```
tukey_fun(data$BMI, data$Diabetes_012)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
## 
## Fit: aov(formula = numericka_var ~ as.factor(grupna_var))
## 
## $`as.factor(grupna_var)`
##                                 diff       lwr      upr p adj
## predijabetes-nema dijabetes 2.981944 2.7577893 3.206100     0
## dijabetes-nema dijabetes    4.201489 4.1148337 4.288145     0
## dijabetes-predijabetes      1.219545 0.9836992 1.455391     0
```

```
ggplot(data, aes(x = Diabetes_012, fill = Smoker))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija Smoker po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Pušač"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault")+
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  )+
  theme(legend.position = "bottom")
```



```
data %>%
  count(Diabetes_012, Smoker) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = Smoker, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija Smoker po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "Smoker",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



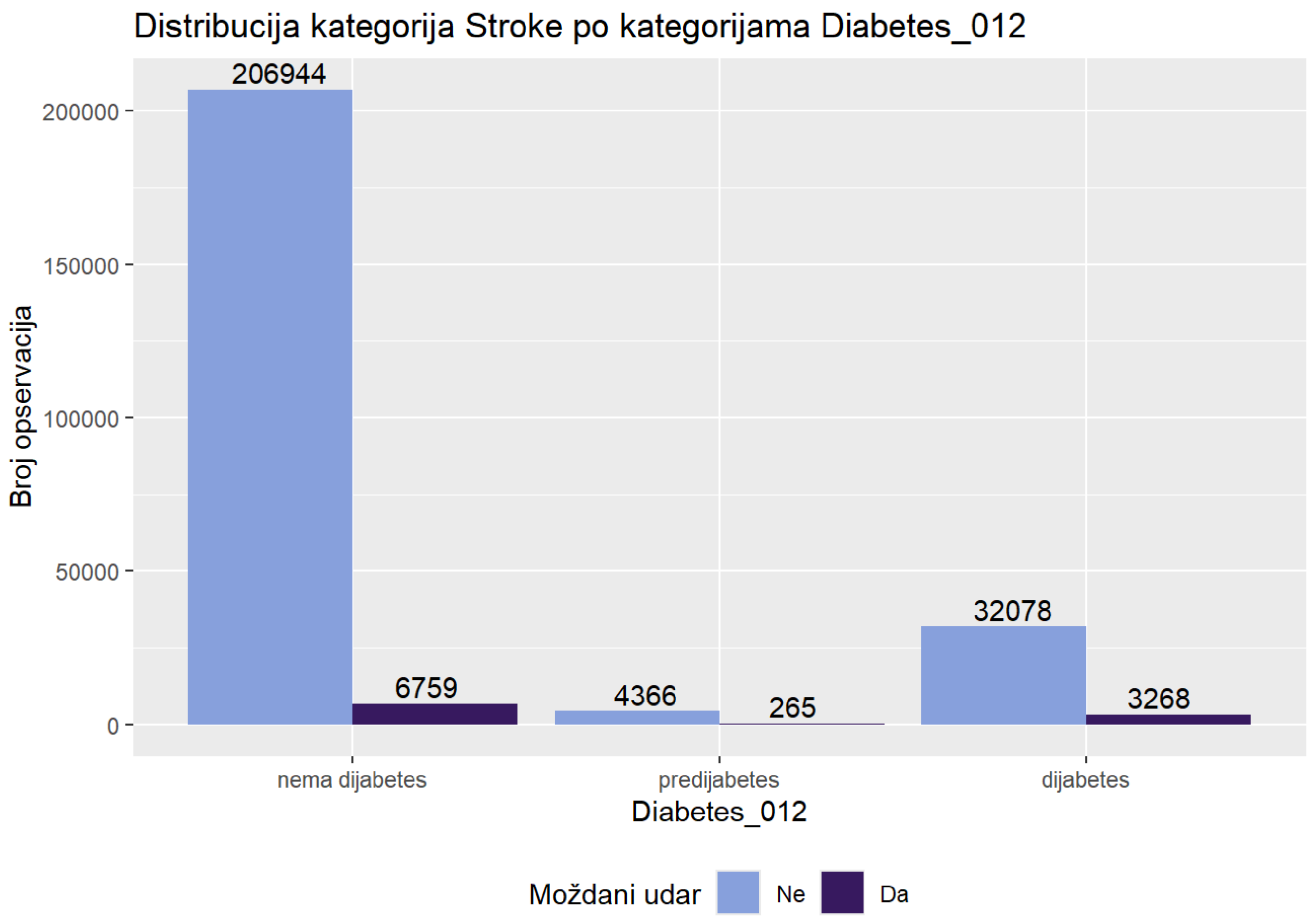

```
chi_sq_test(data$Smoker, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 1010.5, df = 2, p-value < 2.2e-16
```

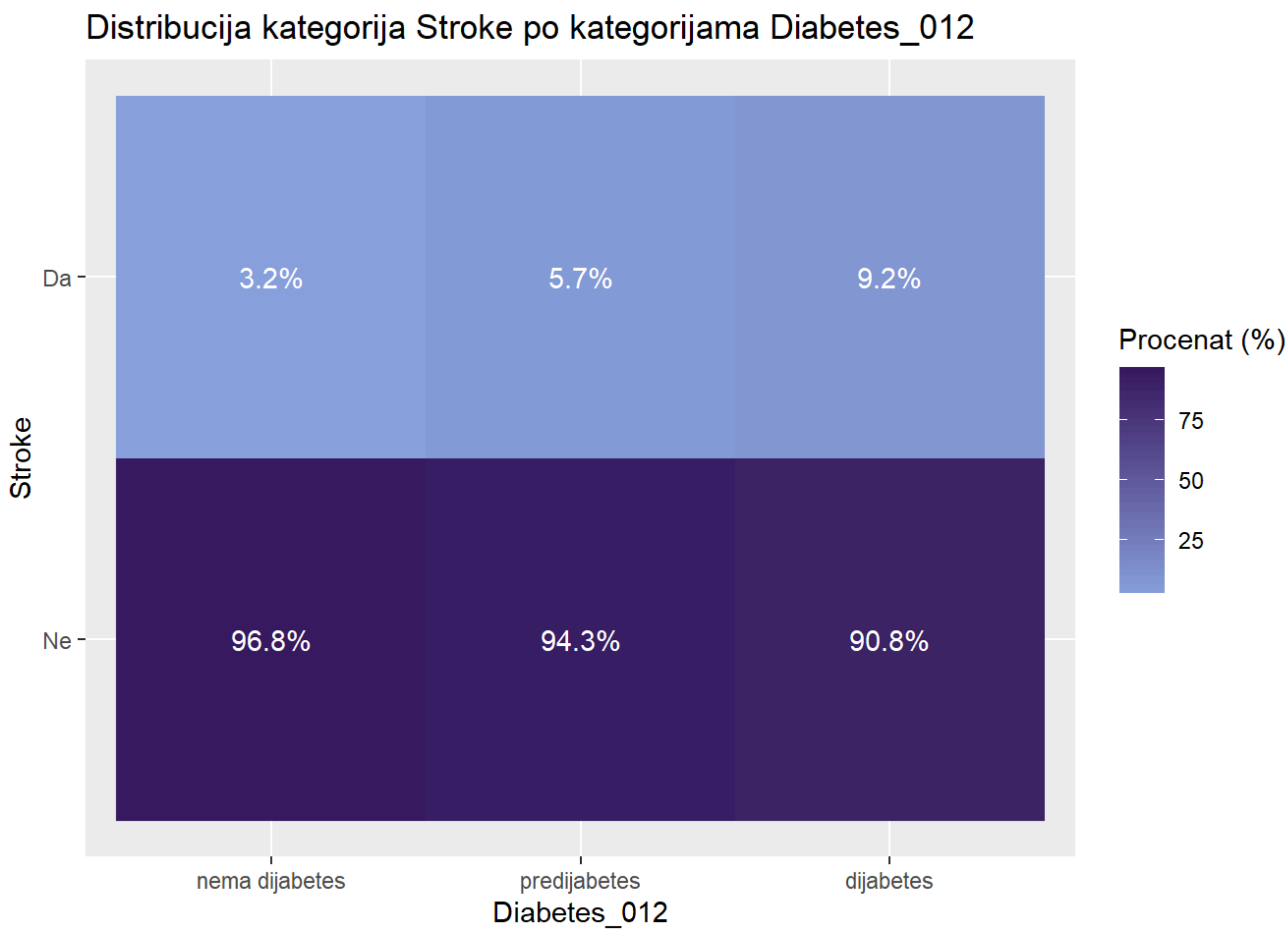
```
cramer_v(data$Smoker, data$Diabetes_012)
```

```
## [1] 0.06311427
```

```
ggplot(data, aes(x = Diabetes_012, fill = Stroke))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija Stroke po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Moždani udar"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault")+
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  )+
  theme(legend.position = "bottom")
```



```
data %>%
  count(Diabetes_012, Stroke) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = Stroke, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija Stroke po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "Stroke",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



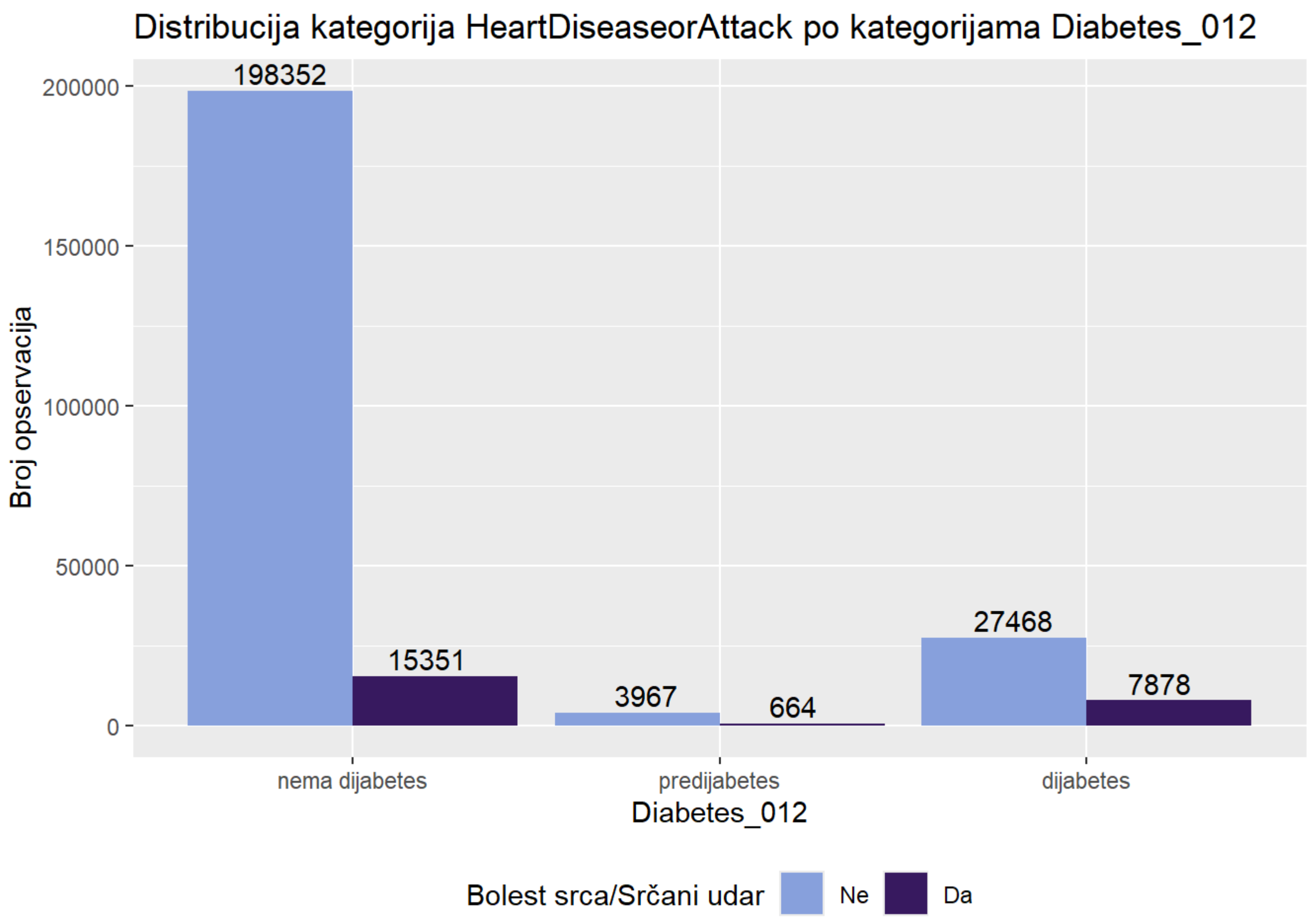

```
chi_sq_test(data$Stroke, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 2916.8, df = 2, p-value < 2.2e-16
```

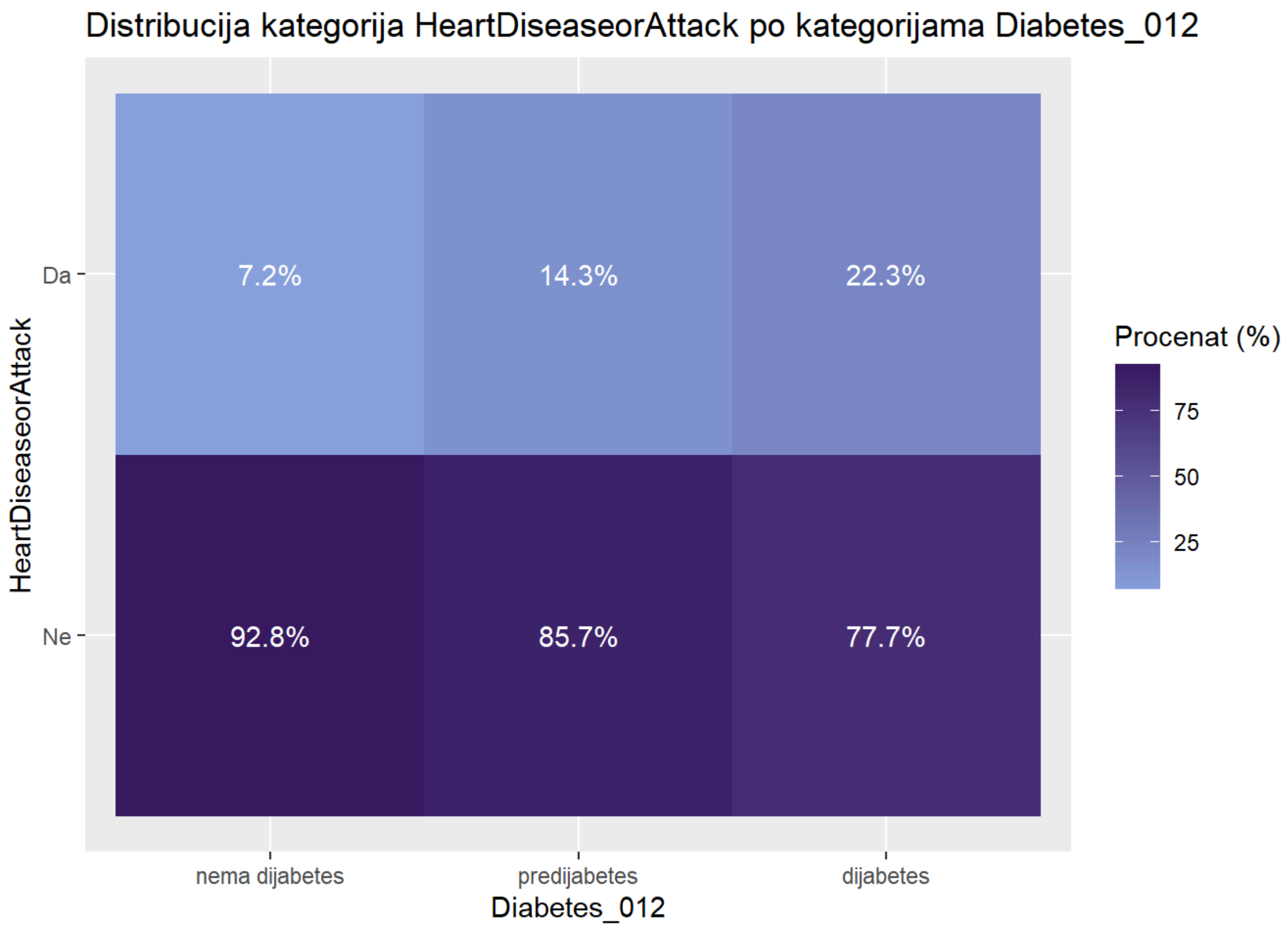
```
cramer_v(data$Stroke, data$Diabetes_012)
```

```
## [1] 0.1072276
```

```
ggplot(data, aes(x = Diabetes_012, fill = HeartDiseaseorAttack))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija HeartDiseaseorAttack po kategorijama Diabetes_01
2",
    y = "Broj opservacija",
    fill = "Bolest srca/Srčani udar"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault")+
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  )+
  theme(legend.position = "bottom")
```



```
data %>%
  count(Diabetes_012, HeartDiseaseorAttack) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = HeartDiseaseorAttack, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija HeartDiseaseorAttack po kategorijama Diabetes_01
2",
    x = "Diabetes_012",
    y = "HeartDiseaseorAttack",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



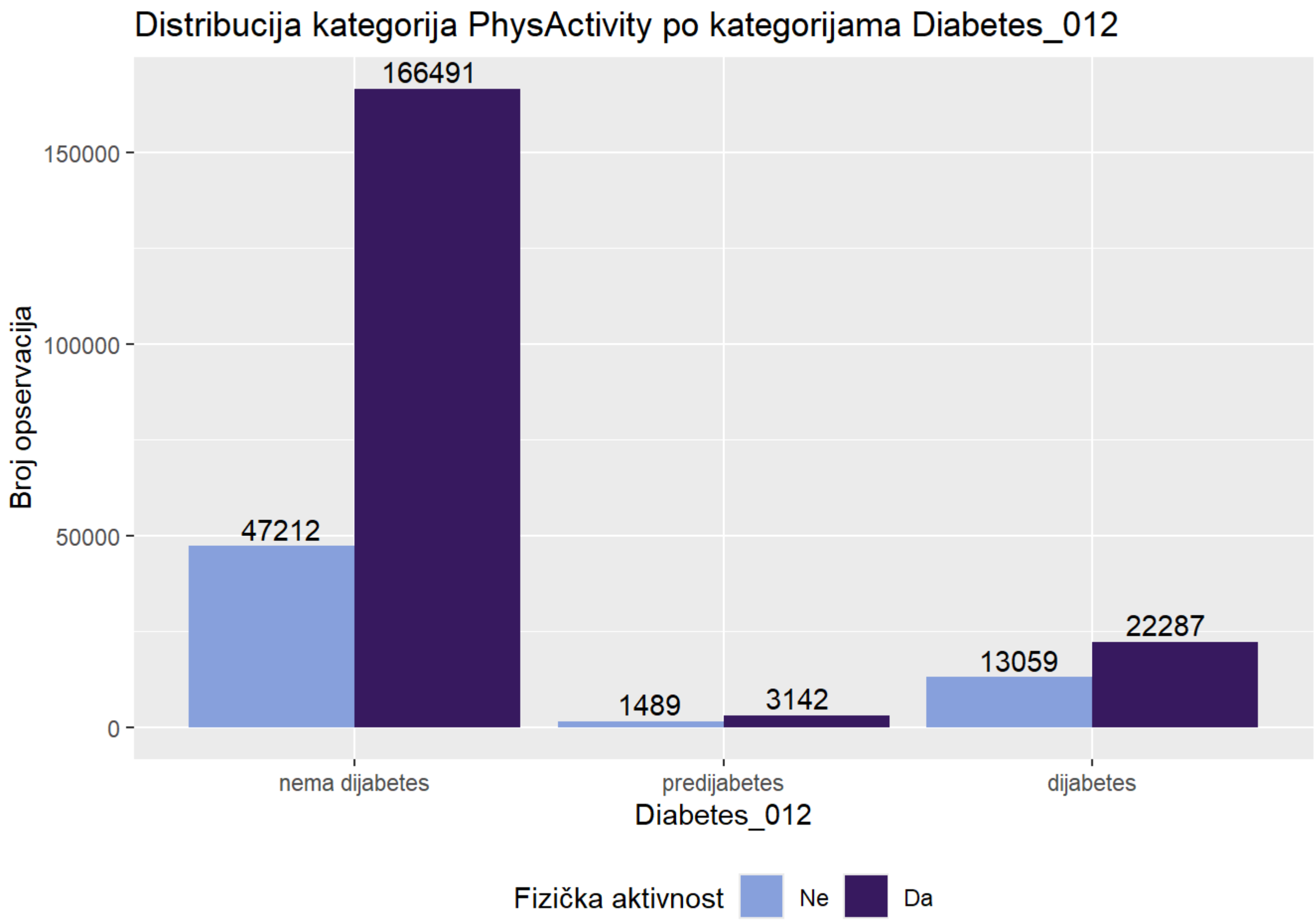

```
chi_sq_test(data$HeartDiseaseorAttack , data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 8244.9, df = 2, p-value < 2.2e-16
```

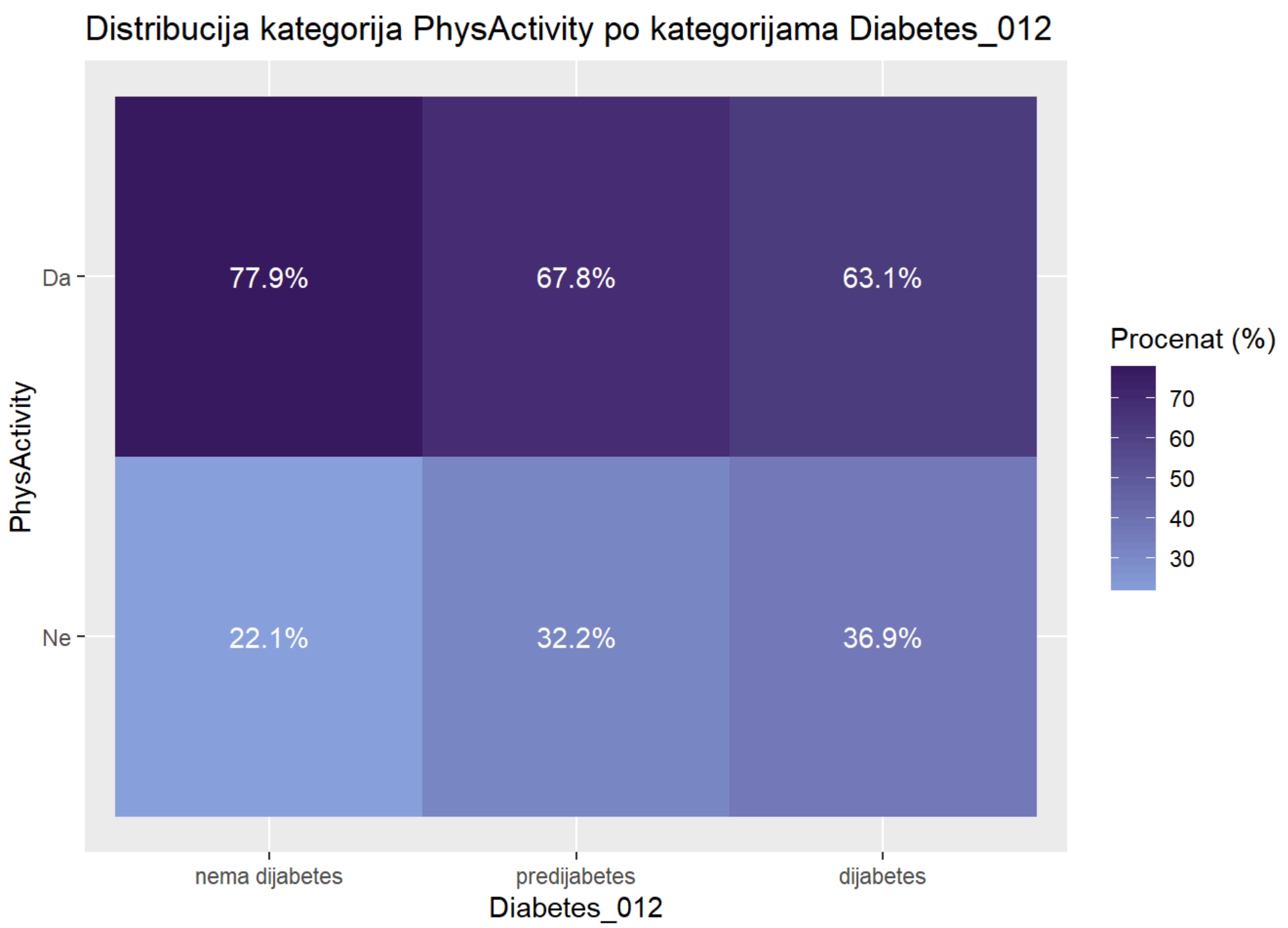
```
cramer_v(data$HeartDiseaseorAttack , data$Diabetes_012)
```

```
## [1] 0.1802807
```

```
ggplot(data, aes(x = Diabetes_012, fill = PhysActivity))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija PhysActivity po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Fizička aktivnost"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault")+
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  )+
  theme(legend.position = "bottom")
```



```
data %>%
  count(Diabetes_012, PhysActivity) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = PhysActivity, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija PhysActivity po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "PhysActivity",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



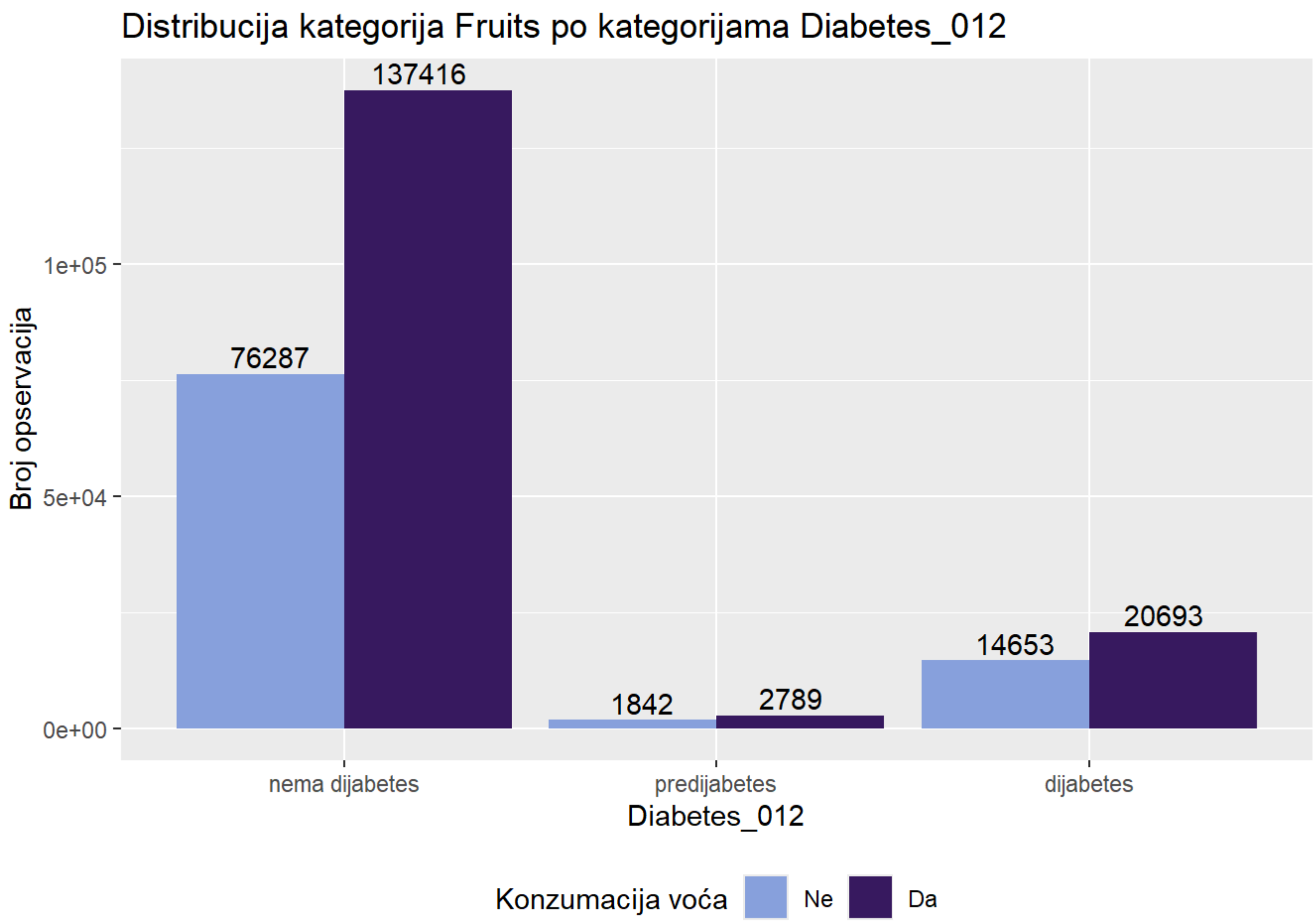

```
chi_sq_test(data$PhysActivity , data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 3789.3, df = 2, p-value < 2.2e-16
```

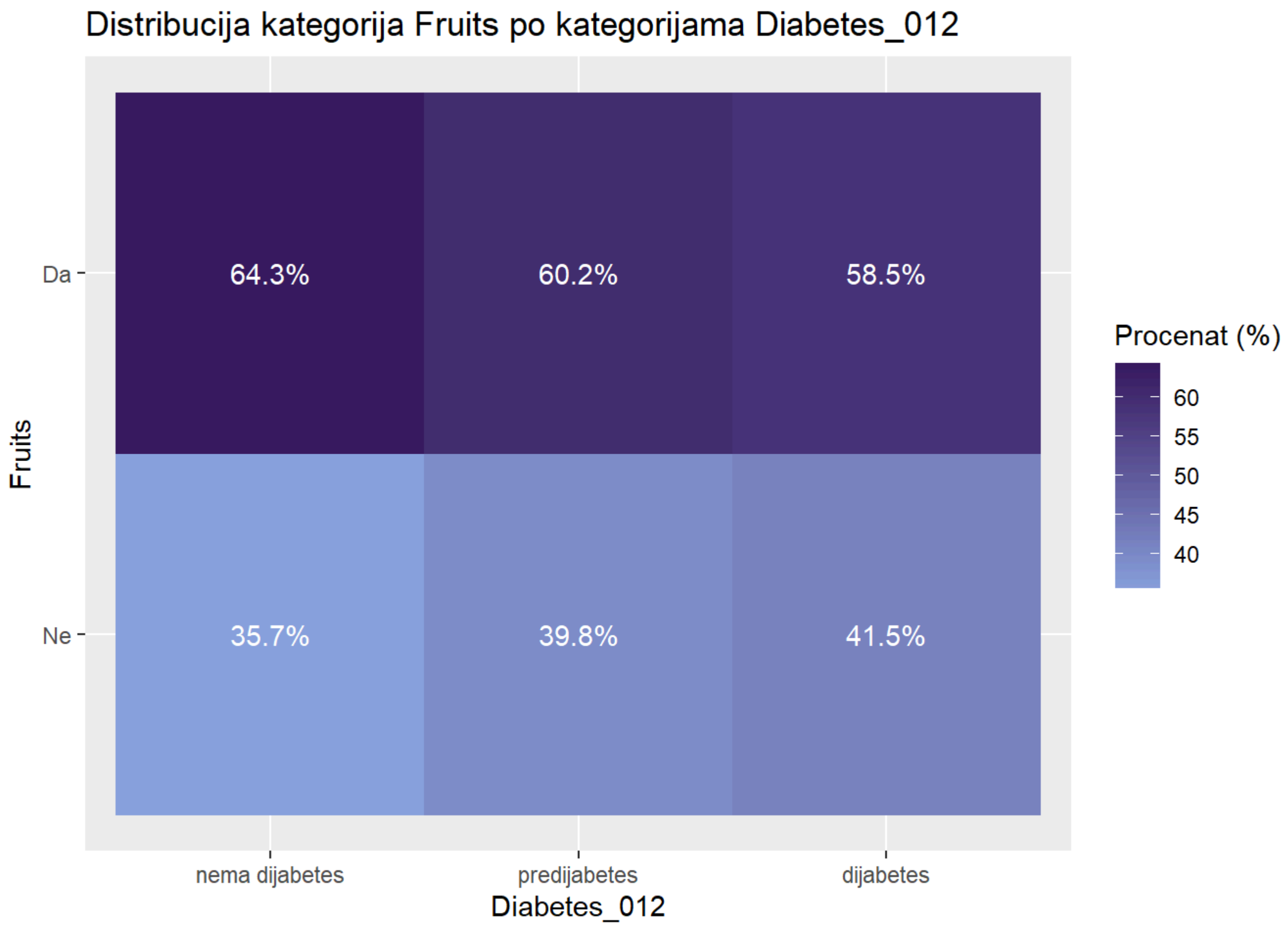
```
cramer_v(data$PhysActivity , data$Diabetes_012)
```

```
## [1] 0.1222184
```

```
ggplot(data, aes(x = Diabetes_012, fill = Fruits))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija Fruits po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Konzumacija voća"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault")+
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  )+
  theme(legend.position = "bottom")
```



```
data %>%
  count(Diabetes_012, Fruits) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = Fruits, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija Fruits po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "Fruits",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



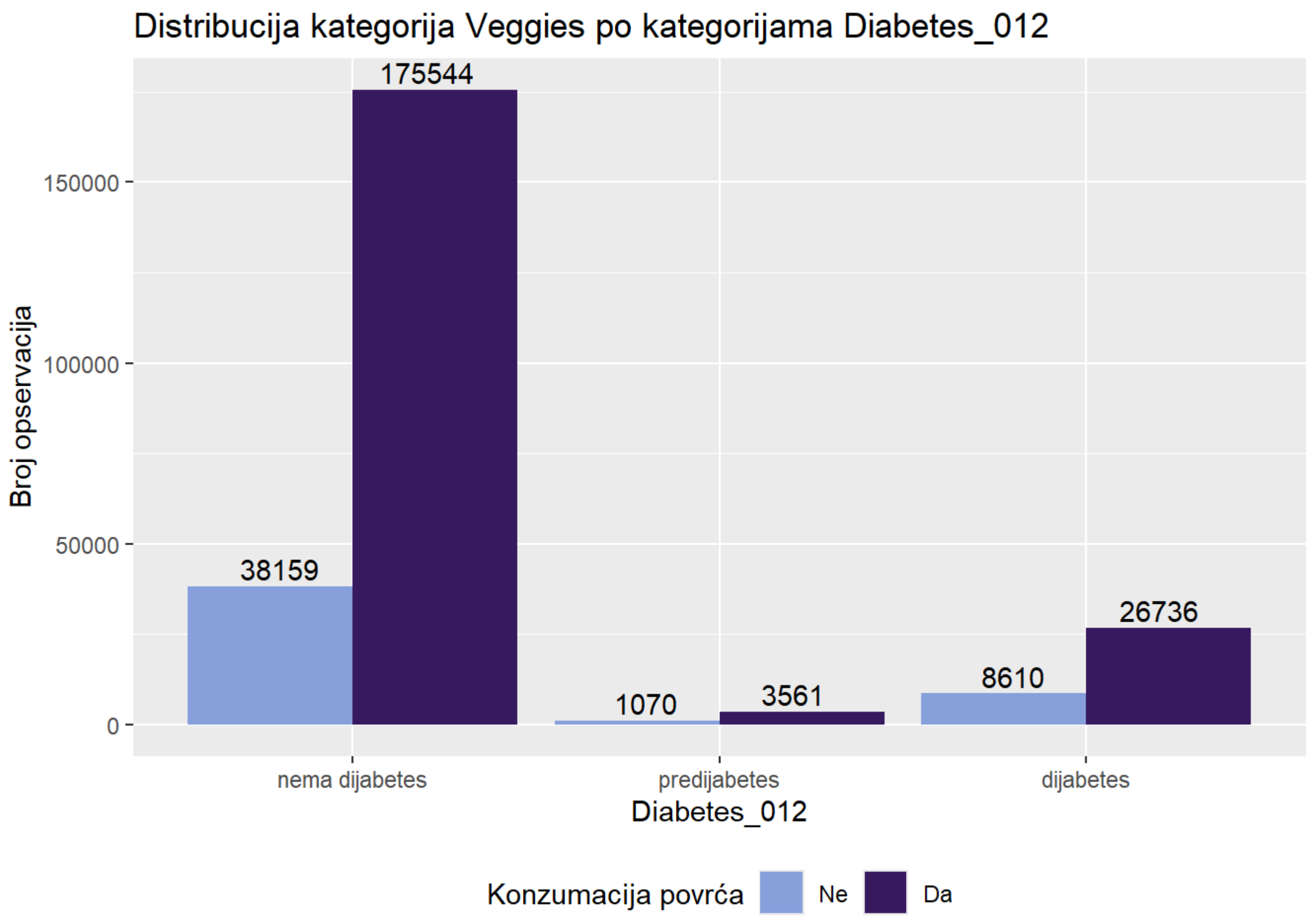

```
chi_sq_test(data$Fruits , data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 454.35, df = 2, p-value < 2.2e-16
```

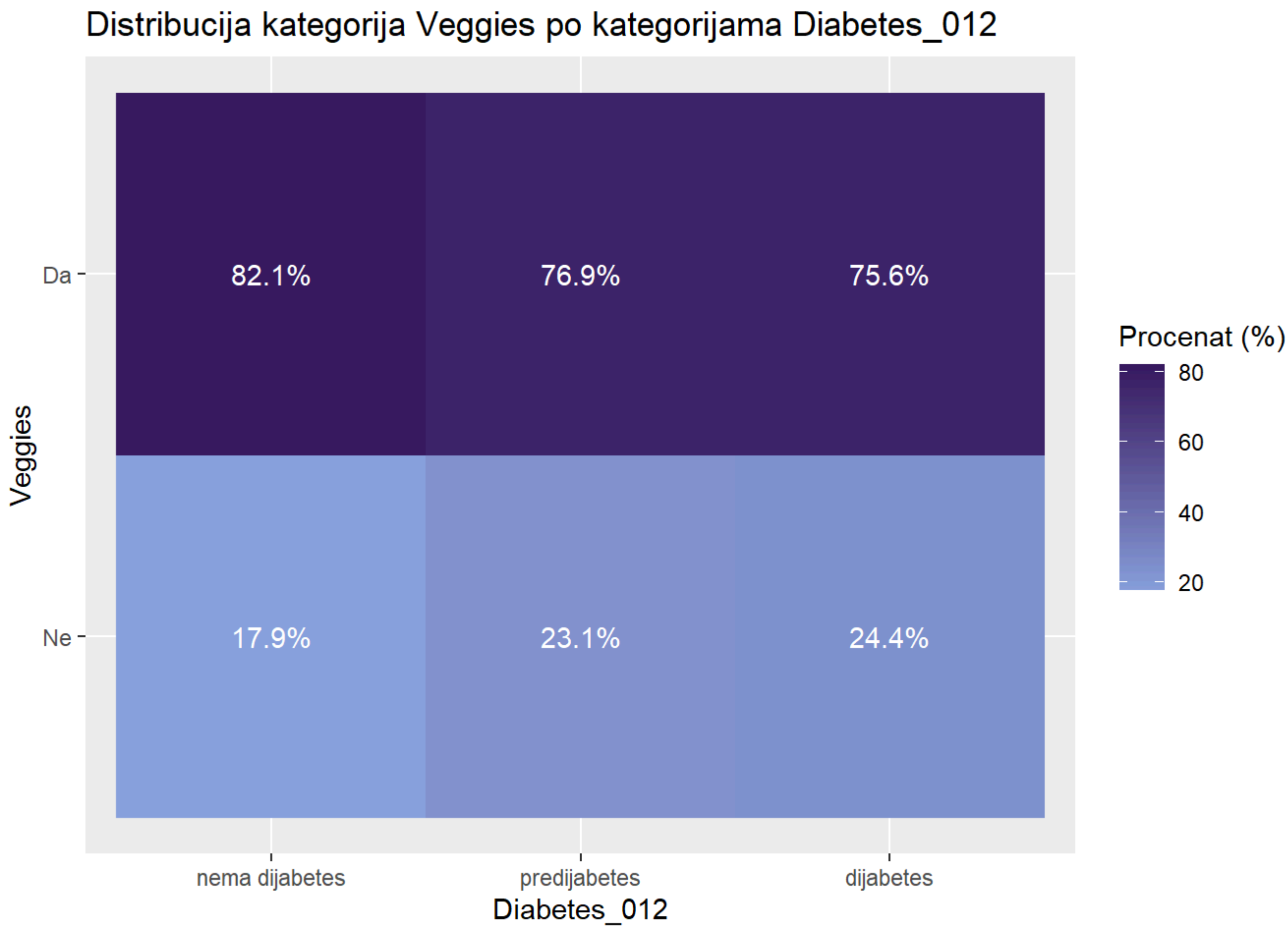
```
cramer_v(data$Fruits , data$Diabetes_012)
```

```
## [1] 0.0423205
```

```
ggplot(data, aes(x = Diabetes_012, fill = Veggies))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija Veggies po kategorijama Diabetes_012",
    y = "Broj opservacija",
    fill = "Konzumacija povrća"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault")+
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  )+
  theme(legend.position = "bottom")
```



```
data %>%
  count(Diabetes_012, Veggies) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = Veggies, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija Veggies po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "Veggies",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



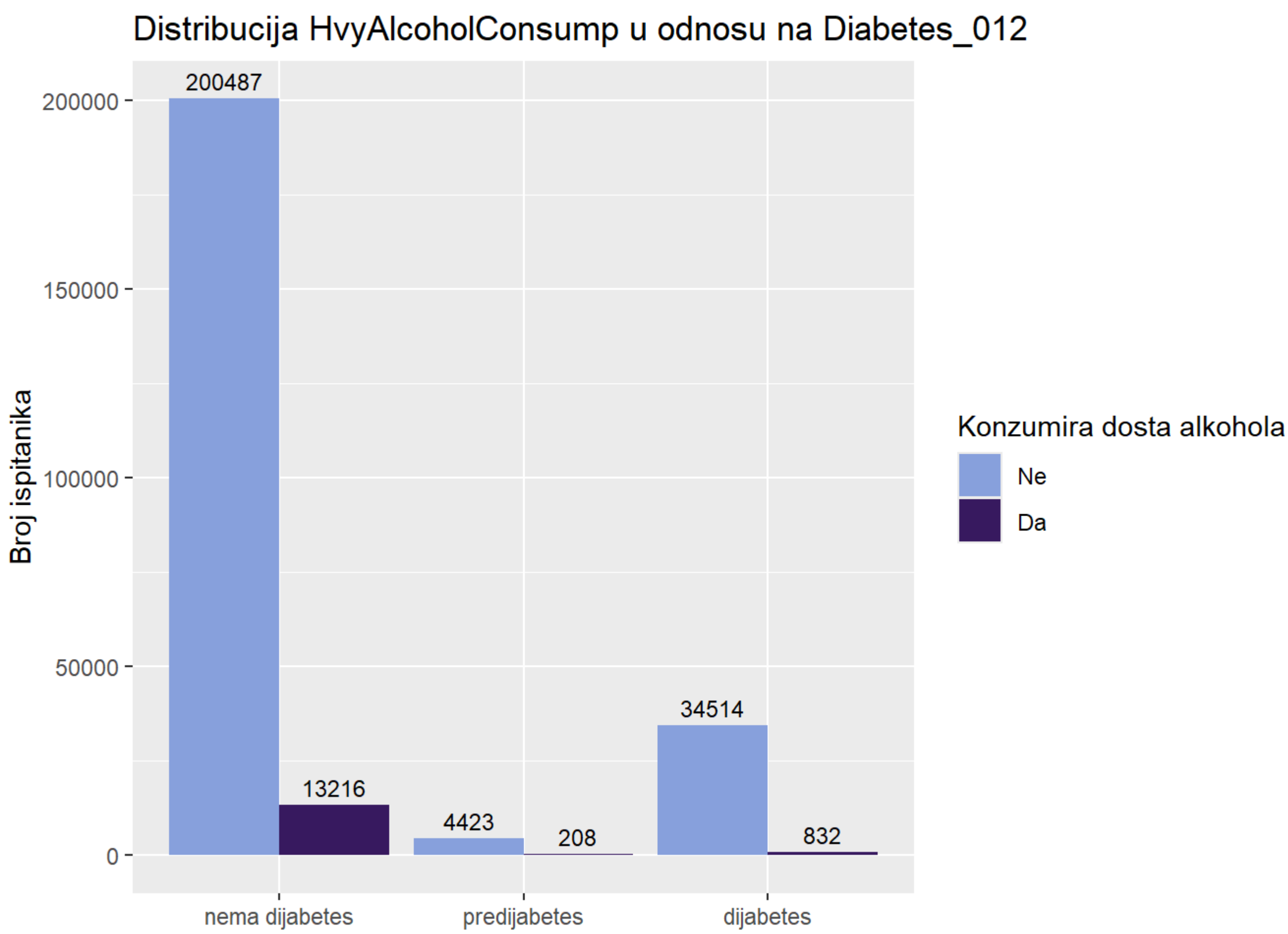

```
chi_sq_test(data$Veggies, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 893.84, df = 2, p-value < 2.2e-16
```

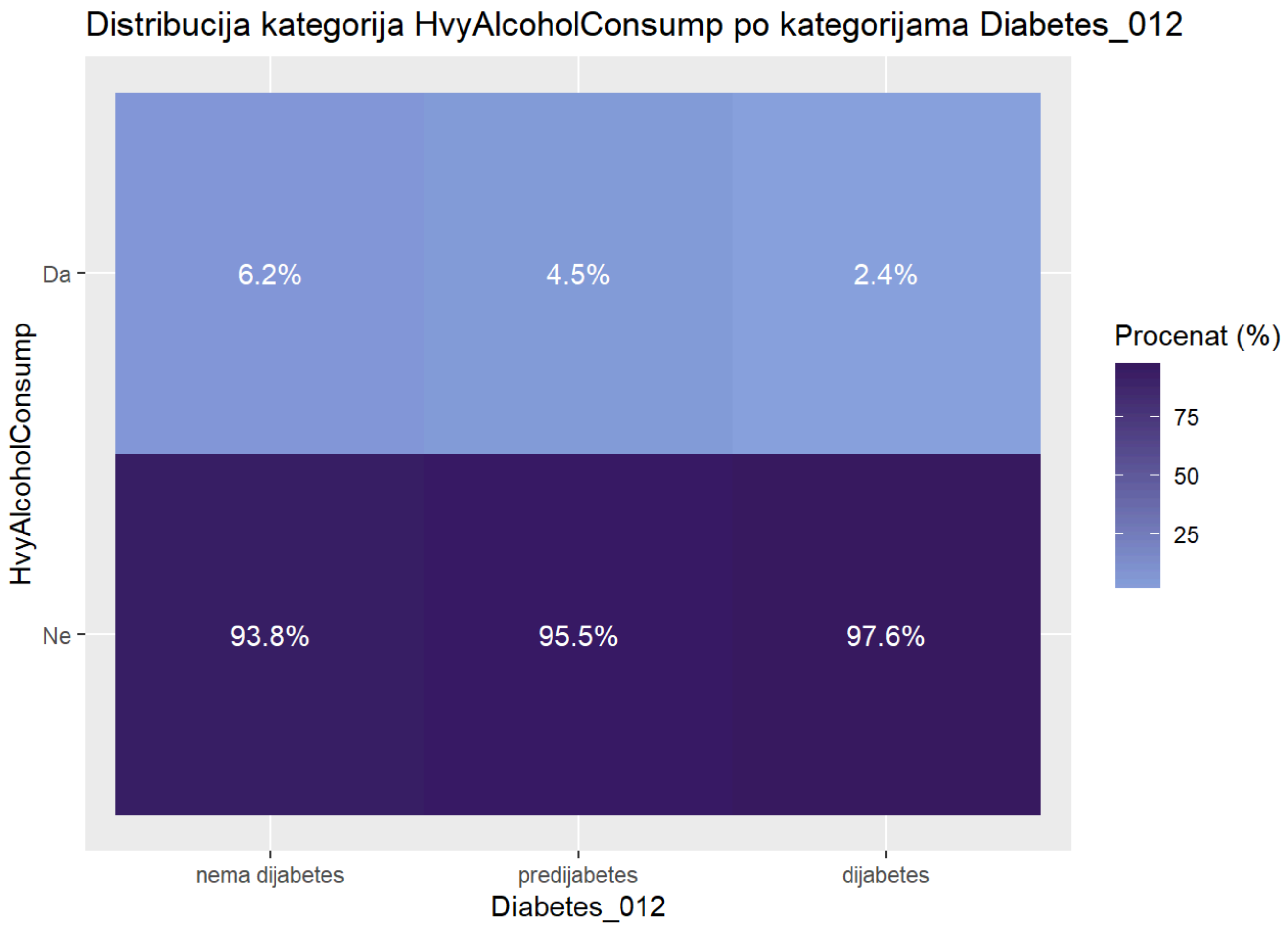
```
cramer_v(data$Veggies, data$Diabetes_012)
```

```
## [1] 0.05935909
```

```
ggplot(data, aes(x = Diabetes_012, fill = HvyAlcoholConsump)) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS)+
  labs(title = "Distribucija HvyAlcoholConsump u odnosu na Diabetes_012",
       x = NULL,
       y = "Broj ispitanika",
       fill ="Konzumira dosta alkohola")
```



```
data %>%
  count(Diabetes_012, HvyAlcoholConsump) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = HvyAlcoholConsump, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija HvyAlcoholConsump po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "HvyAlcoholConsump",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



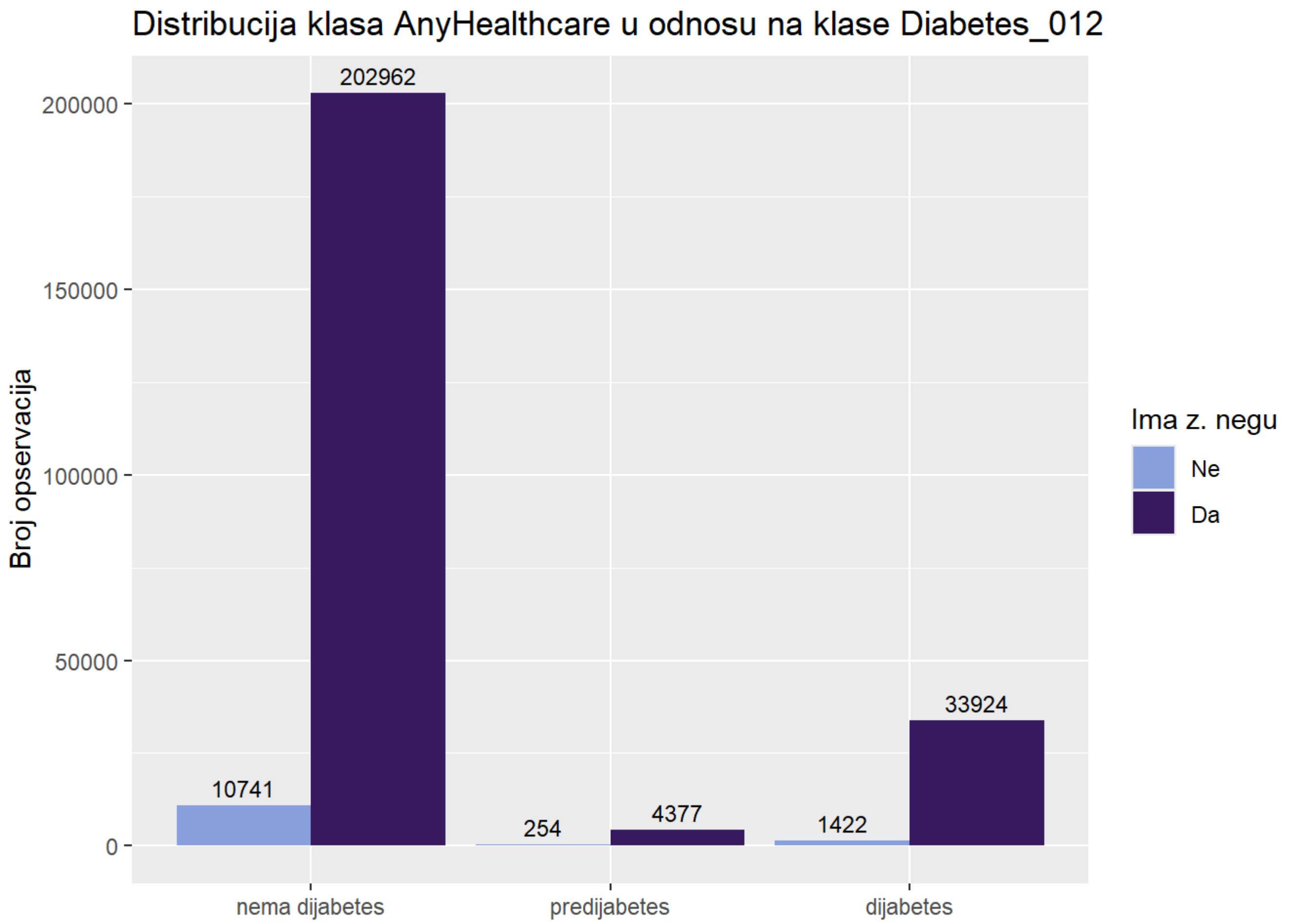

```
chi_sq_test(data$HvyAlcoholConsump, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 850.32, df = 2, p-value < 2.2e-16
```

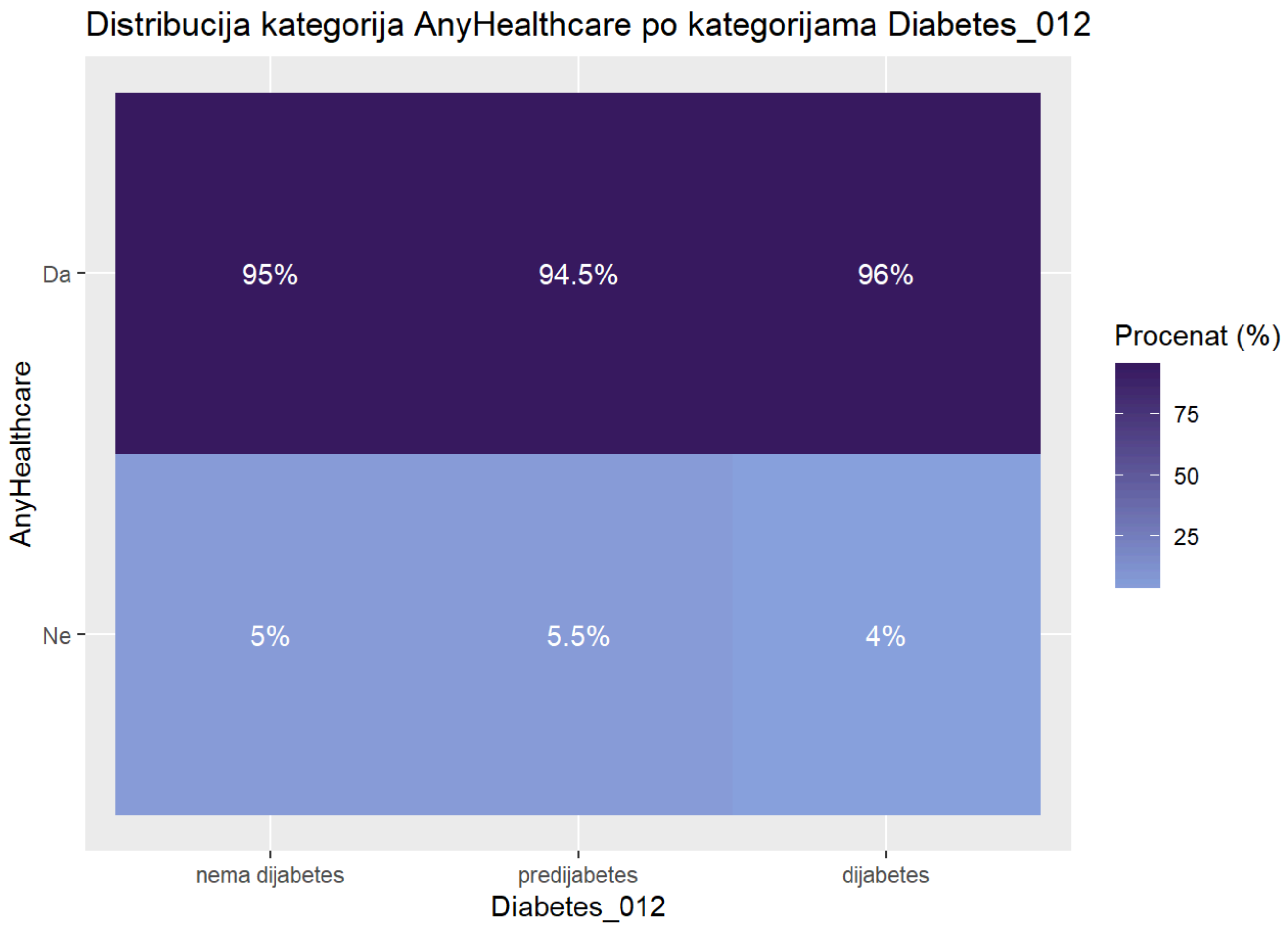
```
cramer_v(data$HvyAlcoholConsump, data$Diabetes_012)
```

```
## [1] 0.05789607
```

```
ggplot(data, aes(x = Diabetes_012, fill = AnyHealthcare)) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija klasa AnyHealthcare u odnosu na klase Diabetes_012",
       x = NULL,
       y = "Broj opservacija",
       fill = "Ima z. negu")
```



```
data %>%
  count(Diabetes_012, AnyHealthcare) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = AnyHealthcare, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija AnyHealthcare po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "AnyHealthcare",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



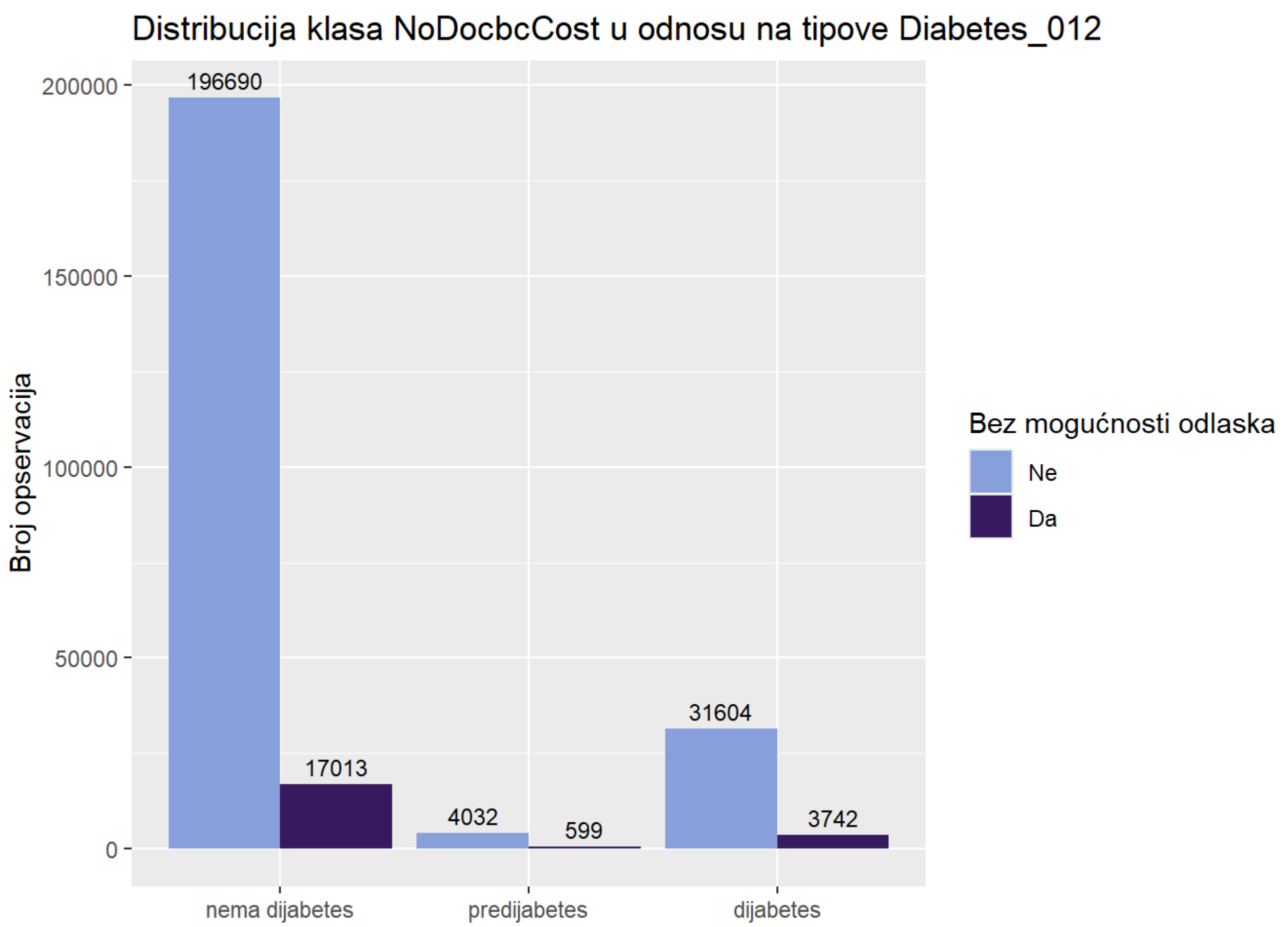

```
chi_sq_test(data$AnyHealthcare, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 69.078, df = 2, p-value = 9.998e-16
```

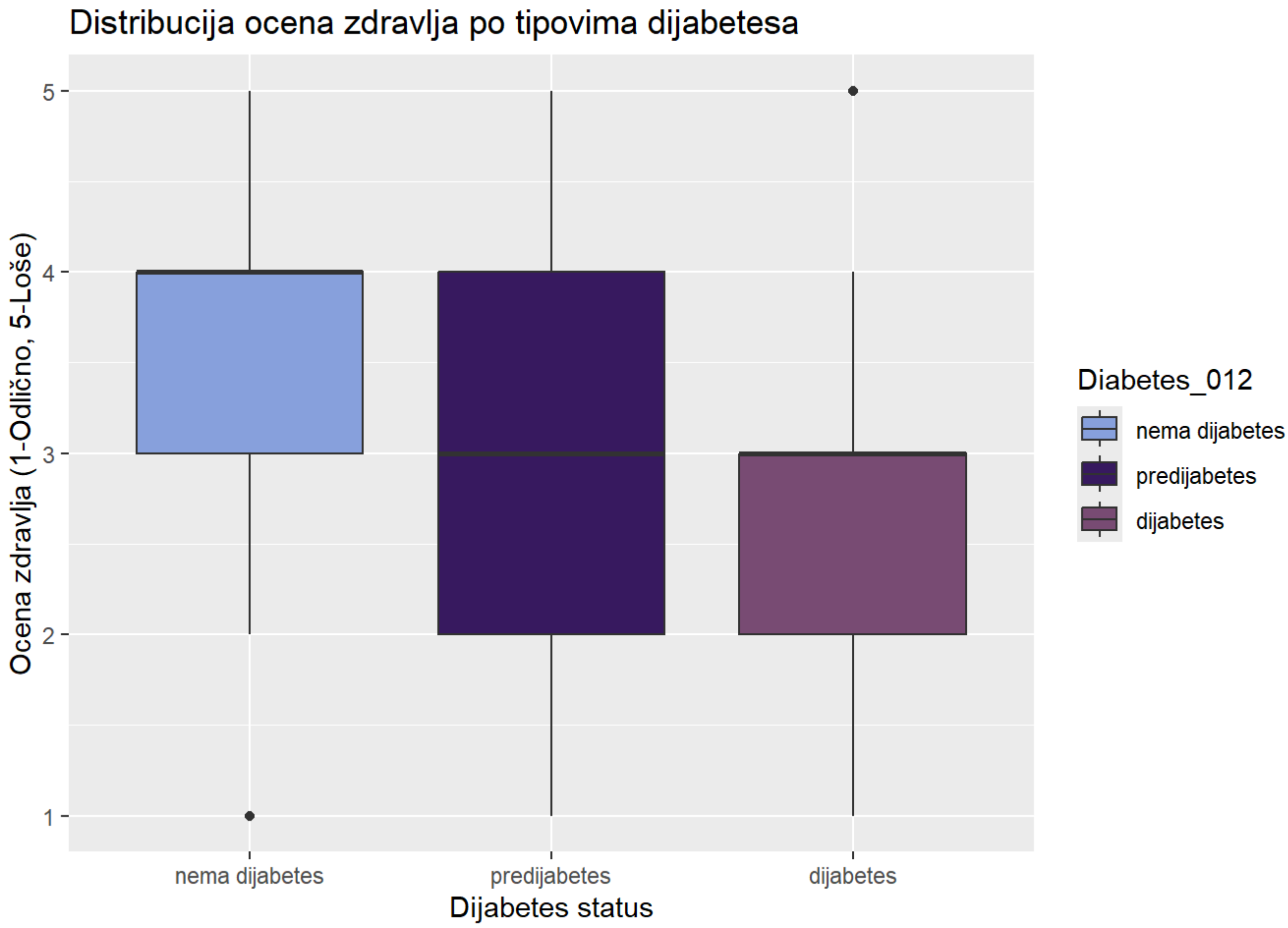
```
cramer_v(data$AnyHealthcare, data$Diabetes_012)
```

```
## [1] 0.01650162
```

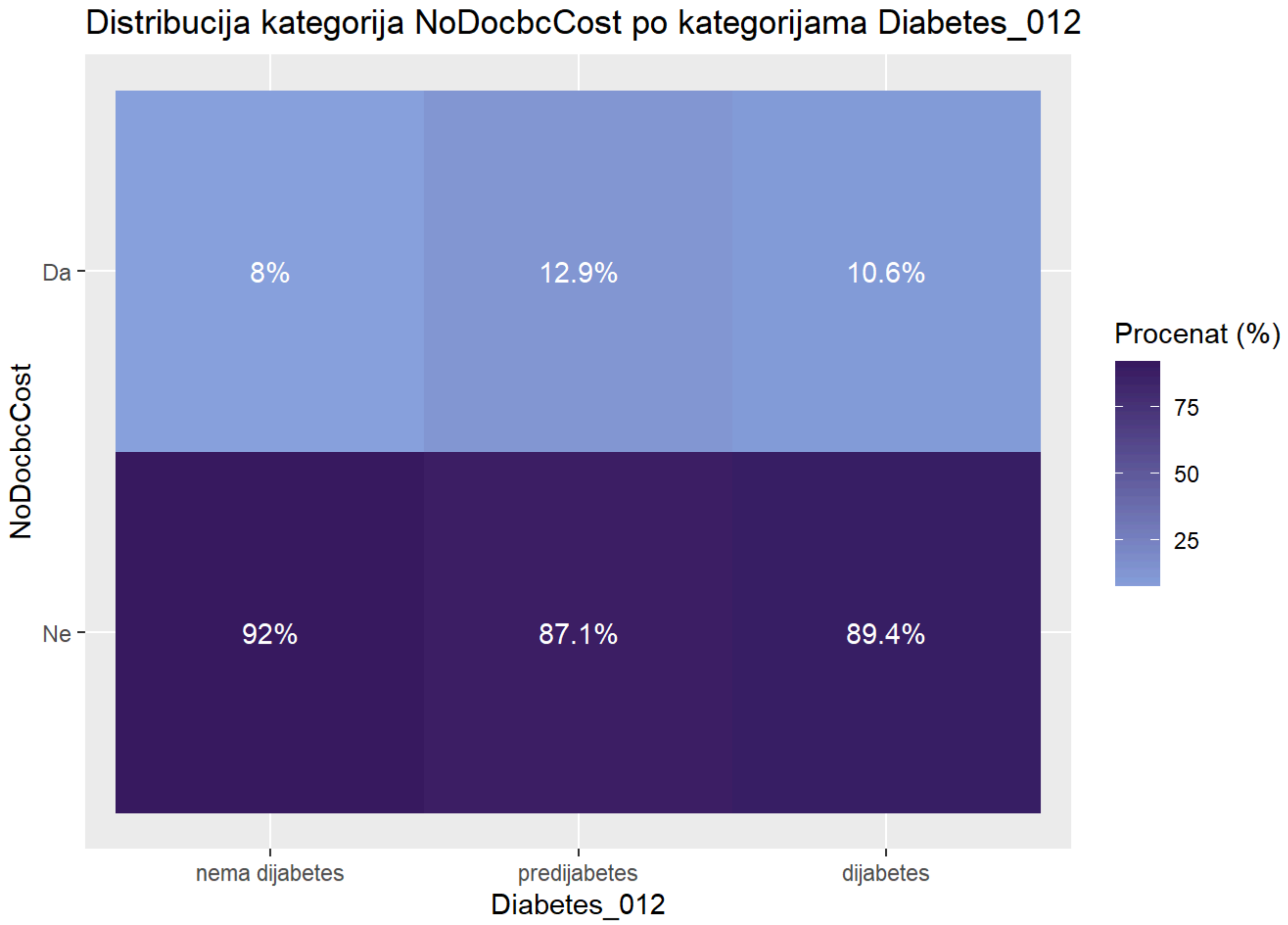
```
ggplot(data, aes(x = Diabetes_012, fill = NoDocbcCost)) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija klasa NoDocbcCost u odnosu na tipove Diabetes_012",
       x = NULL,
       y = "Broj opservacija",
       fill = "Bez mogućnosti odlaska")
```




```
ggplot(data, aes(x = Diabetes_012, y = as.numeric(GenHlth), fill = Diabetes_012)) +
  geom_boxplot() +
  scale_fill_manual(values=colorS)+
  labs(title = "Distribucija ocena zdravlja po tipovima dijabetesa",
       y = "Ocena zdravlja (1-Odlično, 5-Loše)",
       x = "Dijabetes status")
```




```
data %>%
  count(Diabetes_012, NoDocbcCost) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = NoDocbcCost, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija NoDocbcCost po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "NoDocbcCost",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



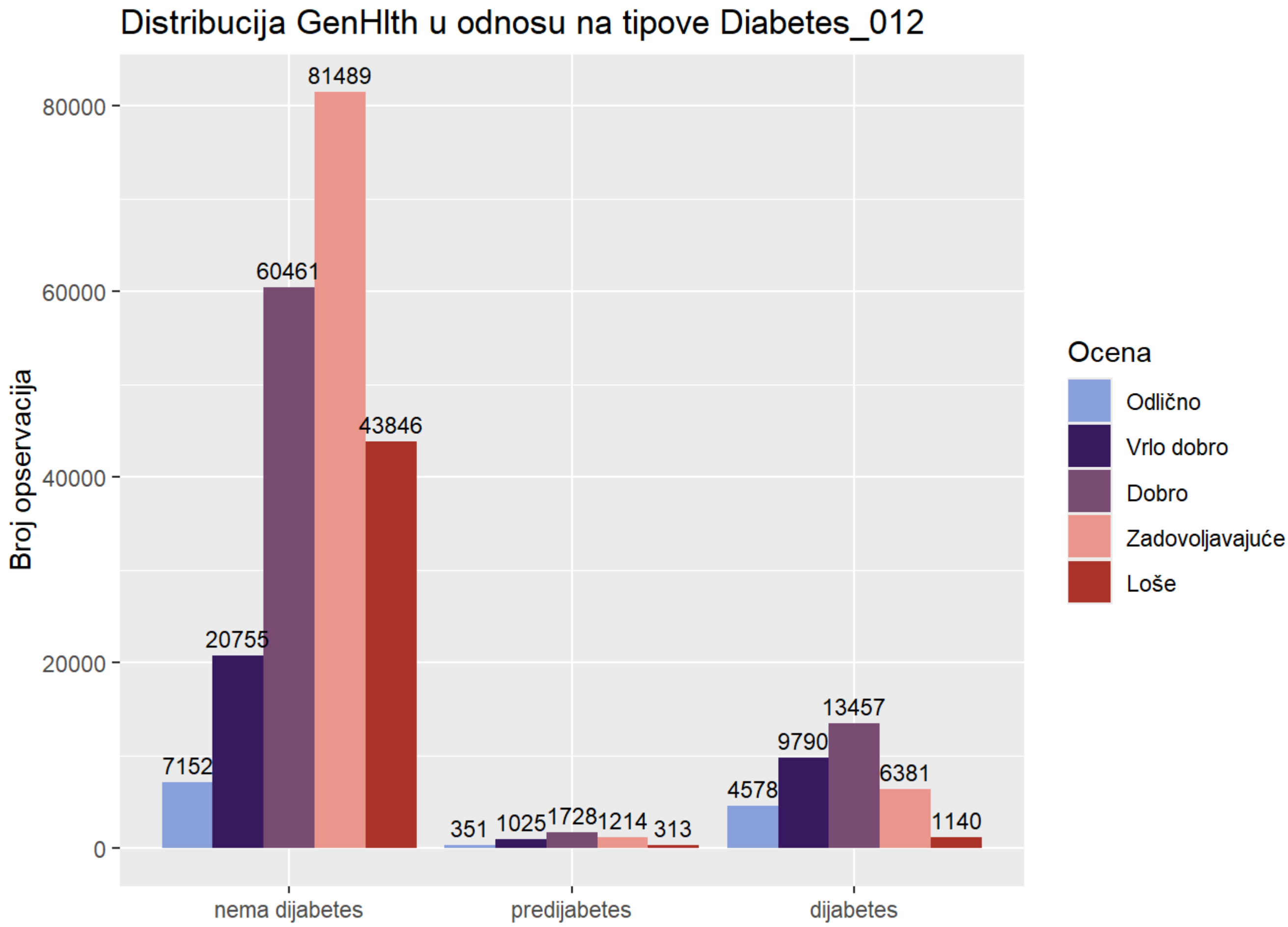

```
chi_sq_test(data$NoDocbcCost, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 396.08, df = 2, p-value < 2.2e-16
```

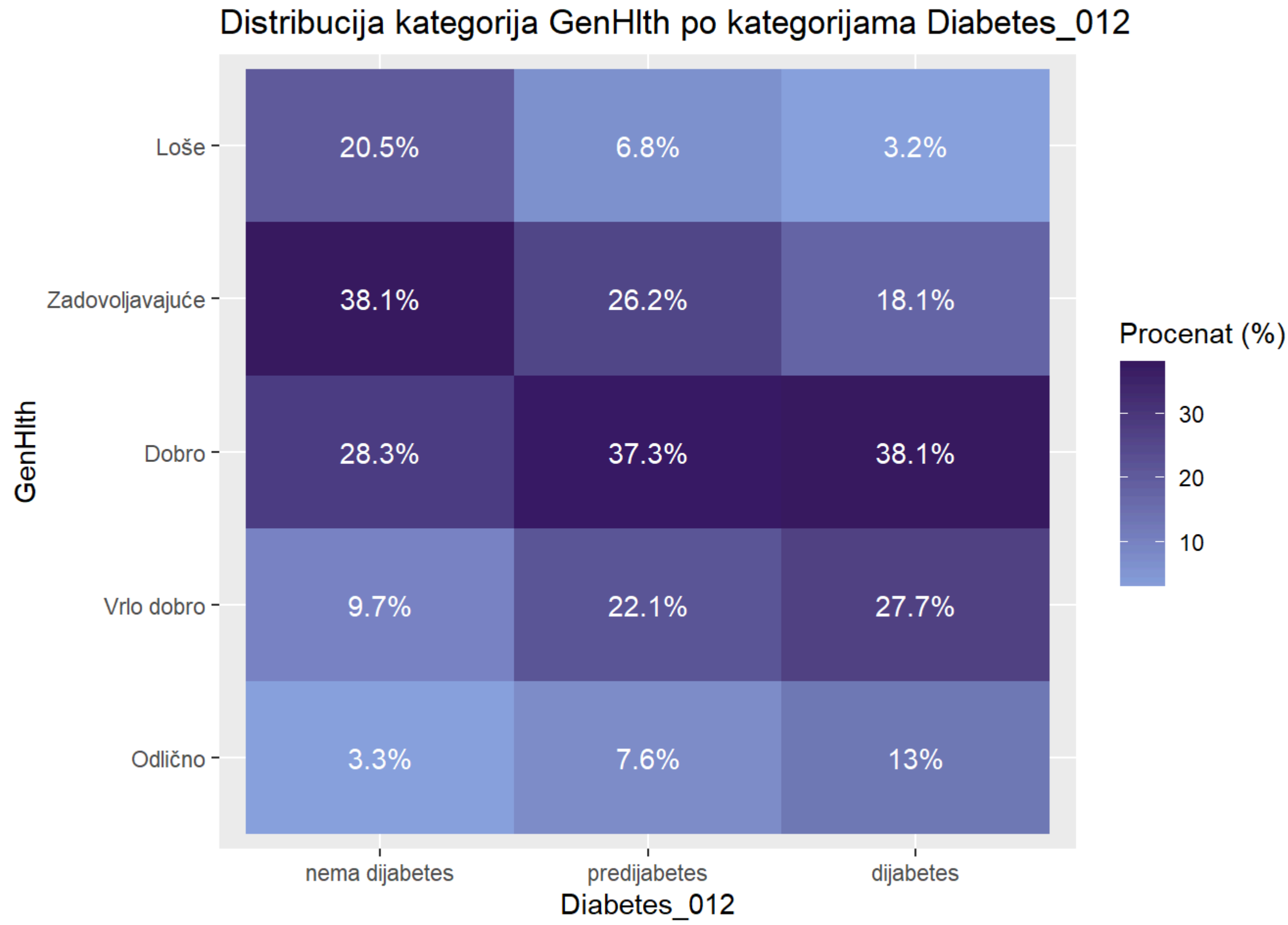
```
cramer_v(data$NoDocbcCost, data$Diabetes_012)
```

```
## [1] 0.03951385
```

```
ggplot(data, aes(x = Diabetes_012, fill = GenHlth)) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija GenHlth u odnosu na tipove Diabetes_012",
       x = NULL,
       y = "Broj opservacija",
       fill = "Ocena")
```




```
data %>%
  count(Diabetes_012, GenHlth) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = GenHlth, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija GenHlth po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "GenHlth",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



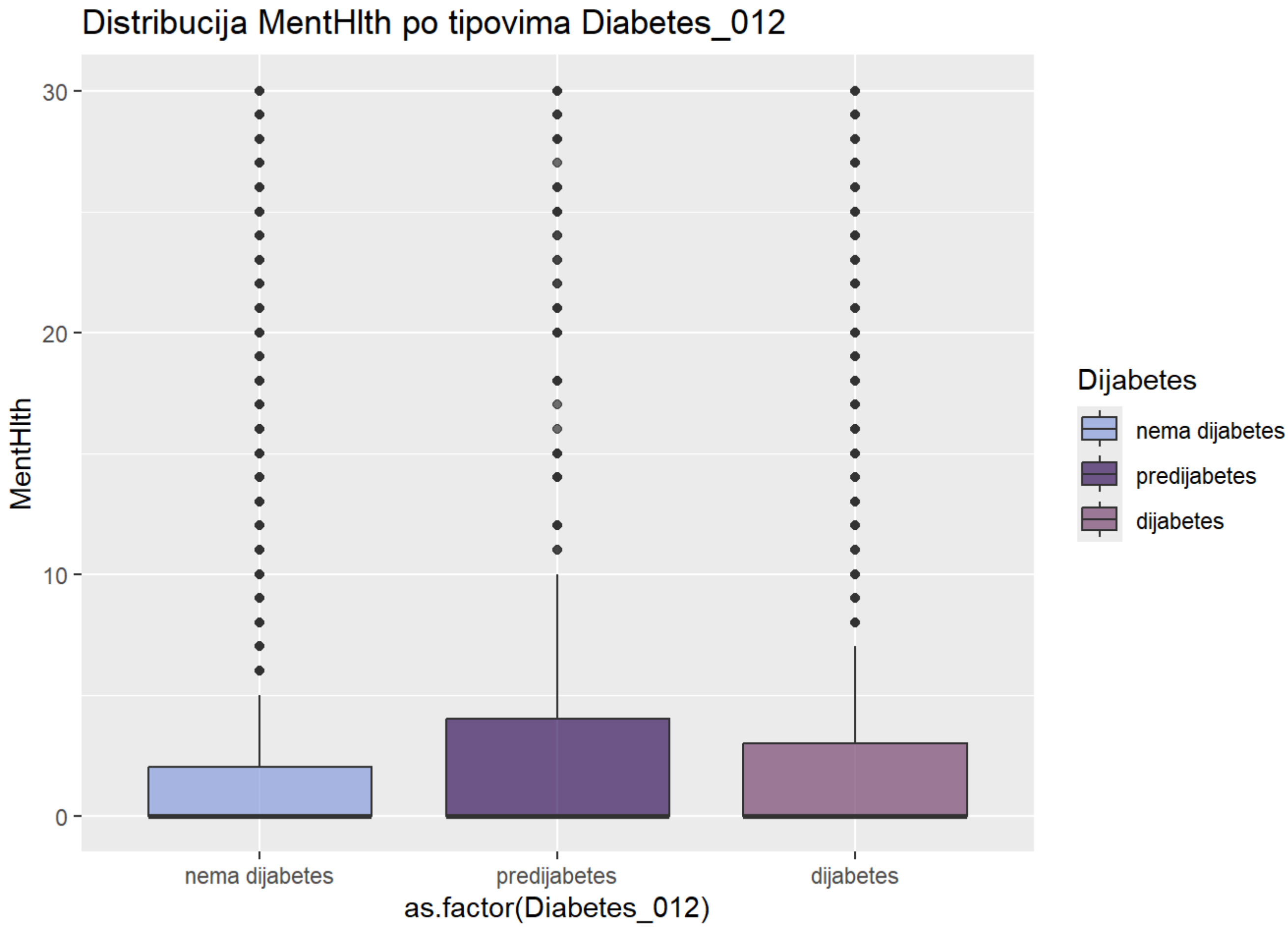

```
chi_sq_test(data$GenHlth, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 24248, df = 8, p-value < 2.2e-16
```

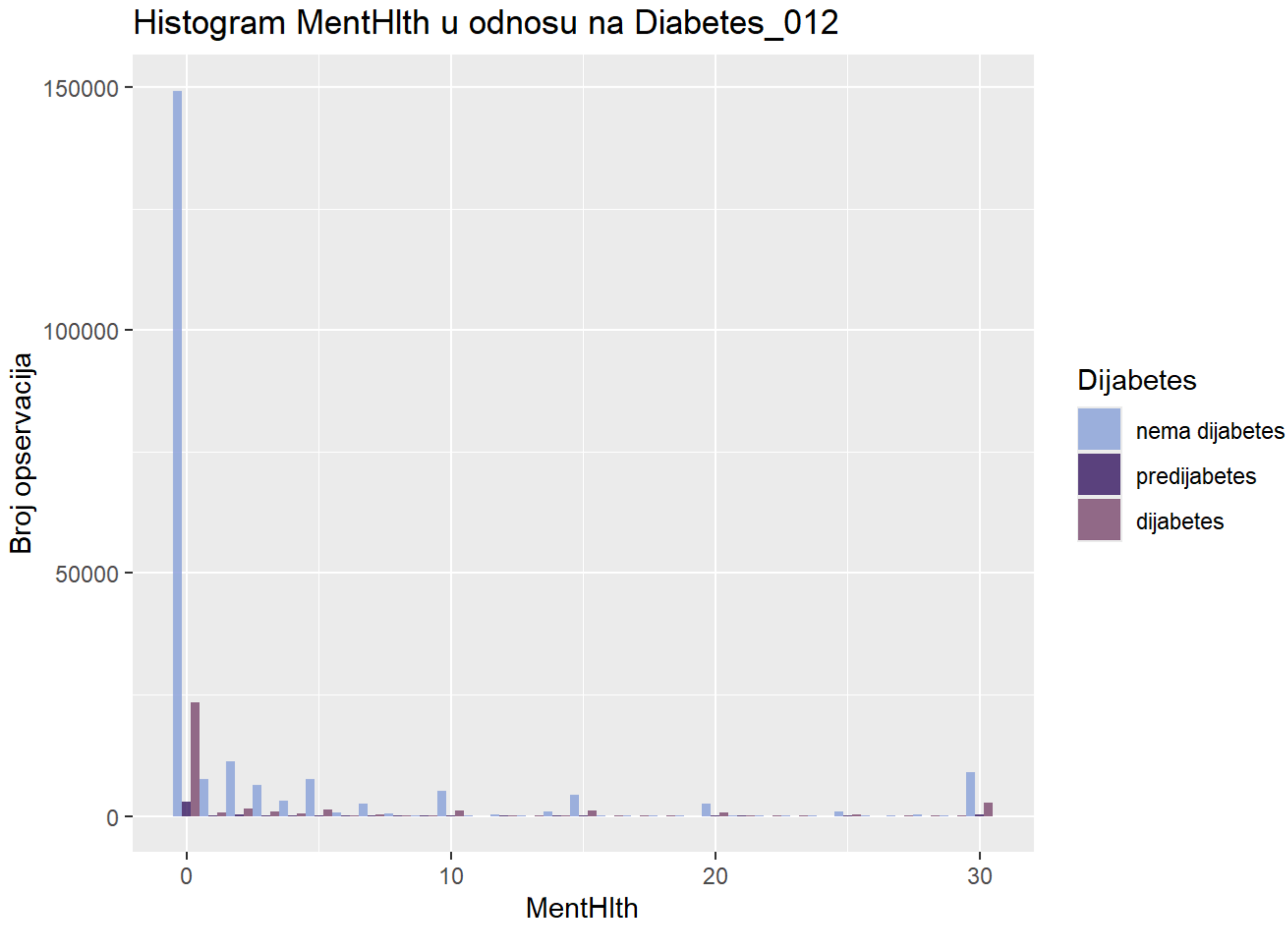
```
cramer_v(data$GenHlth, data$Diabetes_012)
```

```
## [1] 0.2186154
```

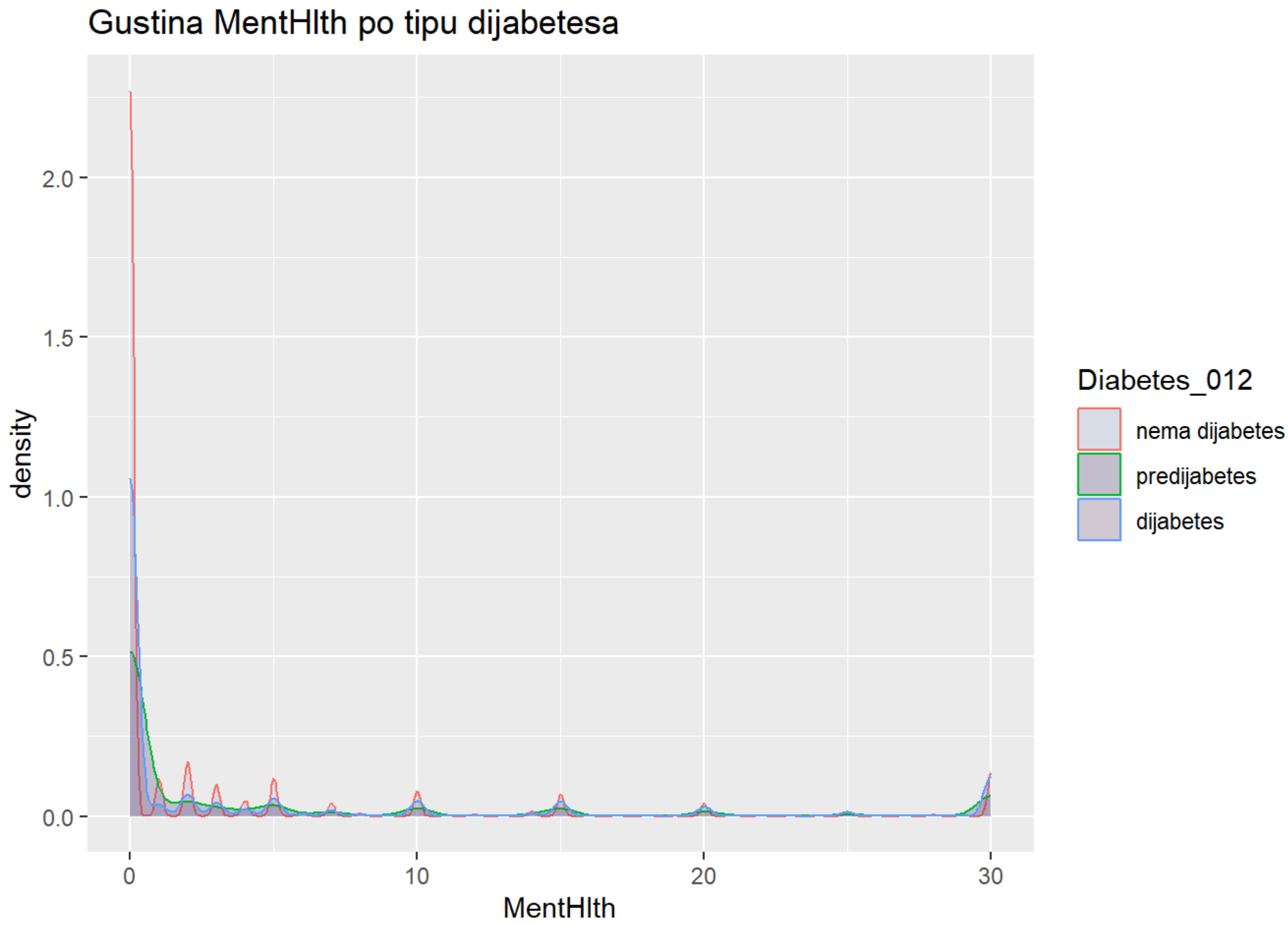
```
ggplot(data, aes(x = as.factor(Diabetes_012), y = MentHlth, fill = as.factor(Diabetes_
012))) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values=colorS) +
  labs(title = "Distribucija MentHlth po tipovima Diabetes_012", fill = "Dijabetes")
```




```
ggplot(data, aes(x = MentHlth, fill = Diabetes_012)) +
  geom_histogram(binwidth = 1, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values =colorS) +
  labs(title = "Histogram MentHlth u odnosu na Diabetes_012",
       y = "Broj opservacija",
       fill = "Dijabetes")
```




```
ggplot(data, aes(x = MentHlth, color = Diabetes_012, fill = Diabetes_012)) +
  geom_density(alpha = 0.2) +
  scale_fill_manual(values=colorS)+
  labs(title = "Gustina MentHlth po tipu dijabetesa")
```



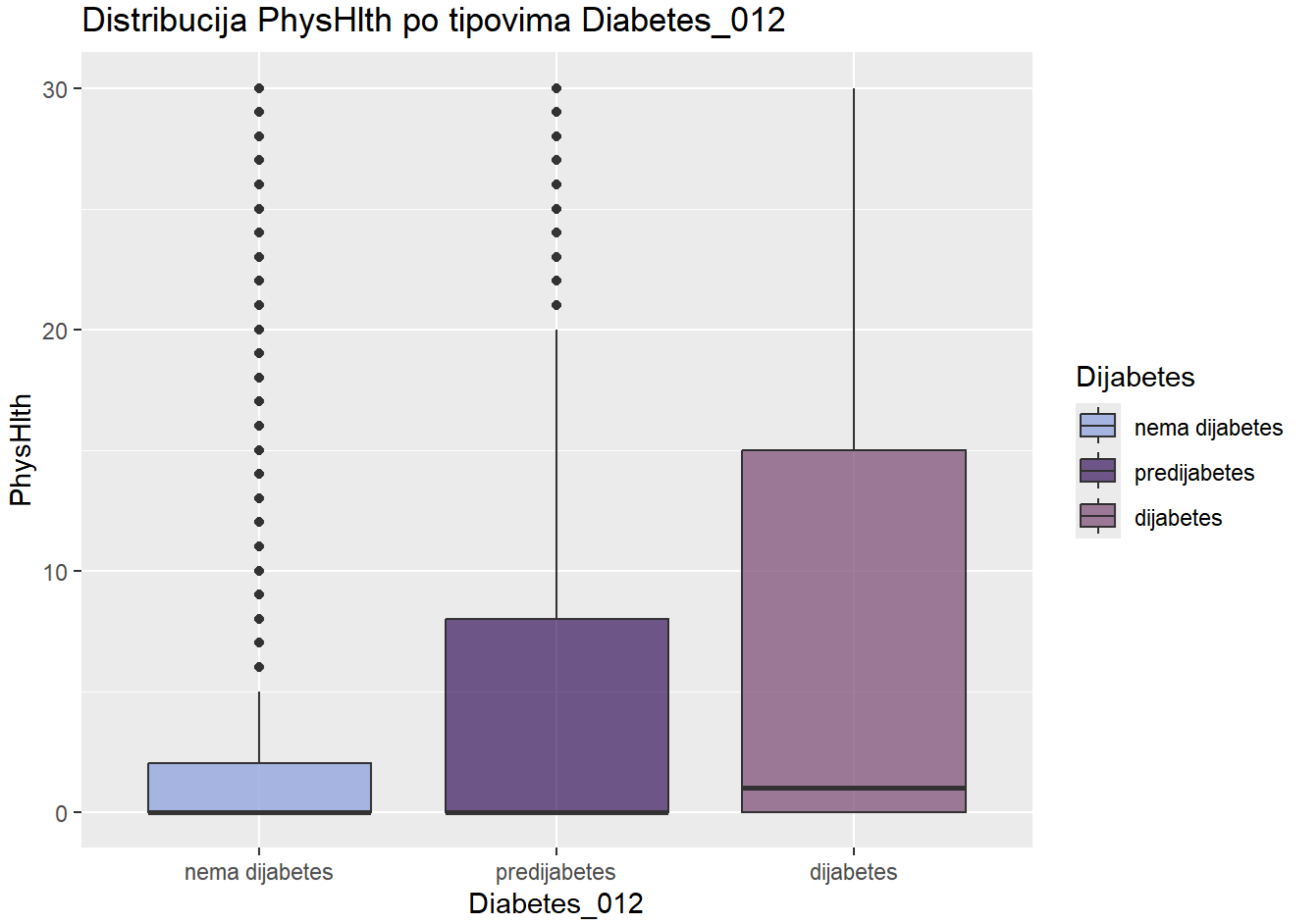

```
anova_test(data$MentHlth,data$Diabetes_012)
```

```
##                             Df   Sum Sq Mean Sq F value Pr(>F)    
## as.factor(grupna_var)        2    78369   39185   717.1 <2e-16 ***
## Residuals               253677 13861367      55                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

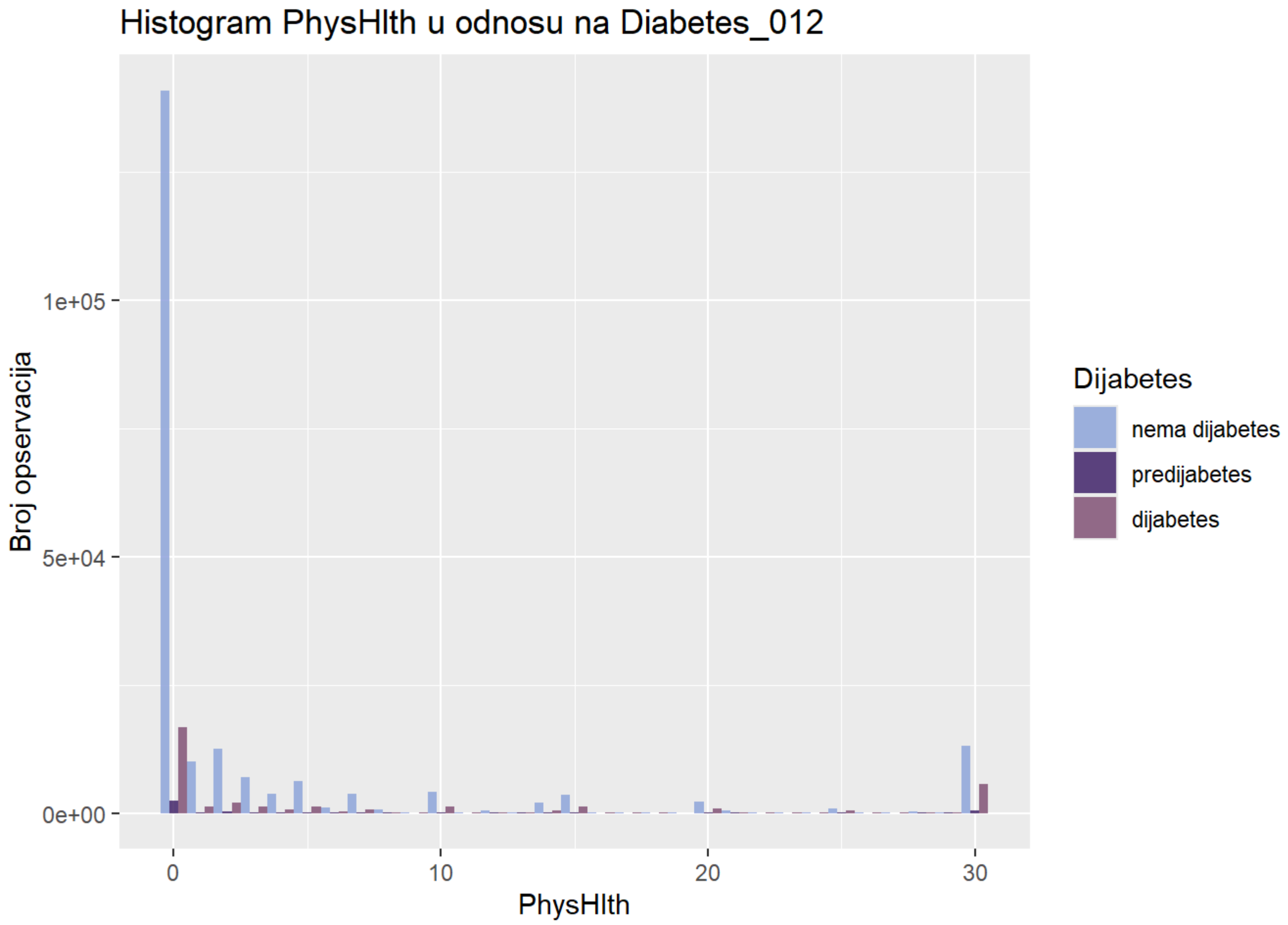
```
tukey_fun(data$MentHlth,data$Diabetes_012)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
## 
## Fit: aov(formula = numericka_var ~ as.factor(grupna_var))
## 
## $`as.factor(grupna_var)`
##                                  diff        lwr       upr     p adj
## predijabetes-nema dijabetes  1.585503  1.3281776 1.8428284 0.000000
## dijabetes-nema dijabetes     1.517402  1.4179230 1.6168810 0.000000
## dijabetes-predijabetes      -0.068101 -0.3388471 0.2026451 0.825752
```

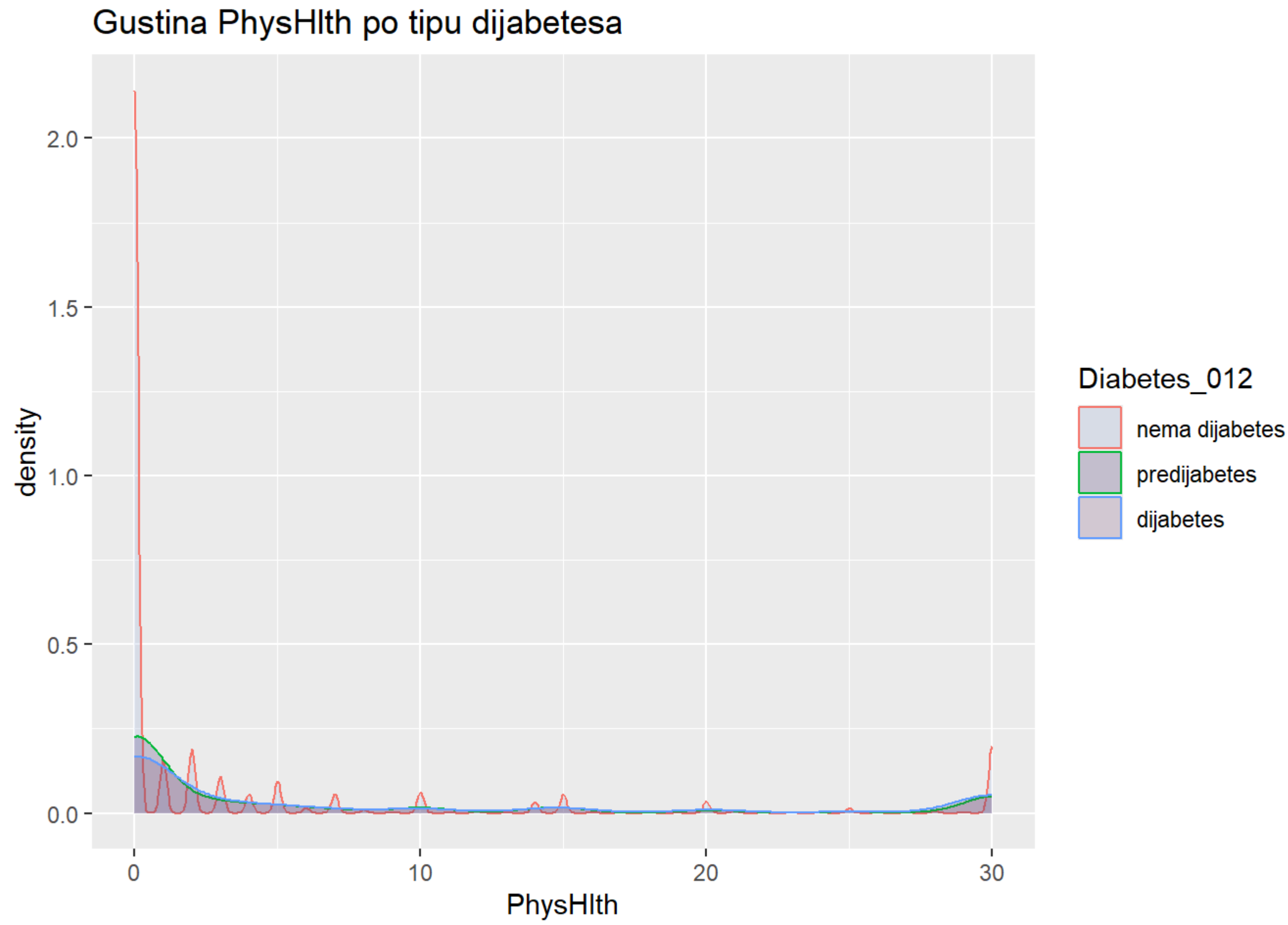
```
ggplot(data, aes(x = Diabetes_012, y = PhysHlth, fill = Diabetes_012)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values=colorS) +
  labs(title = "Distribucija PhysHlth po tipovima Diabetes_012",
       fill = "Dijabetes")
```




```
ggplot(data, aes(x = PhysHlth, fill = Diabetes_012)) +
  geom_histogram(binwidth = 1, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values =colorS) +
  labs(title = "Histogram PhysHlth u odnosu na Diabetes_012",
       y = "Broj opservacija",
       fill = "Dijabetes")
```




```
ggplot(data, aes(x = PhysHlth, color = Diabetes_012, fill = Diabetes_012)) +
  geom_density(alpha = 0.2) +
  scale_fill_manual(values=colorS)+
  labs(title = "Gustina PhysHlth po tipu dijabetesa")
```



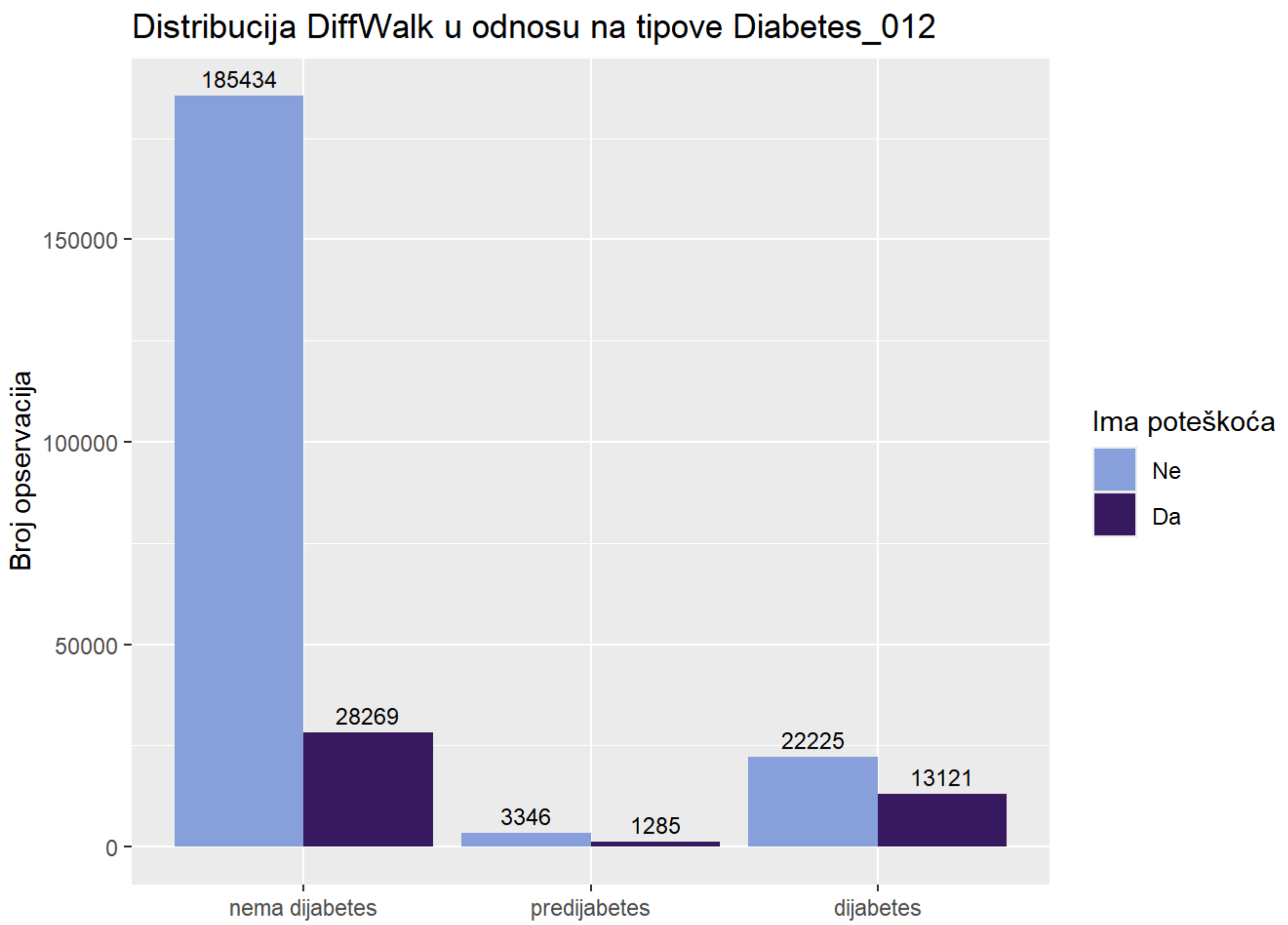

```
anova_test(data$PhysHlth,data$Diabetes_012)
```

```
##                            Df   Sum Sq Mean Sq F value Pr(>F)
## as.factor(grupna_var)       2   600673  300337    4079 <2e-16 ***
## Residuals              253677 18679609      74
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

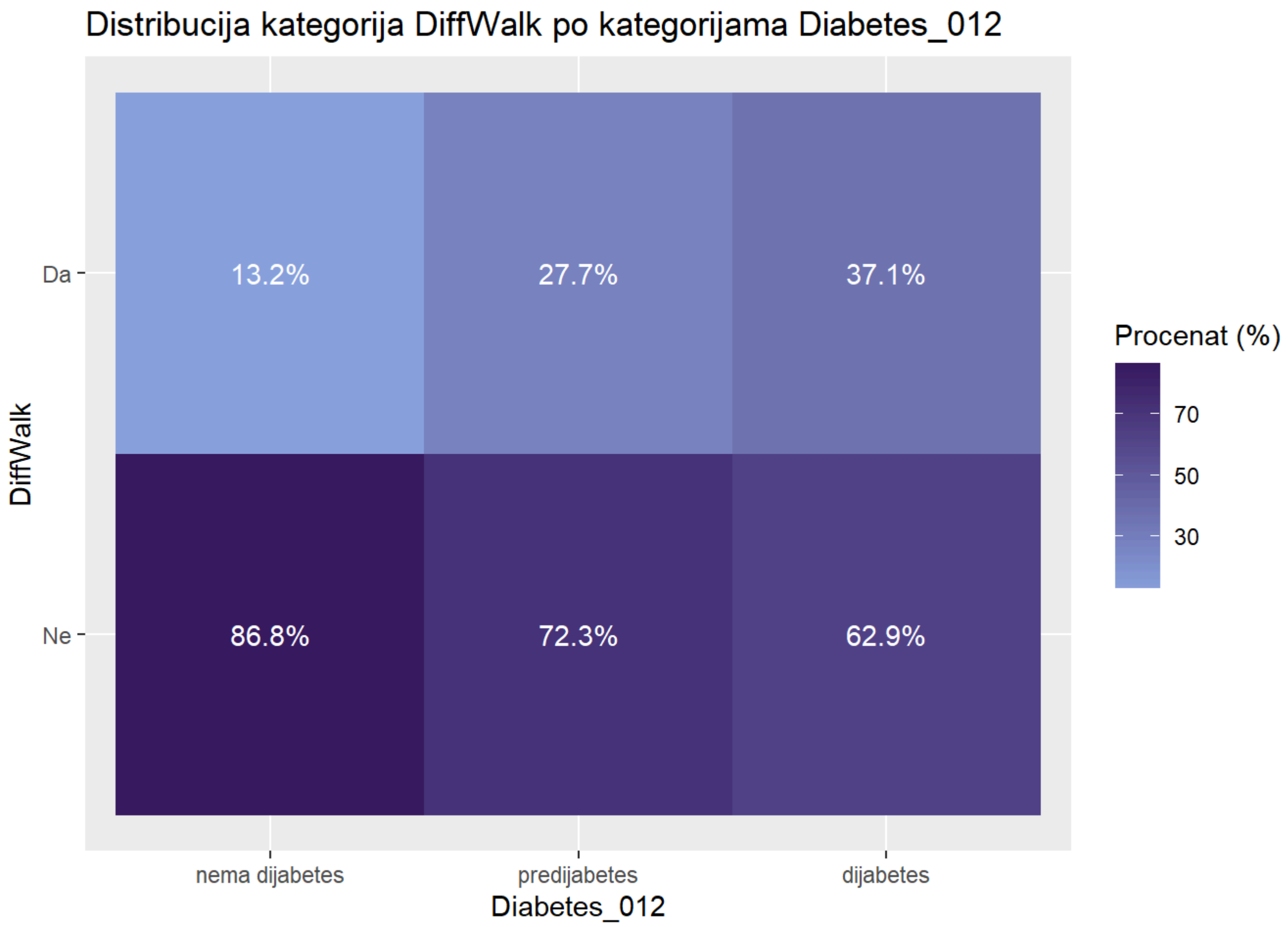
```
tukey_fun(data$PhysHlth,data$Diabetes_012)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = numericka_var ~ as.factor(grupna_var))
##
## $`as.factor(grupna_var)`
##                                diff      lwr      upr p adj
## predijabetes-nema dijabetes 2.765889 2.467170 3.064609     0
## dijabetes-nema dijabetes    4.372063 4.256581 4.487544     0
## dijabetes-predijabetes      1.606174 1.291875 1.920473     0
```

```
ggplot(data, aes(x = Diabetes_012, fill =DiffWalk )) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija DiffWalk u odnosu na tipove Diabetes_012",
       x = NULL,
       y = "Broj opservacija",
       fill = "Ima poteškoća")
```



```
data %>%
  count(Diabetes_012, DiffWalk) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = DiffWalk, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija DiffWalk po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "DiffWalk",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



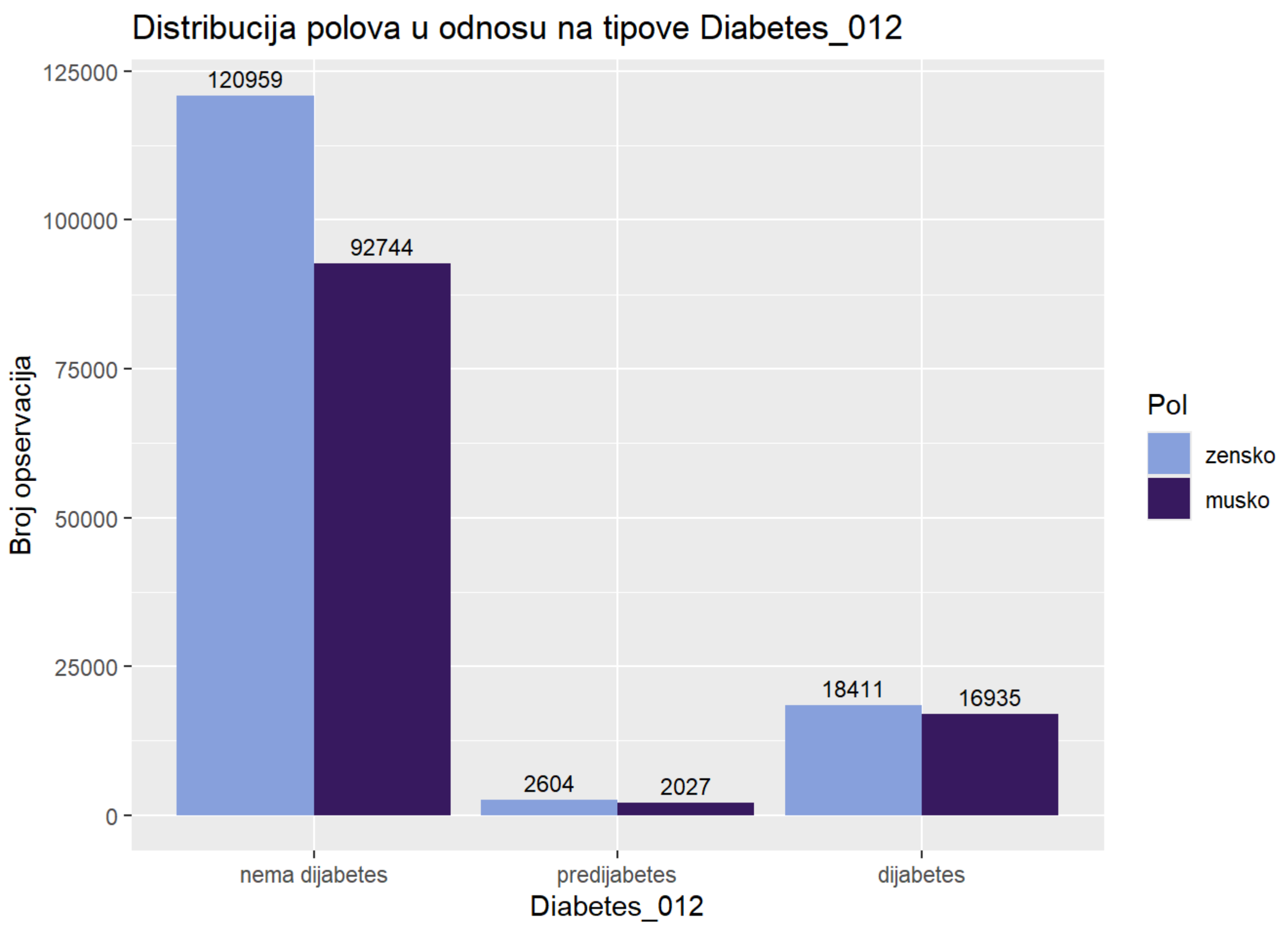

```
chi_sq_test(data$DiffWalk, data$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 12777, df = 2, p-value < 2.2e-16
```

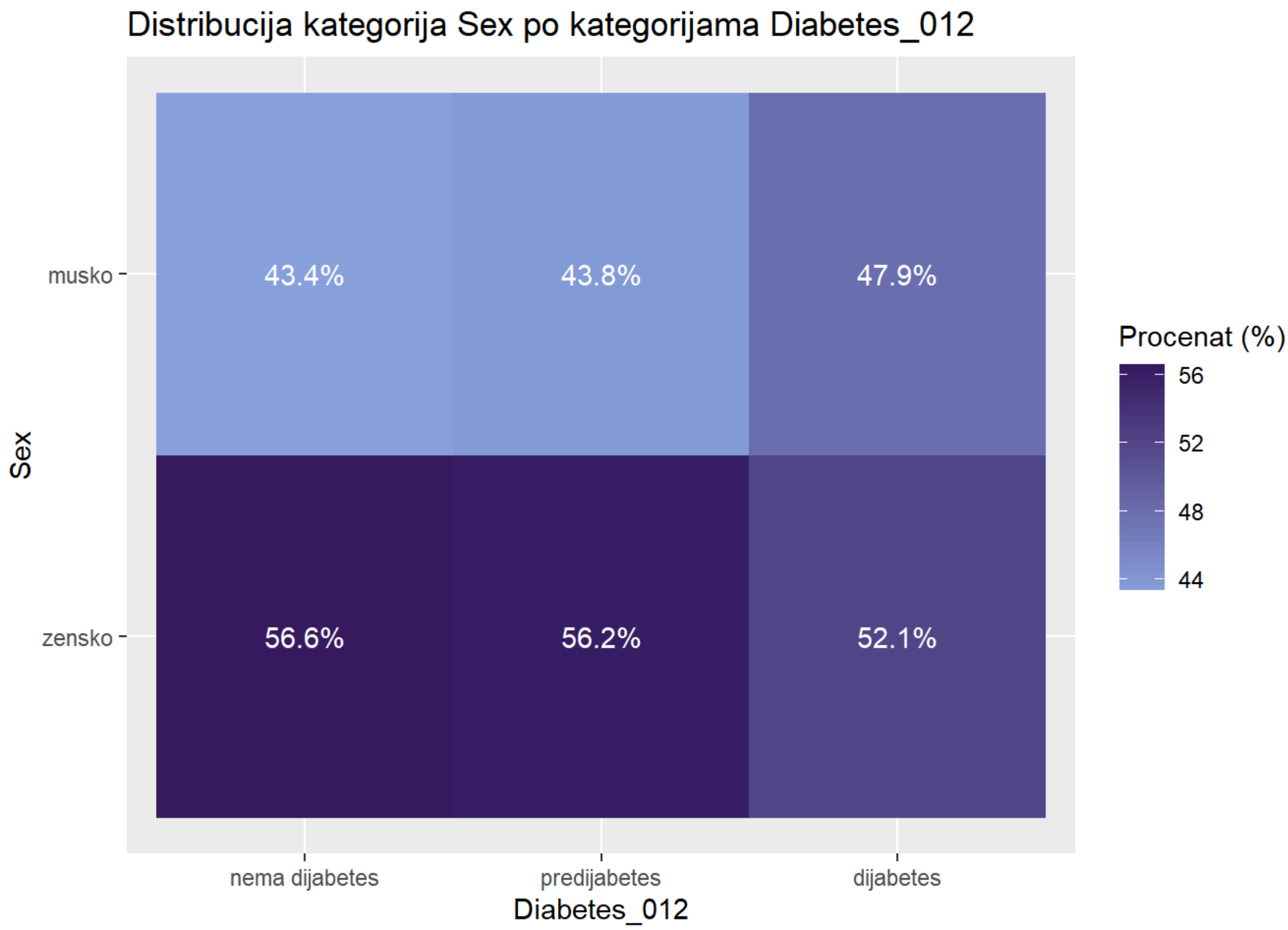
```
cramer_v(data$DiffWalk, data$Diabetes_012)
```

```
## [1] 0.2244245
```

```
ggplot(data, aes(x = Diabetes_012, fill =Sex )) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija polova u odnosu na tipove Diabetes_012",
       y = "Broj opservacija",
       fill = "Pol")
```



```
data %>%
  count(Diabetes_012, Sex) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = Sex, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija Sex po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "Sex",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



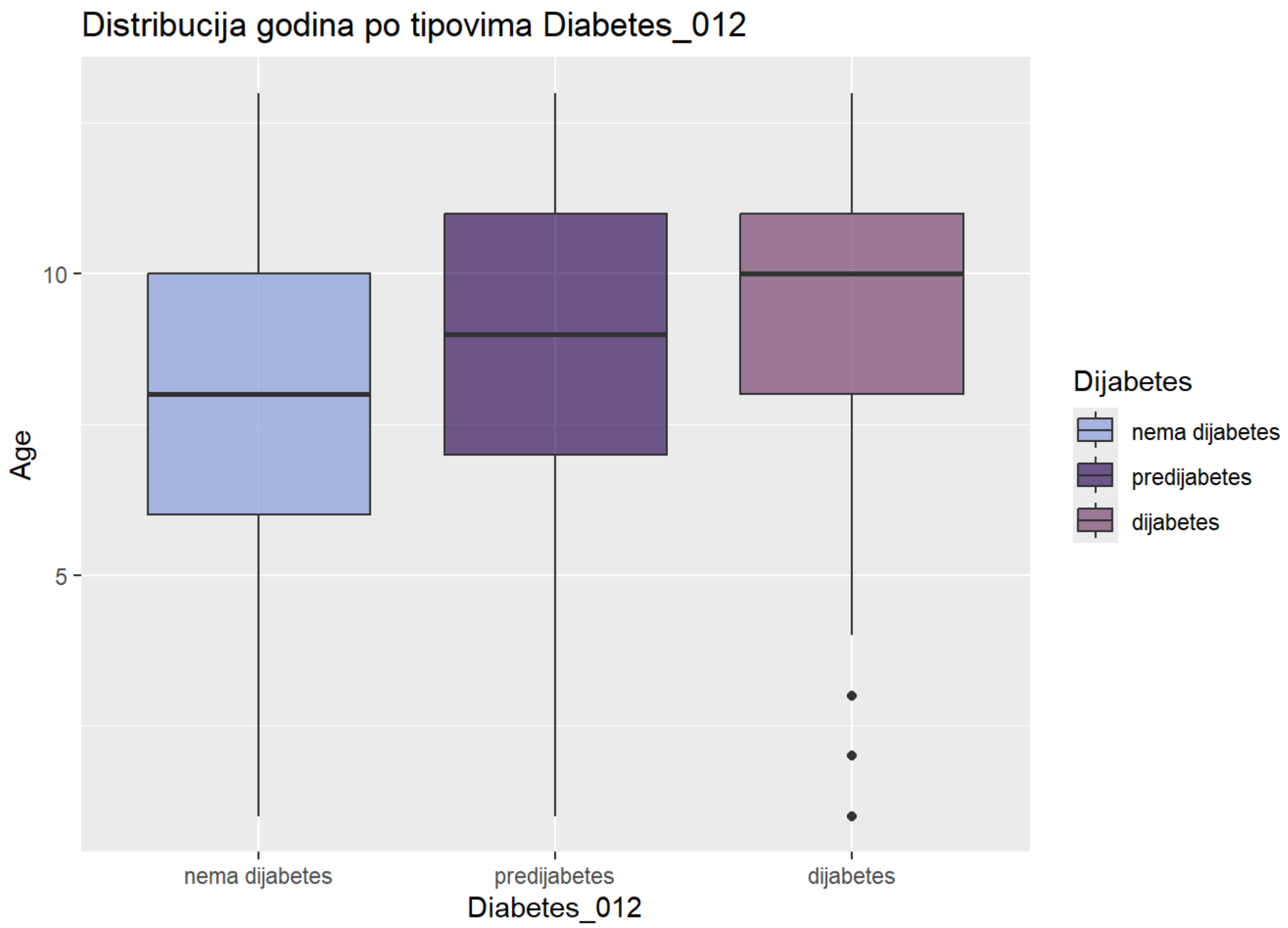

```
chi_sq_test(data$Diabetes_012,data$Sex)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 250.85, df = 2, p-value < 2.2e-16
```

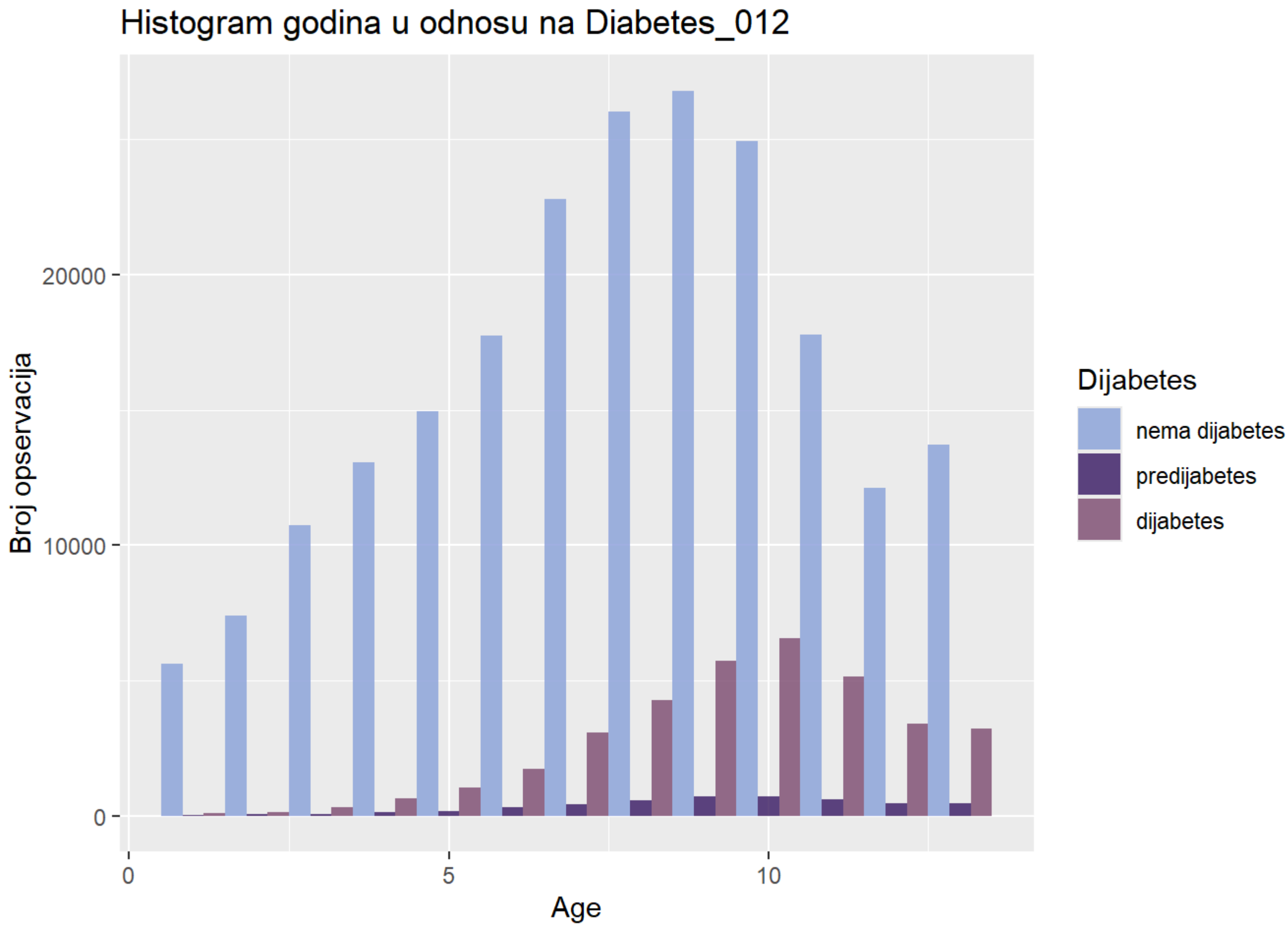
```
cramer_v(data$Diabetes_012,data$Sex)
```

```
## [1] 0.03144593
```

```
ggplot(data, aes(x = Diabetes_012, y = Age, fill = Diabetes_012)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values=colorS) +
  labs(title = "Distribucija godina po tipovima Diabetes_012",
       fill = "Dijabetes")
```




```
ggplot(data, aes(x = Age, fill = Diabetes_012)) +
  geom_histogram(binwidth = 1, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values =colorS) +
  labs(title = "Histogram godina u odnosu na Diabetes_012",
       y = "Broj opservacija",
       fill = "Dijabetes")
```



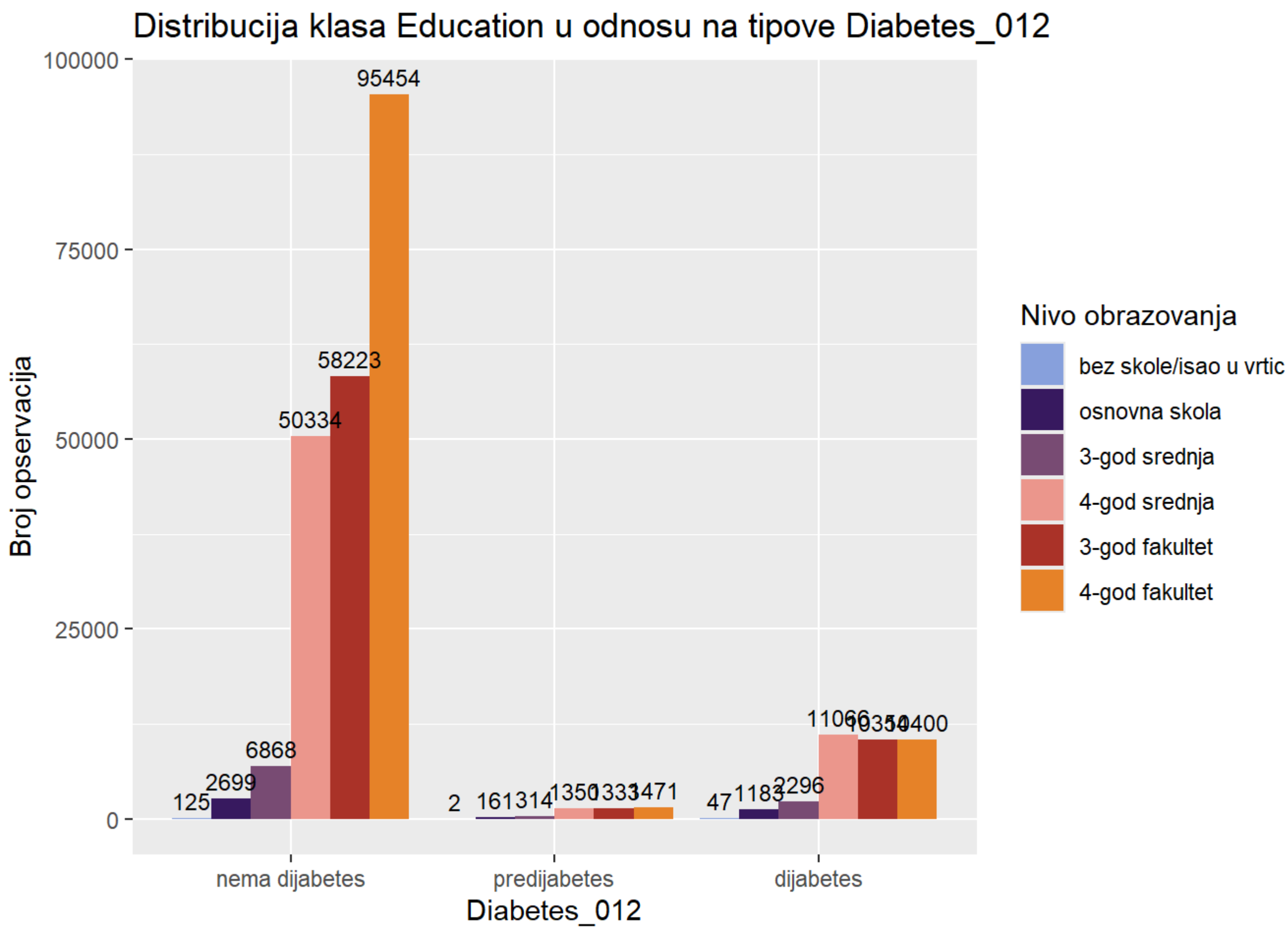

```
anova_test(data$Age,data$Diabetes_012)
```

```
##                          Df  Sum Sq Mean Sq F value Pr(>F)    
## as.factor(grupna_var)     2   82130   41065    4560 <2e-16 ***
## Residuals            253677 2284255       9                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

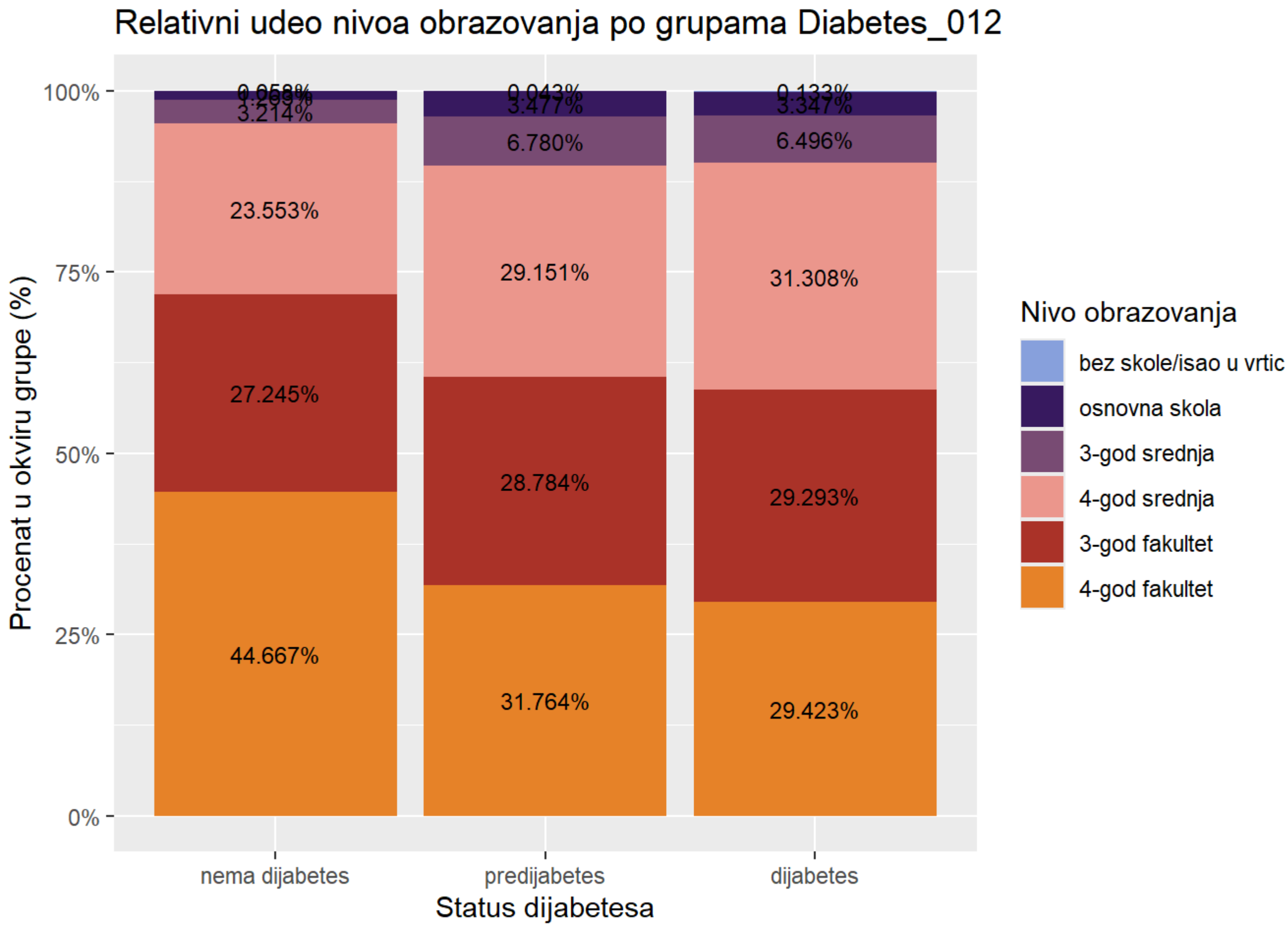
```
tukey_fun(data$Age,data$Diabetes_012)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
## 
## Fit: aov(formula = numericka_var ~ as.factor(grupna_var))
## 
## $`as.factor(grupna_var)`
##                                 diff       lwr      upr p adj
## predijabetes-nema dijabetes 1.2967924 1.1923320 1.401253     0
## dijabetes-nema dijabetes    1.5924939 1.5521107 1.632877     0
## dijabetes-predijabetes      0.2957015 0.1857929 0.405610     0
```

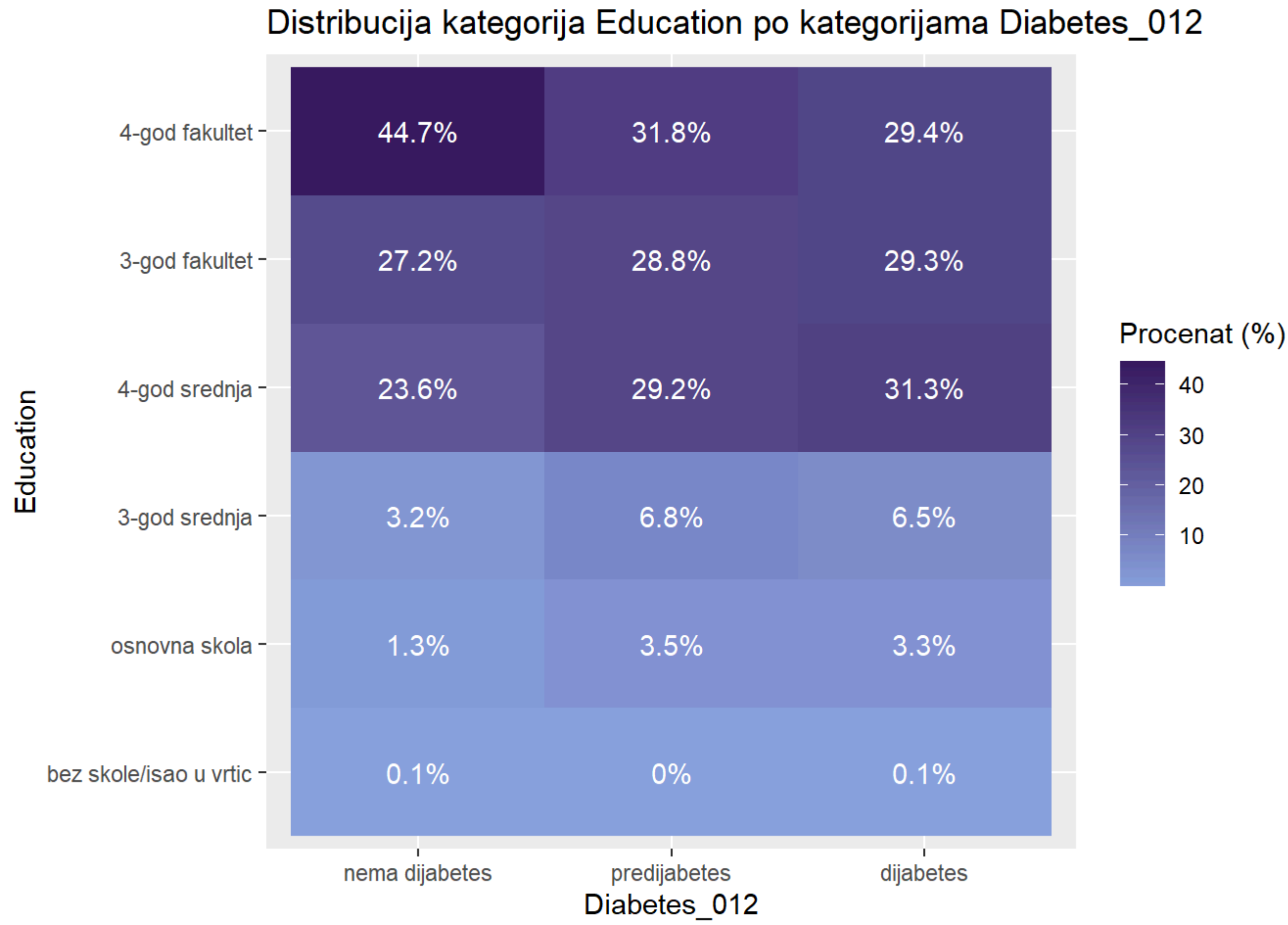
```
ggplot(data, aes(x = Diabetes_012, fill =Education )) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija klasa Education u odnosu na tipove Diabetes_012",
       y = "Broj opservacija",
       fill = "Nivo obrazovanja")
```




```
ggplot(data, aes(x = Diabetes_012, fill = Education)) +
  geom_bar(position = "fill") +
  geom_text(stat = "count",
            aes(label = scales::percent(after_stat(count) / ave(after_stat(count), aft
er_stat(x), FUN = sum))),
            position = position_fill(vjust = 0.5),
            size = 3) +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = colorS) +
  labs(title = "Relativni udeo nivoa obrazovanja po grupama Diabetes_012",
       y = "Procenat u okviru grupe (%)",
       x = "Status dijabetesa",
       fill = "Nivo obrazovanja")
```




```
data %>%
  count(Diabetes_012, Education) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = Education, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija Education po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "Education",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```




```
chi_sq_test(data$Diabetes_012,data$Education)
```

```
## Warning in chisq.test(tabela): Chi-squared approximation may be incorrect
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 4560.6, df = 10, p-value < 2.2e-16
```

```
cramer_v(data$Diabetes_012,data$Education)
```

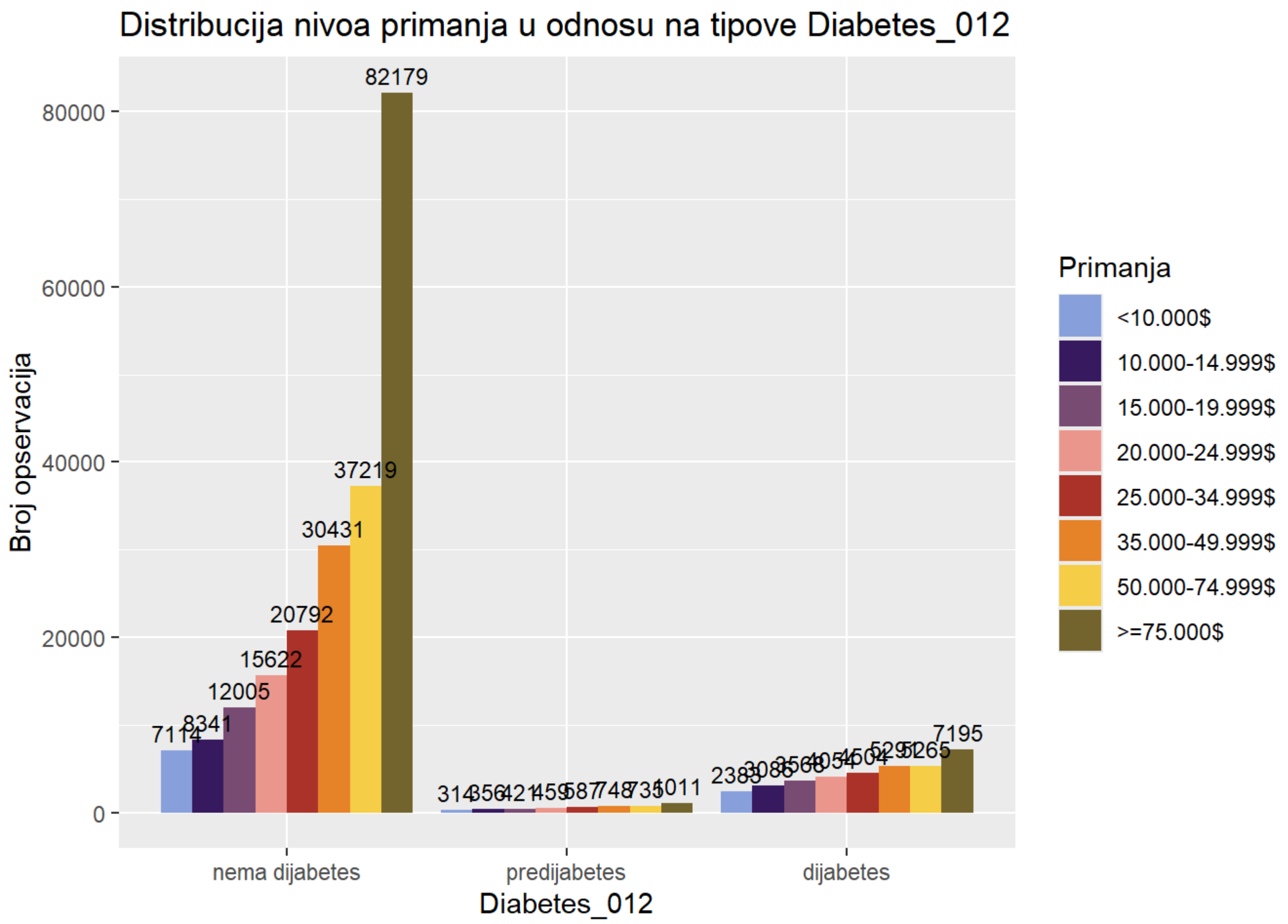
```
## Warning in chisq.test(tabela): Chi-squared approximation may be incorrect
```

```
## [1] 0.09481014
```

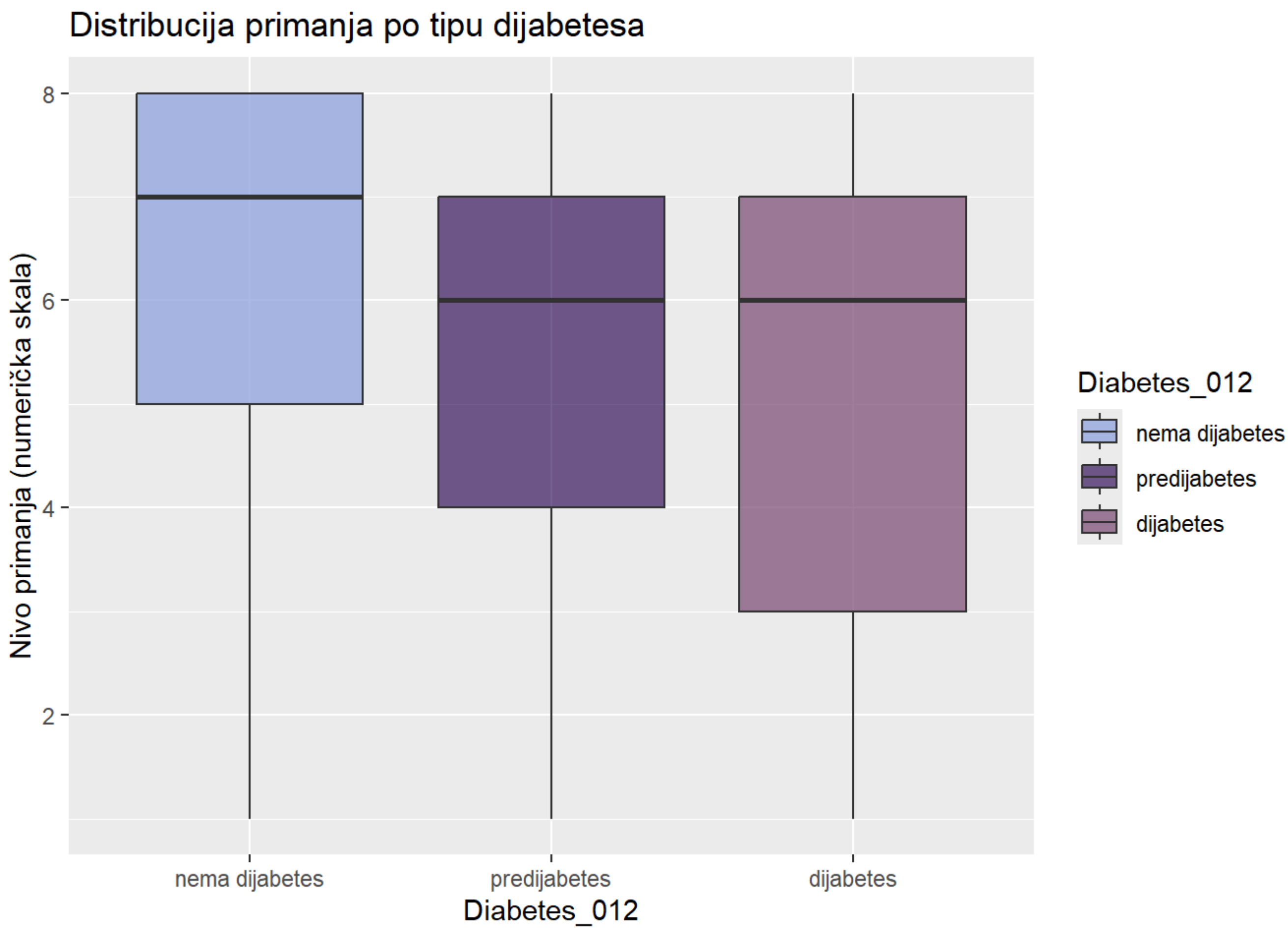
```
prop.table(table(data$Education,data$Diabetes_012), margin = 2)*100
```

```
##
##                              nema dijabetes predijabetes   dijabetes
##   bez skole/isao u vrtic      0.05849239   0.04318722  0.13297120
##   osnovna skola               1.26296776   3.47657094  3.34691337
##   3-god srednja               3.21380608   6.78039300  6.49578453
##   4-god srednja              23.55324914  29.15137119 31.30764443
##   3-god fakultet             27.24482108  28.78427985 29.29327222
##   4-god fakultet             44.66666355  31.76419780 29.42341425
```

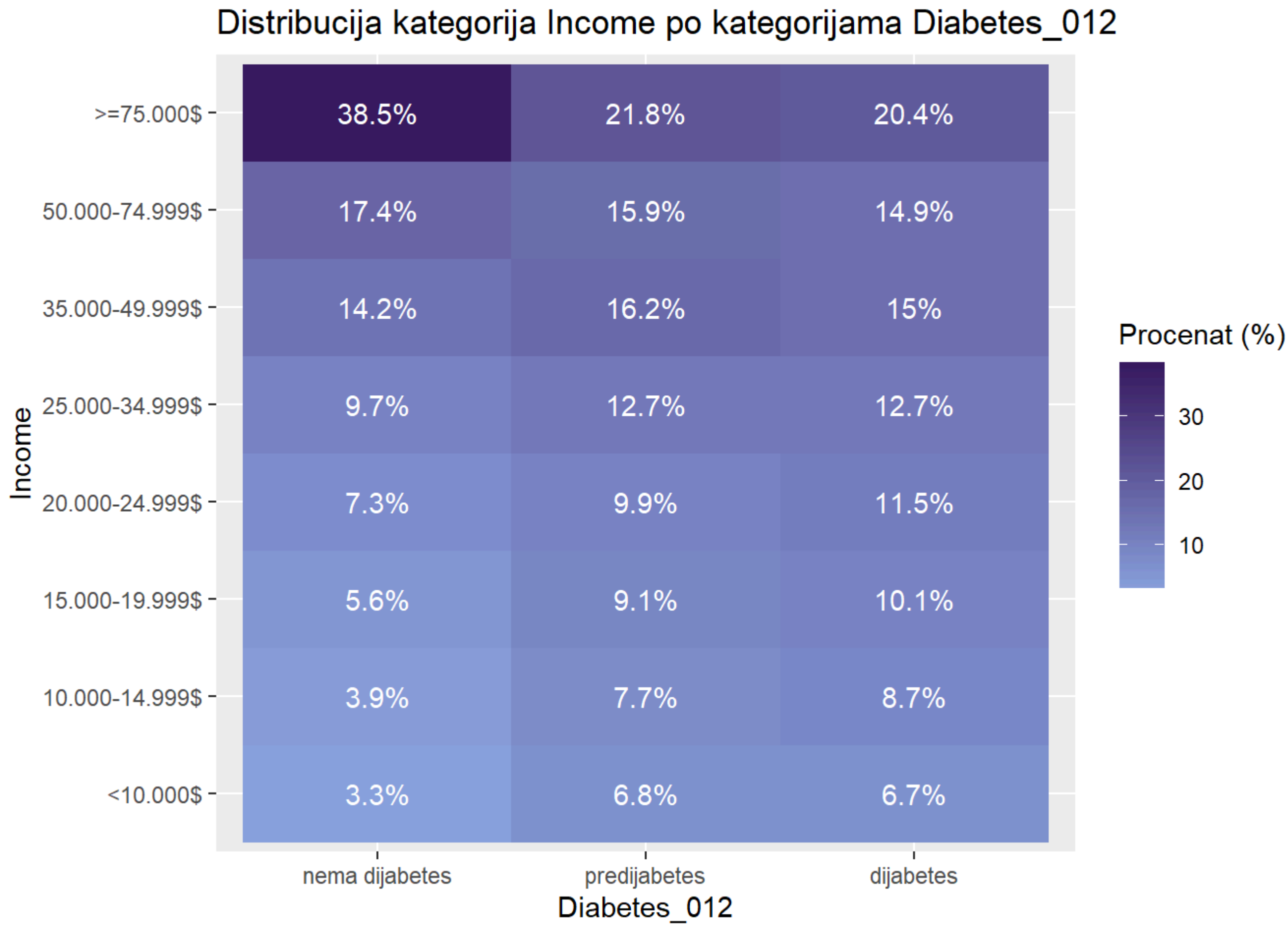
```
ggplot(data, aes(x = Diabetes_012, fill =Income )) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count",
            aes(label = ..count..),
            vjust = -0.5,
            position = position_dodge(width = 0.9),
            size = 3) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija nivoa primanja u odnosu na tipove Diabetes_012",
       y = "Broj opservacija",
       fill = "Primanja")
```



```
ggplot(data, aes(x = Diabetes_012, y = as.numeric(Income), fill = Diabetes_012)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija primanja po tipu dijabetesa",
       y = "Nivo primanja (numerička skala)")
```




```
data %>%
  count(Diabetes_012, Income) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = Income, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija Income po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "Income",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```




```
chi_sq_test(data$Diabetes_012,data$Income)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 7816.5, df = 14, p-value < 2.2e-16
```

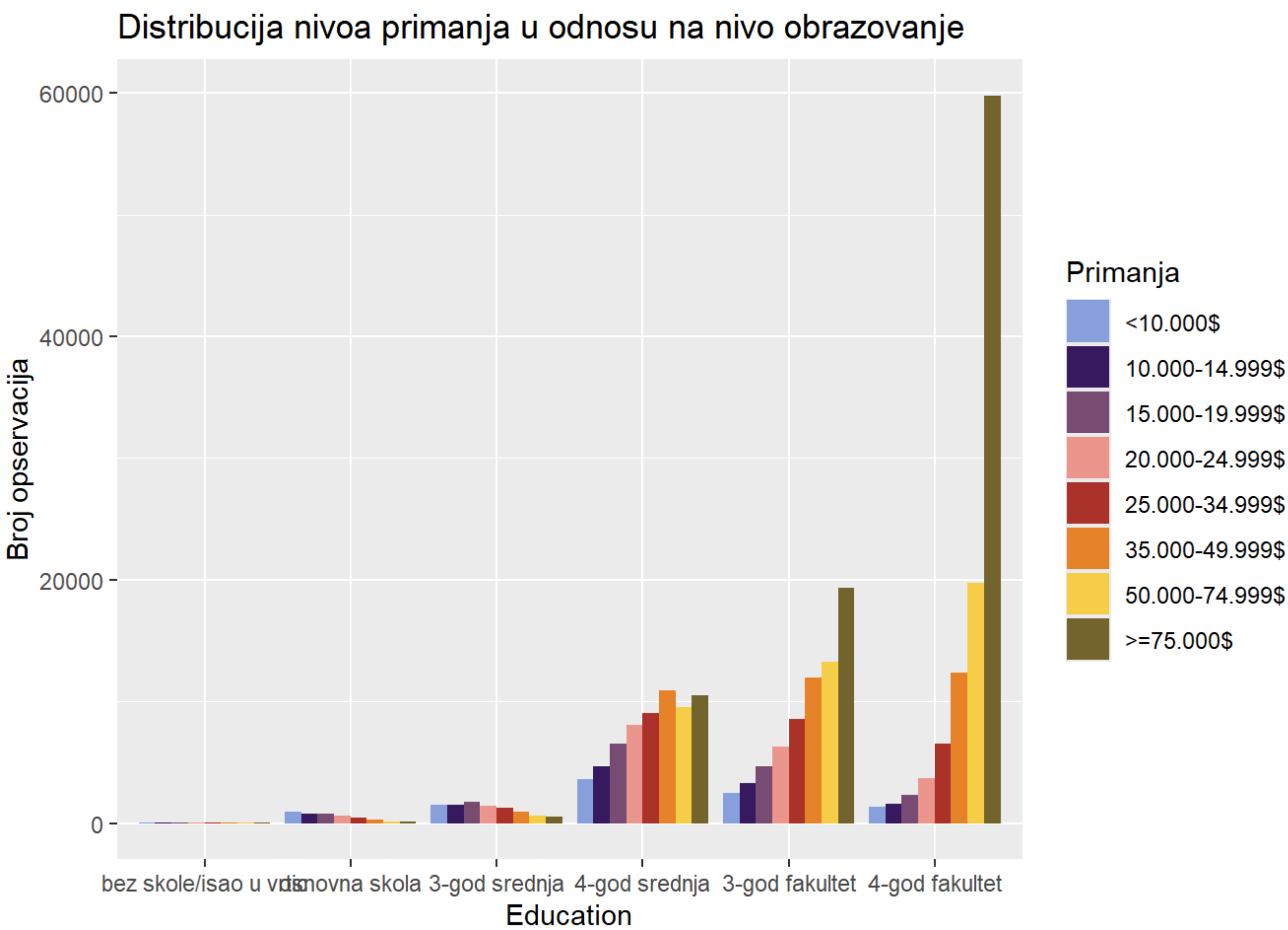
```
cramer_v(data$Diabetes_012,data$Income)
```

```
## [1] 0.1241215
```

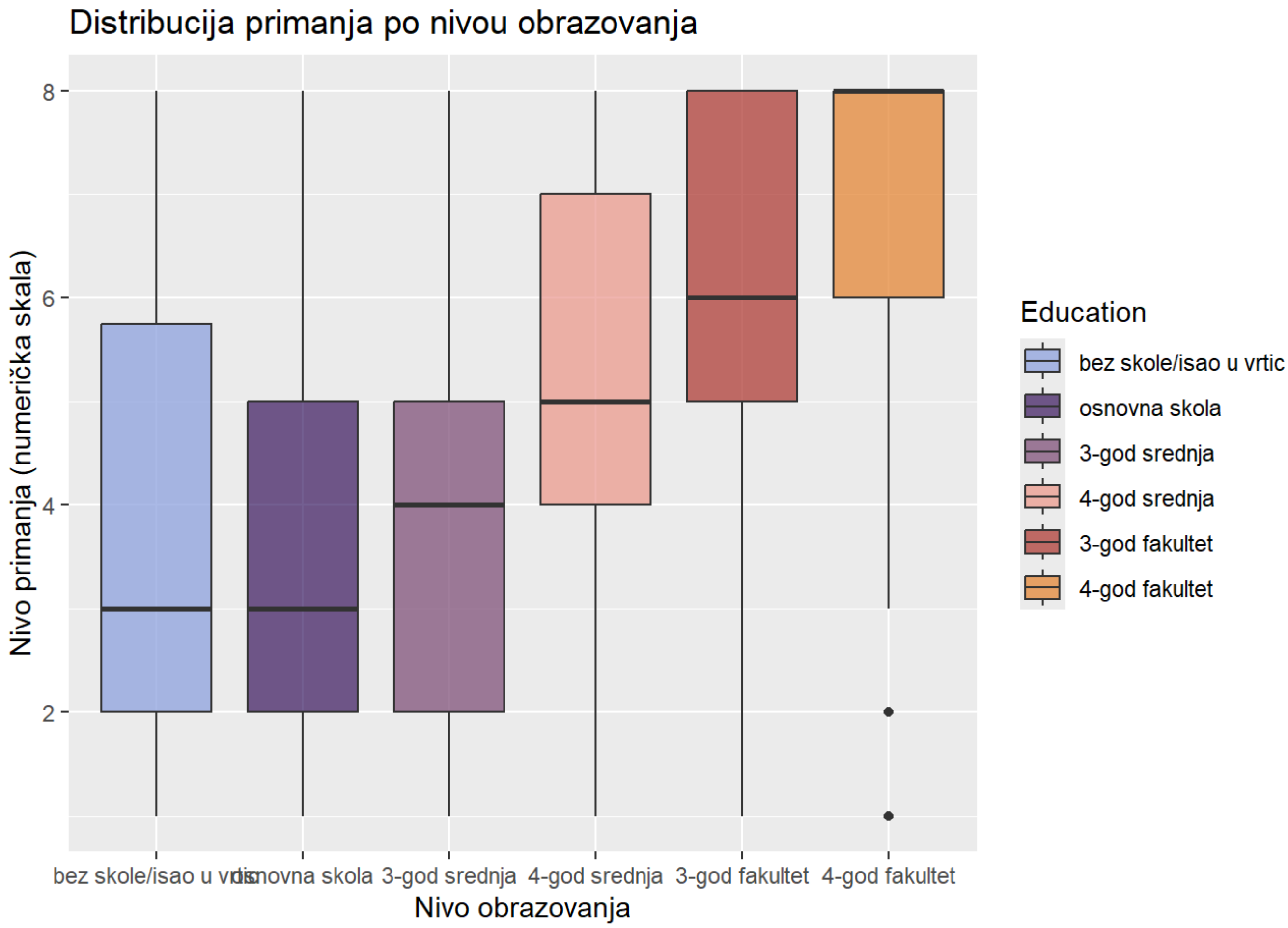
```
tabla_Incom_Diabetes_broj <- table(data$Income, data$Diabetes_012)
prop.table(tabla_Incom_Diabetes_broj, margin = 2) * 100
```

```
##
##                   nema dijabetes predijabetes dijabetes
##   <10.000$             3.328919     6.780393  6.741923
##   10.000-14.999$       3.903080     7.687325  8.730832
##   15.000-19.999$       5.617609     9.090909 10.094494
##   20.000-24.999$       7.310145     9.911466 11.469473
##   25.000-34.999$       9.729391    12.675448 12.742602
##   35.000-49.999$      14.239856    16.152019 14.969162
##   50.000-74.999$      17.416227    15.871302 14.895603
##   >=75.000$           38.454771    21.831138 20.355910
```

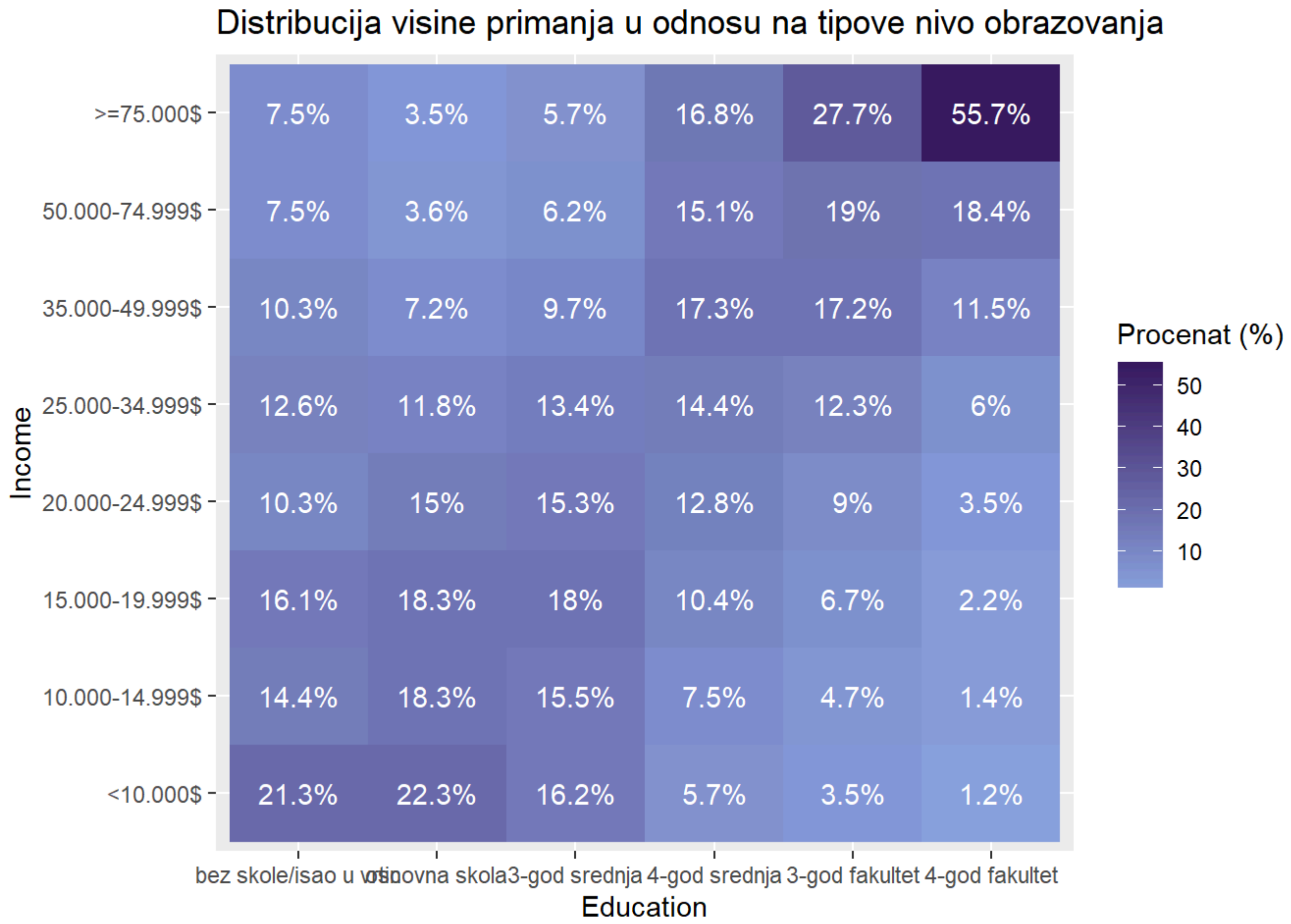
```
ggplot(data, aes(x = Education, fill =Income )) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija nivoa primanja u odnosu na nivo obrazovanje",
       y = "Broj opservacija",
       fill = "Primanja")
```




```
ggplot(data, aes(x = Education, y = as.numeric(Income), fill = Education)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija primanja po nivou obrazovanja",
       x = "Nivo obrazovanja",
       y = "Nivo primanja (numerička skala)")
```




```
data %>%
  count(Education, Income) %>%
  group_by(Education) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Education, y = Income, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija visine primanja u odnosu na tipove nivo obrazovanja",
    x = "Education",
    y = "Income",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



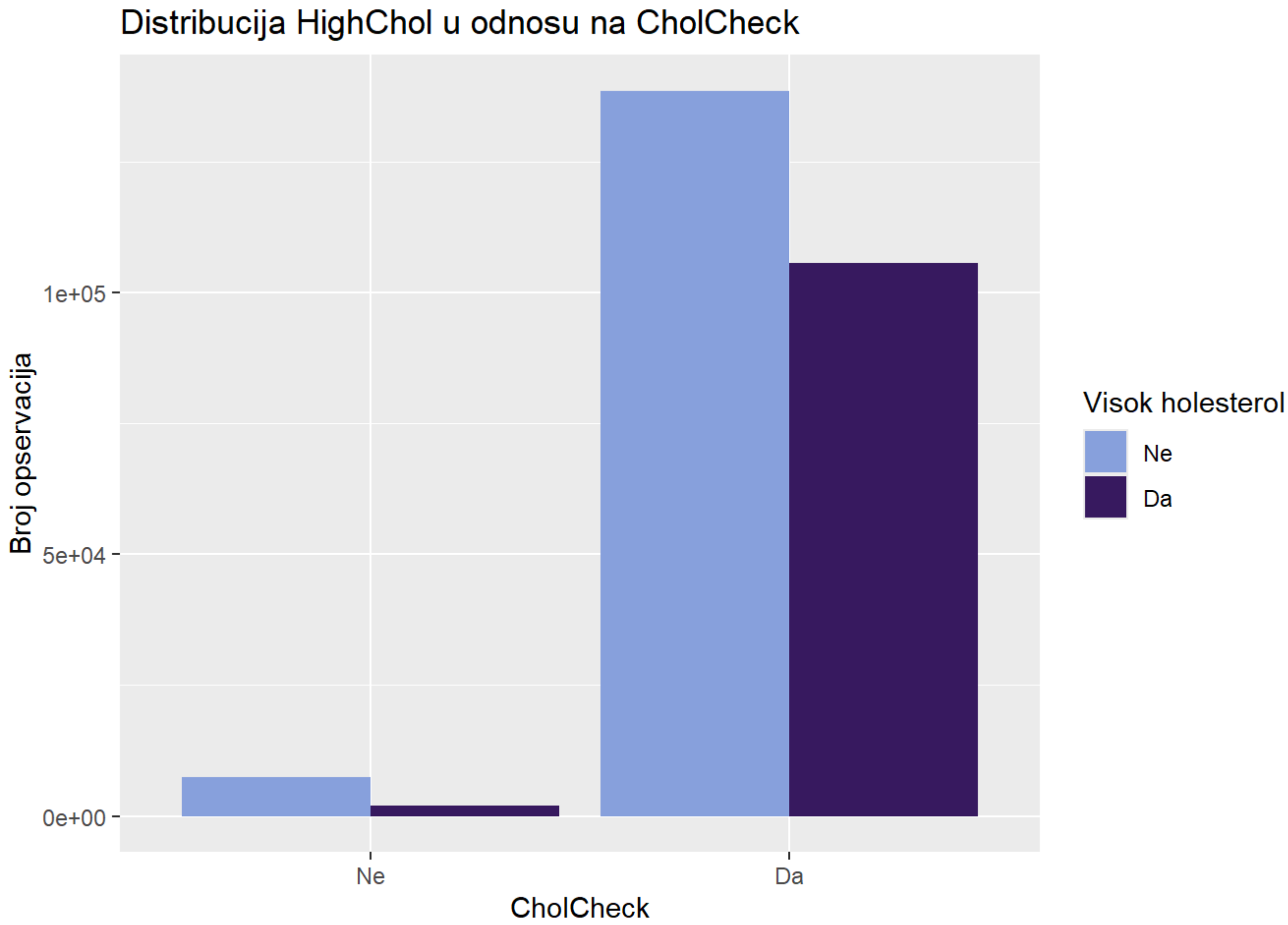

```
chi_sq_test(data$Education,data$Income)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 60337, df = 35, p-value < 2.2e-16
```

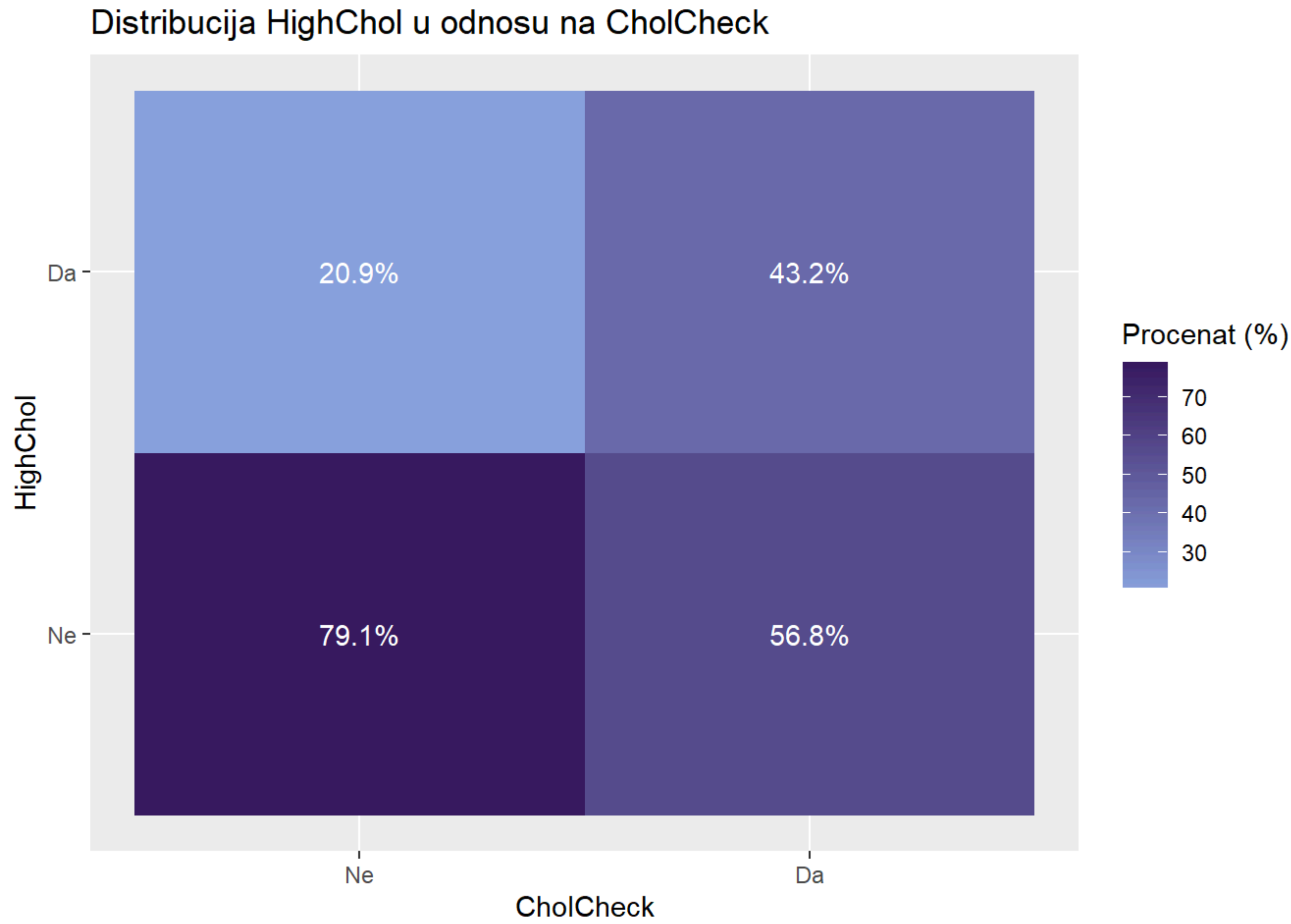
```
cramer_v(data$Education,data$Income)
```

```
## [1] 0.2181037
```

```
ggplot(data, aes(x = CholCheck, fill =HighChol )) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija HighChol u odnosu na CholCheck",
       y = "Broj opservacija",
       fill = "Visok holesterol")
```




```
data %>%
  count(CholCheck,HighChol) %>%
  group_by(CholCheck) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = CholCheck, y = HighChol, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija HighChol u odnosu na CholCheck",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



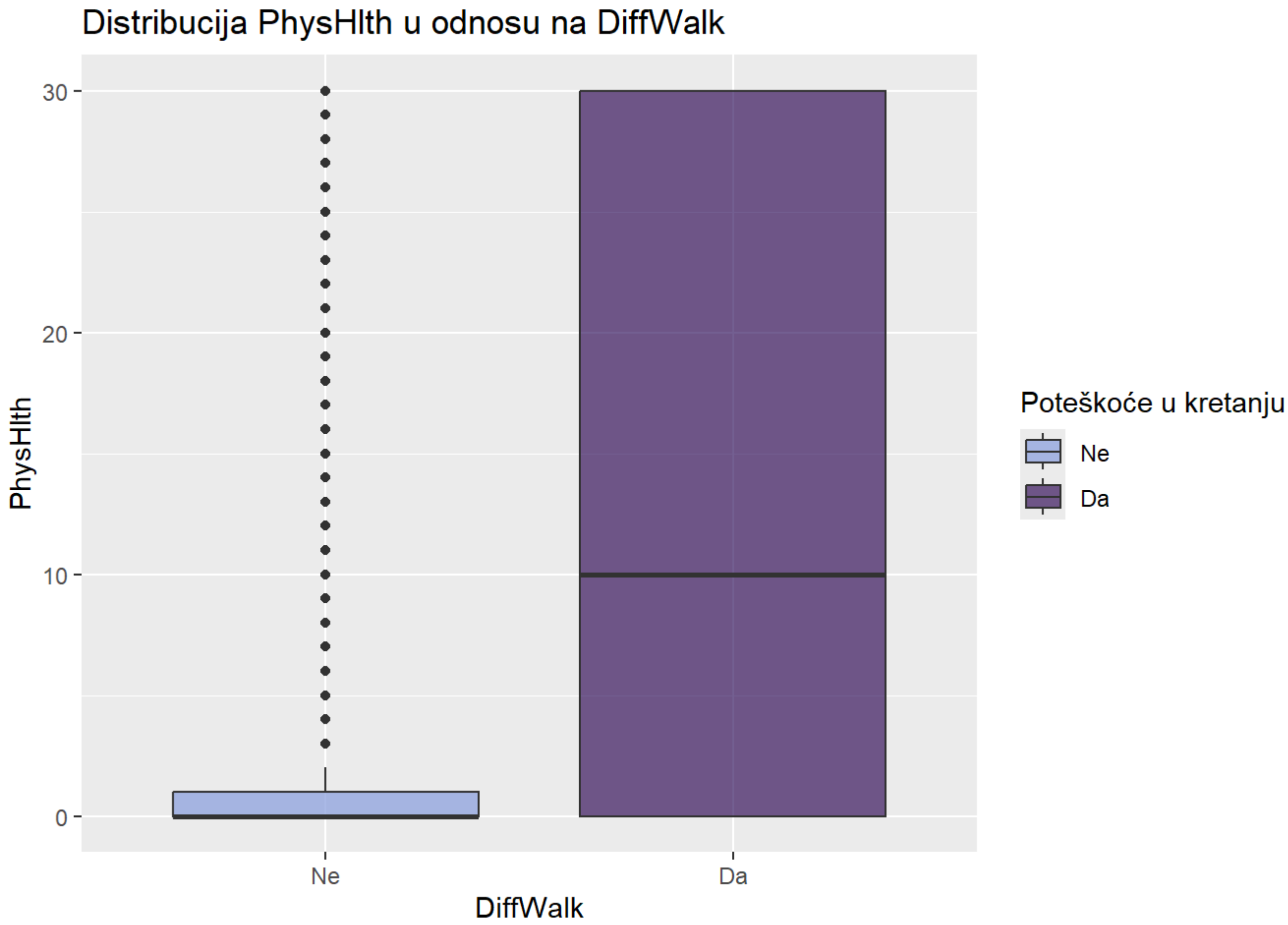

```
chi_sq_test(data$CholCheck,data$HighChol)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tabela
## X-squared = 1859.7, df = 1, p-value < 2.2e-16
```

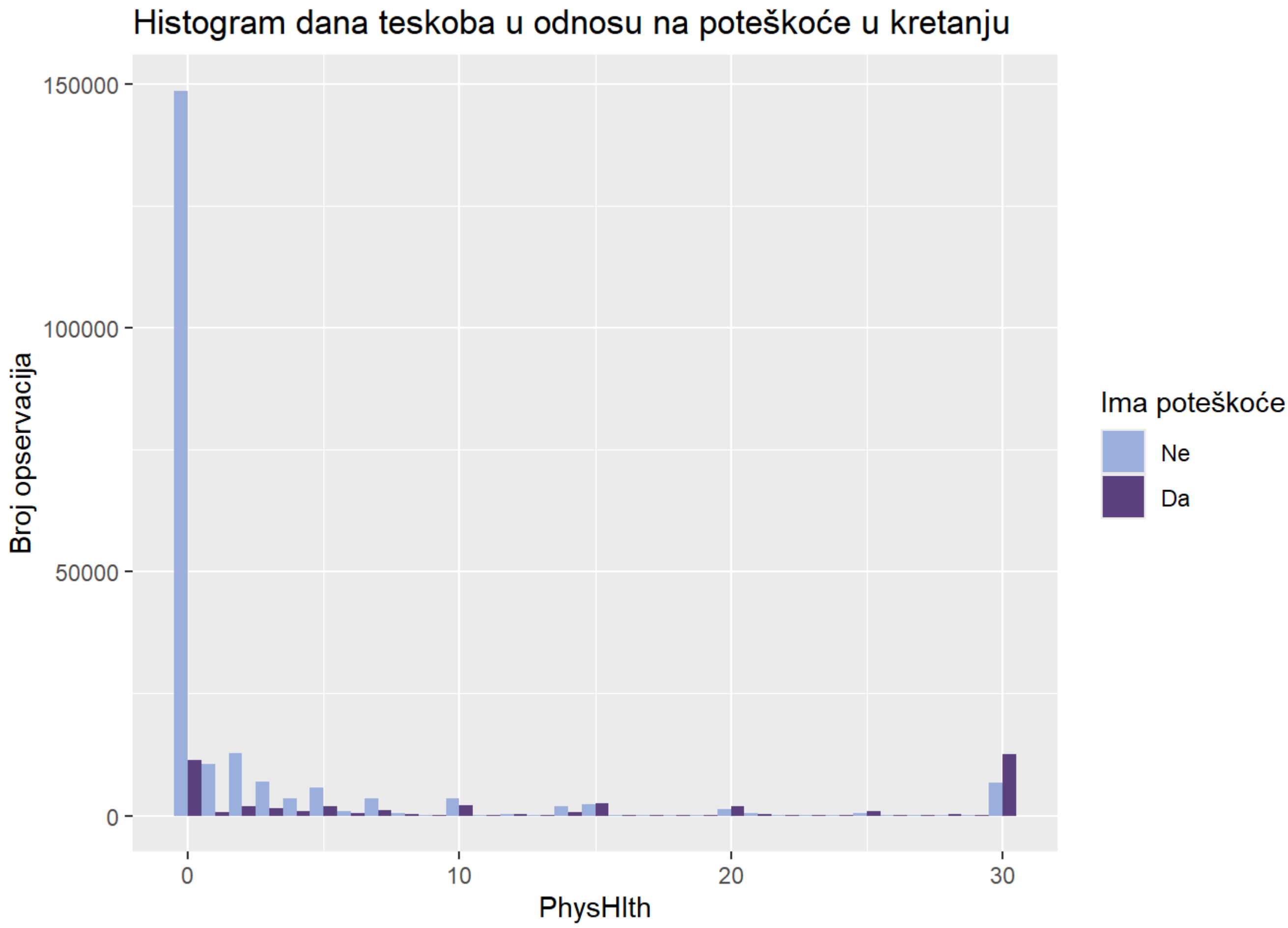
```
cramer_v(data$CholCheck,data$HighChol)
```

```
## [1] 0.08562119
```

```
ggplot(data, aes(x = DiffWalk, y = PhysHlth, fill = DiffWalk)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values=colorS) +
  labs(title = "Distribucija PhysHlth u odnosu na DiffWalk",
       fill = "Poteškoće u kretanju")
```




```
ggplot(data, aes(x = PhysHlth, fill = DiffWalk)) +
  geom_histogram(binwidth = 1, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values =colorS) +
  labs(title = "Histogram dana teskoba u odnosu na poteškoće u kretanju",
       y = "Broj opservacija",
       fill = "Ima poteškoće")
```



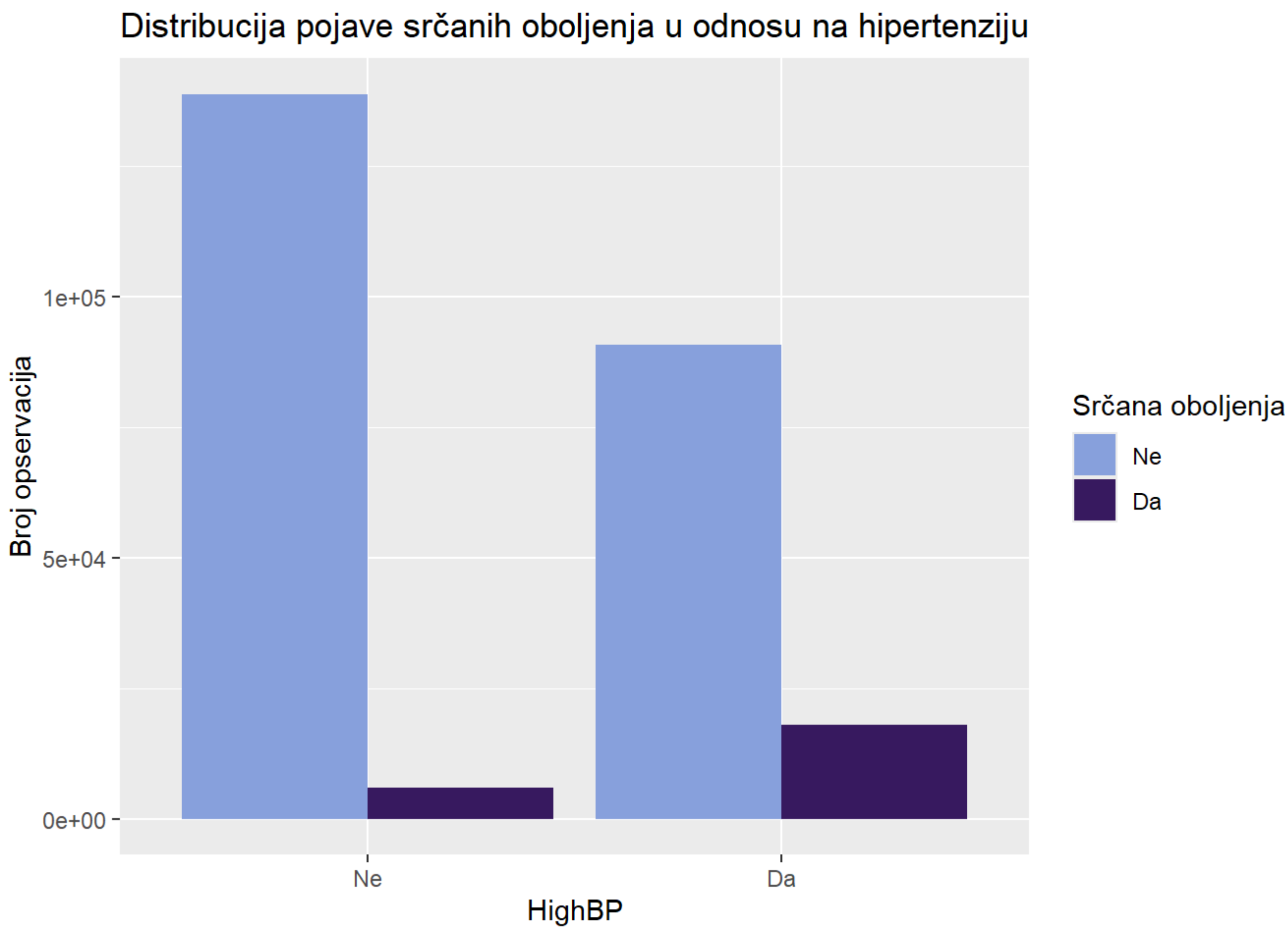

```
anova_test(data$PhysHlth,data$DiffWalk)
```

```
##                             Df   Sum Sq Mean Sq F value Pr(>F)    
## as.factor(grupna_var)       1  4412919 4412919   75296 <2e-16 ***
## Residuals              253678 14867364      59                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

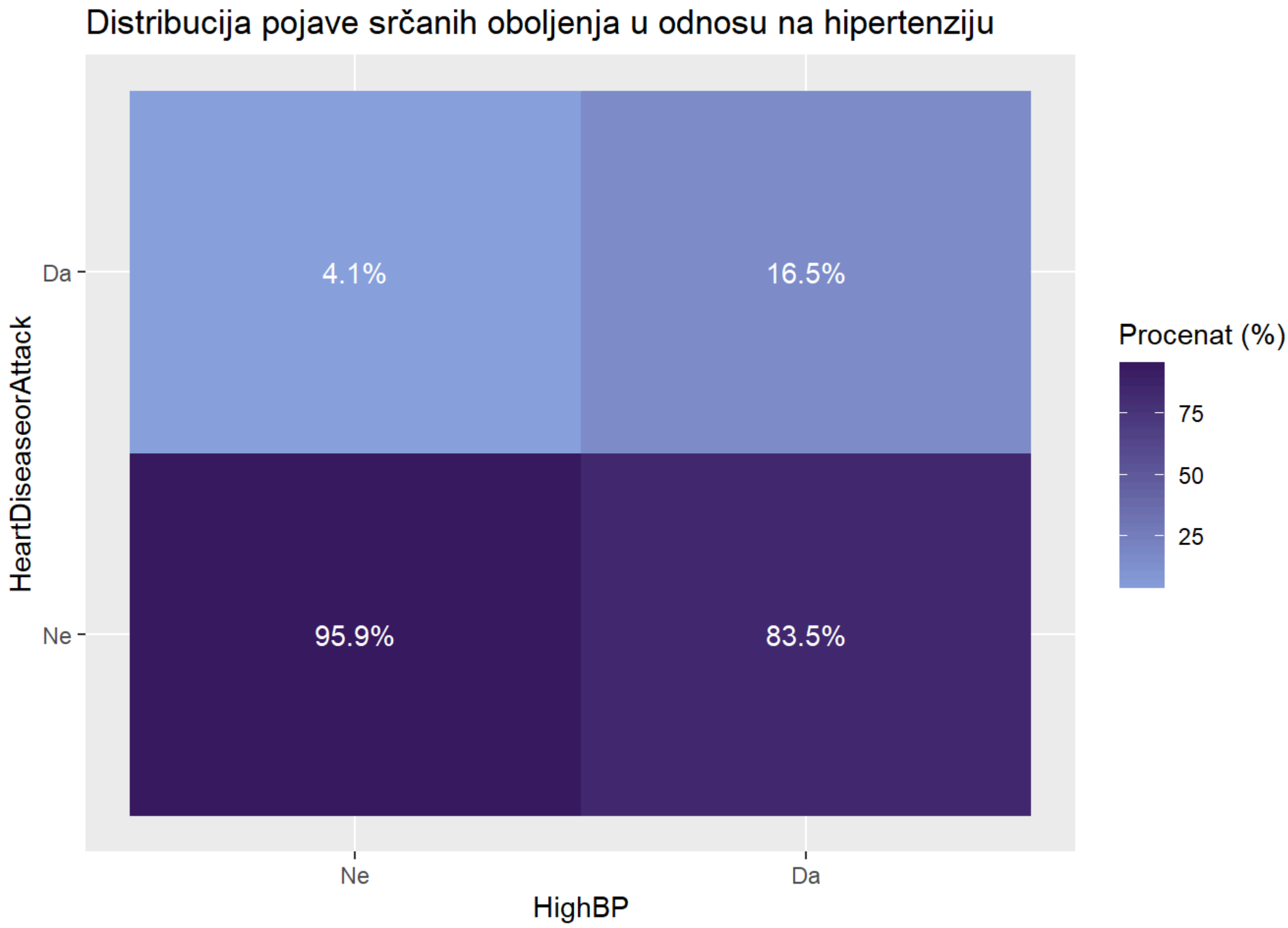
```
tukey_fun(data$PhysHlth,data$DiffWalk)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
## 
## Fit: aov(formula = numericka_var ~ as.factor(grupna_var))
## 
## $`as.factor(grupna_var)`
##           diff      lwr      upr p adj
## Da-Ne 11.14995 11.07031 11.22959     0
```

```
ggplot(data, aes(x = HighBP, fill =HeartDiseaseorAttack )) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija pojave srčanih oboljenja u odnosu na hipertenziju",
       y = "Broj opservacija",
       fill = "Srčana oboljenja")
```




```
data %>%
  count(HighBP, HeartDiseaseorAttack) %>%
  group_by(HighBP) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = HighBP, y = HeartDiseaseorAttack, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija pojave srčanih oboljenja u odnosu na hipertenziju",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



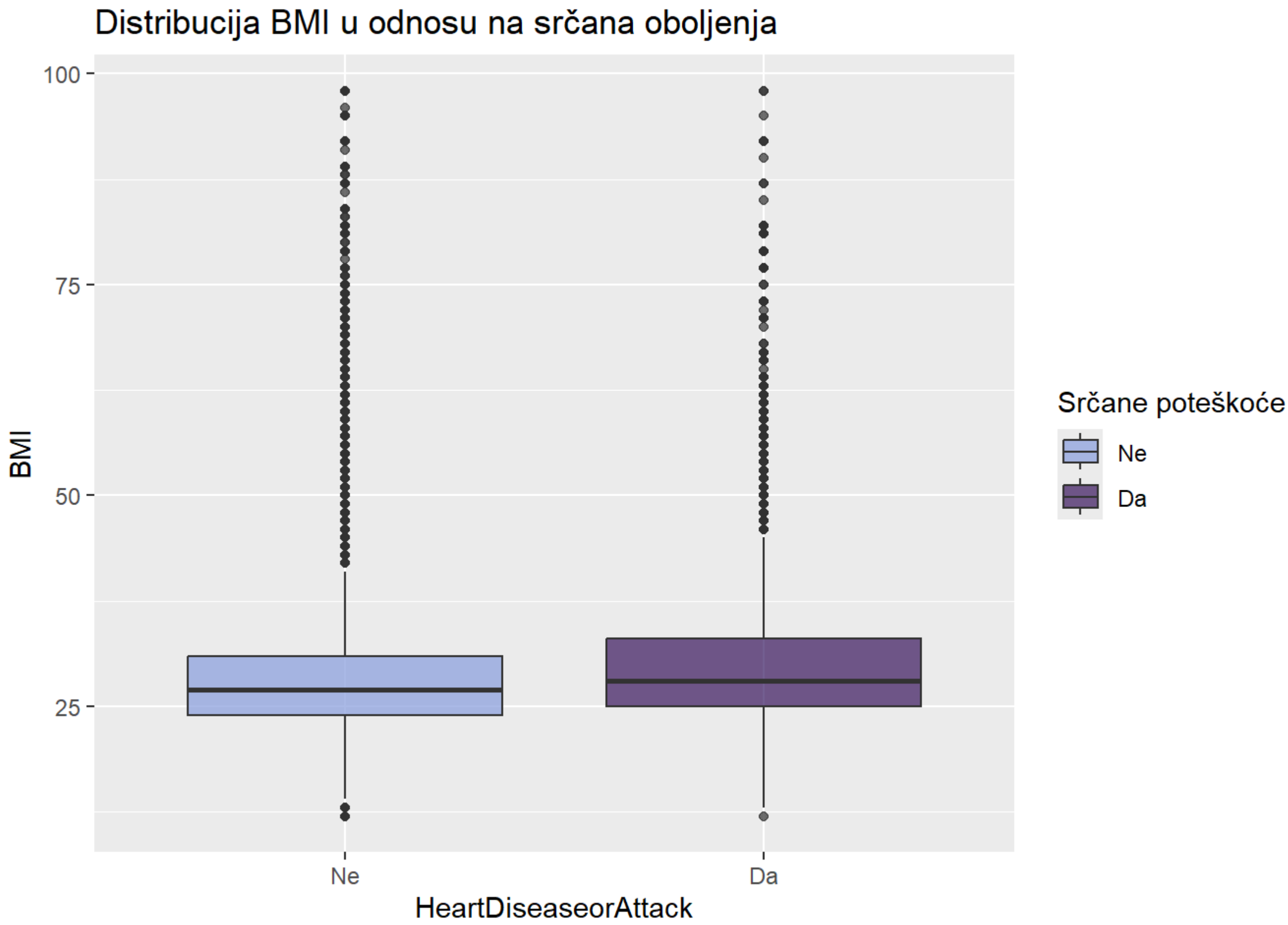

```
chi_sq_test(data$HighBP,data$HeartDiseaseorAttack)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tabela
## X-squared = 11118, df = 1, p-value < 2.2e-16
```

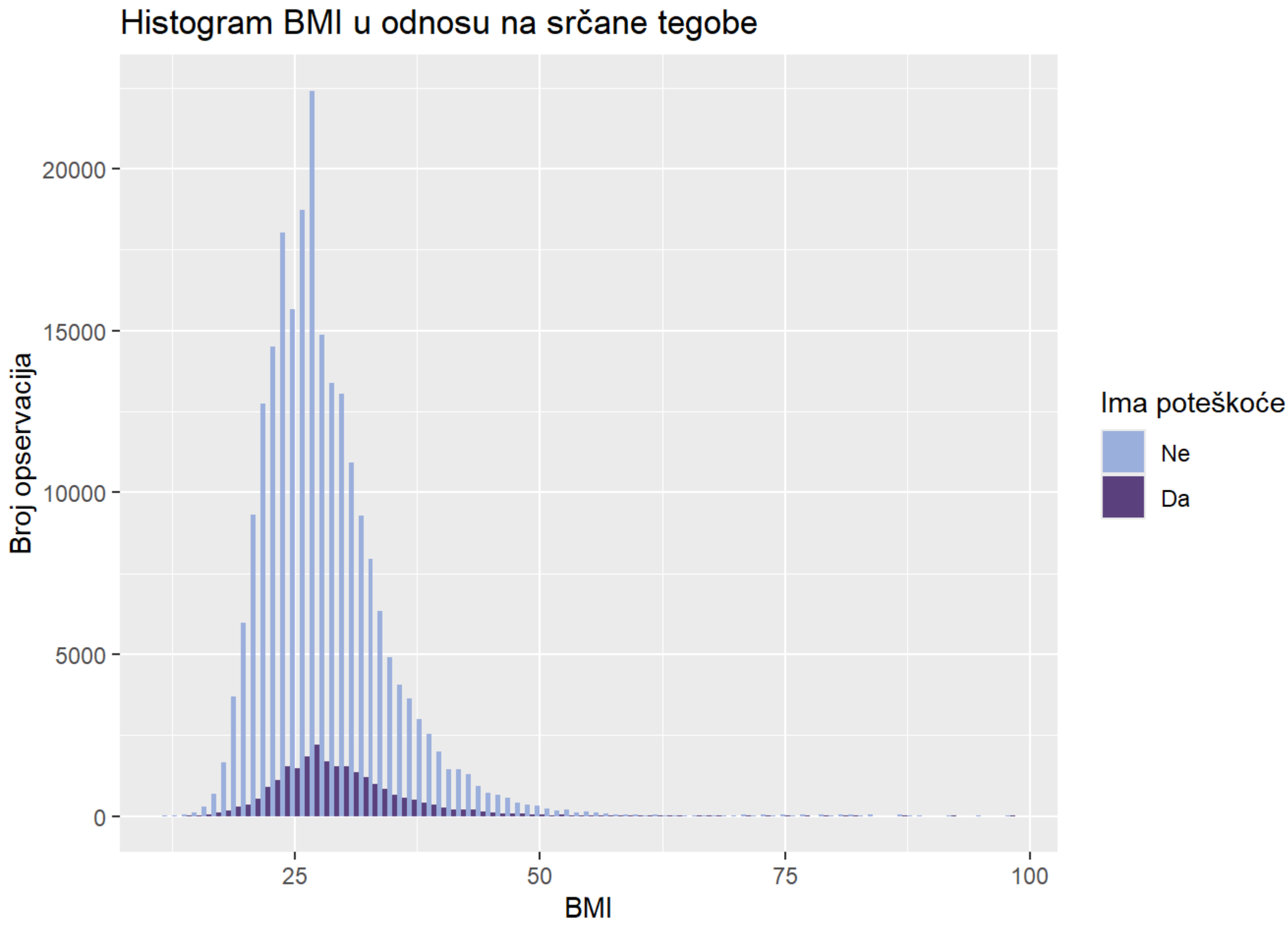
```
cramer_v(data$HighBP,data$HeartDiseaseorAttack)
```

```
## [1] 0.2093476
```

```
ggplot(data, aes(x = HeartDiseaseorAttack, y = BMI, fill = HeartDiseaseorAttack)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values=colorS) +
  labs(title = "Distribucija BMI u odnosu na srčana oboljenja",
       fill = "Srčane poteškoće")
```




```
ggplot(data, aes(x =BMI, fill = HeartDiseaseorAttack)) +
  geom_histogram(binwidth = 1, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values =colorS) +
  labs(title = "Histogram BMI u odnosu na srčane tegobe",
       y = "Broj opservacija",
       fill = "Ima poteškoće")
```



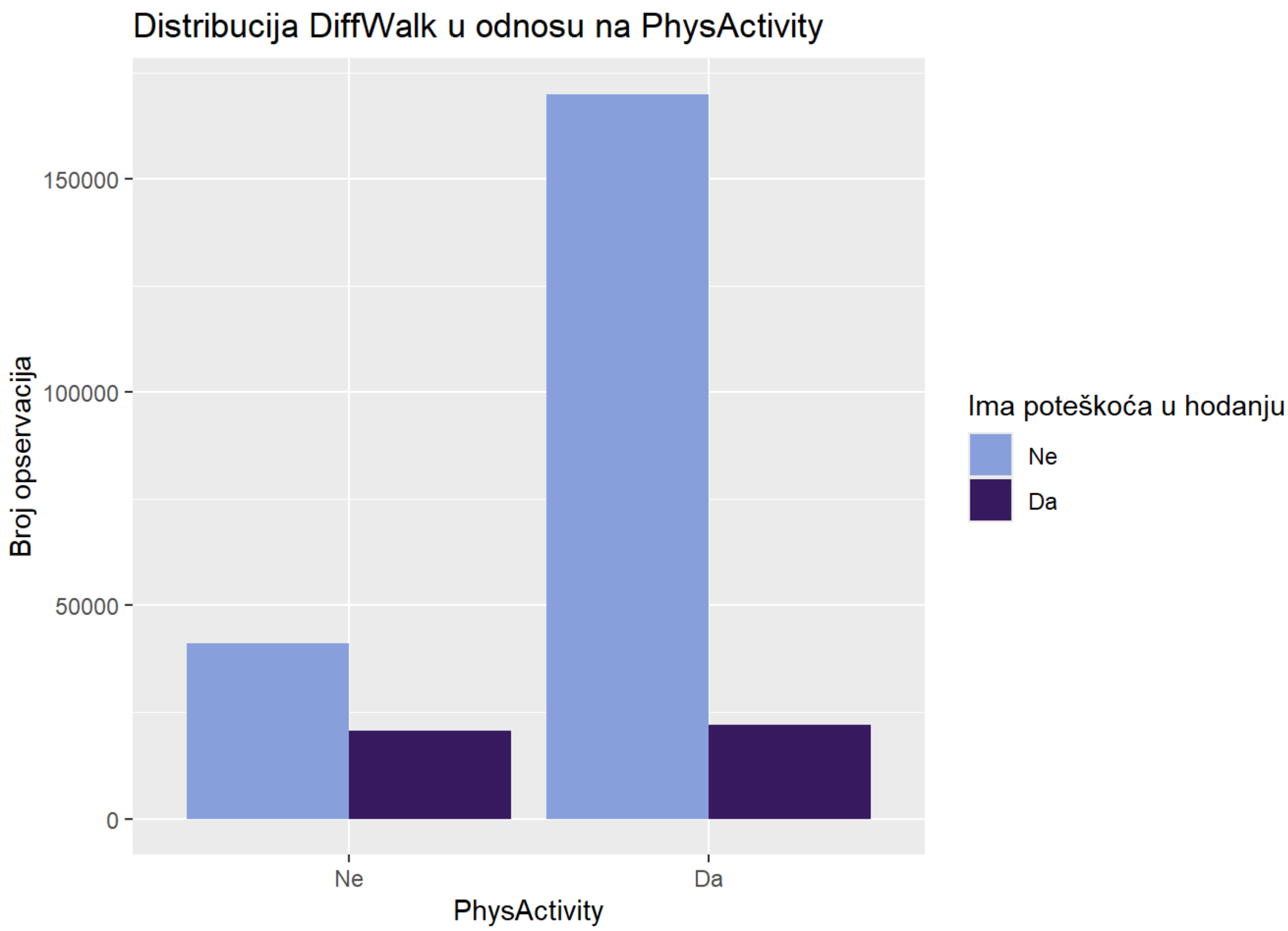

```
anova_test(data$BMI,data$HeartDiseaseorAttack)
```

```
##                              Df   Sum Sq Mean Sq F value Pr(>F)
## as.factor(grupna_var)         1    31010   31010     712 <2e-16 ***
## Residuals                253678 11048380      44
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

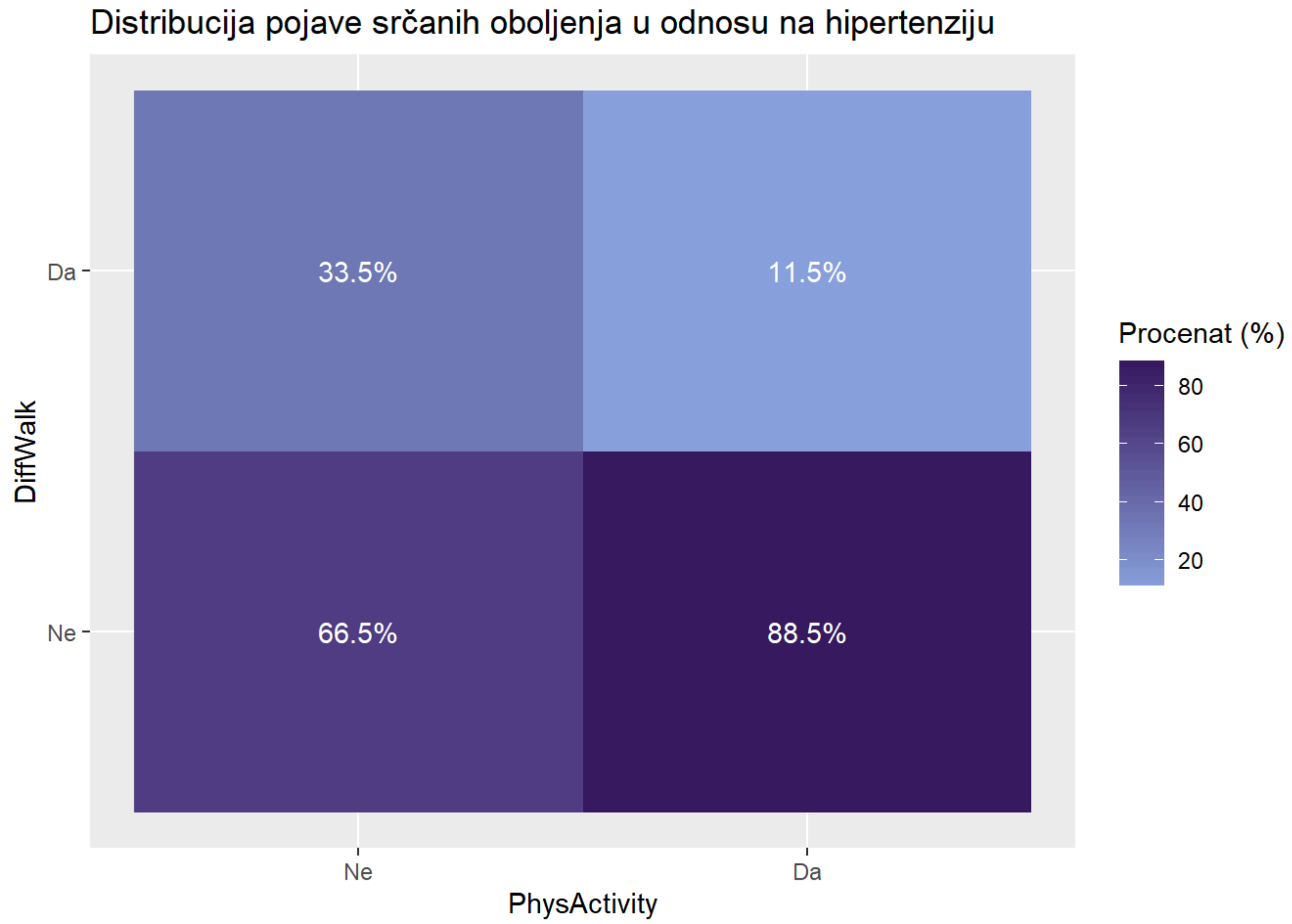
```
tukey_fun(data$BMI,data$HeartDiseaseorAttack)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = numericka_var ~ as.factor(grupna_var))
##
## $`as.factor(grupna_var)`
##           diff      lwr      upr p adj
## Da-Ne 1.196998 1.109076 1.284921     0
```

```
ggplot(data, aes(x = PhysActivity, fill =DiffWalk )) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija DiffWalk u odnosu na PhysActivity",
       y = "Broj opservacija",
       fill = "Ima poteškoća u hodanju")
```




```
data %>%
  count(PhysActivity, DiffWalk) %>%
  group_by(PhysActivity) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = PhysActivity, y = DiffWalk, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija pojave srčanih oboljenja u odnosu na hipertenziju",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



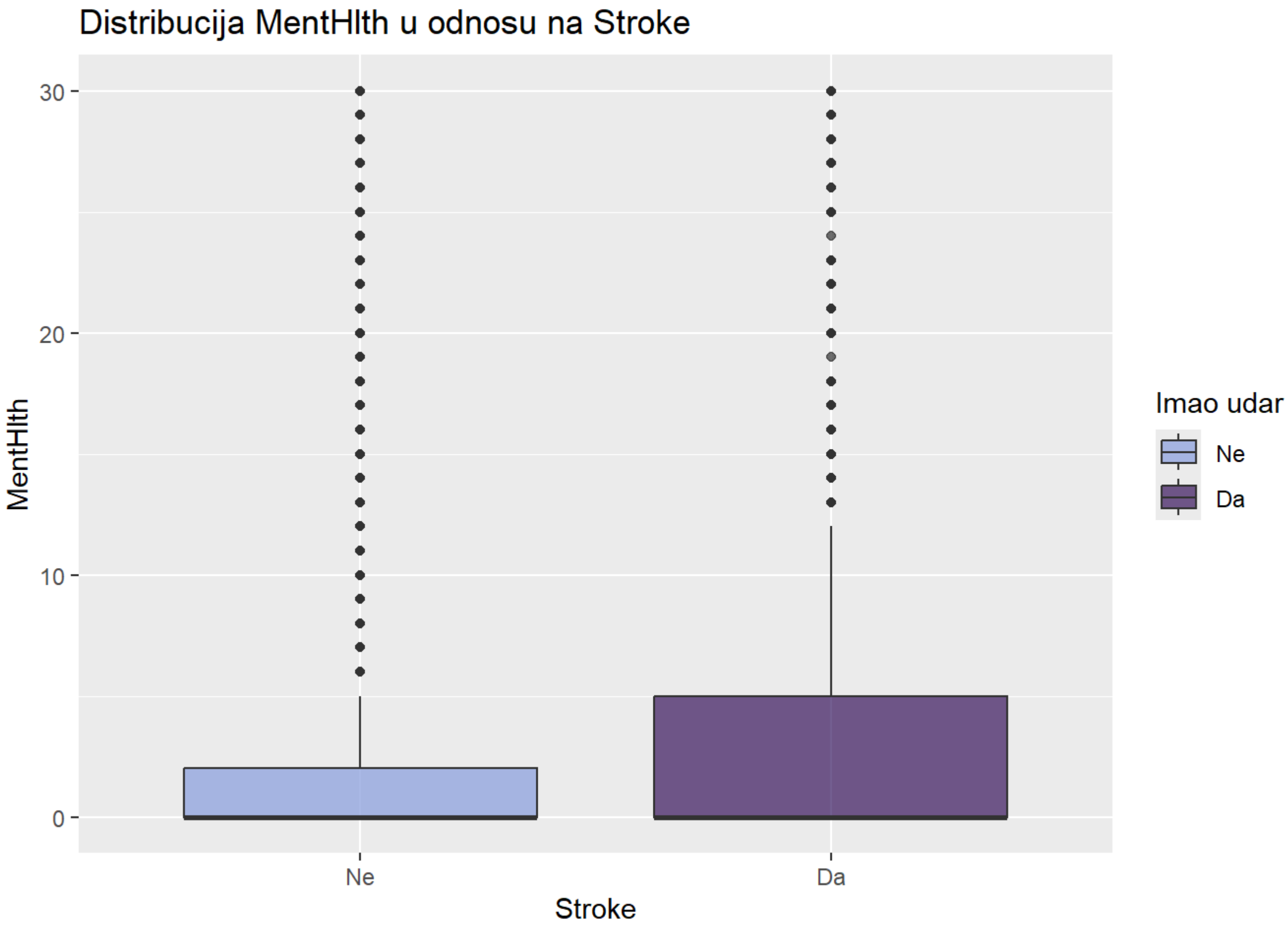

```
chi_sq_test(data$PhysActivity,data$DiffWalk)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tabela
## X-squared = 16259, df = 1, p-value < 2.2e-16
```

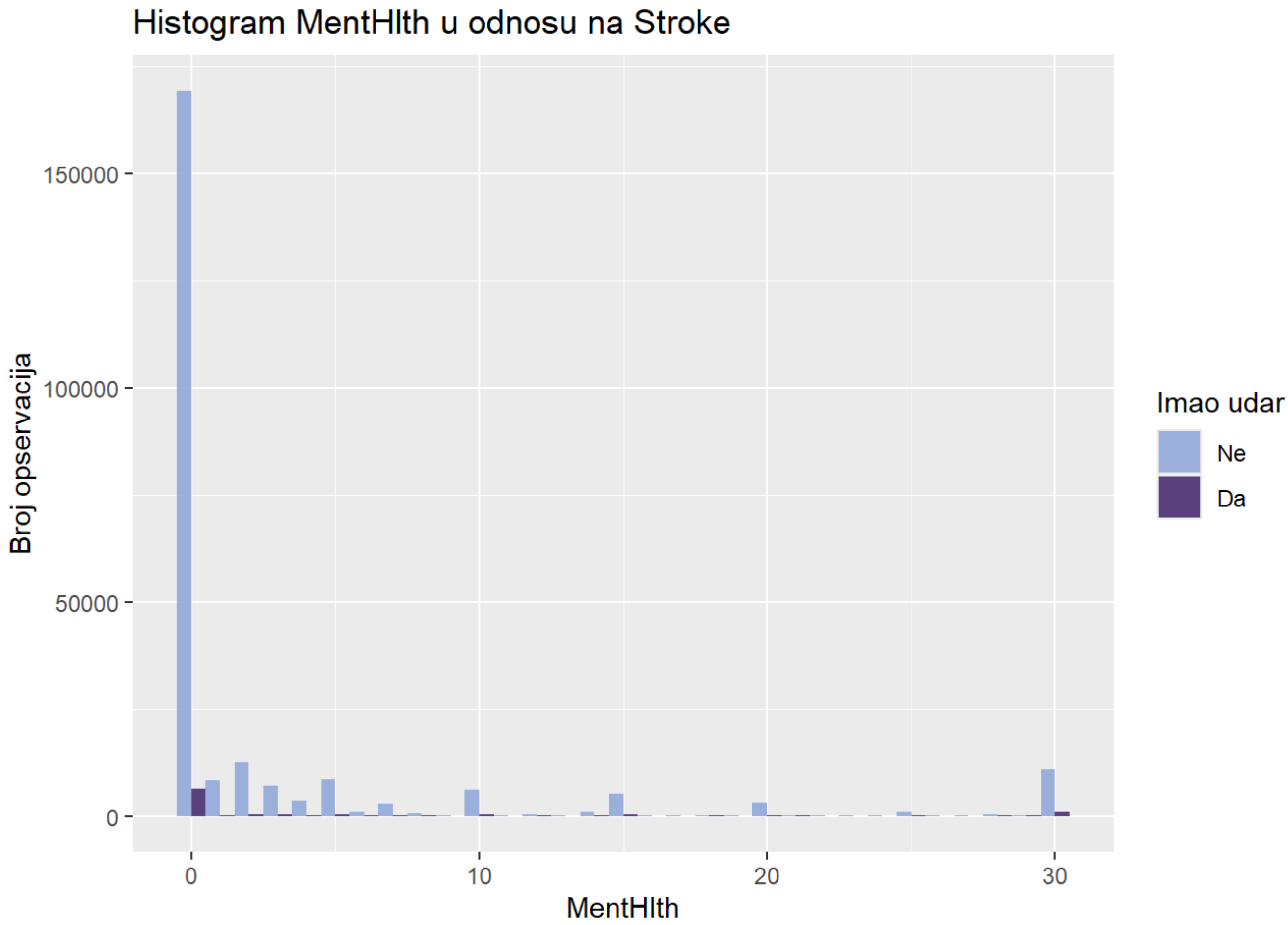
```
cramer_v(data$PhysActivity,data$DiffWalk)
```

```
## [1] 0.2531617
```

```
ggplot(data, aes(x = Stroke, y = MentHlth, fill = Stroke)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values=colorS) +
  labs(title = "Distribucija MentHlth u odnosu na Stroke",
       fill = "Imao udar")
```




```
ggplot(data, aes(x =MentHlth, fill = Stroke)) +
  geom_histogram(binwidth = 1, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values =colorS) +
  labs(title = "Histogram MentHlth u odnosu na Stroke",
       y = "Broj opservacija",
       fill = "Imao udar")
```



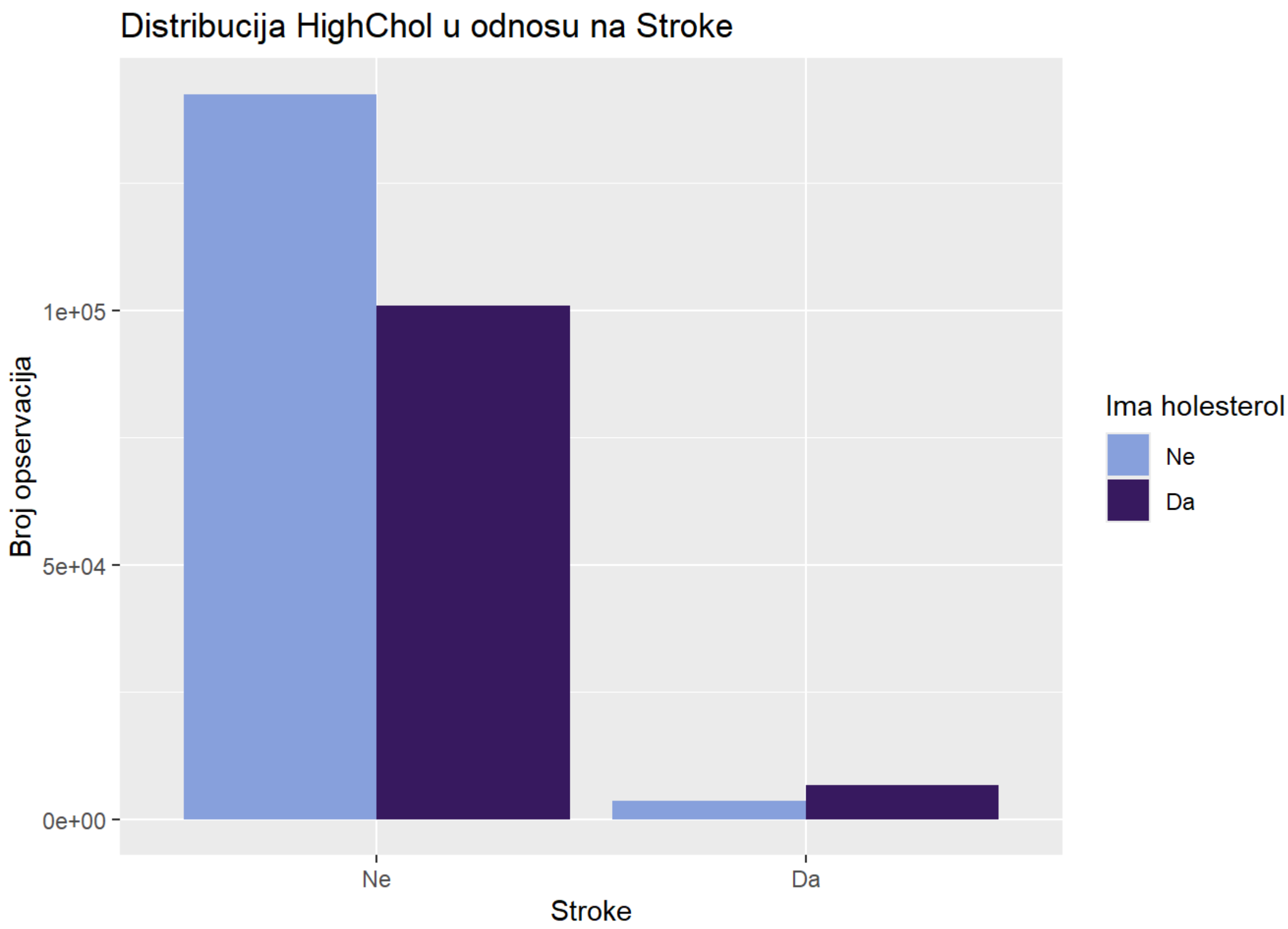

```
anova_test(data$MentHlth,data$Stroke)
```

```
##                              Df    Sum Sq Mean Sq F value Pr(>F)    
## as.factor(grupna_var)         1     68640   68640    1255 <2e-16 ***
## Residuals                253678 13871096      55                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

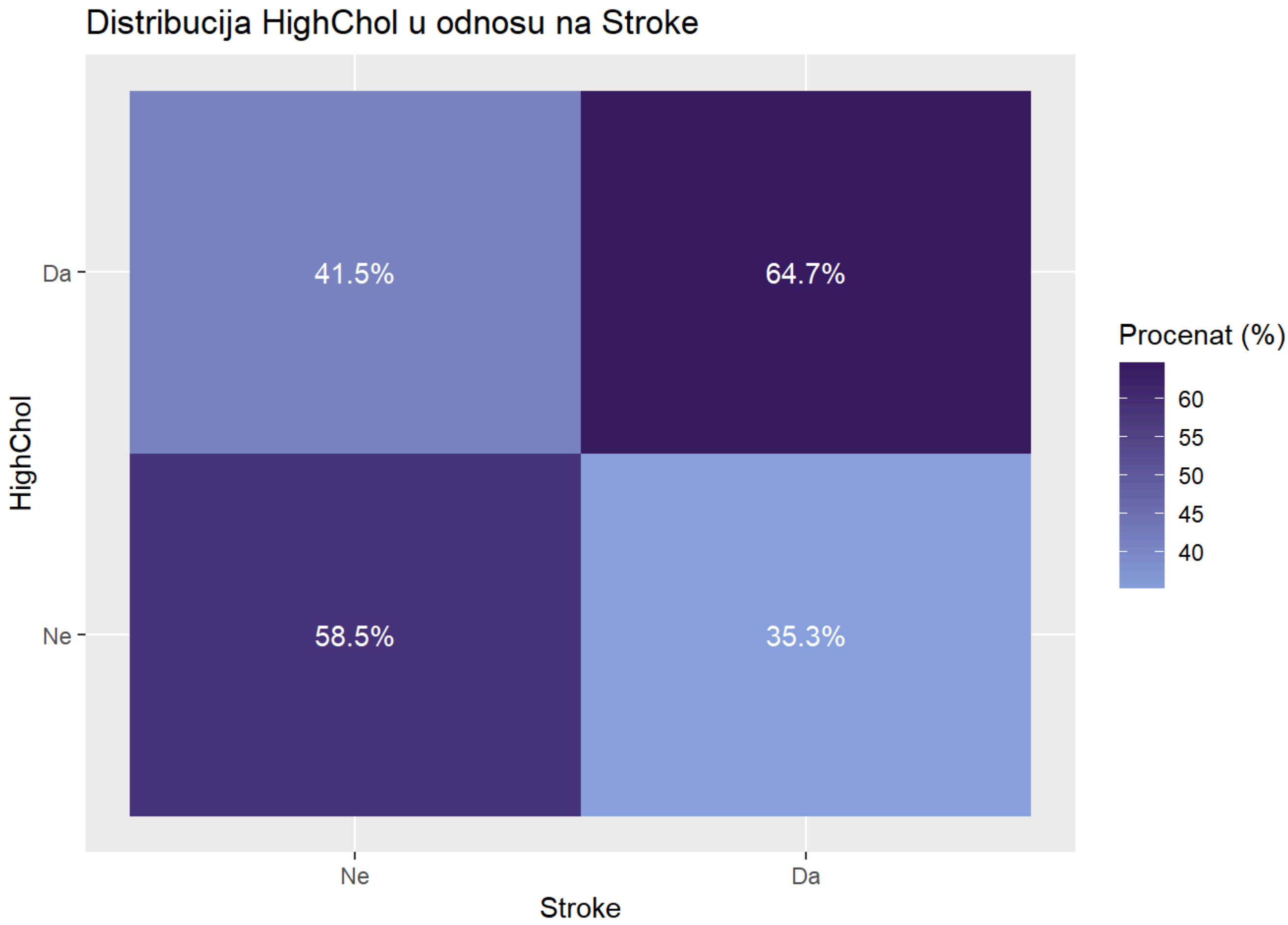
```
tukey_fun(data$MentHlth,data$Stroke)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = numericka_var ~ as.factor(grupna_var))
##
## $`as.factor(grupna_var)`
##           diff      lwr      upr p adj
## Da-Ne 2.636535 2.490686 2.782385     0
```

```
ggplot(data, aes(x = Stroke, fill =HighChol )) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija HighChol u odnosu na Stroke",
       y = "Broj opservacija",
       fill = "Ima holesterol")
```




```
data %>%
  count(Stroke, HighChol) %>%
  group_by(Stroke) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Stroke, y = HighChol, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija HighChol u odnosu na Stroke",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



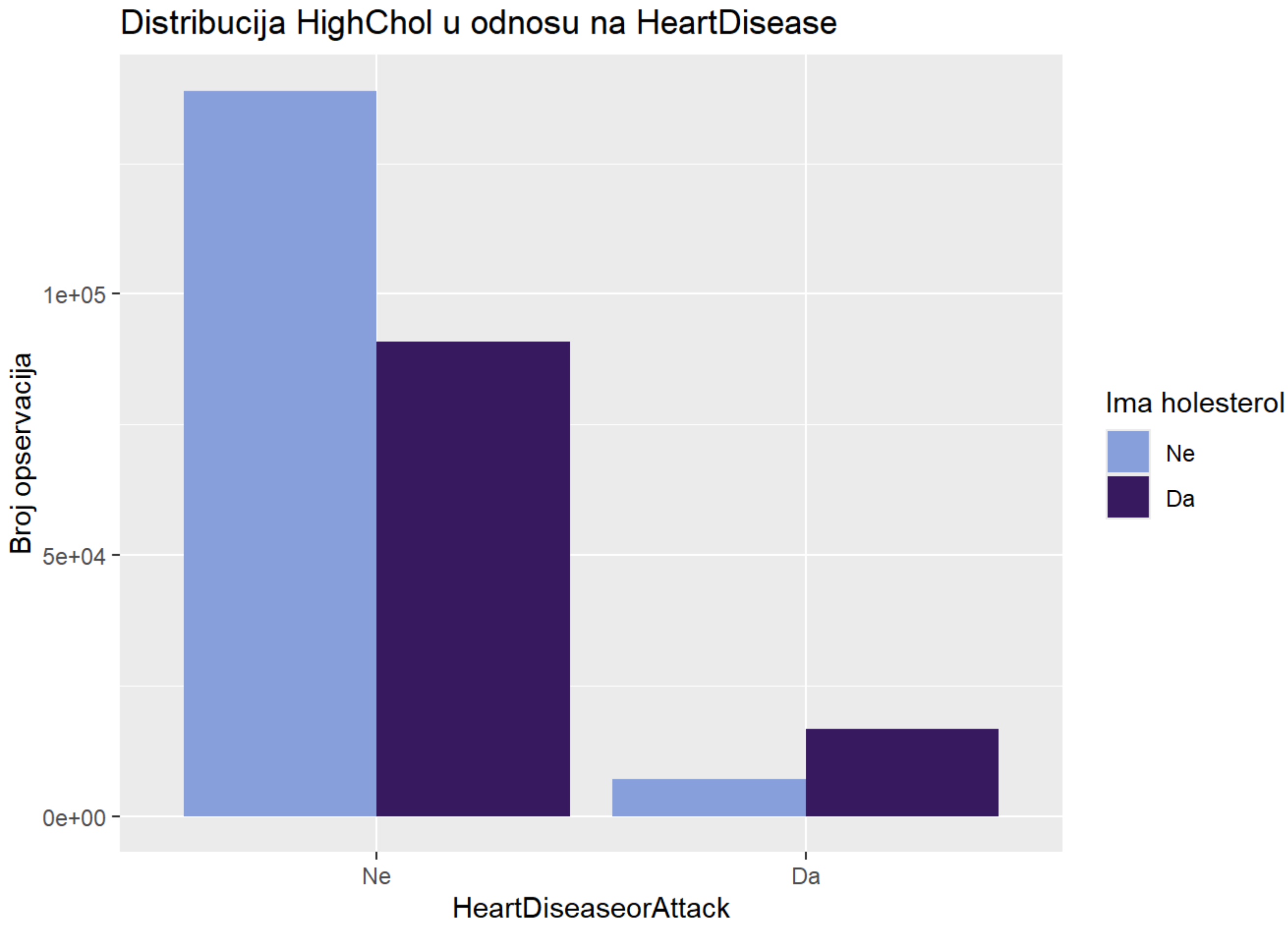

```
chi_sq_test(data$Stroke,data$HighChol)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tabela
## X-squared = 2175.2, df = 1, p-value < 2.2e-16
```

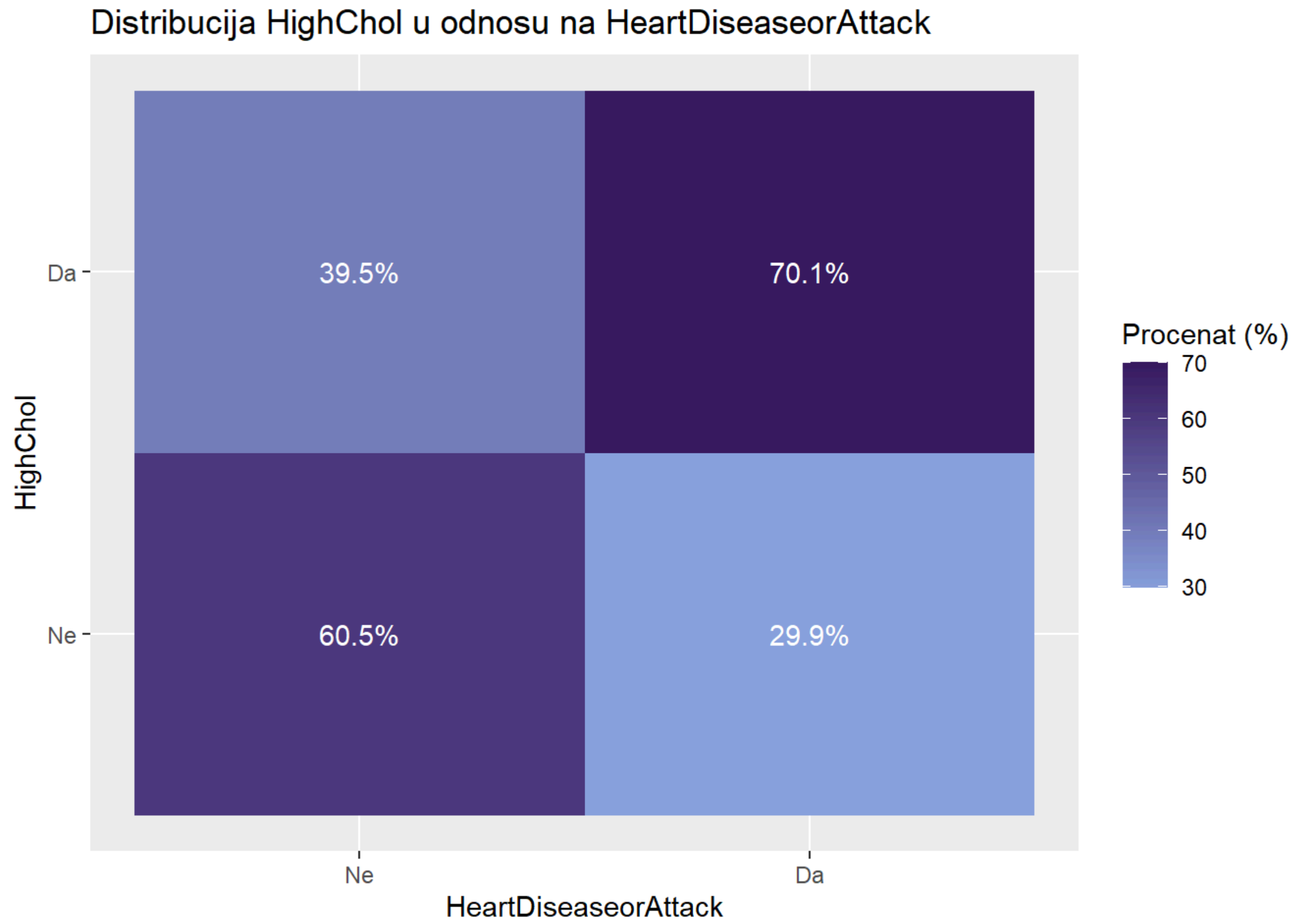
```
cramer_v(data$Stroke,data$HighChol)
```

```
## [1] 0.09259986
```

```
ggplot(data, aes(x = HeartDiseaseorAttack, fill =HighChol )) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija HighChol u odnosu na HeartDisease",
       y = "Broj opservacija",
       fill = "Ima holesterol")
```




```
data %>%
  count(HeartDiseaseorAttack, HighChol) %>%
  group_by(HeartDiseaseorAttack) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = HeartDiseaseorAttack, y = HighChol, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija HighChol u odnosu na HeartDiseaseorAttack",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



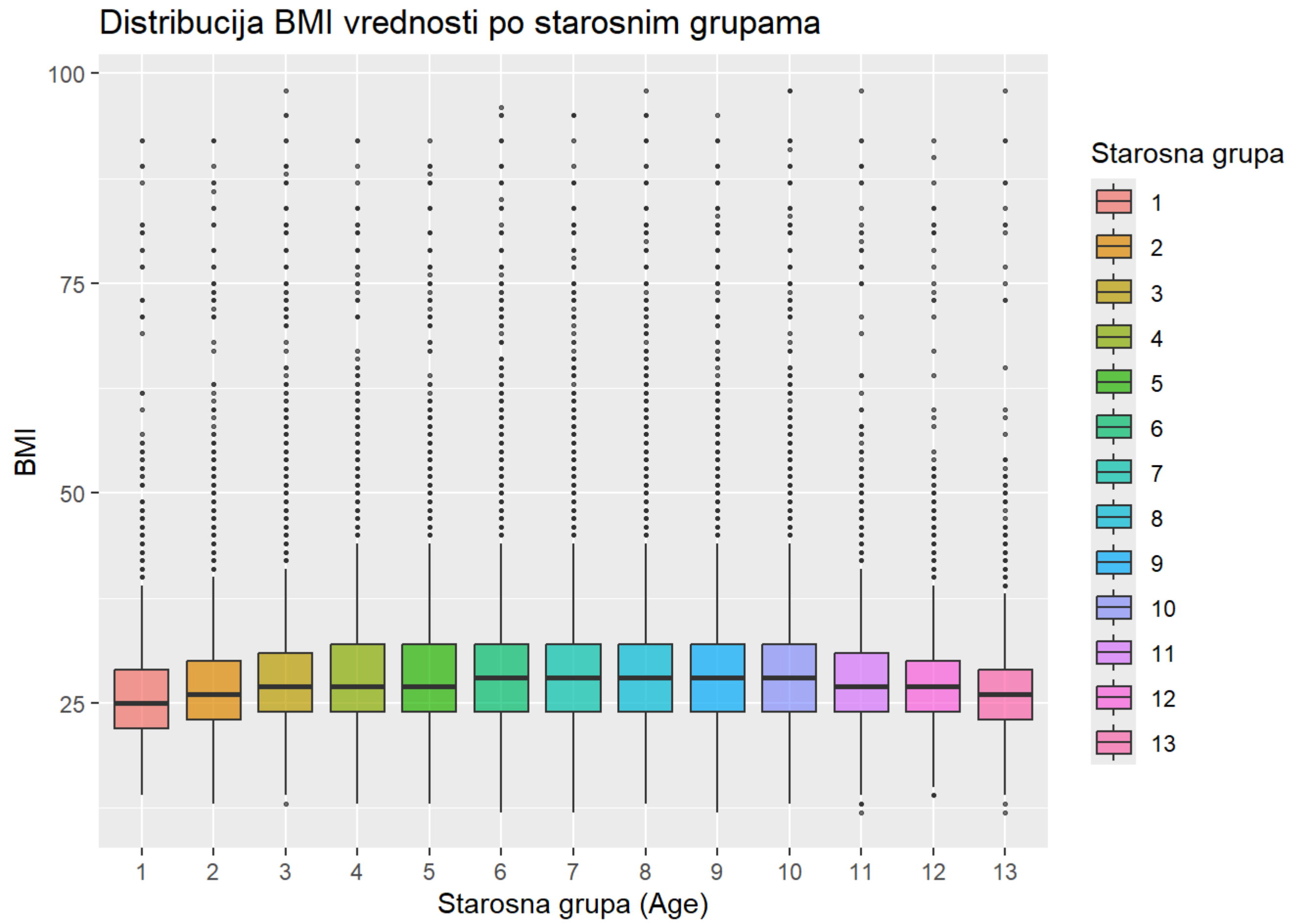

```
chi_sq_test(data$HeartDiseaseorAttack,data$HighChol)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tabela
## X-squared = 8288, df = 1, p-value < 2.2e-16
```

```
cramer_v(data$HeartDiseaseorAttack,data$HighChol)
```

```
## [1] 0.1807517
```

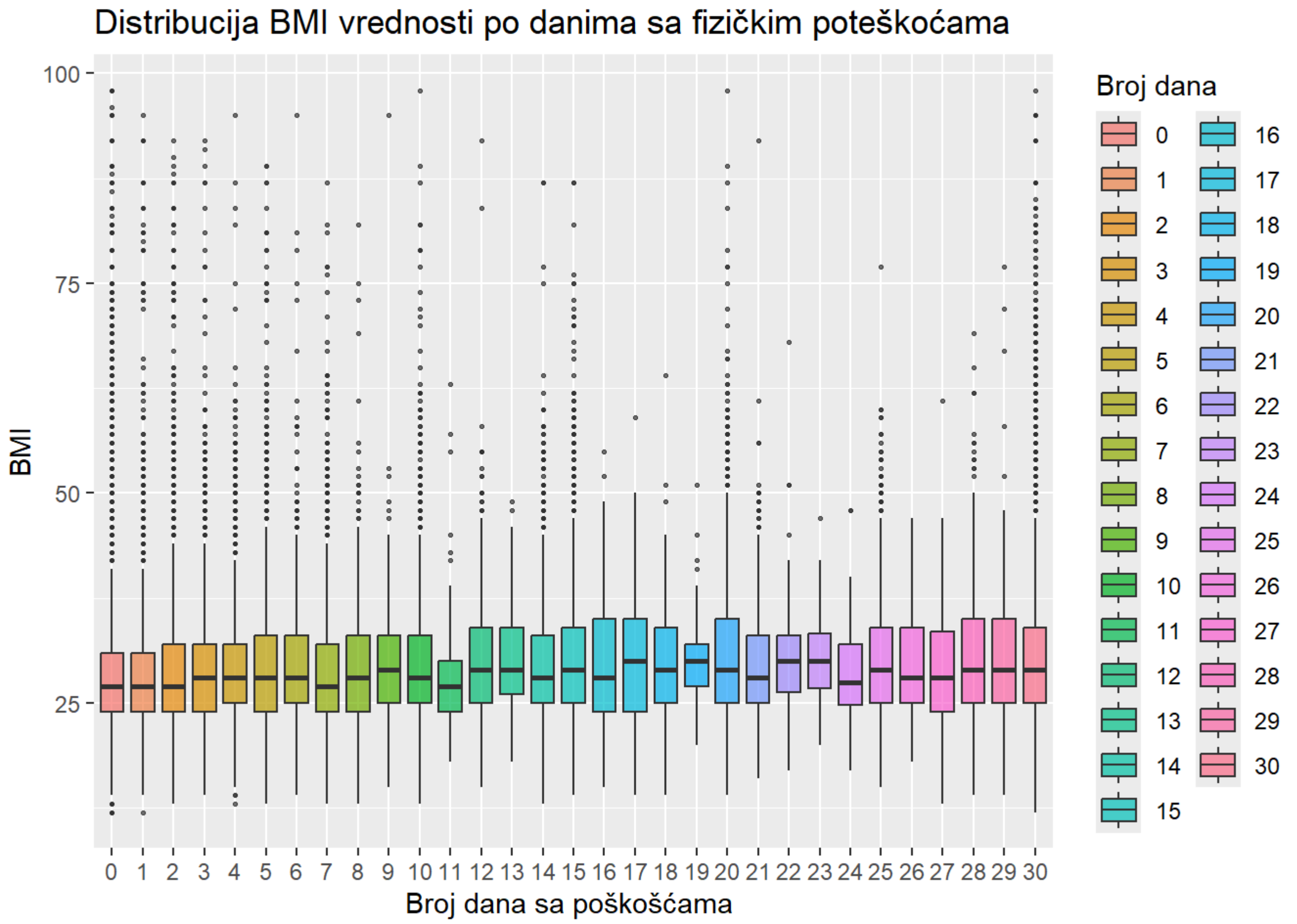
```
library(ggplot2)
ggplot(data, aes(x = as.factor(Age), y = BMI, fill = as.factor(Age))) +
  geom_boxplot(outlier.size = 0.5, alpha = 0.7) +
  labs(title = "Distribucija BMI vrednosti po starosnim grupama",
       x = "Starosna grupa (Age)",
       y = "BMI",
       fill="Starosna grupa")
```




```
pearson_funkcija(data$Age,data$BMI)
```

```
## [1] -0.03661764
```

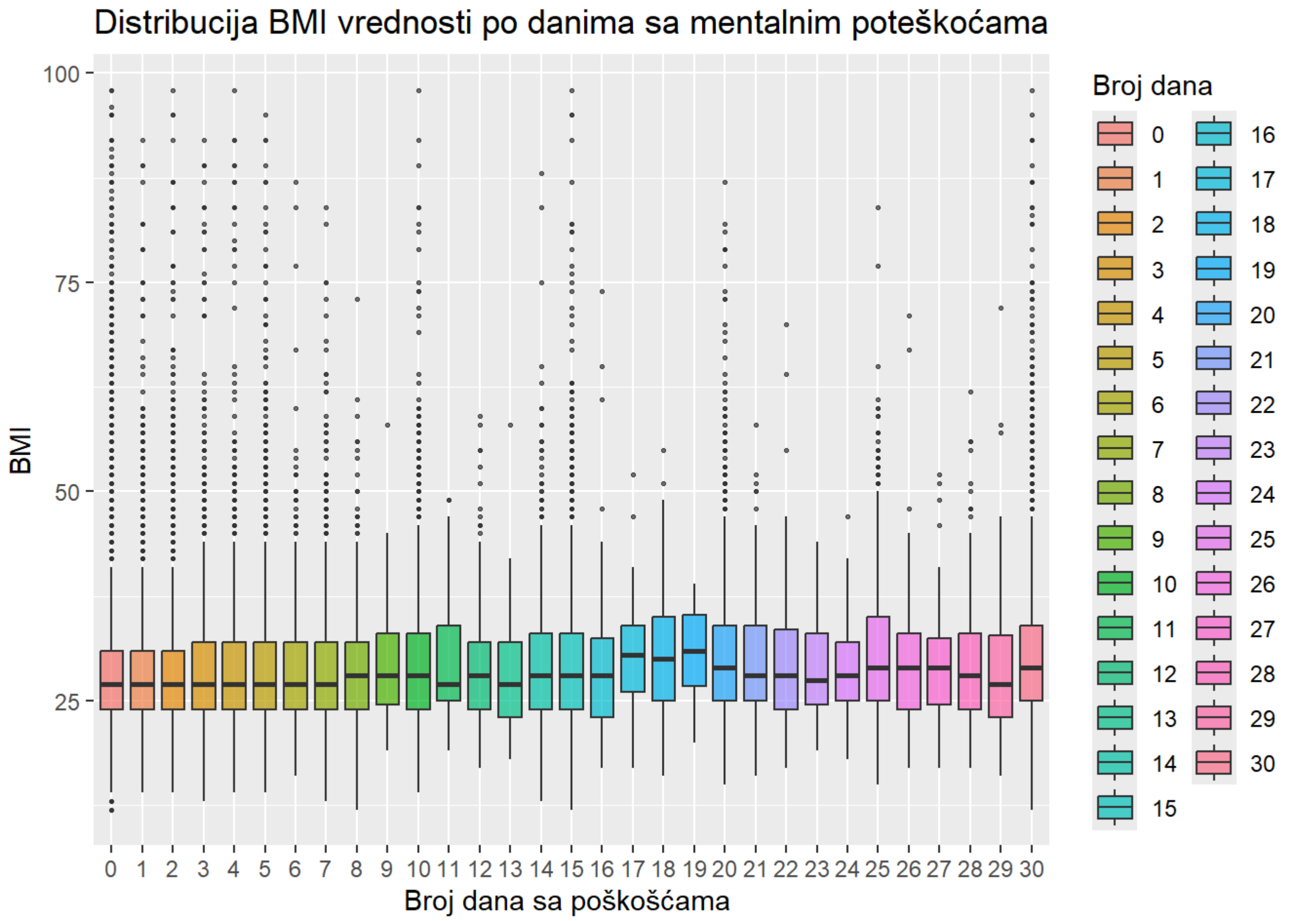
```
ggplot(data, aes(x = as.factor(PhysHlth), y = BMI, fill = as.factor(PhysHlth))) +
  geom_boxplot(outlier.size = 0.5, alpha = 0.7) +
  labs(title = "Distribucija BMI vrednosti po danima sa fizičkim poteškoćama",
       x = "Broj dana sa poškošćama ",
       y = "BMI",
       fill="Broj dana")
```




```
pearson_funkcija(data$PhysHlth,data$BMI)
```

```
## [1] 0.1211411
```

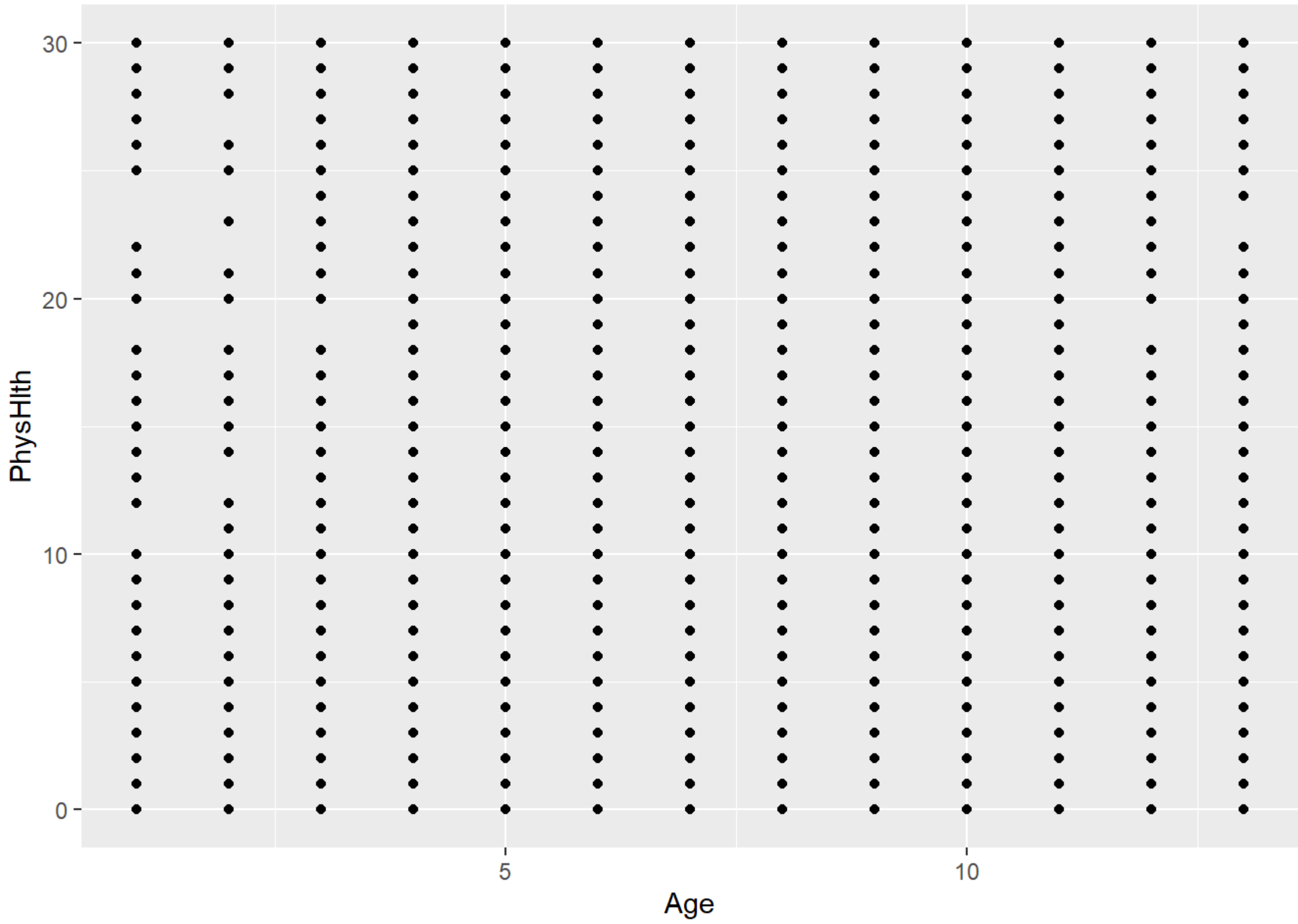
```
ggplot(data, aes(x = as.factor(MentHlth), y = BMI, fill = as.factor(MentHlth))) +
  geom_boxplot(outlier.size = 0.5, alpha = 0.7) +
  labs(title = "Distribucija BMI vrednosti po danima sa mentalnim poteškoćama",
       x = "Broj dana sa poškošćama ",
       y = "BMI",
       fill="Broj dana")
```



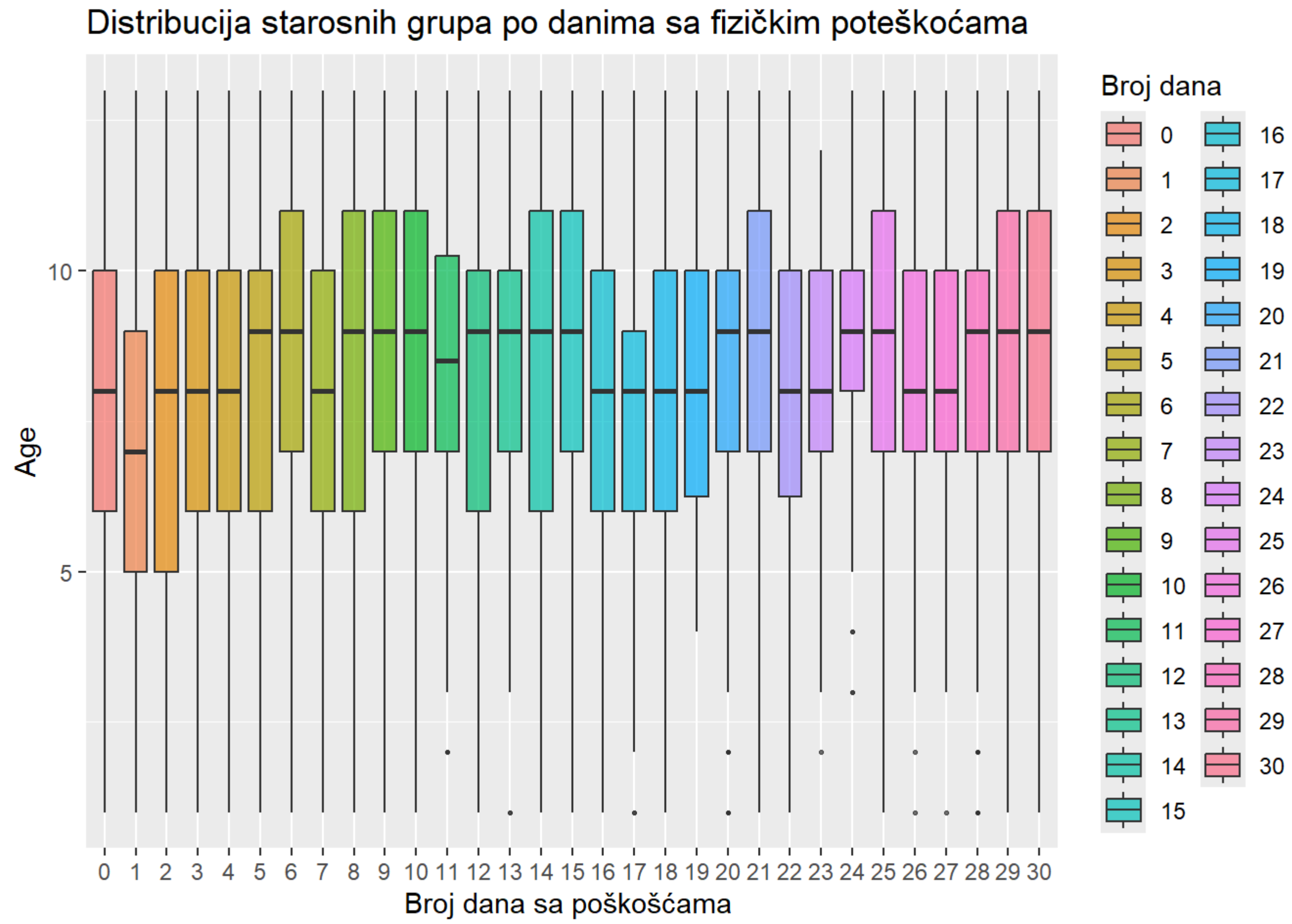

```
pearson_funkcija(data$MentHlth,data$BMI)
```

```
## [1] 0.08531016
```

```
ggplot(data,aes(x=Age,y=PhysHlth))+geom_point()
```




```
ggplot(data, aes(x = as.factor(PhysHlth), y = Age, fill = as.factor(PhysHlth))) +
  geom_boxplot(outlier.size = 0.5, alpha = 0.7) +
  labs(title = "Distribucija starosnih grupa po danima sa fizičkim poteškoćama",
       x = "Broj dana sa poškošćama ",
       y = "Age",
       fill="Broj dana")
```




```
pearson_funkcija(data$Age,data$PhysHlth)
```

```
## [1] 0.09912993
```

##Čišćenje podataka

```
data_clean <- data[data$BMI >= 12 & data$BMI <= 70, ]

print(data.frame(
  Skup = c("Originalni podaci", "Nakon čišćenja BMI"),
  Broj_opservacija = c(nrow(data), nrow(data_clean))
))
```

```
##                  Skup Broj_opservacija
## 1  Originalni podaci           253680
## 2 Nakon čišćenja BMI           253096
```

##Transformacija podataka

```
nivoi_Age = sort(unique(data_clean$Age))
data_clean$Age = factor(data_clean$Age, levels = nivoi_Age, ordered = TRUE)
str(data_clean$Age)
```

```
##  Ord.factor w/ 13 levels "1"<"2"<"3"<"4"<..: 9 7 9 11 11 10 9 11 9 8 ...
```

##Feature Engeenering

```
category_physHlth = c("nema problema", "blagi problemi", "umereni problemi", "teski pr
oblemi")
interval_physHlth = c(0, 5, 15, 30)
data_clean$PhysHlthCat = NA
data_clean$PhysHlthCat[data_clean$PhysHlth == interval_physHlth[1]] = category_physHlt
h[1]
data_clean$PhysHlthCat[data_clean$PhysHlth > interval_physHlth[1] & interval_physHlth
[2] <= 5] = category_physHlth[2]
data_clean$PhysHlthCat[data_clean$PhysHlth > interval_physHlth[2] & interval_physHlth
[3] <= 15] = category_physHlth[3]
data_clean$PhysHlthCat[data_clean$PhysHlth > interval_physHlth[3] & interval_physHlth
[4] <= 30] = category_physHlth[4]

data_clean$PhysHlthCat <- factor(data_clean$PhysHlthCat, levels = category_physHlth,or
dered = TRUE)
str(data_clean$PhysHlthCat)
```

```
##  Ord.factor w/ 4 levels "nema problema"<..: 3 1 4 1 1 2 3 1 4 1 ...
```

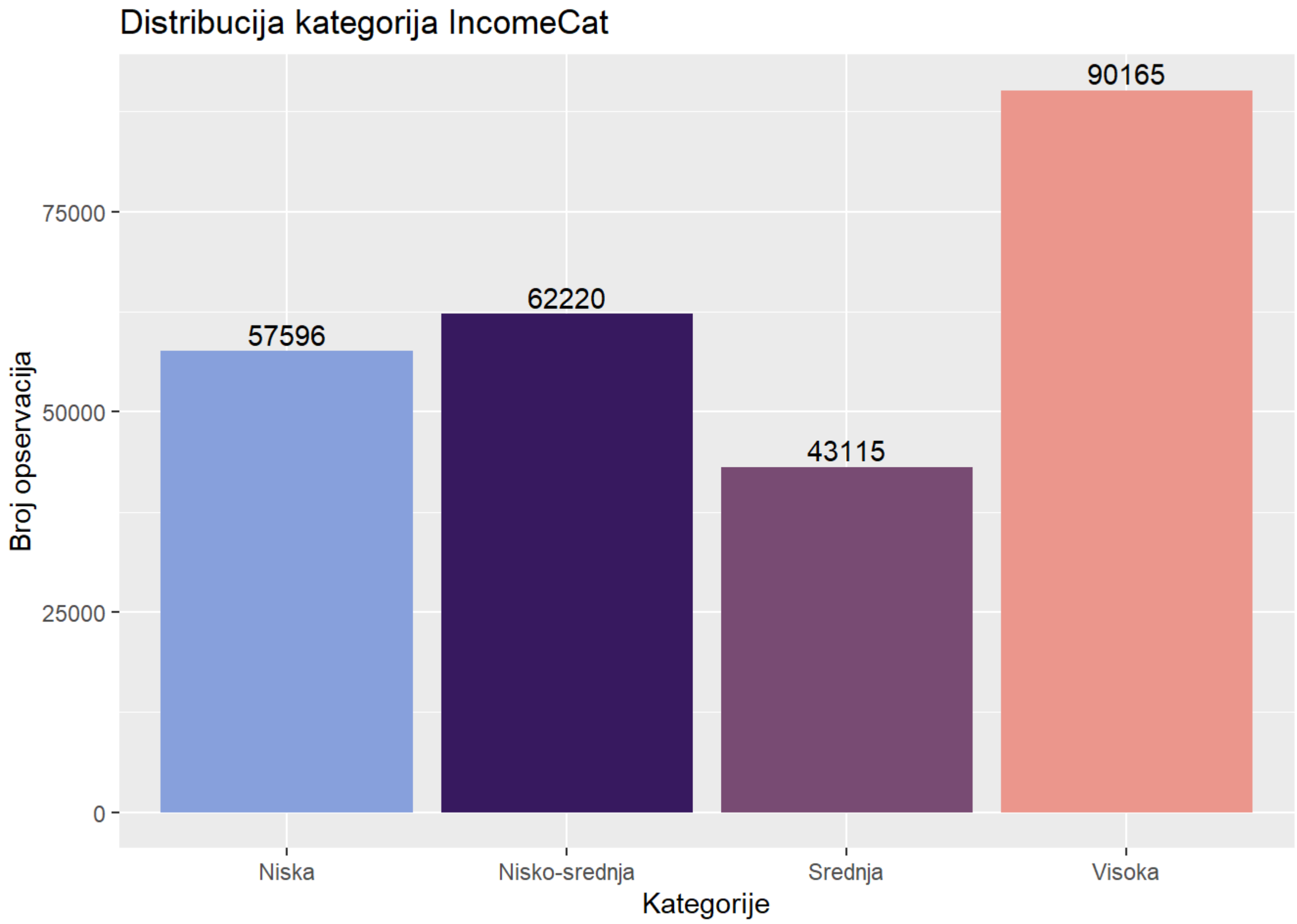
```
category_mentHlth = c("nema problema", "blagi problemi", "umereni problemi", "teski pr
oblemi")
interval_mentHlth = c(0, 5, 15, 30)
data_clean$MentHlthCat = NA
data_clean$MentHlthCat[data_clean$MentHlth == interval_mentHlth[1]] = category_mentHlt
h[1]
data_clean$MentHlthCat[data_clean$MentHlth > interval_mentHlth[1] & interval_mentHlth
[2] <= 5] = category_mentHlth[2]
data_clean$MentHlthCat[data_clean$MentHlth > interval_mentHlth[2] & interval_mentHlth
[3] <= 15] = category_mentHlth[3]
data_clean$MentHlthCat[data_clean$MentHlth > interval_mentHlth[3] & interval_mentHlth
[4] <= 30] = category_mentHlth[4]

data_clean$MentHlthCat <- factor(data_clean$MentHlthCat, levels = category_mentHlth,or
dered = TRUE)
str(data_clean$MentHlthCat)
```

```
##  Ord.factor w/ 4 levels "nema problema"<..: 4 1 4 1 2 1 1 1 4 1 ...
```

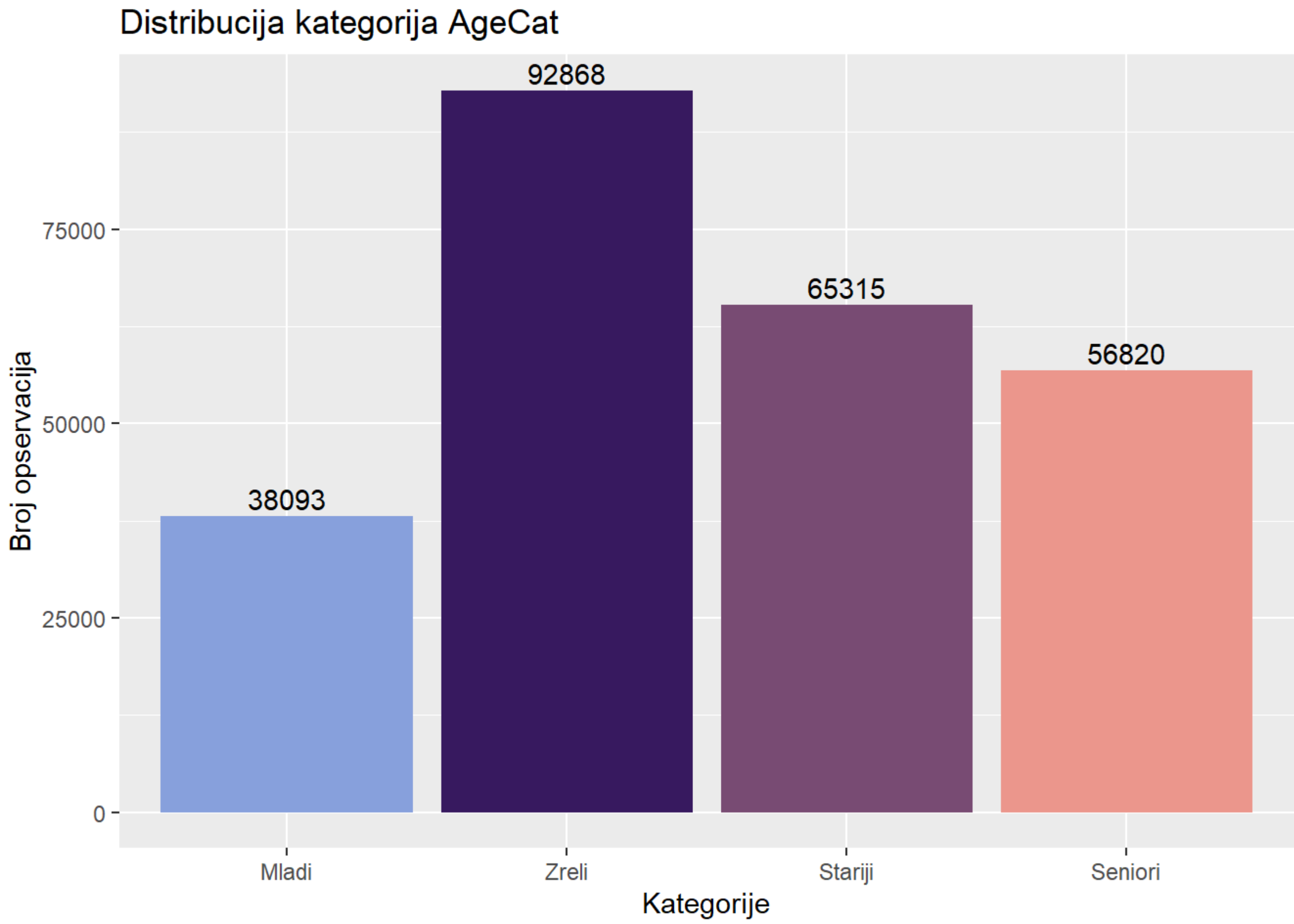
```
prop.table(table(data$Education,data$Diabetes_012), margin = 1)*100
```

```
##
##                          nema dijabetes predijabetes dijabetes
##   bez skole/isao u vrtic      71.839080     1.149425 27.011494
##   osnovna skola               66.757358     3.982191 29.260450
##   3-god srednja               72.462545     3.312935 24.224520
##   4-god srednja               80.213546     2.151394 17.635060
##   3-god fakultet              83.282792     1.906737 14.810471
##   4-god fakultet              88.939203     1.370603  9.690193
```

```
nivoi_Education = levels(data_clean$Education)
data_clean$EducationCat = factor(data_clean$Education,
                             levels = nivoi_Education,
                             labels = c("Nisko",
                                        "Osnovno",
                                        "Srednje",
                                        "Srednje",
                                        "Visoko",
                                        "Visoko")
)
nivoi_Income = levels(data_clean$Income)
data_clean$IncomeCat = factor(data_clean$Income,
                              levels = nivoi_Income,
                              labels = c("Niska",
                                         "Niska",
                                         "Niska",
                                         "Niska",
                                         "Nisko-srednja",
                                         "Nisko-srednja",
                                         "Srednja",
                                         "Visoka"))
ggplot(data_clean, aes(x = IncomeCat, fill = IncomeCat)) +
  geom_bar() +
  labs(title = "Distribucija kategorija IncomeCat", x = "Kategorije", y = "Broj opserv
acija") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3 ) +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



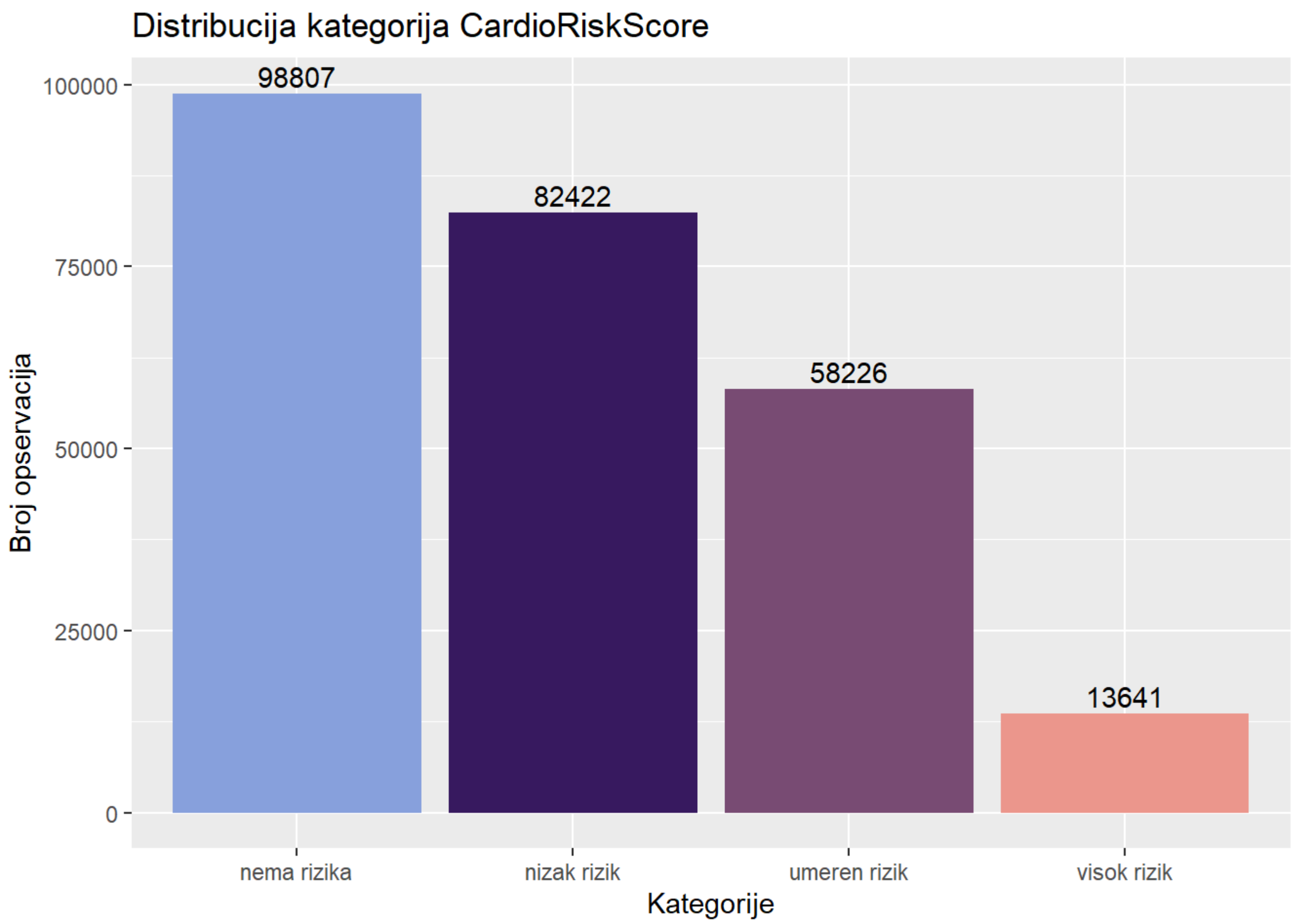

```
data_clean$AgeCat <- factor(data_clean$Age, levels = 1:13)

levels(data_clean$AgeCat) <- c(rep("Mladi", 4), rep("Zreli", 4), rep("Stariji", 2), re
p("Seniori", 3))
ggplot(data_clean, aes(x = AgeCat, fill = AgeCat)) +
  geom_bar() +
  labs(title = "Distribucija kategorija AgeCat", x = "Kategorije", y = "Broj opservaci
ja") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3 ) +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
data_clean$CardioRiskScore = as.numeric(data_clean$HighBP == "Da") +
                             as.numeric(data_clean$HighChol == "Da") +
                             as.numeric(data_clean$HeartDiseaseorAttack == "Da")

nivoi_CardioRiskScore = sort(unique(data_clean$CardioRiskScore))
category_cardioRiskScore = c("nema rizika", "nizak rizik", "umeren rizik", "visok rizi
k")
data_clean$CardioRiskScore = factor(data_clean$CardioRiskScore,
                                    levels = nivoi_CardioRiskScore,
                                    labels = category_cardioRiskScore,
                                    ordered = TRUE)
str(data_clean$CardioRiskScore)
```

```
##  Ord.factor w/ 4 levels "nema rizika"<..: 3 1 3 2 3 3 2 3 4 1 ...
```

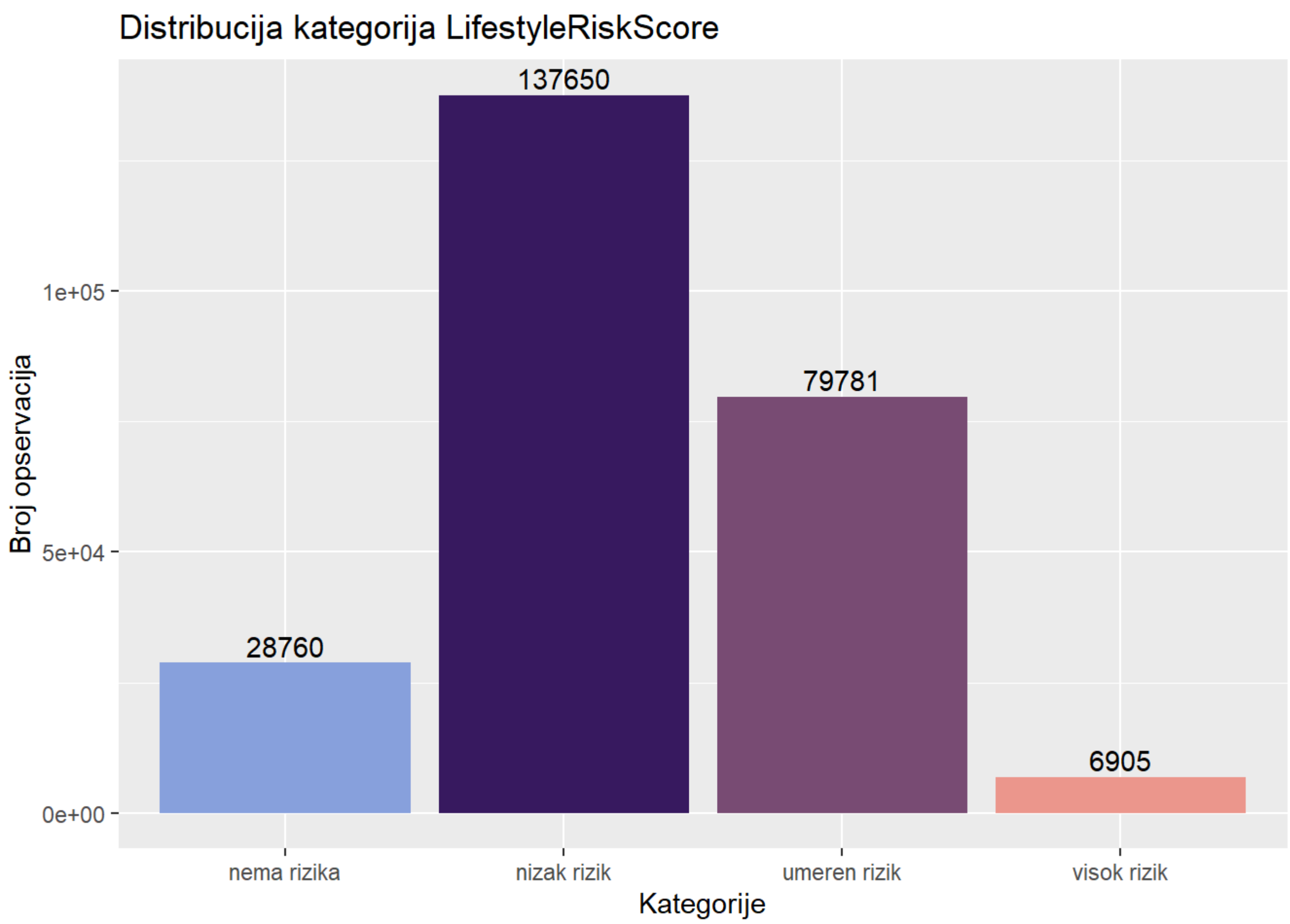
```
ggplot(data_clean, aes(x = CardioRiskScore, fill = CardioRiskScore)) +
  geom_bar() +
  labs(title = "Distribucija kategorija CardioRiskScore", x = "Kategorije", y = "Broj 
opservacija") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



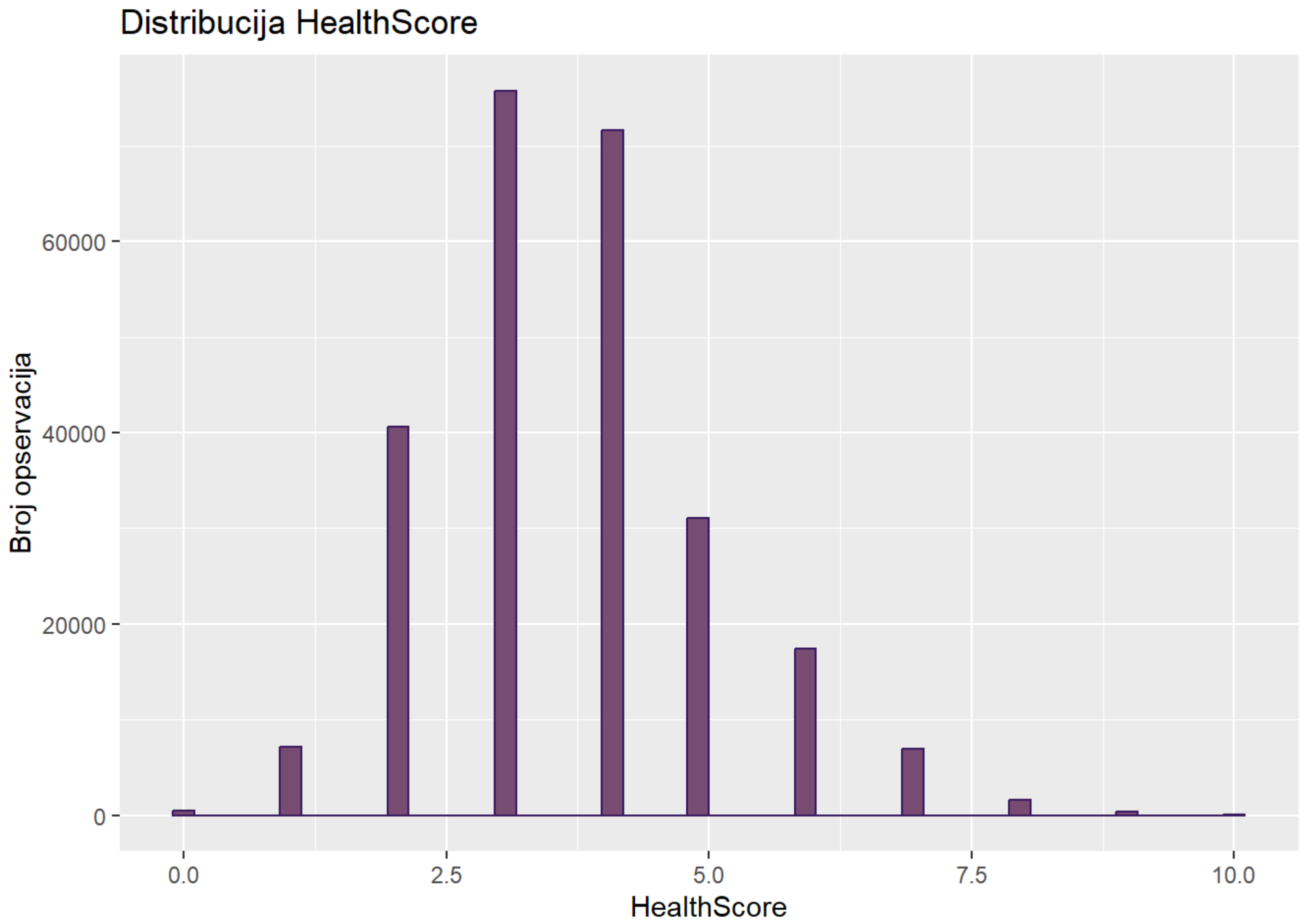

```
data_clean$LifestyleRiskScore = as.numeric(data_clean$Smoker == "Da") +
                                as.numeric(data_clean$HvyAlcoholConsump == "Da") +
                                as.numeric(data_clean$PhysActivity == "Da")

nivoi_LifestyleRiskScore = sort(unique(data_clean$LifestyleRiskScore))
category_lifestyleRiskScore = c("nema rizika", "nizak rizik", "umeren rizik", "visok r
izik")
data_clean$LifestyleRiskScore = factor(data_clean$LifestyleRiskScore,
                                    levels = nivoi_LifestyleRiskScore,
                                    labels = category_lifestyleRiskScore,
                                    ordered = TRUE)
str(data_clean$LifestyleRiskScore)
```

```
##  Ord.factor w/ 4 levels "nema rizika"<..: 2 3 1 2 2 3 2 3 2 1 ...
```

```
ggplot(data_clean, aes(x = LifestyleRiskScore, fill = LifestyleRiskScore)) +
  geom_bar() +
  labs(title = "Distribucija kategorija LifestyleRiskScore", x = "Kategorije", y = "Br
oj opservacija") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



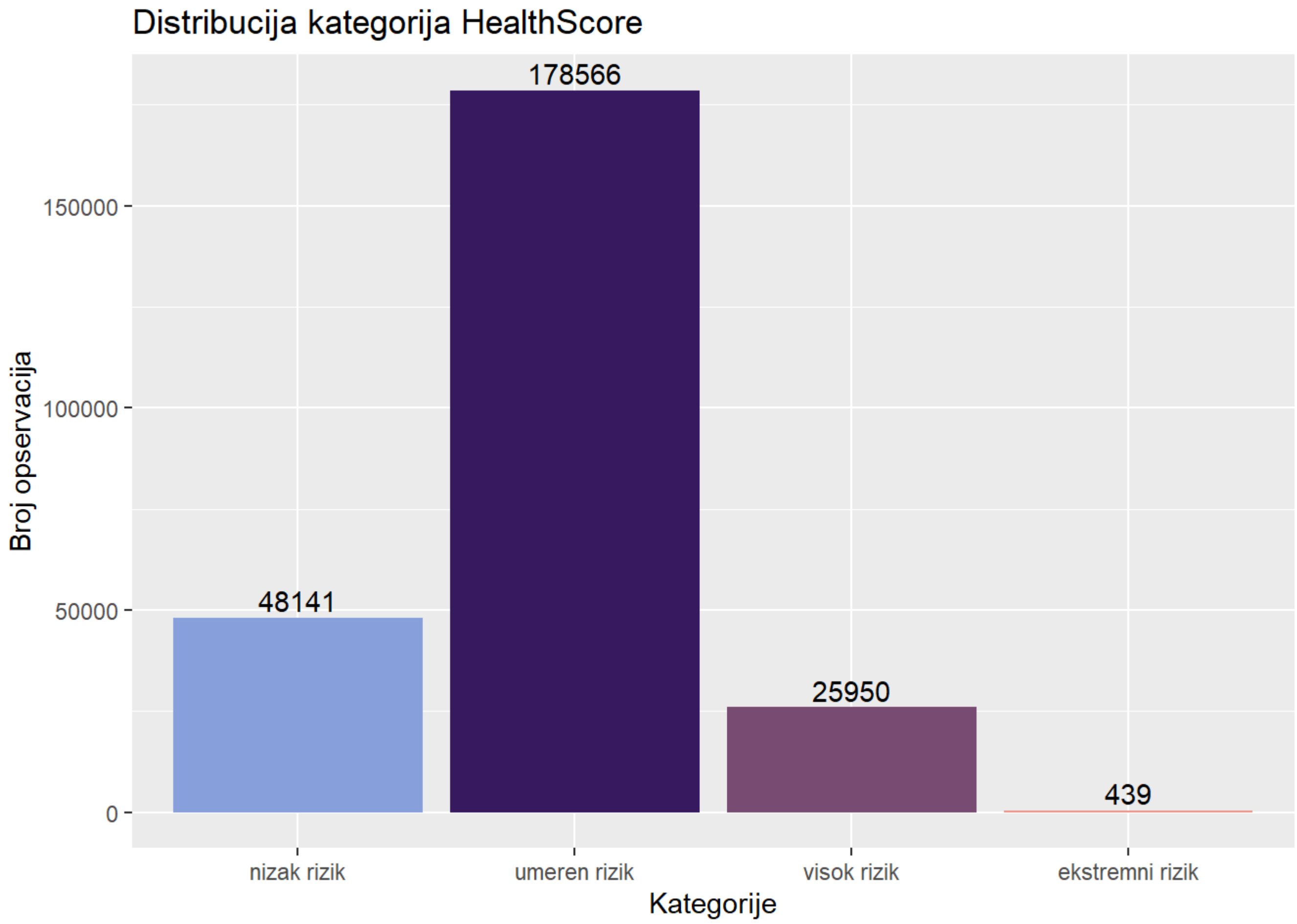

```
data_clean$HealthScore = (as.numeric(data_clean$GenHlth) - 1) +
  (as.numeric(data_clean$PhysHlthCat) - 1) +
  (as.numeric(data_clean$MentHlthCat) - 1)

ggplot(data_clean, aes(x = HealthScore)) +
  geom_histogram(bins = 50, fill = "#7C4B73FF", color="#381A61FF") +
  labs(title = "Distribucija HealthScore", x = "HealthScore", y = "Broj opservacija")
```




```
summary(data_clean$HealthScore)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   3.000   4.000   3.666   4.000  10.000
```

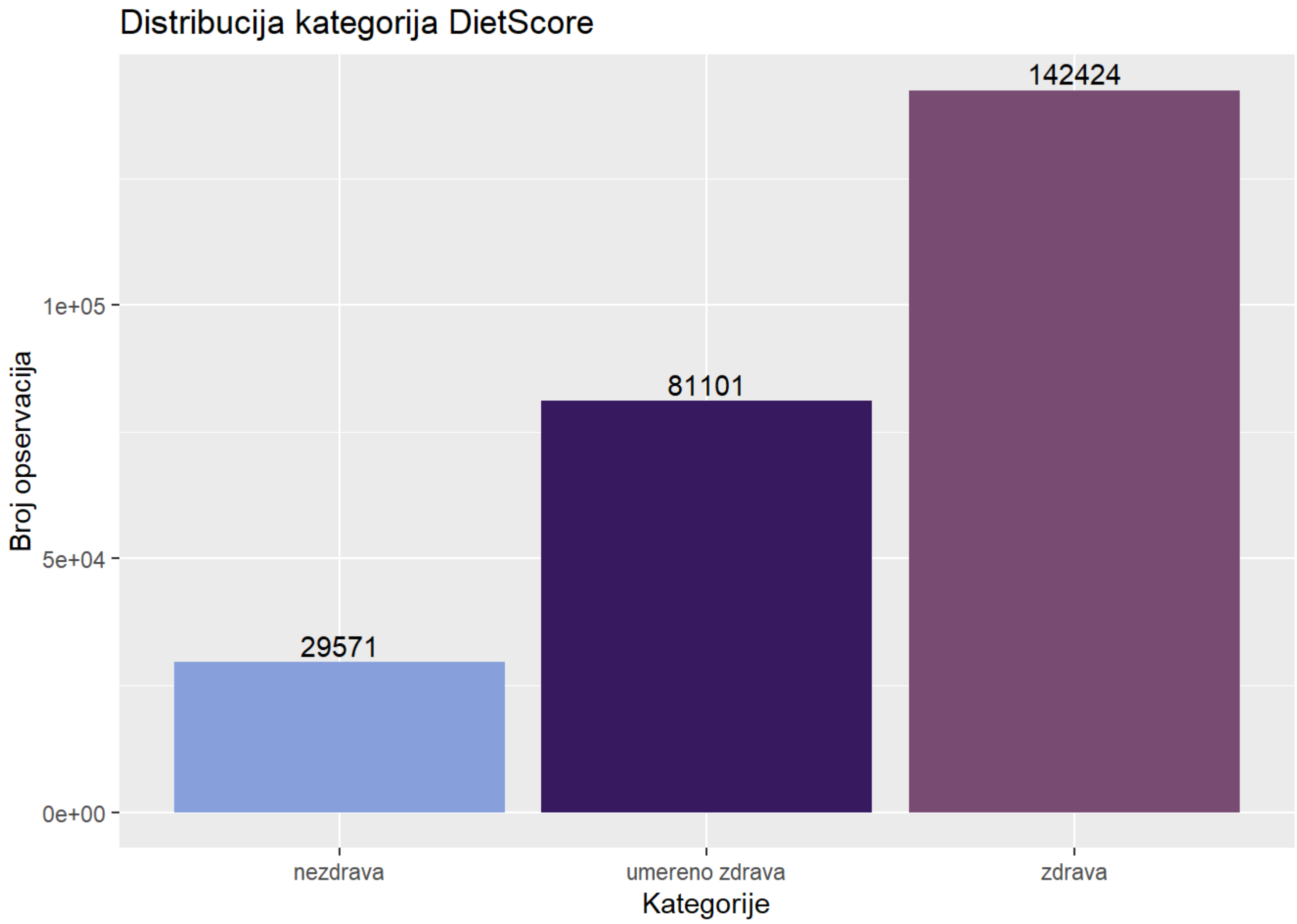
```
category_HealthScore = c("nizak rizik", "umeren rizik", "visok rizik", "ekstremni rizi
k")
interval_HealthScore = c(-1, 2, 5, 8, 10)
data_clean$HealthScore <- cut(data_clean$HealthScore,
                                breaks = interval_HealthScore,
                                labels = category_HealthScore,
                                ordered_result = TRUE)

ggplot(data_clean, aes(x = HealthScore, fill = HealthScore)) +
  geom_bar() +
  labs(title = "Distribucija kategorija HealthScore", x = "Kategorije", y = "Broj opse
rvacija") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```




```
data_clean$DietScore = as.numeric(data_clean$Fruits == "Da") +
                      as.numeric(data_clean$Veggies == "Da")

nivoi_DietScore = sort(unique(data_clean$DietScore))
category_DietScore = c("nezdrava", "umereno zdrava", "zdrava")
data_clean$DietScore = factor(data_clean$DietScore,
                                    levels = nivoi_DietScore,
                                    labels = category_DietScore,
                                    ordered = TRUE)
str(data_clean$DietScore)
```

```
##  Ord.factor w/ 3 levels "nezdrava"<"umereno zdrava"<..: 2 1 2 3 3 3 1 2 3 2 ...
```

```
ggplot(data_clean, aes(x = DietScore, fill = DietScore)) +
  geom_bar() +
  labs(title = "Distribucija kategorija DietScore", x = "Kategorije", y = "Broj opserv
acija") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```



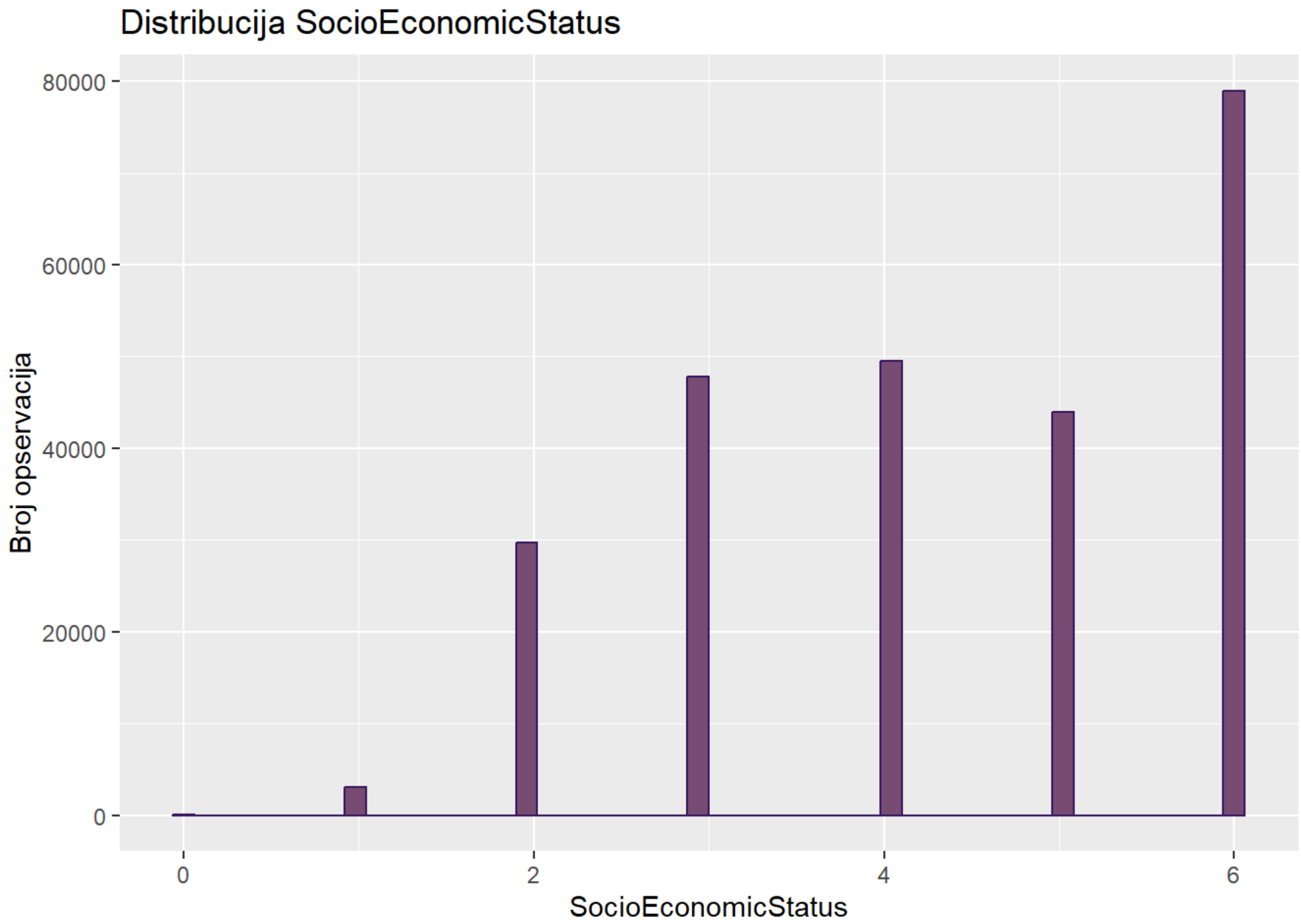

```
unique(data_clean$EducationCat)
```

```
## [1] Srednje Visoko  Osnovno Nisko
## Levels: Nisko < Osnovno < Srednje < Visoko
```

```
unique(data_clean$IncomeCat)
```

```
## [1] Niska         Visoka        Nisko-srednja Srednja
## Levels: Niska < Nisko-srednja < Srednja < Visoka
```

```
data_clean$SocioEconomicStatus  = (as.numeric(data_clean$EducationCat) - 1) +
  (as.numeric(data_clean$IncomeCat) - 1)
ggplot(data_clean, aes(x = SocioEconomicStatus)) +
  geom_histogram(bins = 50, fill = "#7C4B73FF", color="#381A61FF") +
  labs(title = "Distribucija SocioEconomicStatus", x = "SocioEconomicStatus", y = "Bro
j opservacija")
```



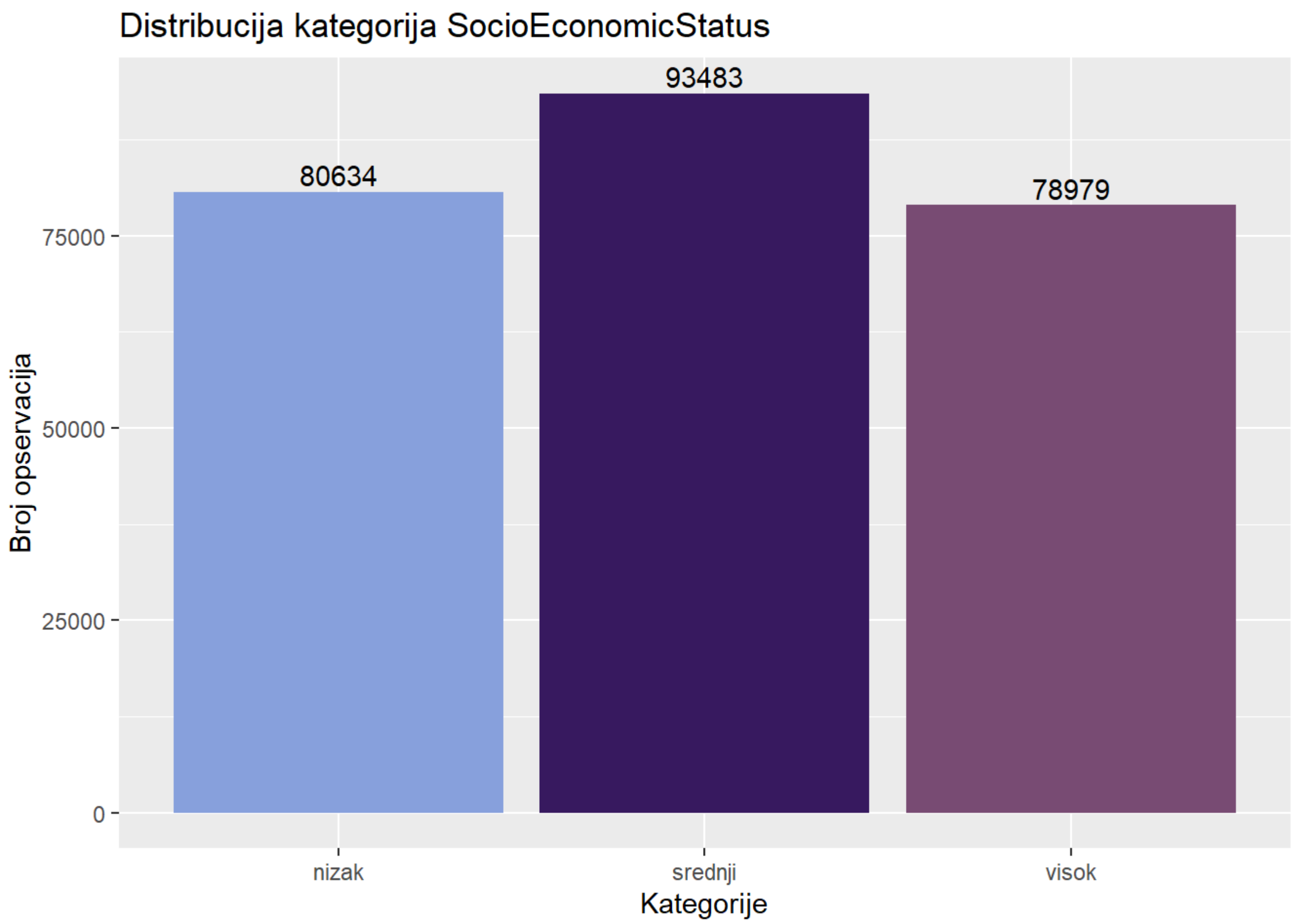

```
summary(data_clean$SocioEconomicStatus)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   3.000   4.000   4.337   6.000   6.000
```

```
category_SocioEconomicStatus = c("nizak", "srednji", "visok")
interval_SocioEconomicStatus = c(-1, 3, 5, 6)
data_clean$SocioEconomicStatus <- cut(data_clean$SocioEconomicStatus,
                                      breaks = interval_SocioEconomicStatus,
                                      labels = category_SocioEconomicStatus,
                                      ordered_result = TRUE)

str(data_clean$SocioEconomicStatus)
```

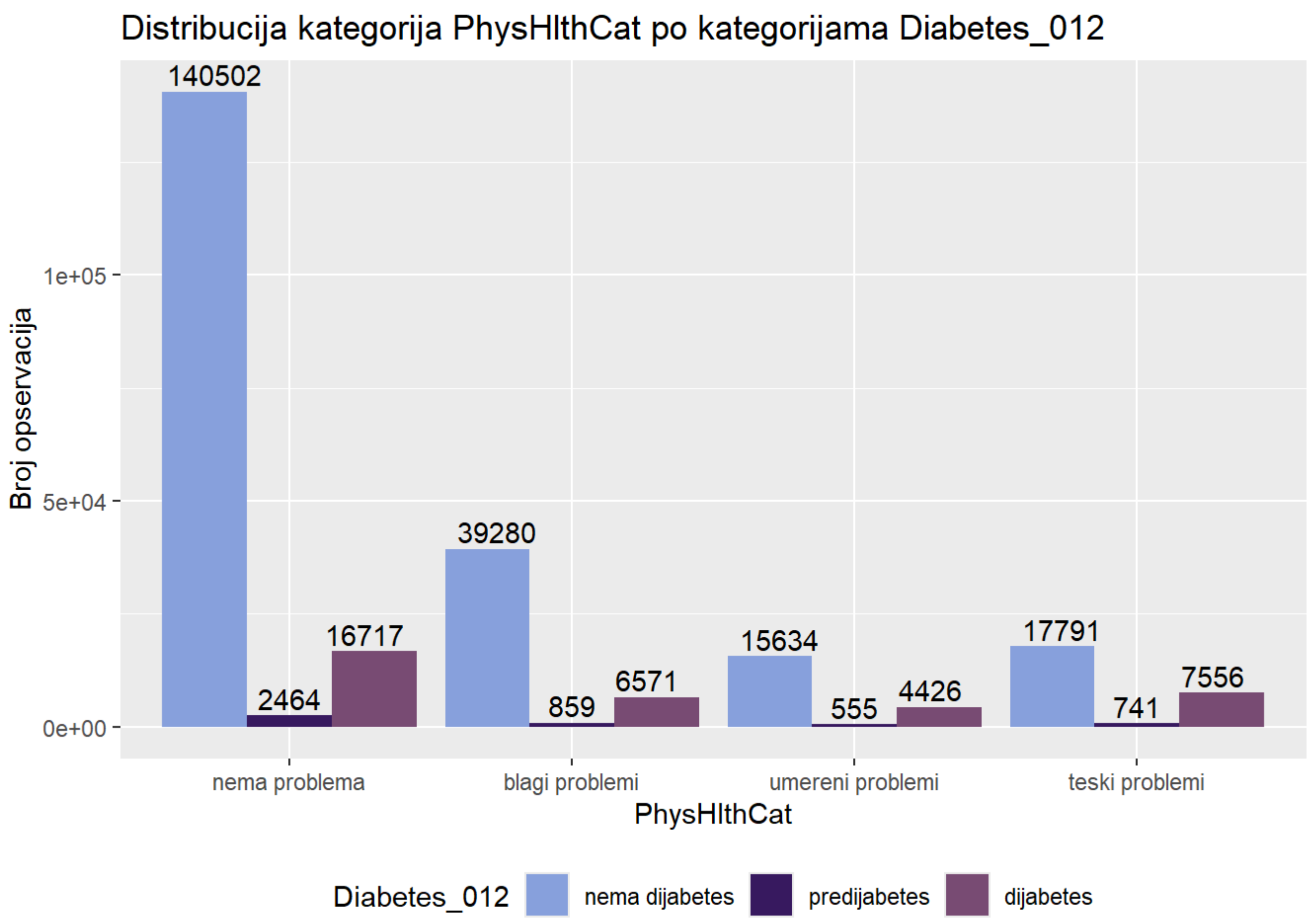
```
##  Ord.factor w/ 3 levels "nizak"<"srednji"<..: 1 1 2 1 1 3 2 1 1 1 ...
```

```
ggplot(data_clean, aes(x = SocioEconomicStatus, fill = SocioEconomicStatus)) +
  geom_bar() +
  labs(title = "Distribucija kategorija SocioEconomicStatus", x = "Kategorije", y = "B
roj opservacija") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  scale_fill_paletteer_d("MetBrewer::Archambault") + theme(legend.position="none")
```

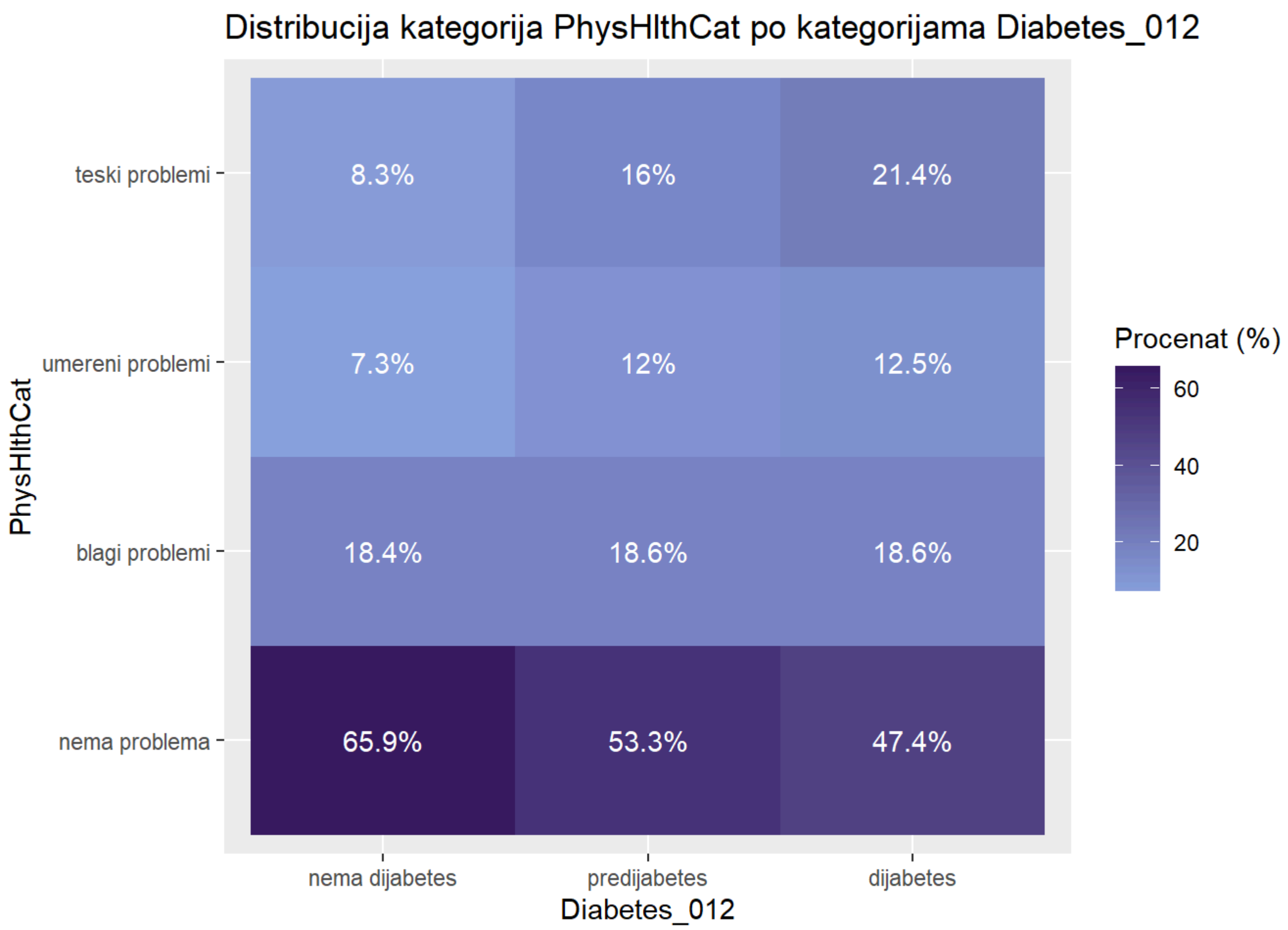


##Bivarijantna analiza transformisanih

```
ggplot(data_clean, aes(x = PhysHlthCat, fill = Diabetes_012 ))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija PhysHlthCat po kategorijama Diabetes_012",
    y = "Broj opservacija"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```



```
data_clean %>%
  count(Diabetes_012, PhysHlthCat) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = PhysHlthCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija PhysHlthCat po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "PhysHlthCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



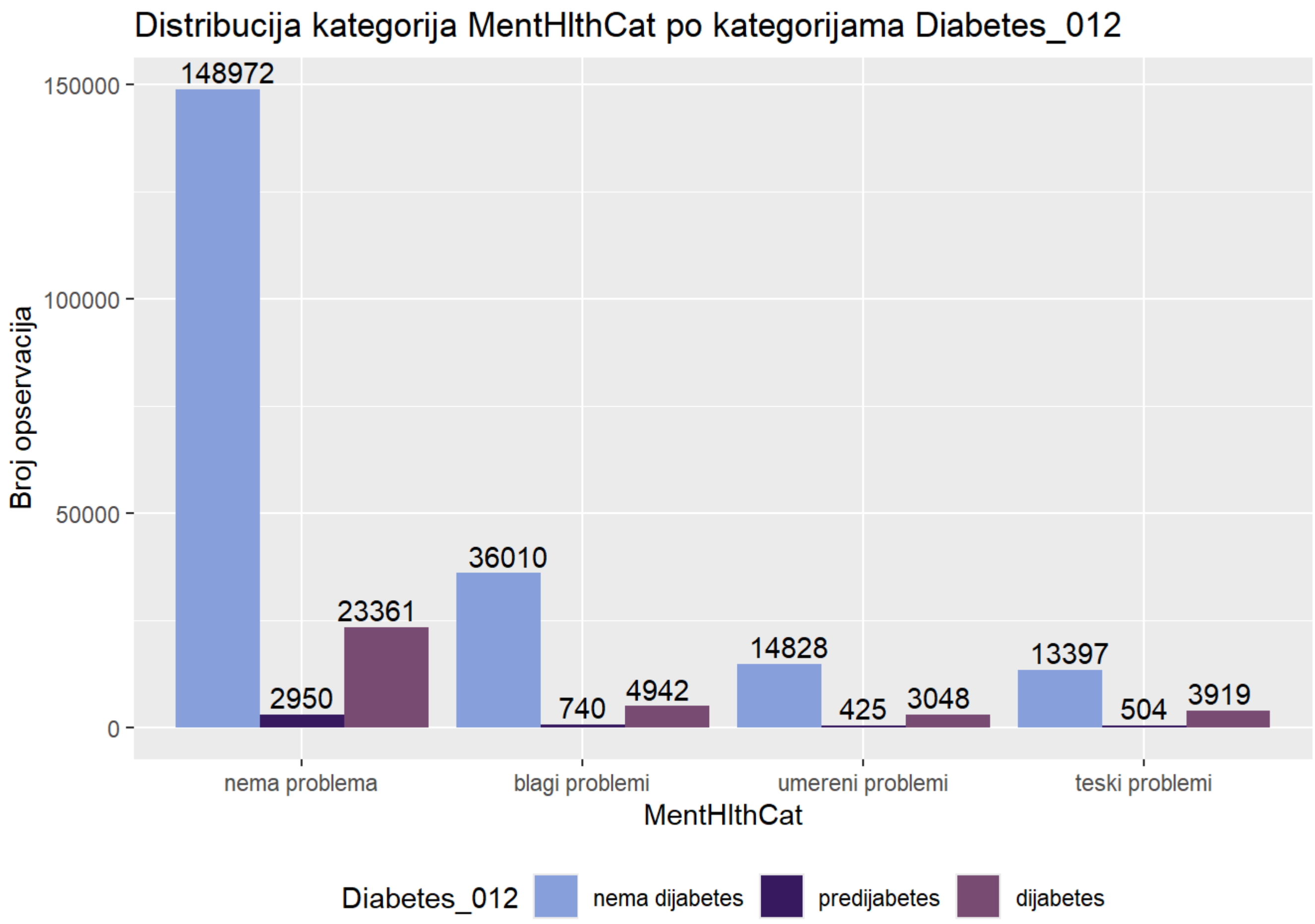

```
chi_sq_test(data_clean$PhysHlthCat, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 7983.5, df = 6, p-value < 2.2e-16
```

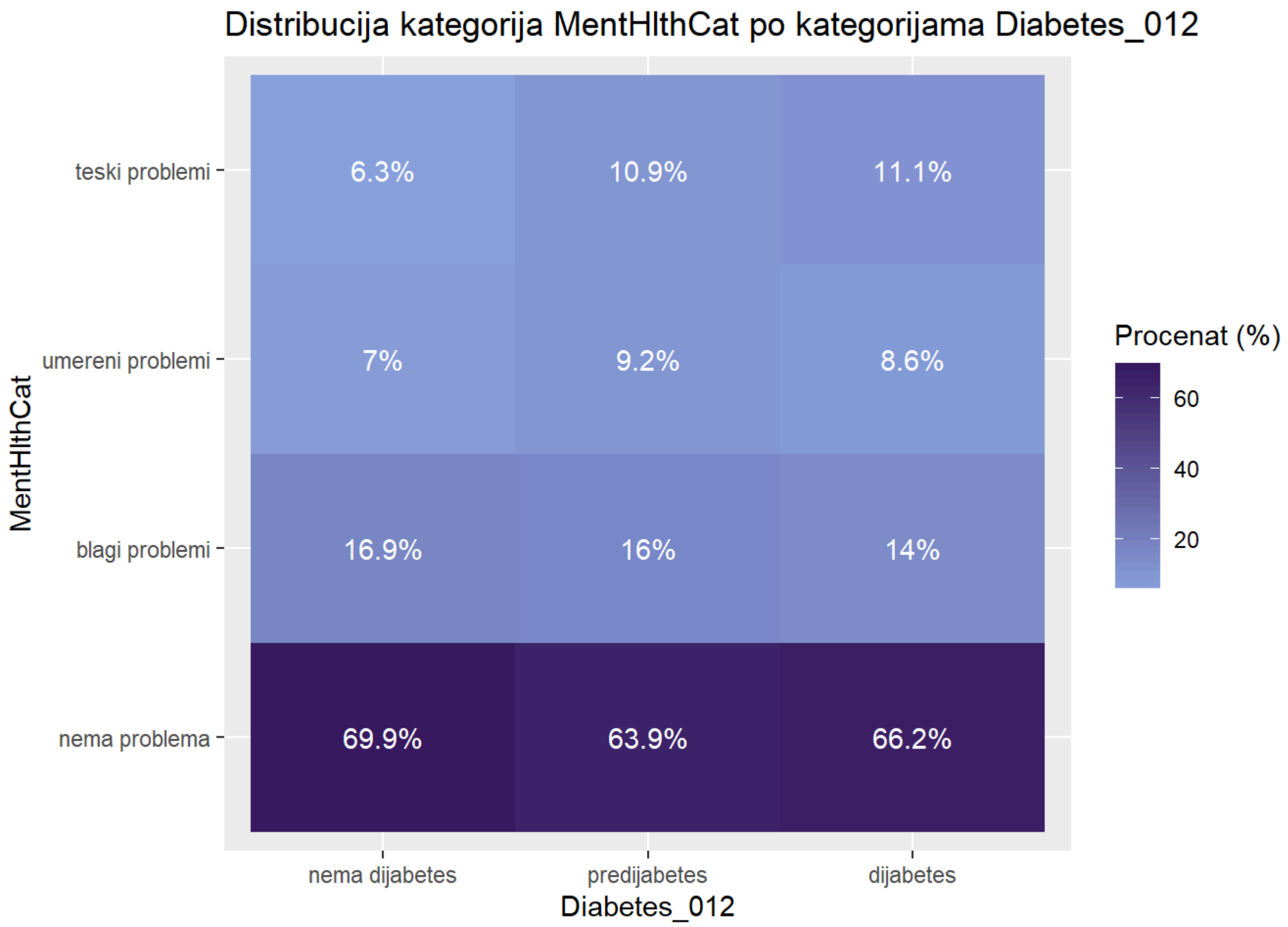
```
cramer_v(data_clean$PhysHlthCat, data_clean$Diabetes_012)
```

```
## [1] 0.1255858
```

```
ggplot(data_clean, aes(x = MentHlthCat, fill = Diabetes_012 ))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija MentHlthCat po kategorijama Diabetes_012",
    y = "Broj opservacija"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```



```
data_clean %>%
  count(Diabetes_012, MentHlthCat) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = MentHlthCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija MentHlthCat po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "MentHlthCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



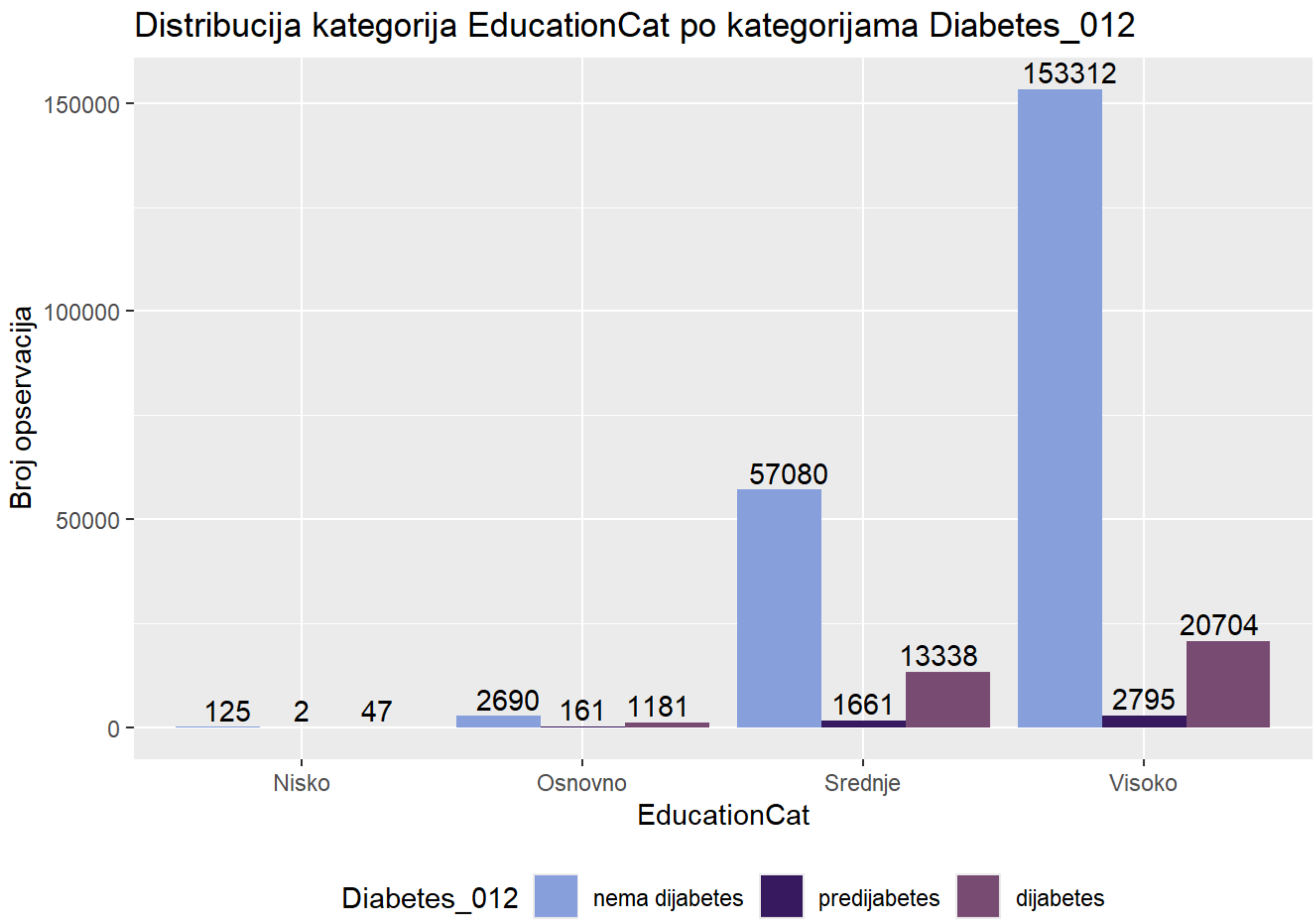

```
chi_sq_test(data_clean$MentHlthCat, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 1476.6, df = 6, p-value < 2.2e-16
```

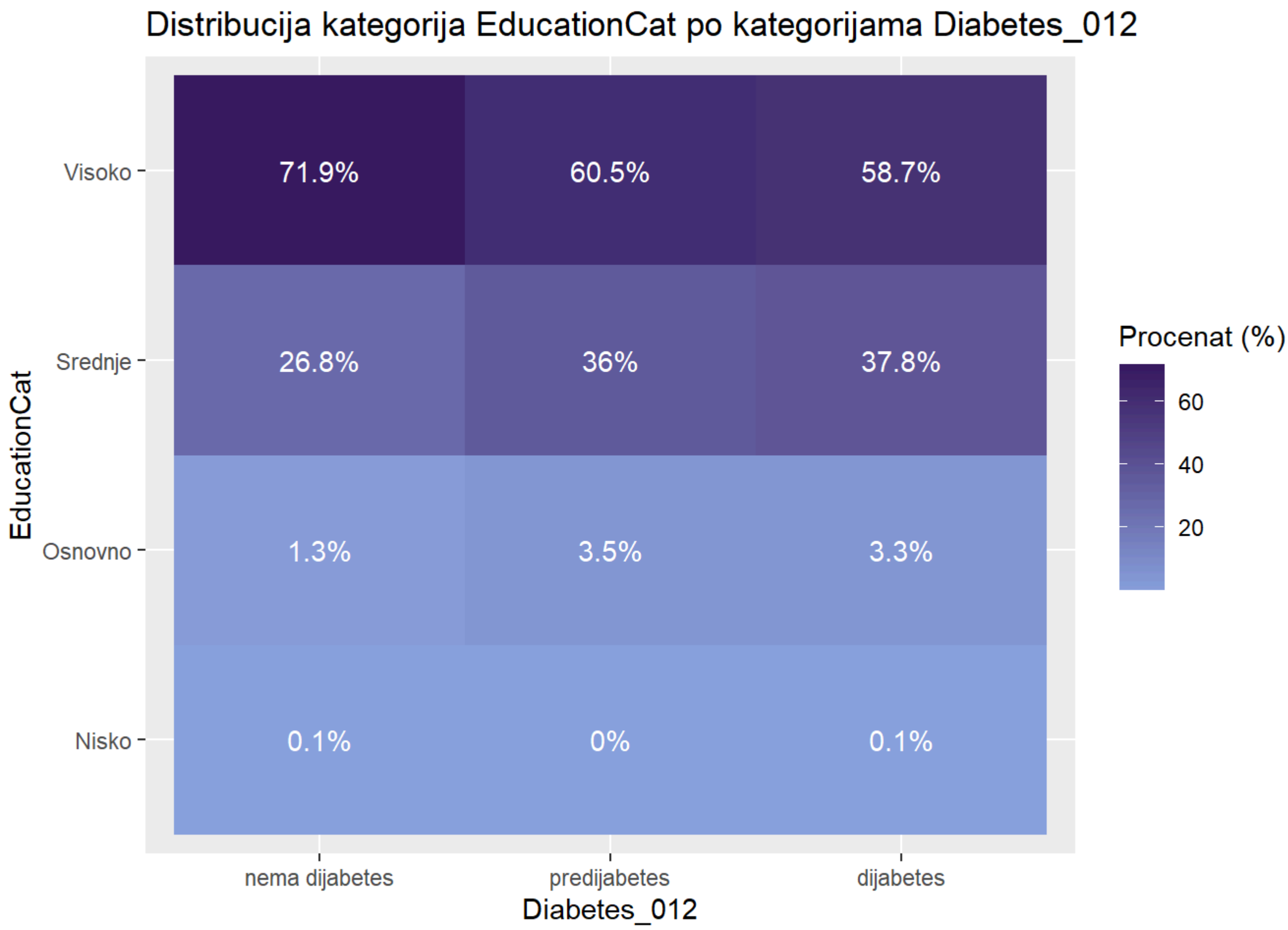
```
cramer_v(data_clean$MentHlthCat, data_clean$Diabetes_012)
```

```
## [1] 0.05401072
```

```
ggplot(data_clean, aes(x = EducationCat, fill = Diabetes_012 ))  +
geom_bar(position = "dodge") +
labs(
  title = "Distribucija kategorija EducationCat po kategorijama Diabetes_012",
  y = "Broj opservacija"
) + scale_fill_paletteer_d("MetBrewer::Archambault") +
geom_text(
  stat = "count",
  aes(label = ..count..),
  position = position_dodge(width = 0.8),
  vjust = -0.3
) +
theme(legend.position="bottom")
```



```
data_clean %>%
  count(Diabetes_012, EducationCat) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = EducationCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija EducationCat po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "EducationCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```




```
chi_sq_test(data_clean$EducationCat, data_clean$Diabetes_012)
```

```
## Warning in chisq.test(tabela): Chi-squared approximation may be incorrect
```

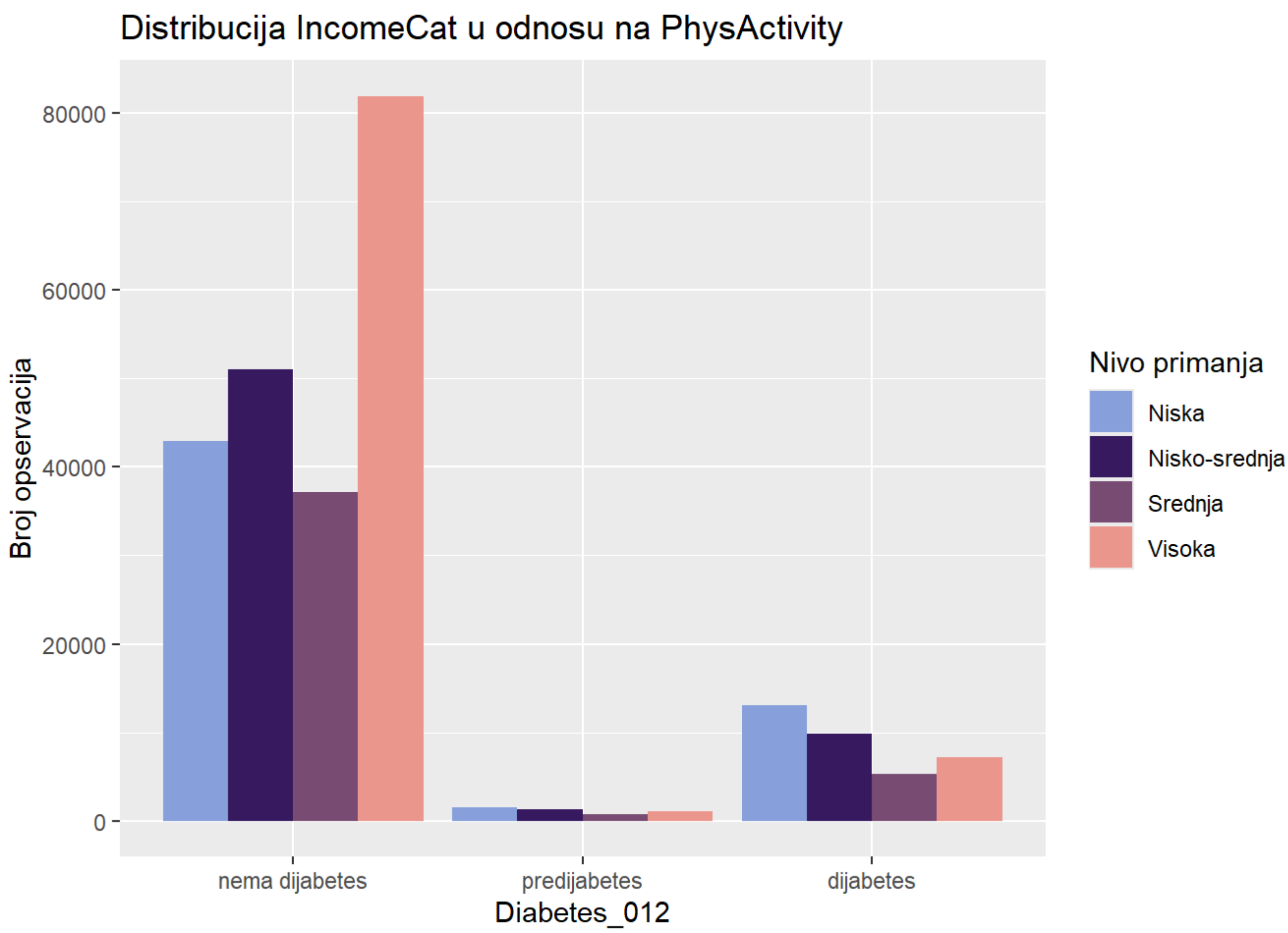
```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 3161.1, df = 6, p-value < 2.2e-16
```

```
cramer_v(data_clean$EducationCat, data_clean$Diabetes_012)
```

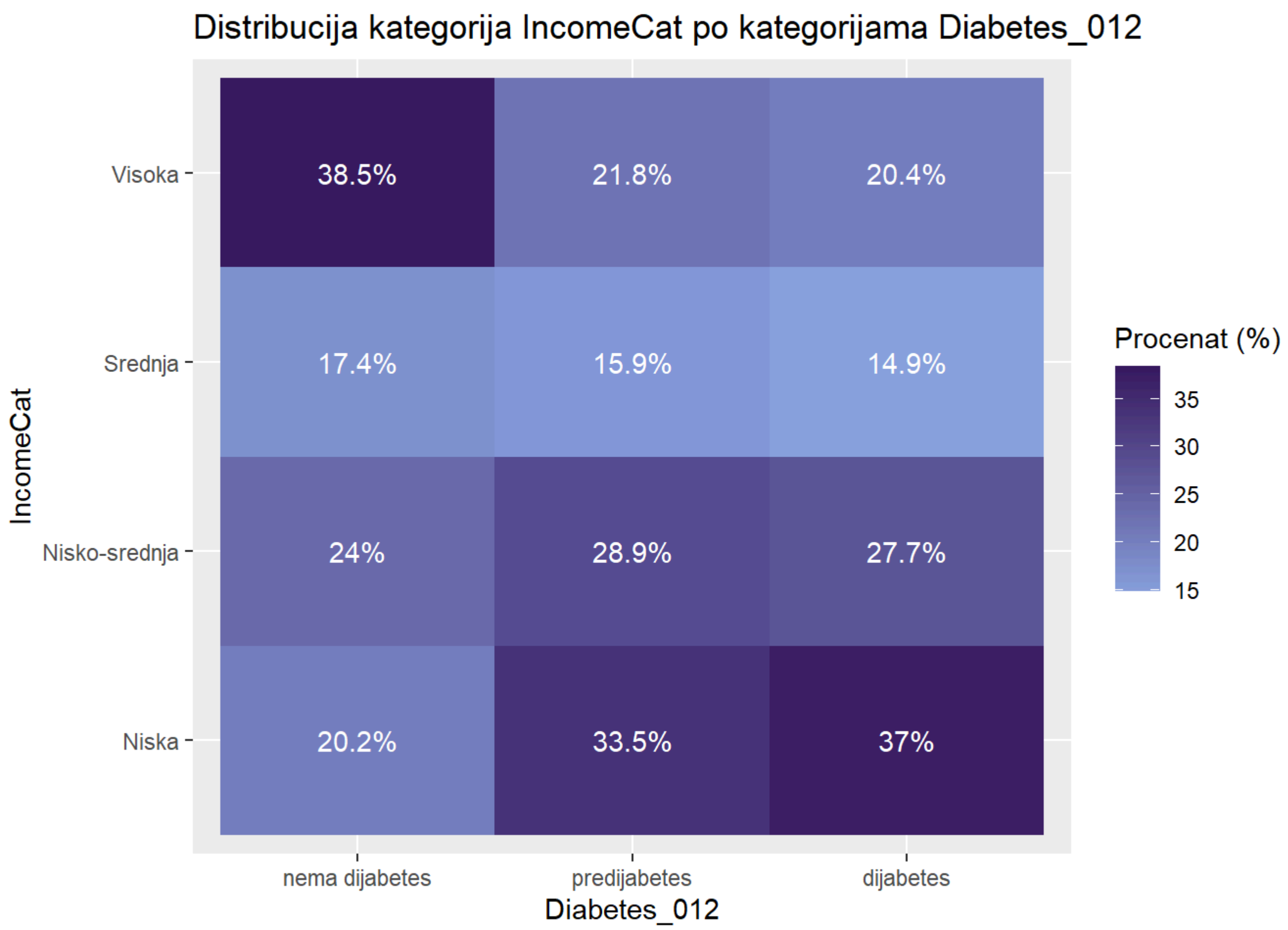
```
## Warning in chisq.test(tabela): Chi-squared approximation may be incorrect
```

```
## [1] 0.07902491
```

```
ggplot(data_clean, aes(x = Diabetes_012, fill =IncomeCat )) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija IncomeCat u odnosu na PhysActivity",
       y = "Broj opservacija",
       fill = "Nivo primanja")
```



```
data_clean %>%
  count(Diabetes_012, IncomeCat) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = IncomeCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija IncomeCat po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "IncomeCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



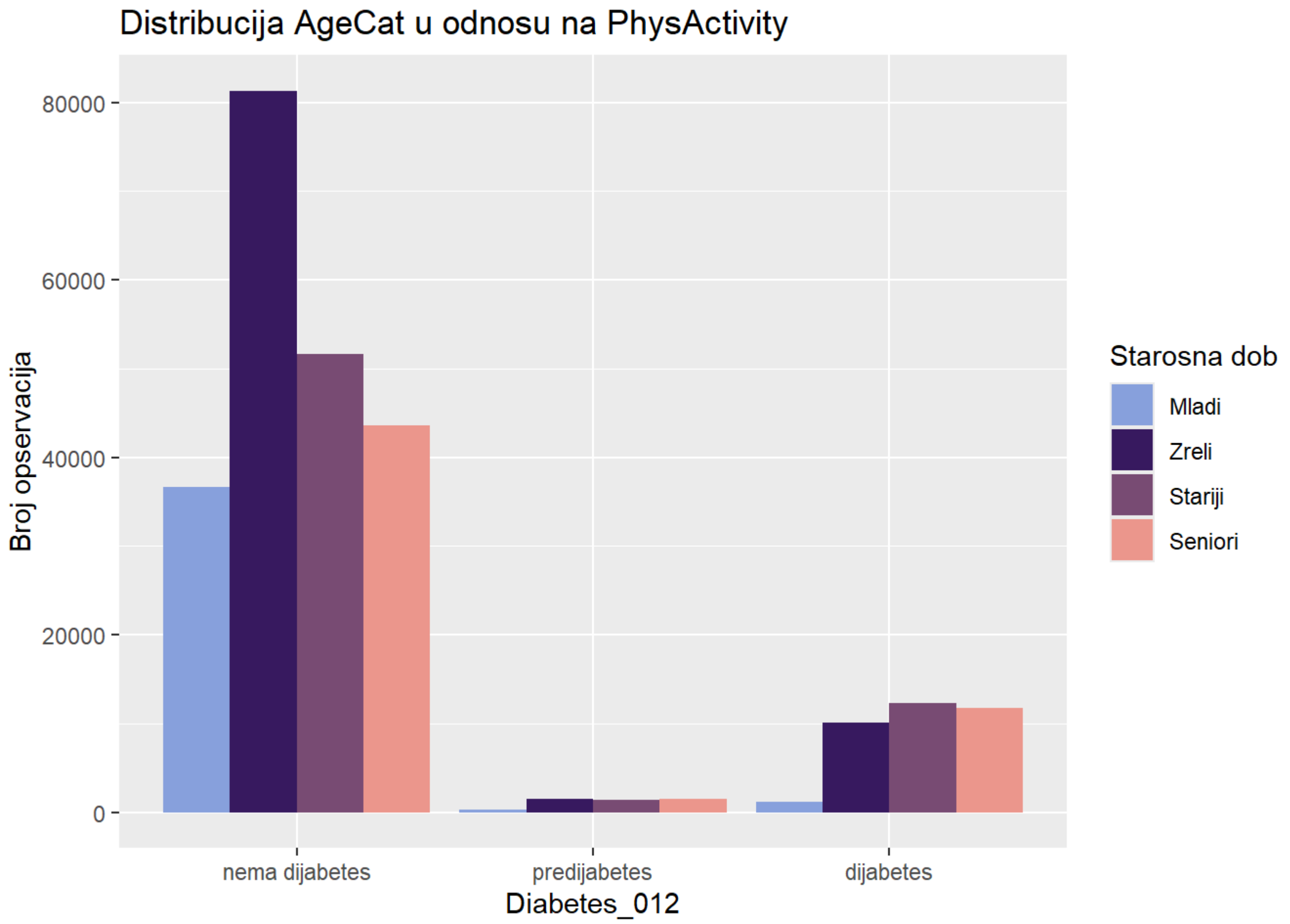

```
chi_sq_test(data_clean$IncomeCat, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 7369.1, df = 6, p-value < 2.2e-16
```

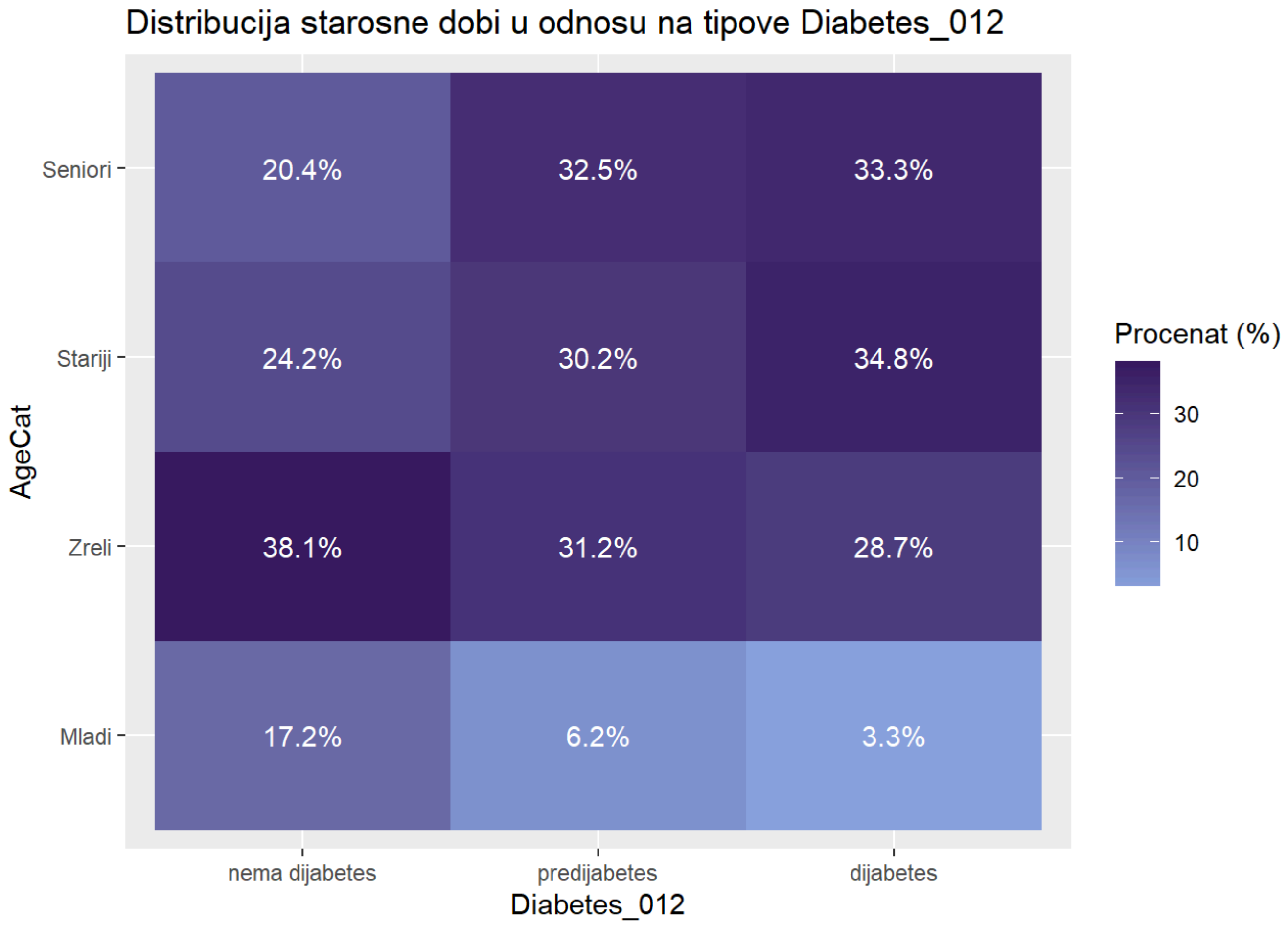
```
cramer_v(data_clean$IncomeCat, data_clean$Diabetes_012)
```

```
## [1] 0.1206558
```

```
ggplot(data_clean, aes(x = Diabetes_012, fill =AgeCat)) +
  geom_bar(position = "dodge")+
  scale_fill_manual(values = colorS) +
  labs(title = "Distribucija AgeCat u odnosu na PhysActivity",
       y = "Broj opservacija",
       fill = "Starosna dob")
```




```
data_clean %>%
  count(Diabetes_012, AgeCat) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = AgeCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija starosne dobi u odnosu na tipove Diabetes_012",
    x = "Diabetes_012",
    y = "AgeCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



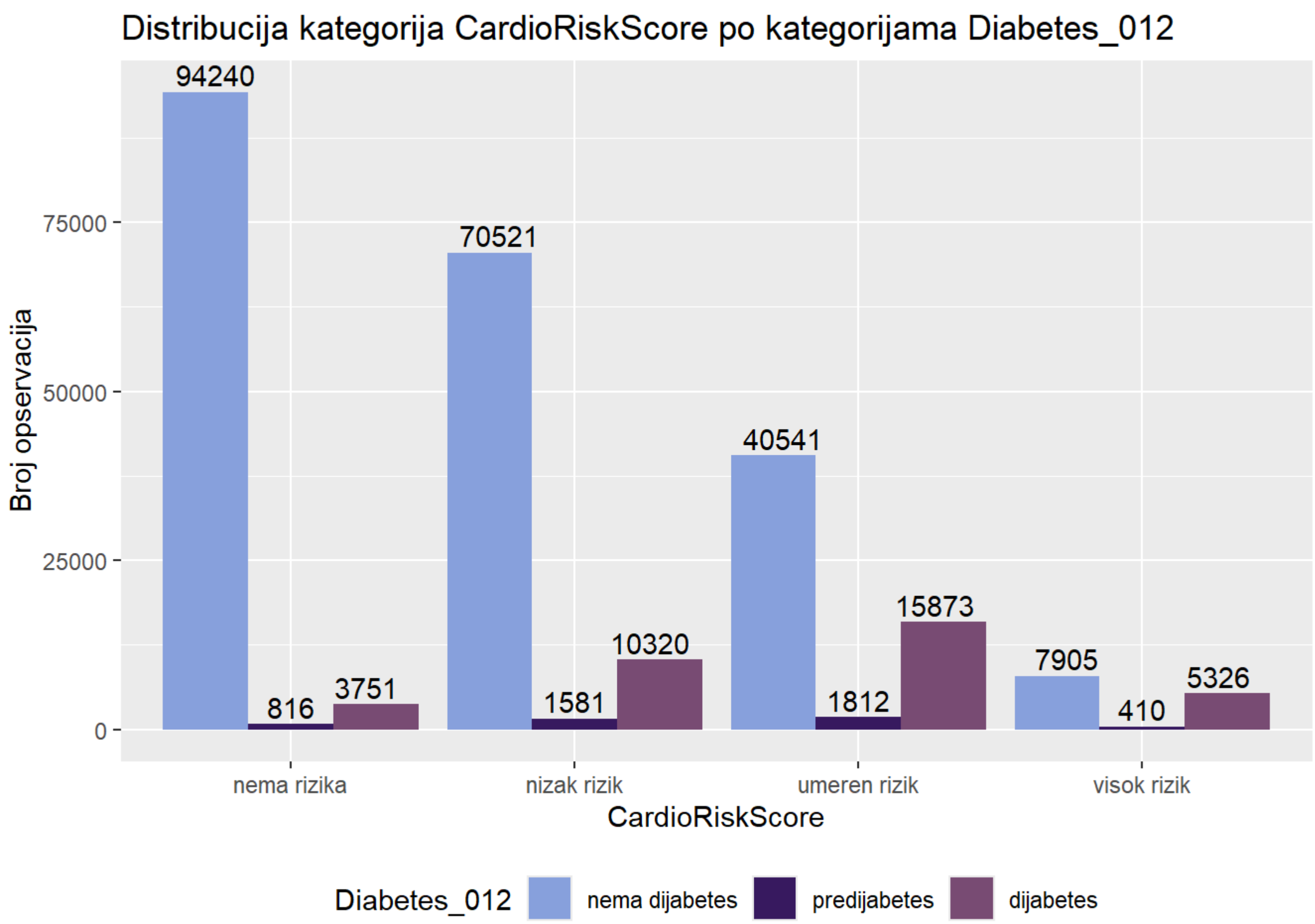

```
chi_sq_test(data_clean$AgeCat, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 8705.2, df = 6, p-value < 2.2e-16
```

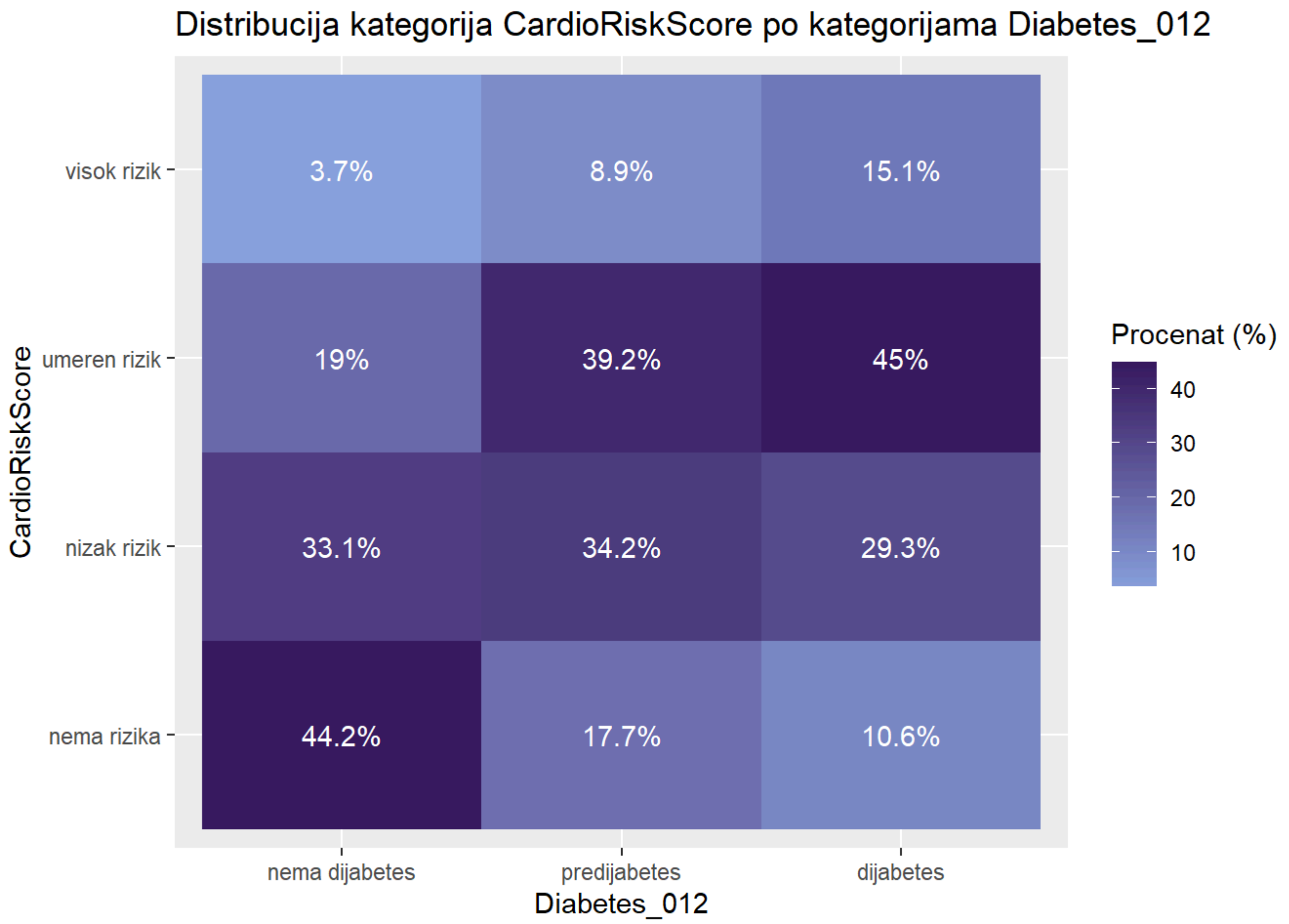
```
cramer_v(data_clean$AgeCat, data_clean$Diabetes_012)
```

```
## [1] 0.131139
```

```
ggplot(data_clean, aes(x = CardioRiskScore, fill = Diabetes_012 ))  +
geom_bar(position = "dodge") +
labs(
  title = "Distribucija kategorija CardioRiskScore po kategorijama Diabetes_012",
  y = "Broj opservacija"
) + scale_fill_paletteer_d("MetBrewer::Archambault") +
geom_text(
  stat = "count",
  aes(label = ..count..),
  position = position_dodge(width = 0.8),
  vjust = -0.3
) +
theme(legend.position="bottom")
```



```
data_clean %>%
  count(Diabetes_012, CardioRiskScore) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = CardioRiskScore, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija CardioRiskScore po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "CardioRiskScore",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



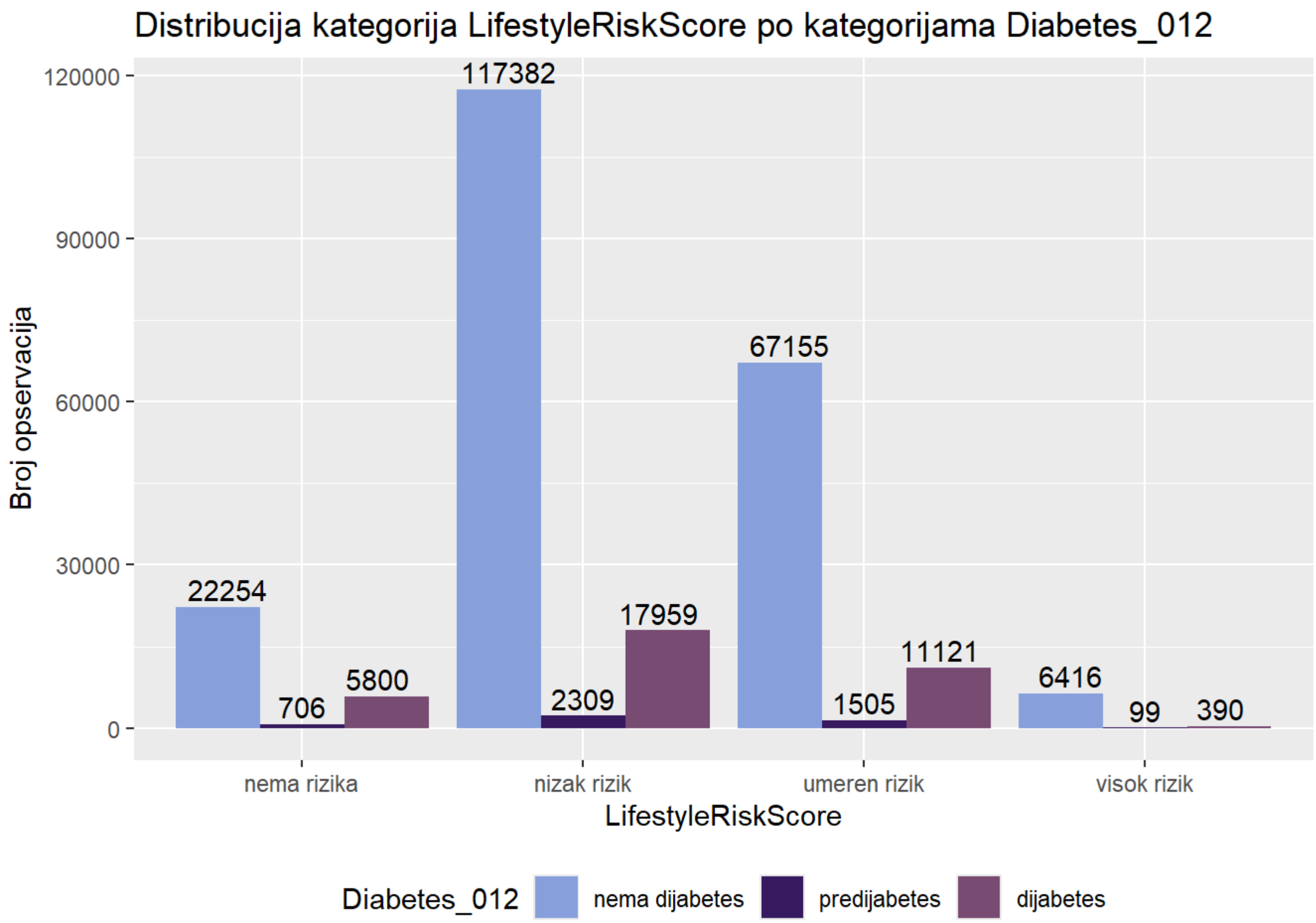

```
chi_sq_test(data_clean$CardioRiskScore, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 26242, df = 6, p-value < 2.2e-16
```

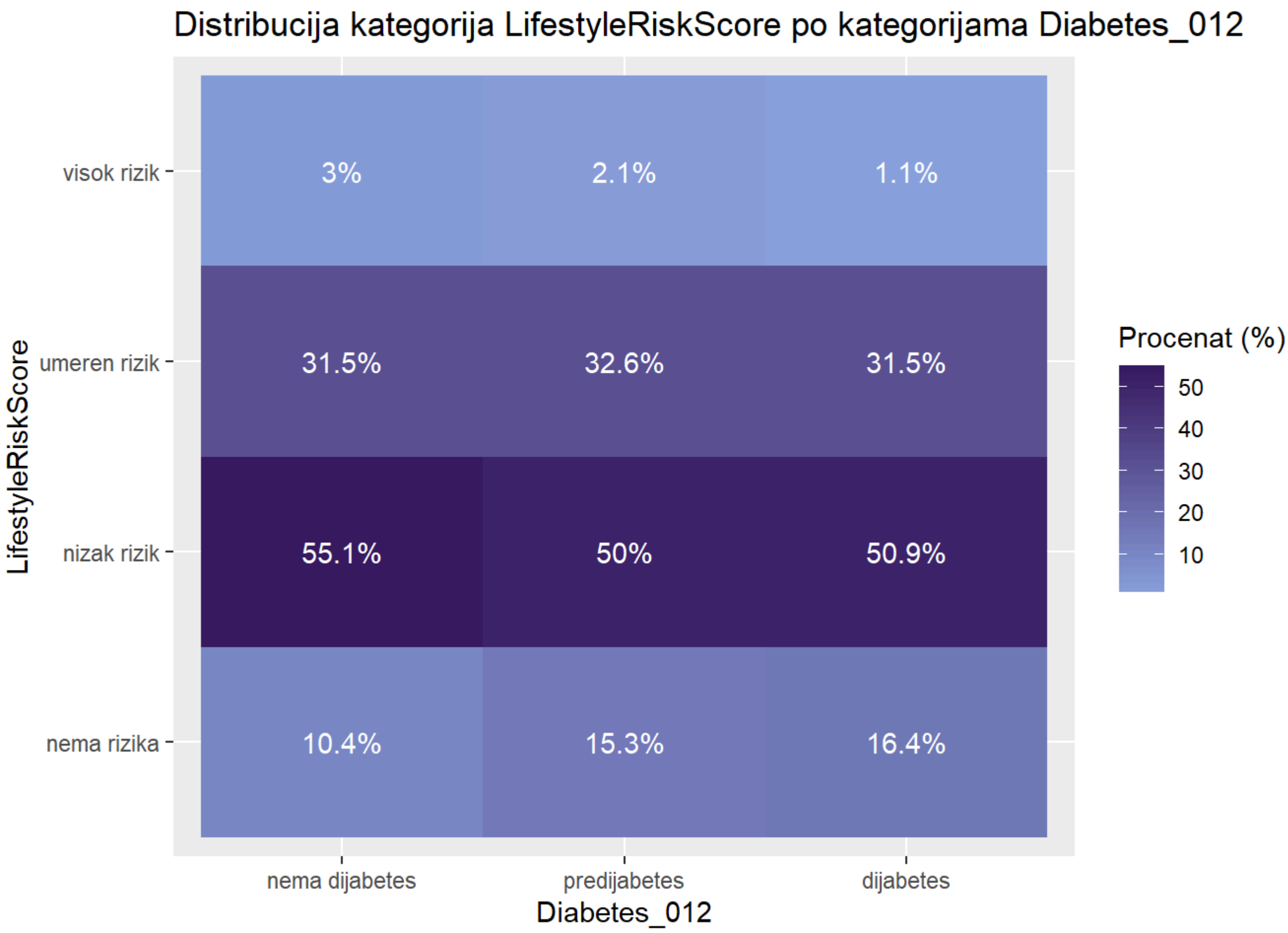
```
cramer_v(data_clean$CardioRiskScore, data_clean$Diabetes_012)
```

```
## [1] 0.2276899
```

```
ggplot(data_clean, aes(x = LifestyleRiskScore, fill = Diabetes_012 ))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija LifestyleRiskScore po kategorijama Diabetes_012",
    y = "Broj opservacija"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```



```
data_clean %>%
  count(Diabetes_012, LifestyleRiskScore) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = LifestyleRiskScore, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija LifestyleRiskScore po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "LifestyleRiskScore",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



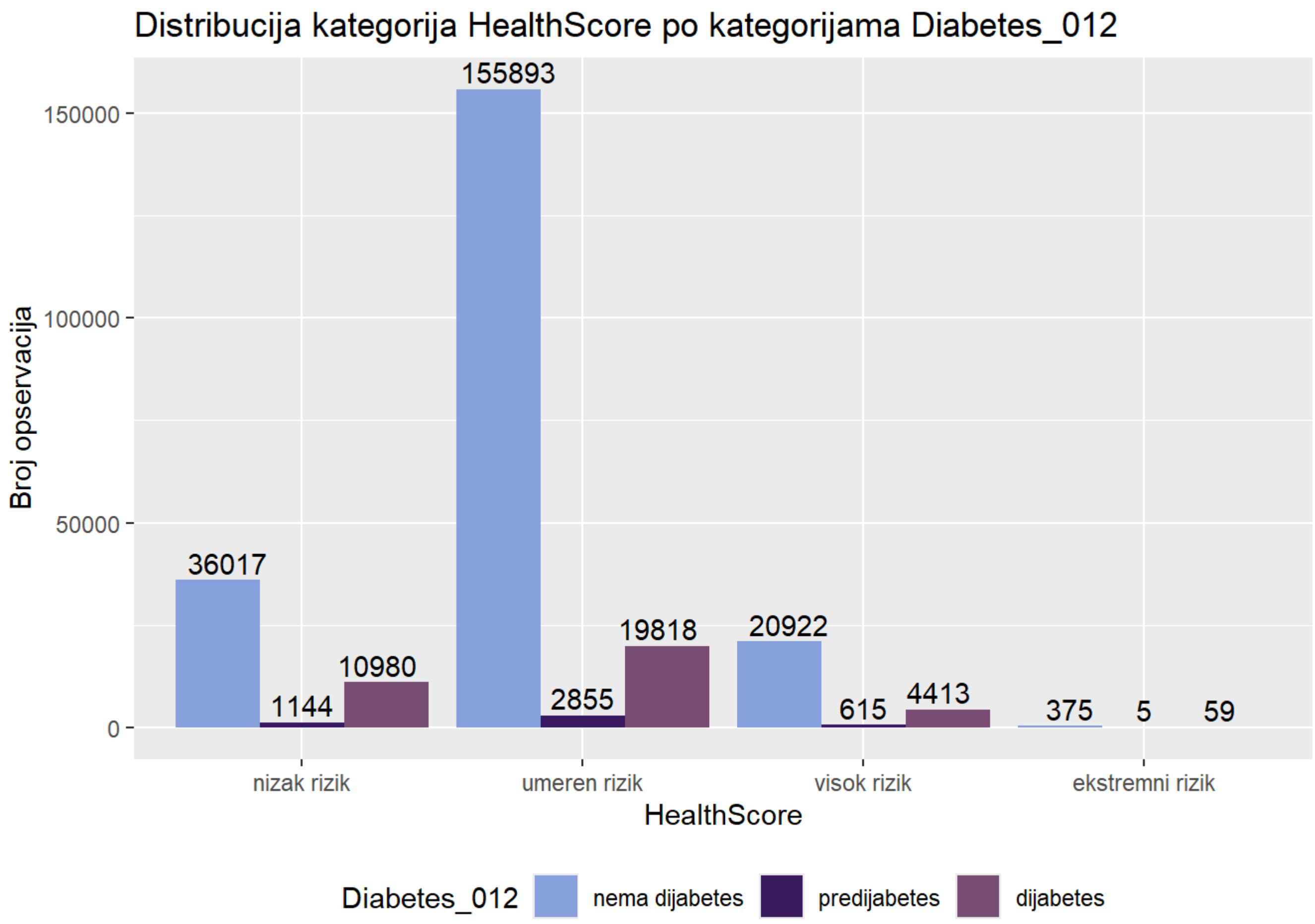

```
chi_sq_test(data_clean$LifestyleRiskScore, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 1546.1, df = 6, p-value < 2.2e-16
```

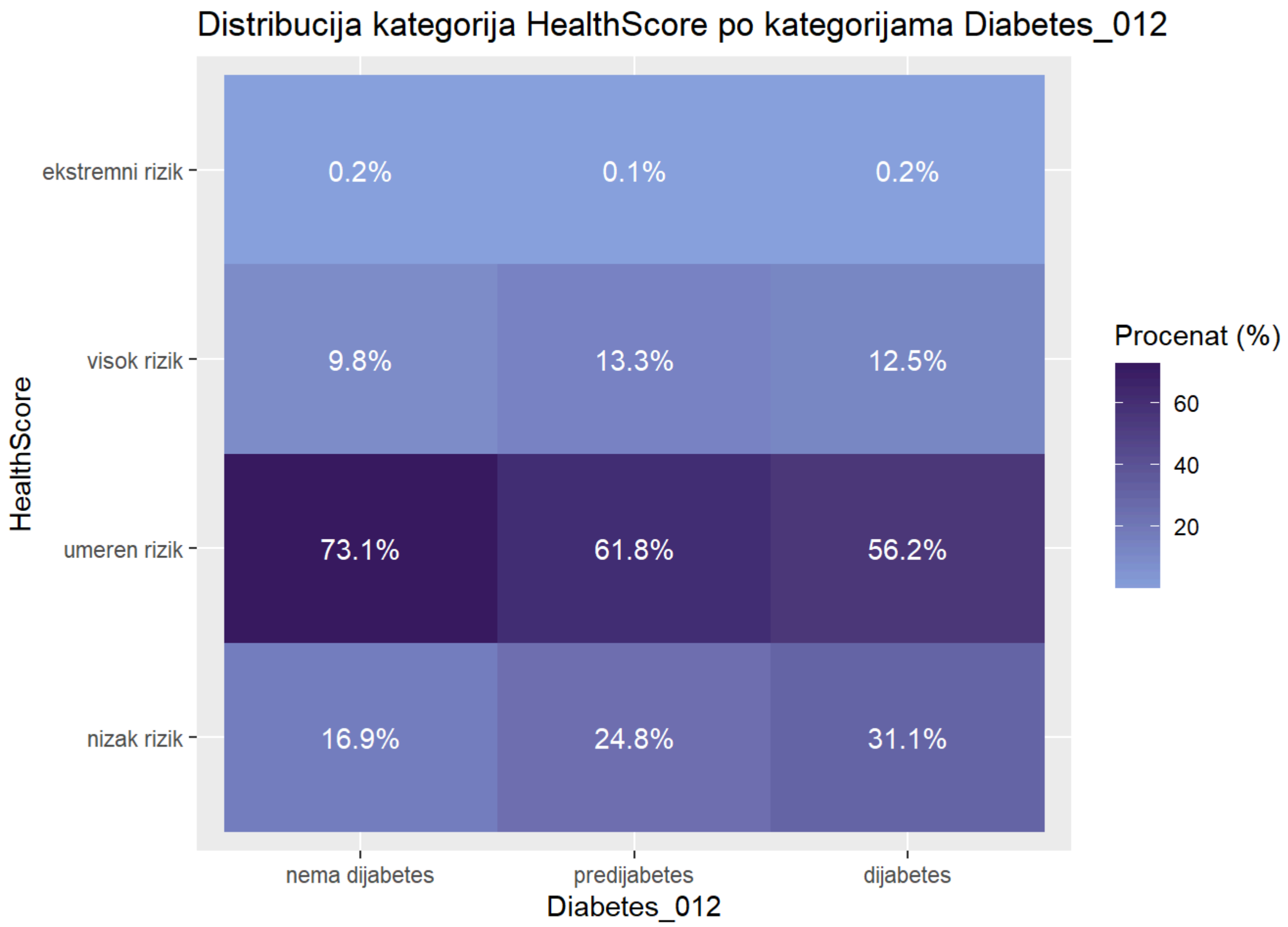
```
cramer_v(data_clean$LifestyleRiskScore, data_clean$Diabetes_012)
```

```
## [1] 0.05526669
```

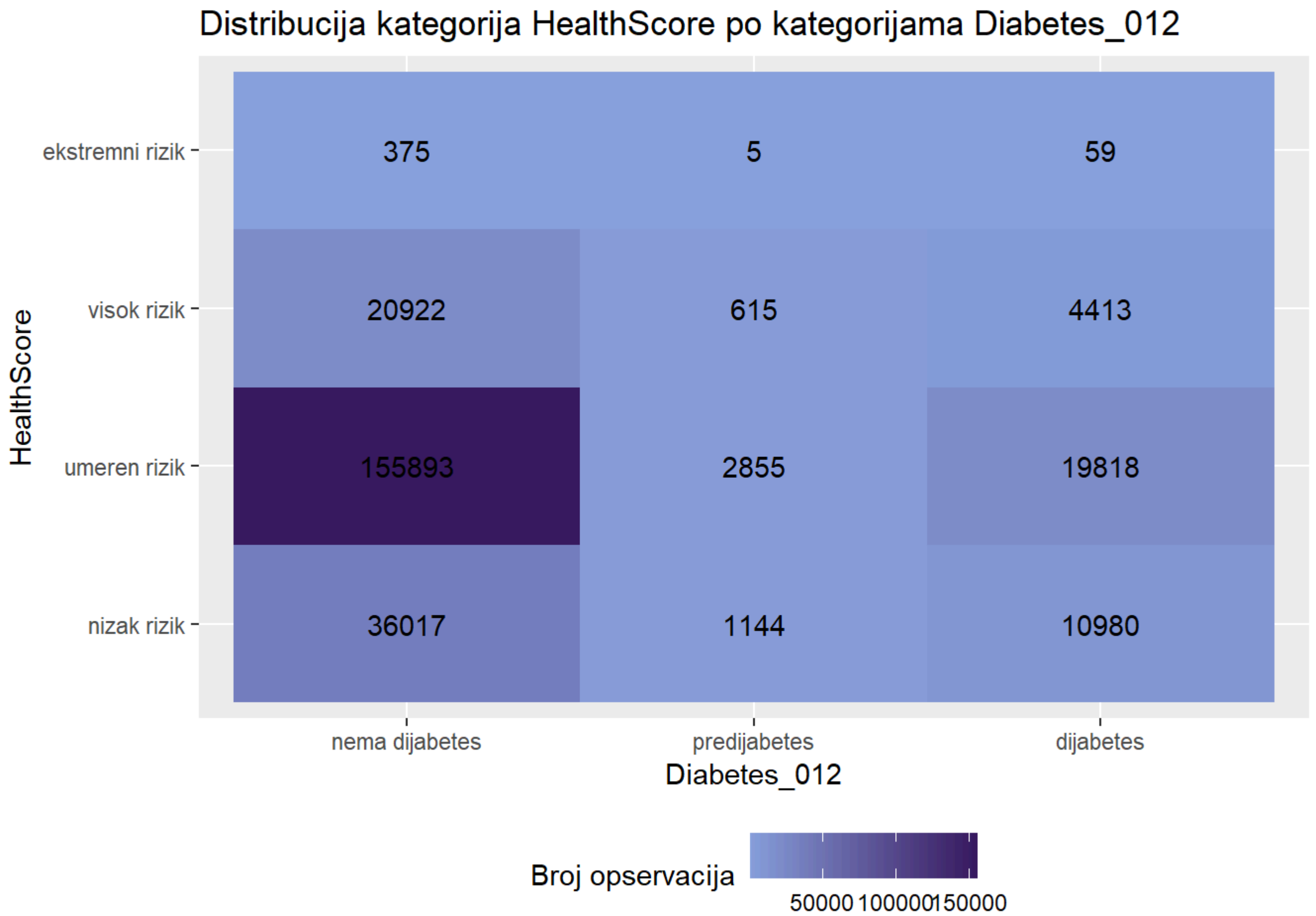
```
ggplot(data_clean, aes(x = HealthScore, fill = Diabetes_012 ))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija HealthScore po kategorijama Diabetes_012",
    y = "Broj opservacija"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```



```
data_clean %>%
  count(Diabetes_012, HealthScore) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = HealthScore, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija HealthScore po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "HealthScore",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



```
ggplot(data_clean %>% count(Diabetes_012, HealthScore),
       aes(x = Diabetes_012, y = HealthScore, fill = n)) +
  geom_tile() +
  geom_text(aes(label = n)) +
  labs(
    title = "Distribucija kategorija HealthScore po kategorijama Diabetes_012",
    x = "Diabetes_012",
    y = "HealthScore",
    fill = "Broj opservacija"

  )+
  scale_fill_gradient(low = colorS[1], high = colorS[2]) +
  theme(legend.position="bottom")
```



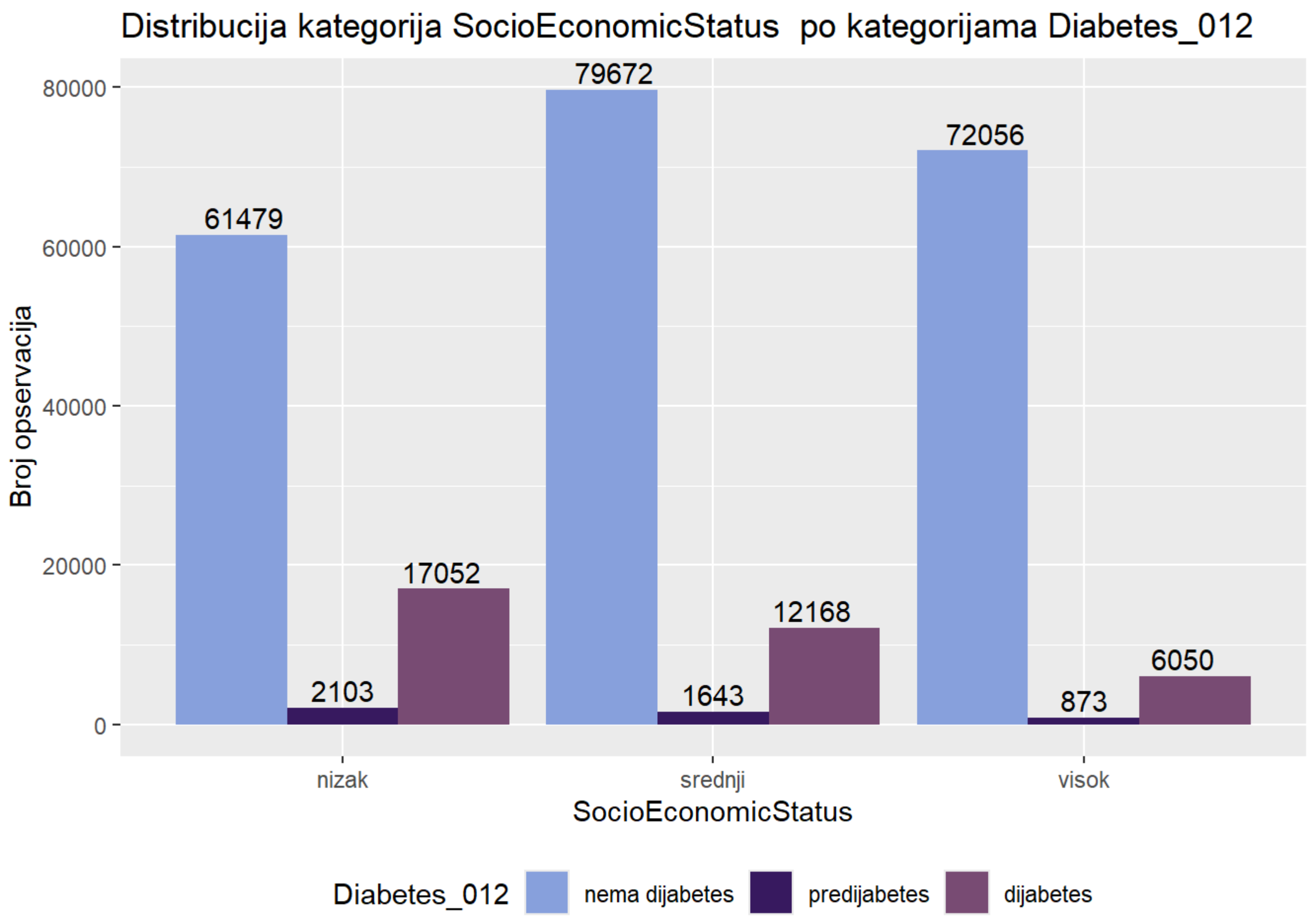

```
chi_sq_test(data_clean$HealthScore, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 4846.8, df = 6, p-value < 2.2e-16
```

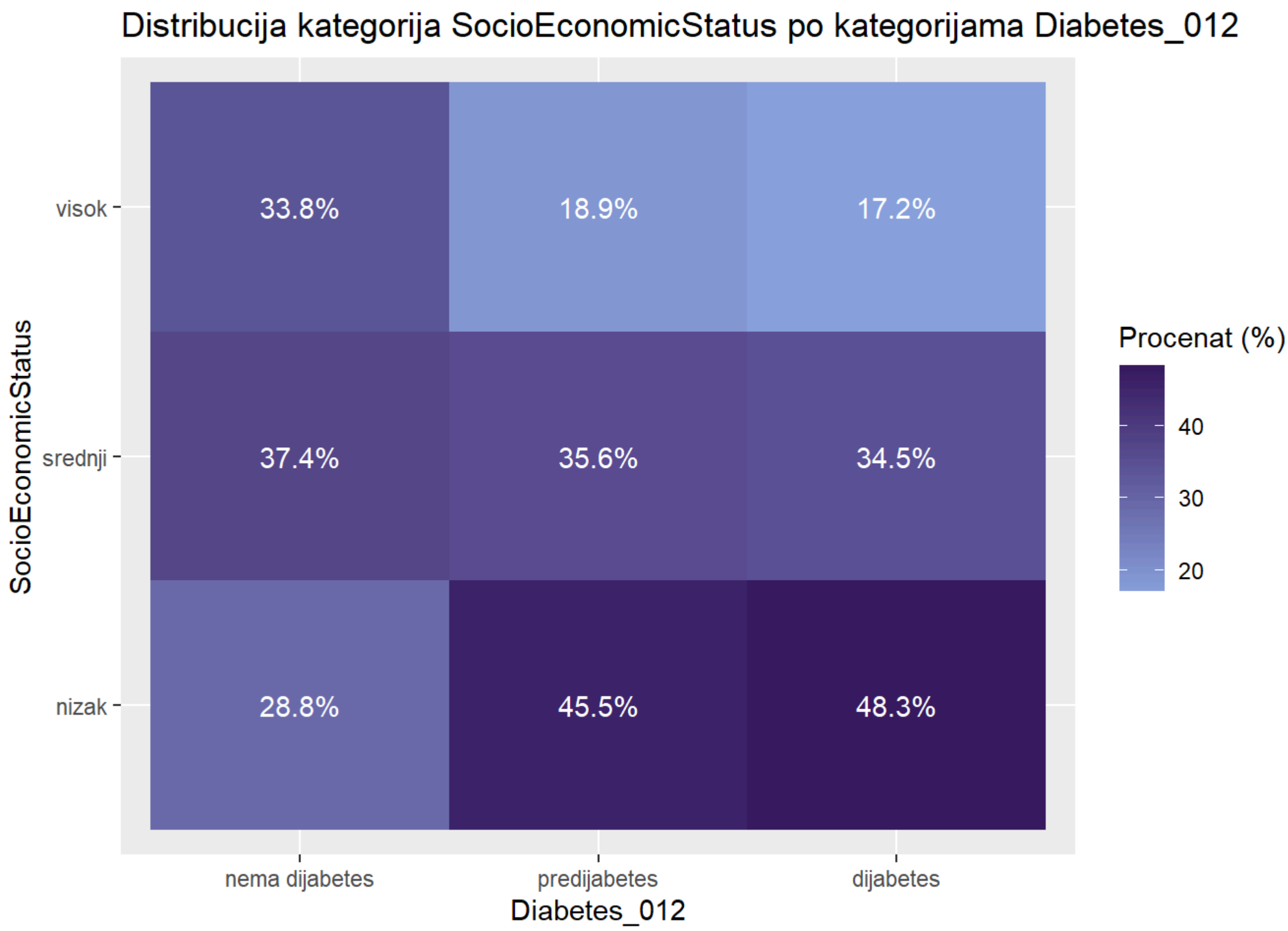
```
cramer_v(data_clean$HealthScore, data_clean$Diabetes_012)
```

```
## [1] 0.09785248
```

```
ggplot(data_clean, aes(x = SocioEconomicStatus , fill = Diabetes_012 ))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija SocioEconomicStatus  po kategorijama Diabetes_01
2",
    y = "Broj opservacija"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```



```
data_clean %>%
  count(Diabetes_012, SocioEconomicStatus) %>%
  group_by(Diabetes_012) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Diabetes_012, y = SocioEconomicStatus, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija SocioEconomicStatus po kategorijama Diabetes_01
2",
    x = "Diabetes_012",
    y = "SocioEconomicStatus",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



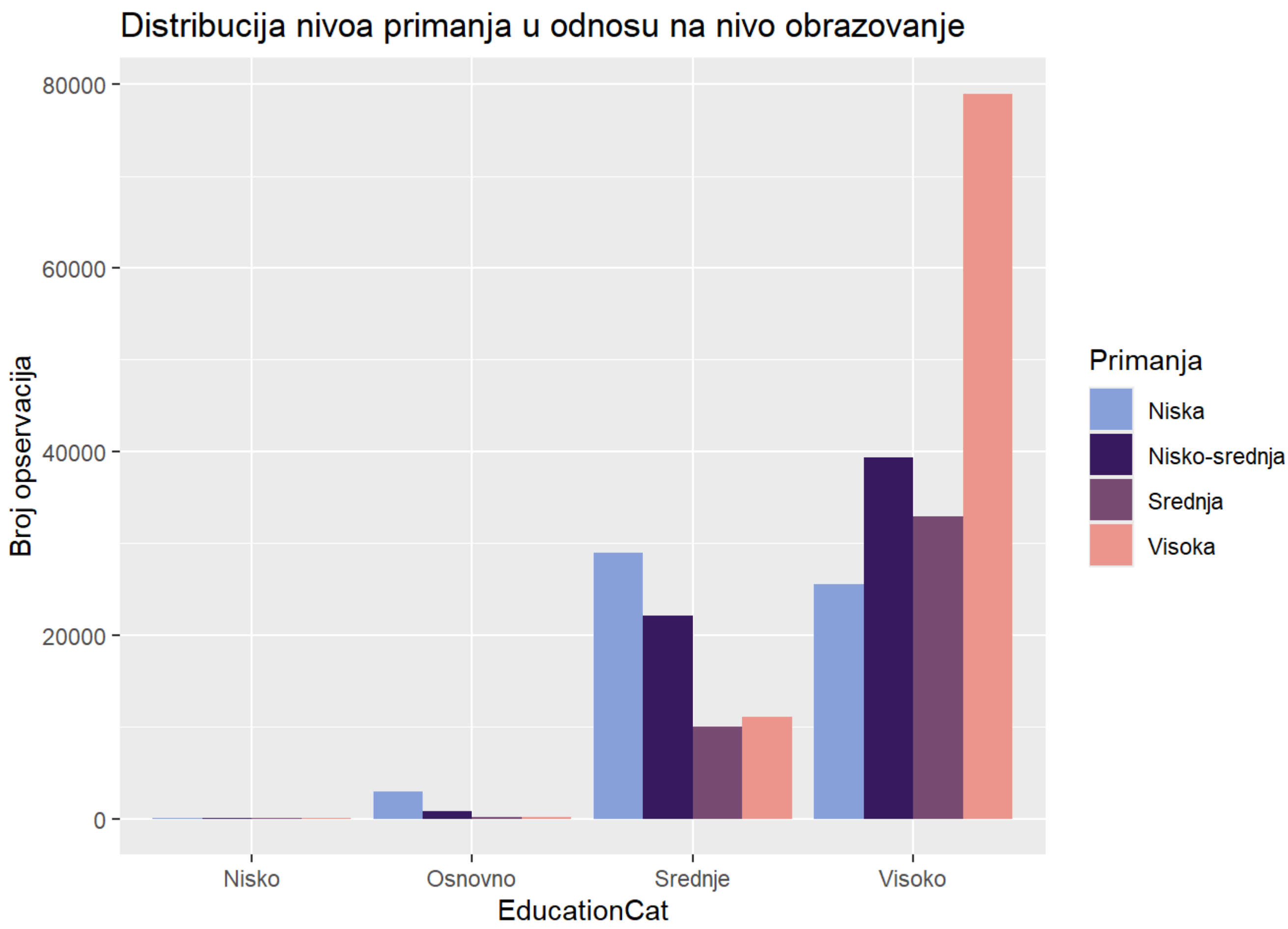

```
chi_sq_test(data_clean$SocioEconomicStatus, data_clean$Diabetes_012)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 6876.8, df = 4, p-value < 2.2e-16
```

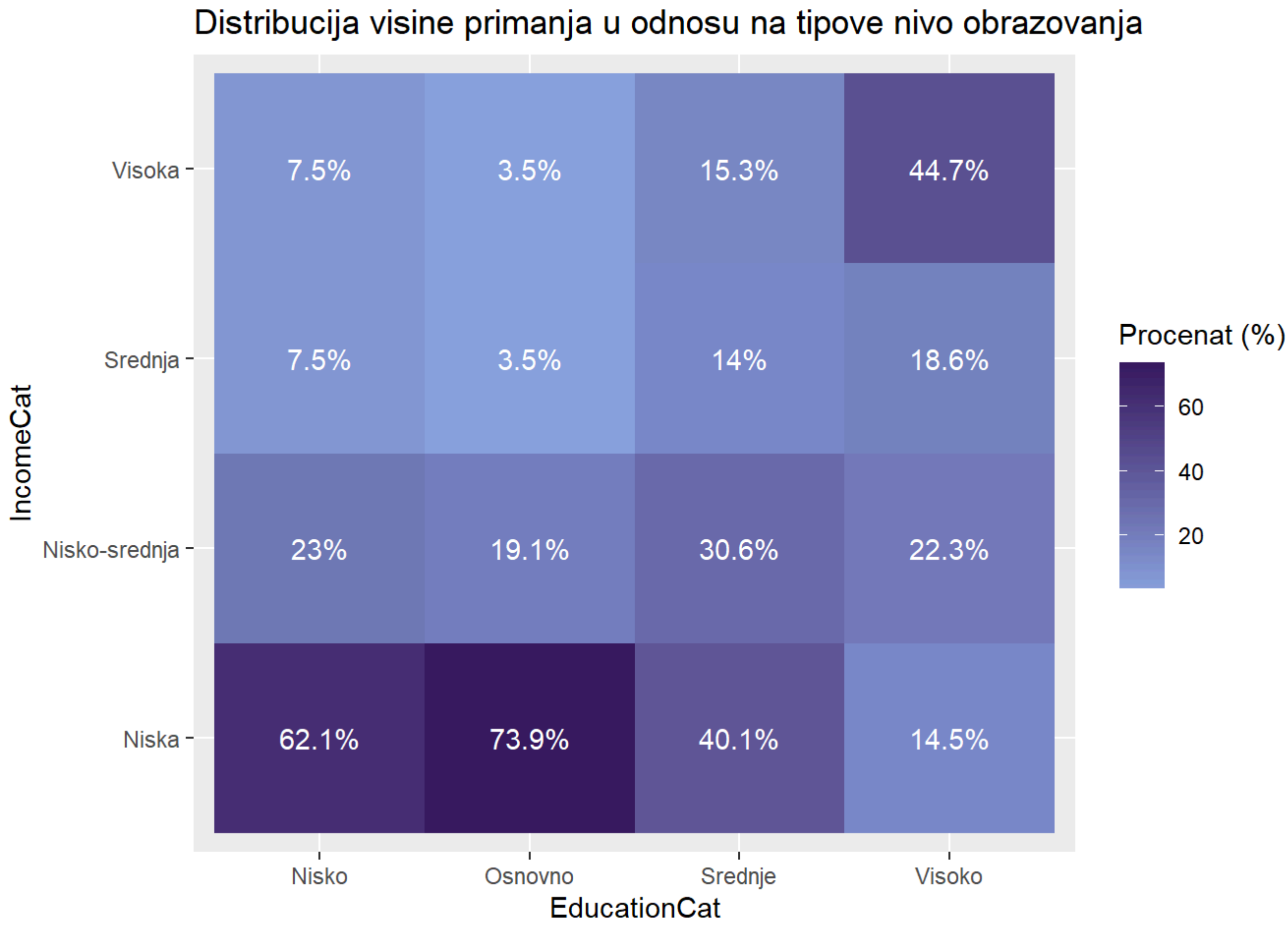
```
cramer_v(data_clean$SocioEconomicStatus, data_clean$Diabetes_012)
```

```
## [1] 0.1165559
```

```
ggplot(data_clean, aes(x = EducationCat, fill =IncomeCat )) +
  geom_bar(position = "dodge")+ scale_fill_manual(values = colorS) +
  labs(title = "Distribucija nivoa primanja u odnosu na nivo obrazovanje", y = "Broj o
pservacija",
       fill = "Primanja")
```




```
data_clean %>%
  count(EducationCat, IncomeCat) %>%
  group_by(EducationCat) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = EducationCat, y = IncomeCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija visine primanja u odnosu na tipove nivo obrazovanja",
    x = "EducationCat",
    y = "IncomeCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



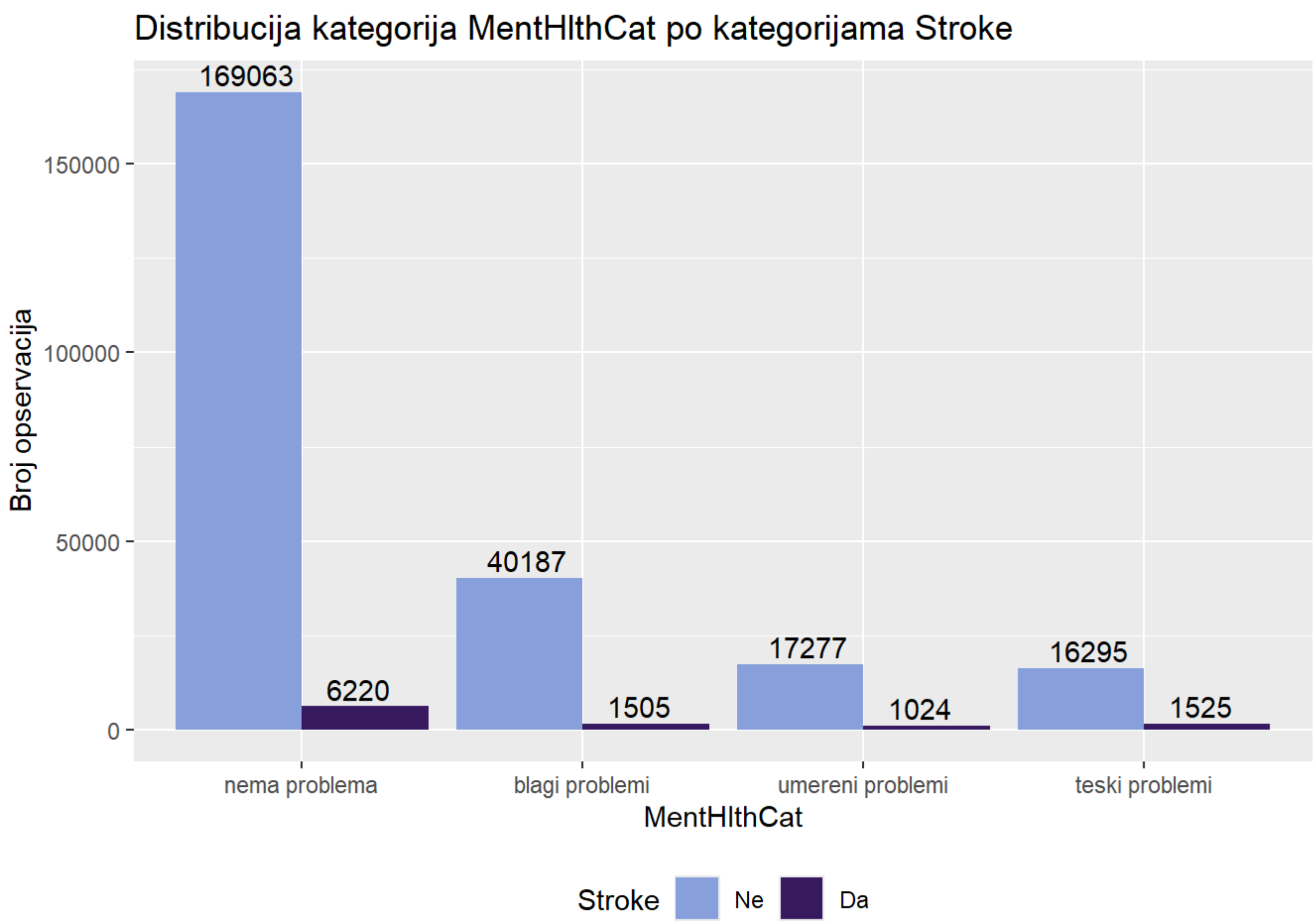

```
chi_sq_test(data_clean$EducationCat,data_clean$IncomeCat)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 35868, df = 9, p-value < 2.2e-16
```

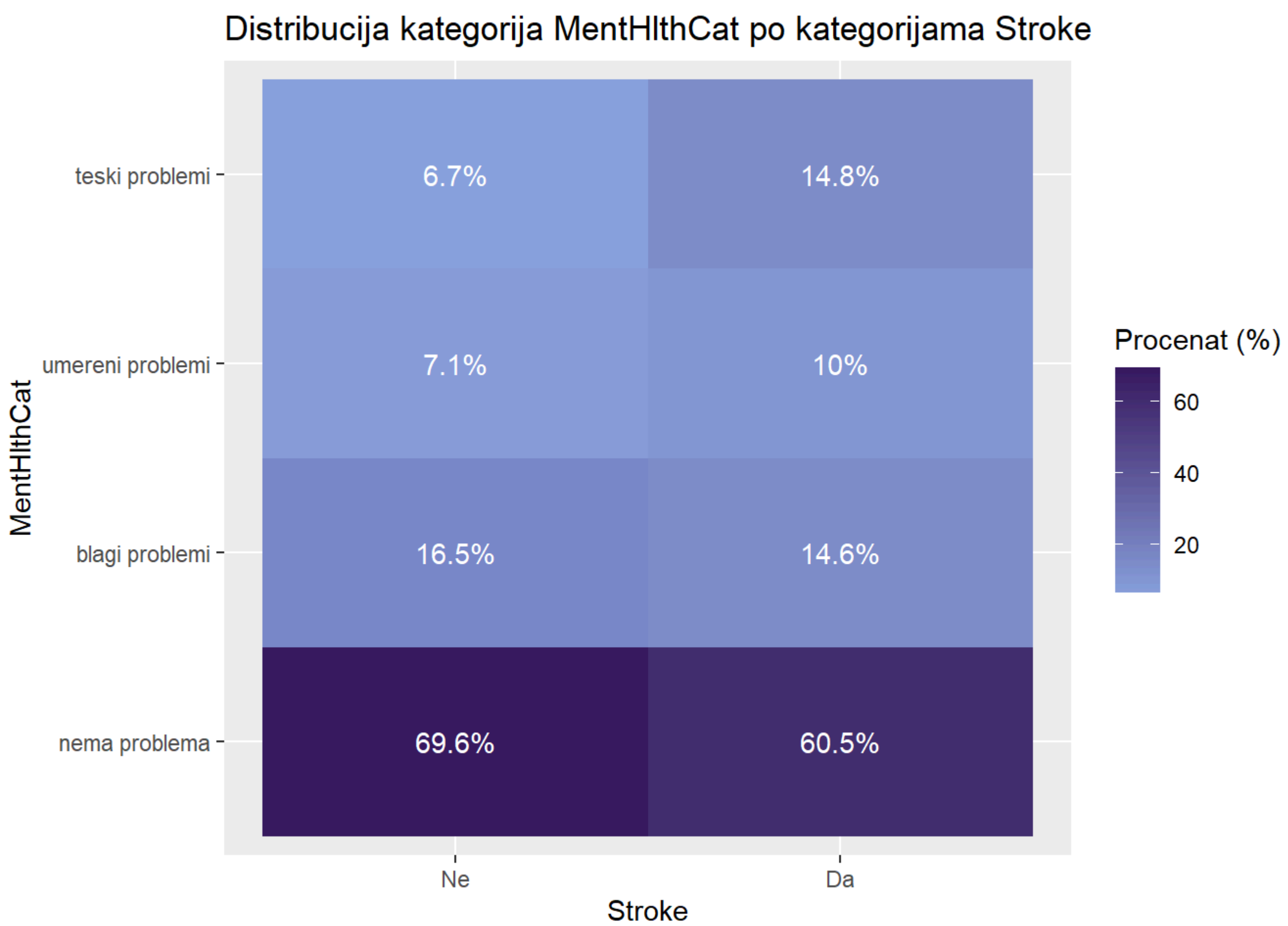
```
cramer_v(data_clean$EducationCat,data_clean$IncomeCat)
```

```
## [1] 0.2173442
```

```
ggplot(data_clean, aes(x = MentHlthCat, fill = Stroke ))  +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribucija kategorija MentHlthCat po kategorijama Stroke",
    y = "Broj opservacija"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault") +
  geom_text(
    stat = "count",
    aes(label = ..count..),
    position = position_dodge(width = 0.8),
    vjust = -0.3
  ) +
  theme(legend.position="bottom")
```



```
data_clean %>%
  count(Stroke, MentHlthCat) %>%
  group_by(Stroke) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = Stroke, y = MentHlthCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija MentHlthCat po kategorijama Stroke",
    x = "Stroke",
    y = "MentHlthCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



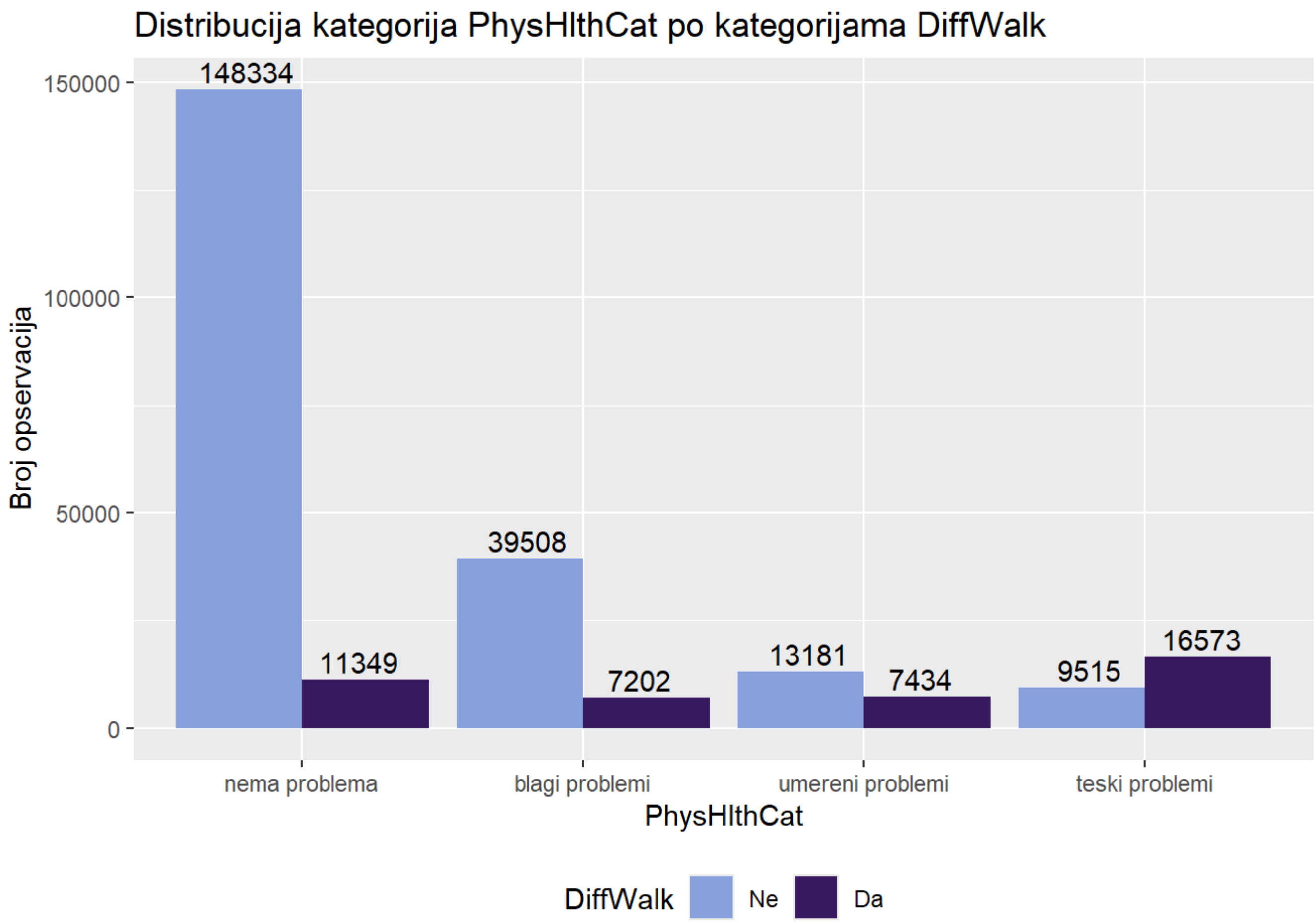

```
chi_sq_test(data_clean$MentHlthCat, data_clean$Stroke)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 1175.9, df = 3, p-value < 2.2e-16
```

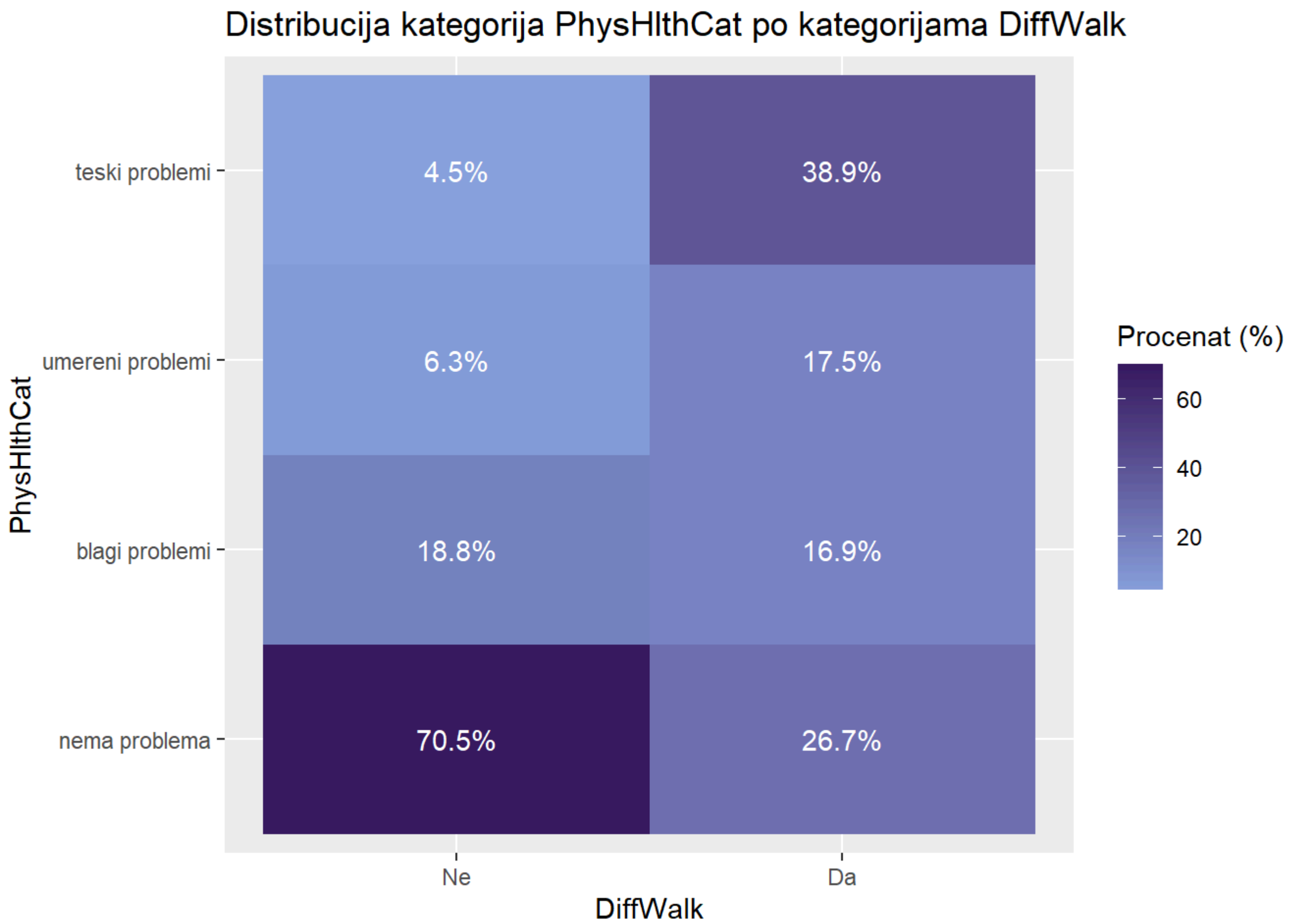
```
cramer_v(data_clean$MentHlthCat, data_clean$Stroke)
```

```
## [1] 0.06816076
```

```
ggplot(data_clean, aes(x = PhysHlthCat, fill = DiffWalk ))  +
geom_bar(position = "dodge") +
labs(
  title = "Distribucija kategorija PhysHlthCat po kategorijama DiffWalk",
  y = "Broj opservacija"
) + scale_fill_paletteer_d("MetBrewer::Archambault") +
geom_text(
  stat = "count",
  aes(label = ..count..),
  position = position_dodge(width = 0.8),
  vjust = -0.3
) +
theme(legend.position="bottom")
```



```
data_clean %>%
  count(DiffWalk, PhysHlthCat) %>%
  group_by(DiffWalk) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = DiffWalk, y = PhysHlthCat, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija kategorija PhysHlthCat po kategorijama DiffWalk",
    x = "DiffWalk",
    y = "PhysHlthCat",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```



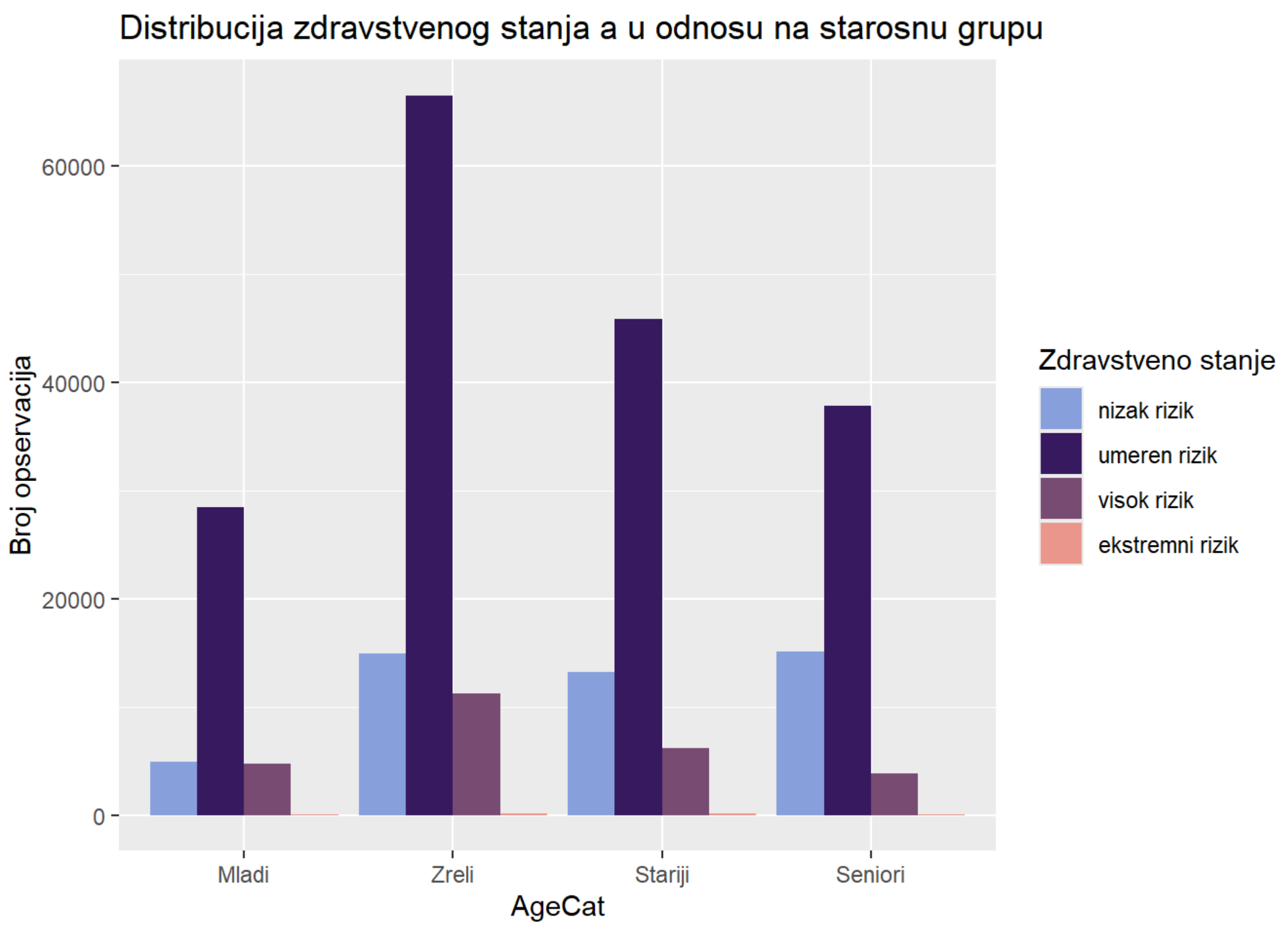

```
chi_sq_test(data_clean$PhysHlthCat, data_clean$DiffWalk)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 56980, df = 3, p-value < 2.2e-16
```

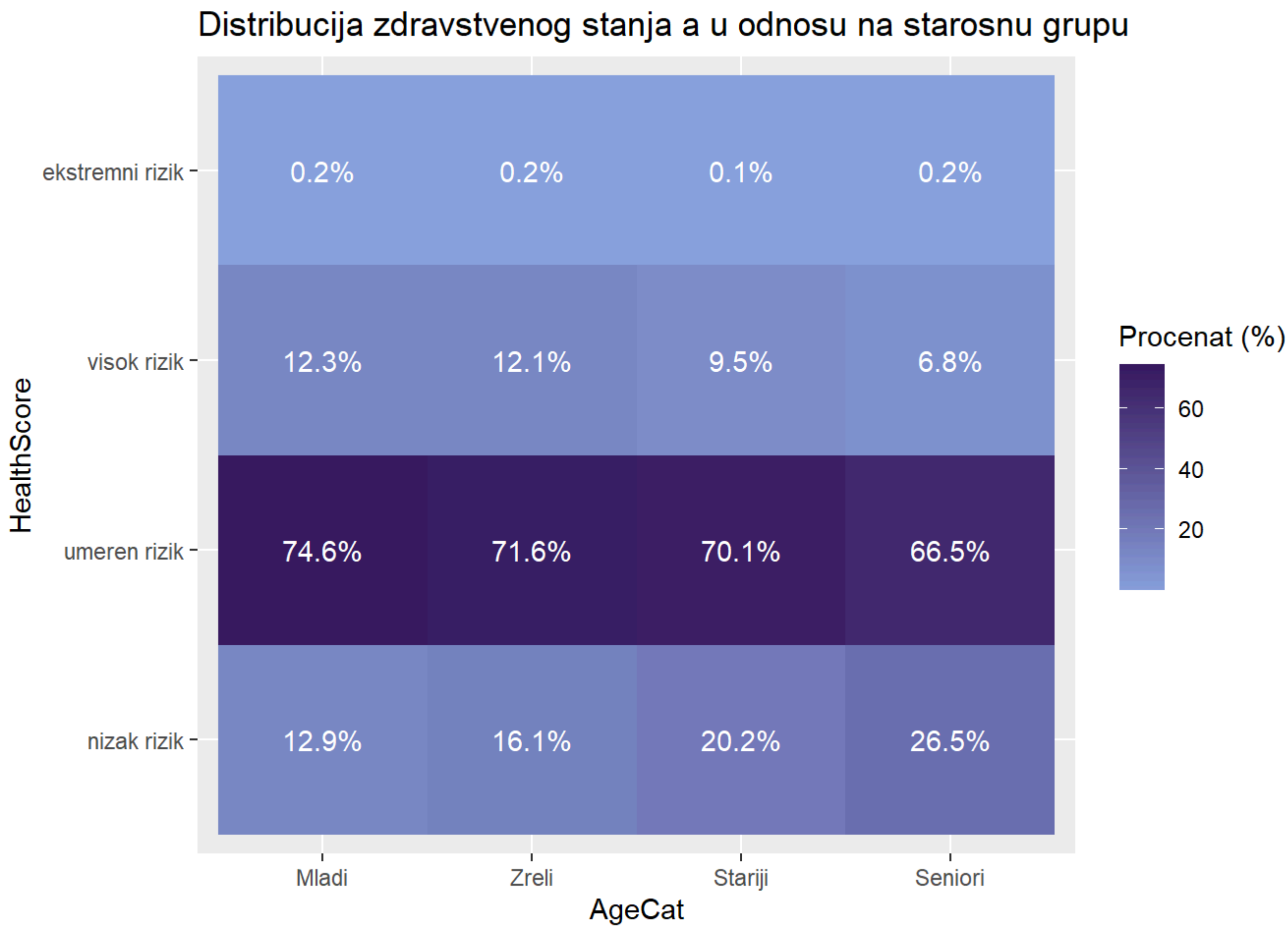
```
cramer_v(data_clean$PhysHlthCat, data_clean$DiffWalk)
```

```
## [1] 0.4744805
```

```
ggplot(data_clean, aes(x = AgeCat, fill =HealthScore )) +

  geom_bar(position = "dodge")+

  scale_fill_manual(values = colorS) +

  labs(title = "Distribucija zdravstvenog stanja a u odnosu na starosnu grupu",

       y = "Broj opservacija",

       fill = "Zdravstveno stanje")
```



```
data_clean %>%
  count(AgeCat, HealthScore) %>%
  group_by(AgeCat) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ggplot(aes(x = AgeCat, y = HealthScore, fill = percent)) +
  geom_tile() +
  geom_text(aes(label = paste0(round(percent, 1), "%")), color = "white") +
  labs(
    title = "Distribucija zdravstvenog stanja a u odnosu na starosnu grupu",
    x = "AgeCat",
    y = "HealthScore",
    fill = "Procenat (%)"
  ) +
  scale_fill_gradient(low = colorS[1], high = colorS[2])
```




```
chi_sq_test(data_clean$AgeCat,data_clean$HealthScore)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabela
## X-squared = 4328.5, df = 9, p-value < 2.2e-16
```

```
cramer_v(data_clean$AgeCat,data_clean$HealthScore)
```

```
## [1] 0.07550288
```

##Feature Selection

```
dim(data_clean)
```

```
## [1] 253096     32
```

```
names(data_clean)
```

```
##  [1] "Diabetes_012"         "HighBP"               "HighChol"
##  [4] "CholCheck"            "BMI"                  "Smoker"
##  [7] "Stroke"               "HeartDiseaseorAttack" "PhysActivity"
## [10] "Fruits"               "Veggies"              "HvyAlcoholConsump"
## [13] "AnyHealthcare"        "NoDocbcCost"          "GenHlth"
## [16] "MentHlth"             "PhysHlth"             "DiffWalk"
## [19] "Sex"                  "Age"                  "Education"
## [22] "Income"               "PhysHlthCat"          "MentHlthCat"
## [25] "EducationCat"         "IncomeCat"            "AgeCat"
## [28] "CardioRiskScore"      "LifestyleRiskScore"   "HealthScore"
## [31] "DietScore"            "SocioEconomicStatus"
```

```
selected_features <- c("BMI", "Stroke", "DiffWalk", "CardioRiskScore",
                       "LifestyleRiskScore", "HealthScore",
                       "SocioEconomicStatus", "AgeCat", "Diabetes_012")

data_selected <- data_clean[, selected_features]
dim(data_selected)
```

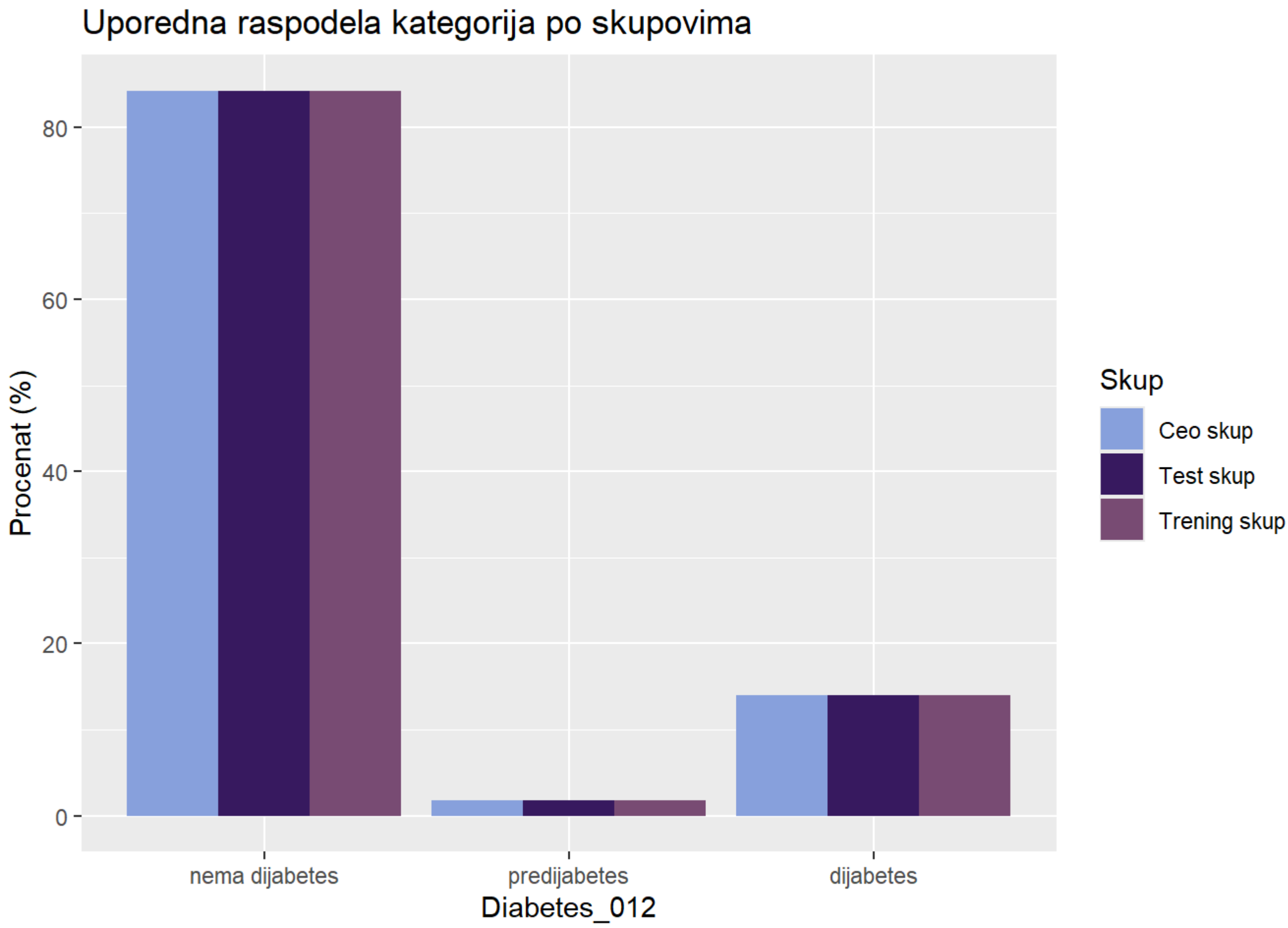
```
## [1] 253096      9
```

##Test/Train split

```
raspodela_Diabetes_012 = data_selected %>%
                          count(Diabetes_012) %>%
                          mutate(Udeo = round(n / sum(n) * 100, 2))

print(raspodela_Diabetes_012)
```

```
##     Diabetes_012      n  Udeo
## 1 nema dijabetes 213207 84.24
## 2   predijabetes   4619  1.82
## 3      dijabetes  35270 13.94
```

```
set.seed(83762021)
train_index <- createDataPartition(
  data_selected$Diabetes_012,
  p = 0.8,
  list = FALSE
)

data_train = data_selected[train_index, ]
data_test  = data_selected[-train_index, ]

raspodela_trening = data_train %>%
              count(Diabetes_012) %>%
              mutate(Udeo = round(n / sum(n) * 100, 2))%>%
              mutate(Skup = "Trening skup")
raspodela_test = data_test %>%
              count(Diabetes_012) %>%
              mutate(Udeo = round(n / sum(n) * 100, 2))%>%
              mutate(Skup = "Test skup")

raspodela_Diabetes_012 = data_selected %>%
              count(Diabetes_012) %>%
              mutate(Udeo = round(n / sum(n) * 100, 2))%>%
              mutate(Skup = "Ceo skup")

raspodela_tabele <- bind_rows(raspodela_trening, raspodela_test, raspodela_Diabetes_01
2)

ggplot(raspodela_tabele, aes(x = Diabetes_012, y = Udeo, fill = Skup)) +
  geom_col(position = "dodge") +
  labs(
    title = "Uporedna raspodela kategorija po skupovima",
    y = "Procenat (%)",
    x = "Diabetes_012"
  ) + scale_fill_paletteer_d("MetBrewer::Archambault")
```




```
#Balansirnje####

print(raspodela_trening)
```

```
##     Diabetes_012      n  Udeo         Skup
## 1 nema dijabetes 170566 84.24 Trening skup
## 2   predijabetes   3696  1.83 Trening skup
## 3      dijabetes  28216 13.94 Trening skup
```

```
predijabetes = subset(data_train, Diabetes_012 == "predijabetes")
dijabetes = subset(data_train, Diabetes_012 == "dijabetes")
nemadijabetes = subset(data_train, Diabetes_012 == "nema dijabetes")
set.seed(83762021)

predijabetes_oversampling = predijabetes[sample(nrow(predijabetes), 15000, replace = T
RUE), ]
nemadijabetes_undersampling = nemadijabetes[sample(nrow(nemadijabetes), 30000), ]

data_train_balanced = rbind(predijabetes_oversampling, dijabetes, nemadijabetes_unders
ampling)



raspodela_data_train_balanced = data_train_balanced %>%
  count(Diabetes_012) %>%
  mutate(Udeo = round(n / sum(n) * 100, 2))

dim(data_train_balanced)
```

```
## [1] 73216     9
```

##Unakrsna validacija

```
set.seed(83762021)
library(nnet)
k = 10
podskupovi = createFolds(data_train_balanced$Diabetes_012, k = k, list = TRUE, returnT
rain = FALSE)

cv_error_multinom_model1 = numeric(k)
cv_error_multinom_model2 = numeric(k)

stepen = 1:4
cv_error_multinom_model3_mean = numeric(length(stepen))
cv_error_multinom_model4_mean = numeric(length(stepen))

cv_error_cart = numeric(k)
cv_error_rf = numeric(k)

for(i in 1:k){
  test_idx = podskupovi[[i]]
  train_idx = setdiff(1:nrow(data_train_balanced), test_idx)

  model_multinom = multinom(Diabetes_012 ~ BMI + Stroke + DiffWalk + CardioRiskScore +
                              LifestyleRiskScore + HealthScore + SocioEconomicStatus
+ AgeCat,
                            data = data_train_balanced[train_idx, ], trace = FALSE)

  preds = predict(model_multinom, newdata = data_train_balanced[test_idx, ])
  cv_error_multinom_model1[i] = mean(preds != data_train_balanced$Diabetes_012[test_id
x])
}

for(i in 1:k){
  test_idx = podskupovi[[i]]
  train_idx = setdiff(1:nrow(data_train_balanced), test_idx)

  model_multinom = multinom(Diabetes_012 ~ BMI * HealthScore + Stroke + DiffWalk + Car
dioRiskScore +
                              LifestyleRiskScore + SocioEconomicStatus + AgeCat,
                            data = data_train_balanced[train_idx, ], trace = FALSE)

  preds = predict(model_multinom, newdata = data_train_balanced[test_idx, ])
  cv_error_multinom_model2[i] = mean(preds != data_train_balanced$Diabetes_012[test_id
x])
}


for (s in stepen) {
  cv_error_multinom_modelS = numeric(k)

  for(i in 1:k){
    test_idx = podskupovi[[i]]
    train_idx = setdiff(1:nrow(data_train_balanced), test_idx)
```

```
    model_multinom = multinom(Diabetes_012 ~ poly(BMI, s) + Stroke + DiffWalk + Cardio
RiskScore +
                                 LifestyleRiskScore + HealthScore + SocioEconomicStatu
s + AgeCat,
                               data = data_train_balanced[train_idx, ], trace = FALSE)

    preds = predict(model_multinom, newdata = data_train_balanced[test_idx, ])
    cv_error_multinom_modelS[i] =  mean(preds != data_train_balanced$Diabetes_012[test
_idx])
  }
  cv_error_multinom_model3_mean[s] = mean(cv_error_multinom_modelS)
}

for (s in stepen) {
  cv_error_multinom_modelS = numeric(k)

  for(i in 1:k){
    test_idx = podskupovi[[i]]
    train_idx = setdiff(1:nrow(data_train_balanced), test_idx)

    model_multinom = multinom(Diabetes_012 ~ poly(BMI, s)*HealthScore + Stroke + DiffW
alk + CardioRiskScore +
                                LifestyleRiskScore  + SocioEconomicStatus + AgeCat,
                              data = data_train_balanced[train_idx, ], trace = FALSE)

    preds = predict(model_multinom, newdata = data_train_balanced[test_idx, ])
    cv_error_multinom_modelS[i] =  mean(preds != data_train_balanced$Diabetes_012[test
_idx])
  }
  cv_error_multinom_model4_mean[s] = mean(cv_error_multinom_modelS)
}

cv_rezultati_logisticka = data.frame(
  Model = c(
    "Model 1: linearni",
    "Model 2: BMI × HealthScore",
    paste0("Model 3: poly(BMI,", stepen, ")"),
    paste0("Model 4: poly(BMI,", stepen, ") × HealthScore")
  ),
  CV_Error = c(
    mean(cv_error_multinom_model1),
    mean(cv_error_multinom_model2),
    cv_error_multinom_model3_mean,
    cv_error_multinom_model4_mean)
)
print(cv_rezultati_logisticka)
```

```
##                                  Model  CV_Error
## 1                   Model 1: linearni 0.4132567
## 2          Model 2: BMI × HealthScore 0.4128606
## 3                Model 3: poly(BMI,1) 0.4132293
## 4                Model 3: poly(BMI,2) 0.4125737
## 5                Model 3: poly(BMI,3) 0.4125328
## 6                Model 3: poly(BMI,4) 0.4124508
## 7  Model 4: poly(BMI,1) × HealthScore 0.4128606
## 8  Model 4: poly(BMI,2) × HealthScore 0.4122050
## 9  Model 4: poly(BMI,3) × HealthScore 0.4125055
## 10 Model 4: poly(BMI,4) × HealthScore 0.4125055
```

```
cv_rezultati_logisticka$Model[cv_rezultati_logisticka$CV_Error == min(cv_rezultati_log
isticka$CV_Error)]
```

```
## [1] "Model 4: poly(BMI,2) × HealthScore"
```

```
for(i in 1:k){
  test_idx <- podskupovi[[i]]
  train_idx <- setdiff(1:nrow(data_train_balanced), test_idx)

  model_cart <- rpart(Diabetes_012 ~ BMI + Stroke + DiffWalk + CardioRiskScore +
                        LifestyleRiskScore + HealthScore + SocioEconomicStatus + AgeCa
t,
                      data = data_train_balanced[train_idx, ], method = "class")

  preds <- predict(model_cart, data_train_balanced[test_idx, ], type = "class")
  cv_error_cart[i] <- mean(preds != data_train_balanced$Diabetes_012[test_idx])
}




for(i in 1:k){
  test_idx <- podskupovi[[i]]
  train_idx <- setdiff(1:nrow(data_train_balanced), test_idx)

  model_rf <- randomForest(Diabetes_012 ~ BMI + Stroke + DiffWalk + CardioRiskScore +
                             LifestyleRiskScore + HealthScore + SocioEconomicStatus +
AgeCat,
                           data = data_train_balanced[train_idx, ],
                           ntree = 500)

  preds <- predict(model_rf, data_train_balanced[test_idx, ])
  cv_error_rf[i] <- mean(preds != data_train_balanced$Diabetes_012[test_idx])
}


mean_cv_error_cart <- mean(cv_error_cart)
mean_cv_error_rf <- mean(cv_error_rf)


results_rf_cart <- data.frame(
  Model = c("CART", "Random Forest"),
  CV_Error = c(mean_cv_error_cart, mean_cv_error_rf)
)
results_rf_cart
```

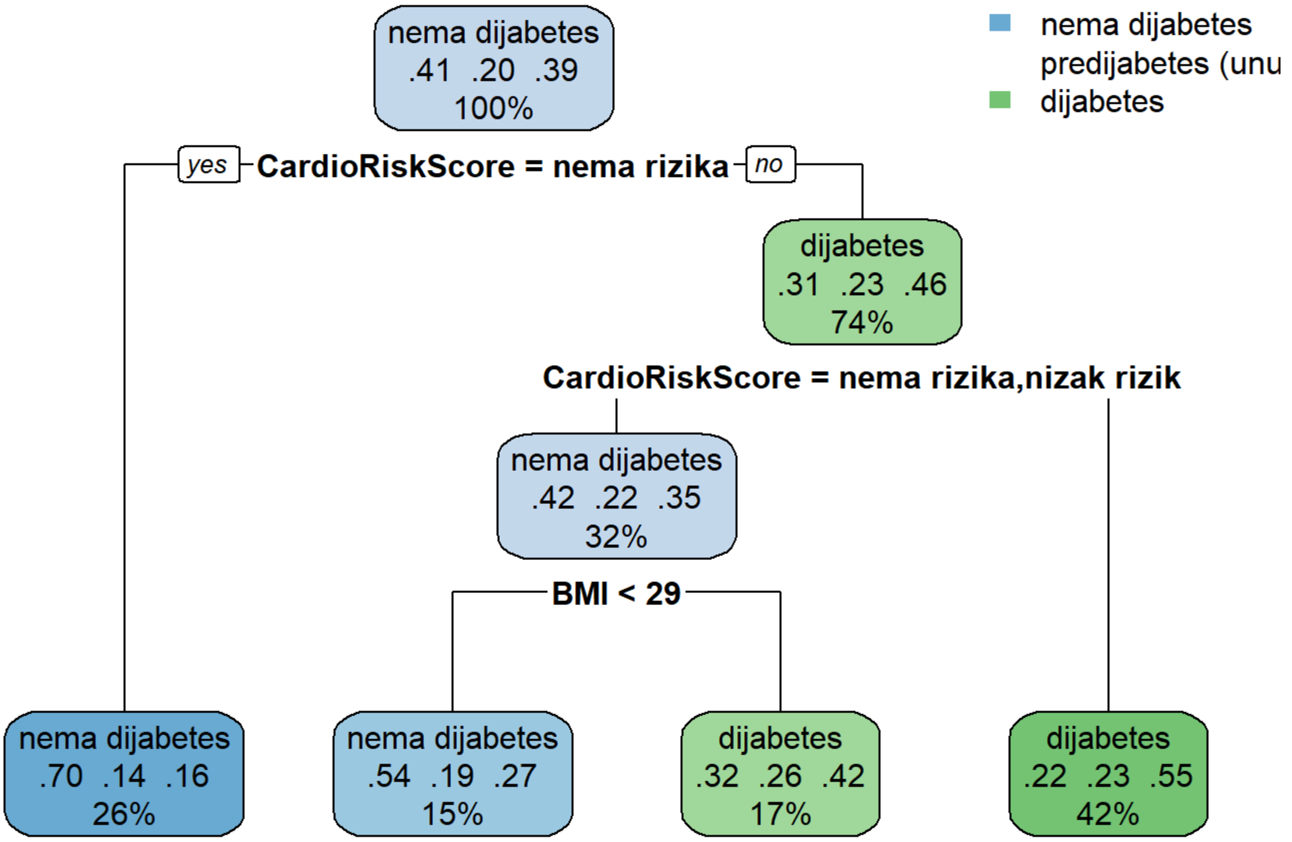
```
##           Model  CV_Error
## 1          CART 0.4355058
## 2 Random Forest 0.4057310
```

##Finalno treniranje

```
lr_model_function = Diabetes_012 ~ BMI + Stroke + DiffWalk + CardioRiskScore + Lifesty
leRiskScore + HealthScore + SocioEconomicStatus + AgeCat
lr_model <- multinom(lr_model_function, data = data_train_balanced, trace = FALSE)

summary(lr_model)
```

```
## Call:
## multinom(formula = lr_model_function, data = data_train_balanced,
##     trace = FALSE)
##
## Coefficients:
##              (Intercept)        BMI   StrokeDa DiffWalkDa CardioRiskScore.L
## predijabetes   -2.965690 0.07108962 0.06304054  0.3038458         0.8564551
## dijabetes      -3.238932 0.09101361 0.32336919  0.5833743         1.4241826
##              CardioRiskScore.Q CardioRiskScore.C LifestyleRiskScore.L
## predijabetes        -0.3027971       -0.05328154           -0.1284486
## dijabetes           -0.2549974       -0.06637704           -0.5694120
##              LifestyleRiskScore.Q LifestyleRiskScore.C HealthScore.L
## predijabetes           0.02891131           -0.1053700    -0.5932912
## dijabetes             -0.28202218           -0.1954065    -0.6400455
##              HealthScore.Q HealthScore.C SocioEconomicStatus.L
## predijabetes   -0.28960348    -0.3385143            -0.4567775
## dijabetes       0.06484718    -0.2981613            -0.4024437
##              SocioEconomicStatus.Q  AgeCat.L   AgeCat.Q    AgeCat.C
## predijabetes           -0.02755193 0.7359606 -0.1430001 0.003067950
## dijabetes               0.02805803 1.0426682 -0.4000698 0.006909903
##
## Std. Errors:
##              (Intercept)         BMI   StrokeDa DiffWalkDa CardioRiskScore.L
## predijabetes  0.09225833 0.001839566 0.05098783 0.02841493        0.03346606
## dijabetes     0.07887305 0.001685225 0.04371865 0.02509414        0.02923302
##              CardioRiskScore.Q CardioRiskScore.C LifestyleRiskScore.L
## predijabetes        0.02682923        0.02027533           0.04995484
## dijabetes           0.02366870        0.01801333           0.05216277
##              LifestyleRiskScore.Q LifestyleRiskScore.C HealthScore.L
## predijabetes           0.03874183           0.02298210     0.1922744
## dijabetes              0.04008435           0.02249111     0.1529255
##              HealthScore.Q HealthScore.C SocioEconomicStatus.L
## predijabetes     0.1441059    0.06805488            0.02085811
## dijabetes        0.1148317    0.05518557            0.01851166
##              SocioEconomicStatus.Q   AgeCat.L   AgeCat.Q   AgeCat.C
## predijabetes            0.01827039 0.03077117 0.02525056 0.02038087
## dijabetes               0.01644015 0.03083992 0.02476230 0.01870481
##
## Residual Deviance: 133653.3
## AIC: 133725.3
```

```
cart_model <- rpart(Diabetes_012 ~ BMI + Stroke + DiffWalk + CardioRiskScore +
                      LifestyleRiskScore + HealthScore + SocioEconomicStatus + AgeCat,
                    data = data_train_balanced,
                    method = "class")

rpart.plot(cart_model)
```




```
rf_model<-randomForest(Diabetes_012 ~ BMI + Stroke + DiffWalk + CardioRiskScore +
                         LifestyleRiskScore + HealthScore + SocioEconomicStatus + AgeC
at,
                       data = data_train_balanced,
                       ntree = 500,
                       importance = TRUE)
print(rf_model$confusion)
```

```
##                  nema dijabetes predijabetes dijabetes class.error
## nema dijabetes            21600           63      8337   0.2800000
## predijabetes               5510          651      8839   0.9566000
## dijabetes                  6879           43     21294   0.2453218
```

##Predikcija i metrike

```
lm_preds <- predict(lr_model, newdata = data_test)
cart_preds <- predict(cart_model, newdata = data_test, type = "class")
rf_preds <- predict(rf_model, newdata = data_test)

nivoi <- levels(as.factor(data_test$Diabetes_012))

stvarni_f <- factor(data_test$Diabetes_012, levels = nivoi)
cart_f    <- factor(cart_preds,             levels = nivoi)
rf_f      <- factor(rf_preds,               levels = nivoi)

lm_f <- factor(lm_preds,  levels = nivoi)

finalna_tabela <- rbind(
  izvuci_sve_metrike(lm_f, stvarni_f, "Logisticka Regresija"),
  izvuci_sve_metrike(cart_f, stvarni_f, "Stablo (CART)"),
  izvuci_sve_metrike(rf_f, stvarni_f, "Random Forest")
)

finalna_tabela[is.na(finalna_tabela)] <- 0
print(finalna_tabela)
```

```
##                         Precision      Recall          F1                Model
## Class: nema dijabetes  0.93445604 0.735231350 0.822958092 Logisticka Regresija
## Class: predijabetes    0.00000000 0.000000000 0.000000000 Logisticka Regresija
## Class: dijabetes       0.30526563 0.738020981 0.431889829 Logisticka Regresija
## Class: nema dijabetes1 0.93902775 0.639642598 0.760946894        Stablo (CART)
## Class: predijabetes1   0.00000000 0.000000000 0.000000000        Stablo (CART)
## Class: dijabetes1      0.25820508 0.789622909 0.389156711        Stablo (CART)
## Class: nema dijabetes2 0.93639480 0.723998030 0.816611559        Random Forest
## Class: predijabetes2   0.01265823 0.001083424 0.001996008        Random Forest
## Class: dijabetes2      0.30142288 0.750779699 0.430149448        Random Forest
##                                 Klasa
## Class: nema dijabetes  nema dijabetes
## Class: predijabetes      predijabetes
## Class: dijabetes            dijabetes
## Class: nema dijabetes1 nema dijabetes
## Class: predijabetes1     predijabetes
## Class: dijabetes1           dijabetes
## Class: nema dijabetes2 nema dijabetes
## Class: predijabetes2     predijabetes
## Class: dijabetes2           dijabetes
```